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This manual and Department of the Air Force (DAF) Manual (DAFMAN) 13-201, *Airspace Management*, implement Air Force Policy Directive (AFPD) 13-2, *Air Traffic, Airfield, Airspace, and Range Management*. This manual also implements Department of Defense Instruction (DoDI) 1322.27, *DoD Urban Training Facilities,* and DoDI 3200.21, *Sustaining Access to the Live Training Domain*. This publication applies to Department of the Air Force civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve, and the Air National Guard (ANG). This publication does not apply to the United States Space Force (USSF). It provides guidance for the planning, operations, management, safety, equipment, facilities, and security of Air Force ranges covered by this manual (see **Attachment 2**). Refer recommended changes and questions about this publication to the office of primary responsibility using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the

appropriate functional chain of command. See **paragraph 1.4** for guidance on supplementing this manual. The authorities to waive wing/unit level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFMAN 90-161, *Publishing Processes and Procedures,* Table A10.1, for a description of the authorities associated with the tier numbers. Waivers to this manual are authorized and are processed in accordance with DAFMAN 90-161. For compliance items not identified with a tier number, waiver authority is delegated to the Headquarters Air Force Operational Training Infrastructure Division (AF/A3TI). Waivers for non-tiered and T-0 compliance items must be coordinated through the major command (MAJCOM) before they are submitted to AF/A3TI for action. T-0 compliance items are those imposed by law, DoD policy, or an external organization (e.g., the FAA). While AF/A3TI will take T-0 waiver requests for action, it is possible the waiver may not be granted. T-1, T-2, and T-3 waivers automatically expire 30 days after a change of command unless the new commander renews the waiver. At the unit level, the range operating authority should monitor implementation of this manual and review approved waivers anytime the circumstances that prompted the waiver or the impacts of the excepted activity change substantially. Submit waiver requests on the DAF Form 679, *Department of the Air Force Publication Compliance Item Waiver Request/Approval*, through the chain of command to the appropriate tier waiver approval authority, or alternately, to the publication office of primary responsibility for non-tiered compliance items (see **paragraph 1.3** for additional information). Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with (IAW) the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Compliance with the attachments is mandatory. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Department of the Air Force.

(EDWARDSAFB) This publication supplements Air Force Manual 13-212, Volume 1 (AFMAN 13-212V1), *Range Planning and Operations*. It provides guidance on responsibilities, policies, and procedures for the planning, maintenance, and operation of Edwards Air Force Base Precision Impact Range Area (PIRA), its associated air-to-ground or air-to-air test and training ranges, or range supplement content. It applies to all Air Force organizations, tenant units, commercial units, contractor units, military personnel, government personnel, and civilian personnel located at Edwards Air Force Base and to all external non-government agencies seeking to use the Edwards Air Force Base PIRA, or 412th Range Squadron (412 RANS) equipment, resources, or services. This supplement does not require tiers at or below the Wing level. Waiver authority for this supplement is the 412th Test Wing Commander (412 TW/CC). This publication may not be supplemented, further implemented, or extended. Refer recommended changes and questions about this publication to the OPR listed above using the AF 847, Recommendation for Change of Publication; route AF Forms 847 from the field through the appropriate chain of command. This Instruction requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by 49 USC, Section 445012(d), General Facilities and Personnel Authority, 32 CFR, Part 855, Civil Aircraft Use of United States Air Force Airfields, by 10 U.S.C. 8013, 10 USC 8012, 44 U.S.C. 3101, Title 37 U.S.C. 301a (Incentive Pay), Public Law 92-204, Section 715 (Appropriations Act for 1973), Public Laws 93-570 (Appropriations Act for 1974), 93-294 (Aviation Career Incentive Act of 1974), and Executive Order 9397 as amended. The applicable

System or Records Notice (SORN) is F010 AFXO A, Civil Aircraft Landing Permit Case Files, F011 AF XO A, Aviation Resource Management Systems (ARMS), F031 AF SP O, Documentation for Identification and Entry Authority, and Integrated Maintenance Data System (IMDS), available at: **http://dpclo.defense.gov/Privacy/SORNs.aspx**. If a written request for DoD records has been made by any person, organization, business, except a Federal Agency, that either explicitly or implicitly invokes the Freedom of Information Act (FOIA), it must be processed by the local FOIA Request Service Center (412 CS/SCOK) in accordance with DoDR 5400.07-R_AFMAN 33-302, DoD/AF Freedom of Information Act Program. Ensure that all records created as a result of processes described in this publication are maintained in accordance with AFI 33-322, Records Management and Information Governance Program, and disposed of in accordance with Air Force Records Information Management System Records Disposition Schedule. The use of the name or mark of any specific manufacturer, commercial mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been substantially revised and requires a complete review. Tiering and compliance statements have been updated IAW Administrative Assistant to the Secretary of the Air Force (SAF/AA) recommendations. Implementing language from DoDI 3200.21 has been incorporated. Additionally, responsibilities for the Air Force Installation and Mission Support Center (AFIMSC) and the Air Force Civil Engineer Center (AFCEC) have been added, and the requirement for individual comprehensive range plans has been eliminated.

(EDWARDSAFB) This supplement is substantially revised from the former Air Force Instruction. This publication now supplements an Air Force Manual and requires an extensive content review.

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Chapter 1

INTRODUCTION

1.1. Purpose. This manual provides guidance for commanders to operate assigned ranges safely, effectively, and efficiently to meet training and test requirements while minimizing potential effects on the environment and the surrounding communities.

1.2. Scope.

1.2.1. This manual applies to all Air Force-operated air and ground ranges listed in **Attachment 2**. These ranges are part of the Air Force's Operational Training and Test Infrastructure. Space-launch activities are not covered by this manual. Small arms training ranges are addressed in AFI 36-2654, *Combat Arms Program*, and are not covered by this manual. Stand-alone Explosive Ordnance Disposal (EOD) ranges are covered under AFMAN 32-3001, *Explosive Ordnance Disposal Program*, and Defense Explosives Safety Regulation (DESR) 6055.09_ AFMAN 91-201, *Explosives Safety Standards*, and are not covered in this publication. Air Force-operated air and ground ranges outside of the United States will comply with the provisions of all applicable international agreements, this manual, Department of Defense Directive (DoDD) 4715.12, *Environmental and Explosives Safety Management on Operational Ranges Outside the United States*, DoDI 4715.22, *Environmental Management Policy for Contingency Locations*, AFI 32-7091, *Environmental Management Outside the United States*, and the Air Force Environmental Impact Analysis Process (EIAP) **(T-0)**

1.2.2. **Attachment 2** is intended to be the definitive list of ranges covered by this manual. Air Force organizations conducting range activities, as defined in Title 10 of United States Code (U.S.C.) Section (§) 101(e) (2), at locations other than those listed in **Attachment 2** must contact the publication office of primary responsibility (through the parent major command [MAJCOM]) for a coverage review unless one or more of the criteria listed in the following subparagraphs are met: **(T-1)**

1.2.2.1. The activity is specifically excluded in **paragraph 1.2.1**.

1.2.2.2. The activity is conducted at an operational range operated by another military service, federal agency, or a host nation, IAW the rules and procedures established by the organization operating the range. **Note:** While this manual only applies to Air Force-operated ranges, ranges operated by non-Air Force entities that are located on Air Force real property (e.g., test sites located on an Air Force installation) shall comply with applicable portions of this manual via a host-tenant agreement to ensure safety and stewardship of Air Force property. IAW AFI 32-9003, *Granting Temporary Use of Air Force Real Property*, when the Air Force grants the right of exclusive or non-exclusive use of Air Force real property to a non-Federal agency a license is required; when granted to another Federal agency a permit is required.

1.2.2.3. The activity requires airspace only (no land or water surface component), and is conducted IAW Air Force, Department of Defense (DoD), and Federal Aviation Administration (FAA) or applicable host nation airspace regulations.

1.2.2.4. The activity consists of training involving drop zones and/or landing zones, and is conducted in accordance with DAFMAN 13-217, *Drop Zone, Landing Zone, and Helicopter Landing Zone Operations*.

1.2.2.5. The activity consists of training conducted off of federal real property. Refer to DoDI 1322.28, *Realistic Military Training (RMT) Off Federal Real Property*, for requirements and procedures for this type of activity.

1.2.3. If the coverage review determines the activity is within the scope and intent of this manual, this AFMAN will apply, and **Attachment 2** will be updated accordingly.

1.3. Waiver Request. Waiver requests will be submitted using the DAF Form 679 and will include the information specified in paragraph 9.4. of DAFMAN 90-161, *Publishing Processes and Procedures*. Additional requirements are at the discretion of the waiver authority. Waiver requests will be staffed through the owning MAJCOM headquarters and then sent to AF/A3TI for coordination.

1.3.1. Non-Tiered and T-0 waiver requests must also include the following:

1.3.1.1. Range name, location, range operating authority and a point of contact.

1.3.1.2. Description of the conditions at issue to include:

1.3.1.2.1. Potential alternatives and their impact on test and training operations, maintenance, cost, and other factors deemed appropriate by the requesting agency.

1.3.1.2.2. Risk management assessment. See **paragraph 4.18**.

1.3.1.2.3. Supporting maps, charts, graphics, or other illustrations as appropriate.

1.3.1.3. A detailed plan to alleviate the condition.

1.3.1.4. Previously granted waiver.

1.3.2. AF/A3TI will coordinate non-Tiered, T-0 and T-1 waiver requests with the Air Force Civil Engineering Readiness Division (AF/A4CX) for Explosive Ordnance Disposal (EOD), unexploded ordnance (UXO), and range clearance matters and with the Air Force Test and Evaluation Directorate (AF/TE) for test and evaluation range matters when appropriate.

1.3.3. AF/A3TI will coordinate all waiver requests with Air Combat Command (ACC), Airspace, Ranges and Airfield Operations Division (ACC/A3A) so they can maintain appropriate situational awareness as the lead MAJCOM.

1.4. Supplements. This publication may be supplemented at any level. All supplements are routed as indicated in paragraphs **1.4.1** and **1.4.2** for coordination prior to certification and approval. Supplements may change, or add procedures, as applicable, to this manual, but changes can be no less restrictive than this manual. All supplements are published in accordance with DAFMAN 90-161 on the Air Force electronics publications website.

1.4.1. For MAJCOM supplements, MAJCOMs must submit a copy to AF/A3TI for review and coordination prior to publication.

1.4.2. Installation supplements shall be submitted to the owning MAJCOM headquarters for review and coordination prior to publication.

1.5. Environmental Impact Analysis Process and Decision Making. The National Environmental Policy Act of 1969 (NEPA) (42 U.S.C. §§ 4321 – 4347, Council on Environmental Quality (CEQ) NEPA Regulations (40 CFR Parts 1500 – 1508), and the Air Force Environmental Impact Analysis Process (EIAP) (32 C.F.R. **Part 989**) apply inside the United States and its Territories. **(T-0)** This AFMAN provides general NEPA and EIAP guidance in support of range planning, operations, and decision making in **paragraph 7.8**.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. The Director of Global Power Programs (SAF/AQP) will:

2.1.1. Act as the focal point for development and procurement of subscale and full-scale aerial target systems and Combat Training Range Systems, aircraft tracking equipment, threat simulators and engagement scoring systems.

2.1.2. Perform all Program Element Monitor (PEM) responsibilities for Aerial Targets and Combat Training Range development and procurement (program elements [PE] 64735, 27429, and 35116).

2.1.3. Coordinate range-related acquisition programs and PE changes with AF/A3TI.

2.2. The Deputy Under Secretary of the Air Force, International Affairs (SAF/IA) will:

2.2.1. Manage (in conjunction with Air Force Security Assistance and Cooperation Directorate) Foreign Military Sales cases, and other security assistance cases, involving major systems or requiring special management consideration of range-related issues.

2.2.2. Serve as the focal point for negotiations with foreign nations on issues related to the use of Air Force-owned or operated ranges and range equipment.

2.3. The Assistant Secretary of the Air Force for Energy, Installations, and Environment (SAF/IE) will:

2.3.1. Provide policy and oversight of all operational range environmental support such as operational range assessments and responses, natural and cultural resource management, hazardous material and hazardous waste management, and compliance with applicable environmental requirements.

2.3.2. Represent the Air Force in meetings with the Office of the Secretary of Defense and environmental regulatory agencies involving discussion of Air Force range environmental compliance activities and environmental policy.

2.3.3. Oversee the acquisition, creation, modification, transfer, or withdrawal of range land IAW AFI 32-9001, *Acquisition of Real Property*, the Federal Land Policy and Management Act and Title 43, U.S.C., Sections 155-158.

2.4. The Deputy Chief of Staff for Operations (AF/A3). IAW Headquarters Air Force Mission Directive 1-54, *Deputy Chief of Staff, Operations*, AF/A3 is responsible for matters concerning ranges. AF/A3, through the Director of Training and Readiness (AF/A3T), establishes range policy, programming, and requirements.

2.4.1. Training and Readiness (AF/A3T) will:

2.4.1.1. Designate AF/A3TI as the Office of Primary Responsibility for Air Force range policy and management.

2.4.1.1.1. Operational Training Infrastructure Division (AF/A3TI) will:

2.4.1.1.1.1. Develop range policy, programming, and requirements IAW AFPD 13-2.

2.4.1.1.1.2. Develop, coordinate, and maintain a Range Capabilities Development Plan (formerly known as the Enterprise Range Plan (ERP)).

2.4.1.1.1.3. Review and coordinate Test/Training Space Needs Statements (T/TSNS) for ranges.

2.4.1.1.1.4. Advocate for Air Force range development and procurement funding in coordination with the Director for Global Power Programs (SAF/AQP) PEM.

2.4.1.1.1.5. Advocate for Air Force range sustainment funding.

2.4.1.1.1.6. Sponsor Air Force PEM for Readiness Training Ranges, Operations and Maintenance (O&M) (PE 27604F).

2.4.1.1.1.7. Sponsor Air Force PEM for the Nevada Test and Training Range and the Utah Test and Training Range (PE 27428F).

2.4.1.1.1.8. Designate the proponent responsible for execution of the Environmental Impact Analysis Process (EIAP) for range-related actions. Support range proponents in meeting their EIAP responsibilities per Title 32 Code of Federal Regulations (CFR) §989.3(d) as documented in the T/TSNS, ensure headquarters review prior to public announcement and preparation of the description of proposed action and alternatives (32 CFR §989.28), and participate early in the scoping and review of all EIAP issues and documents concerning Air Force-operated ranges. All EIAP work associated with ANG ranges will be overseen by the National Guard Bureau (NGB).

2.4.1.1.1.9. Coordinate with the DoD Urban Training Facilities Review Group to ensure Air Force urban training facility proposals are properly evaluated and/or certified IAW DoDI 1322.27.

2.4.1.1.1.10. Coordinate with other public, private, and tribal interests and agencies as required in support of Air Force range actions. The NGB will coordinate with the states on behalf of the ANG.

2.4.1.1.1.11. Provide range-related information to the Strategic Basing Panel and the Strategic Basing Executive Steering Group as requested.

2.5. The Director of Civil Engineers (AF/A4C) will:

2.5.1. Oversee implementation of policy and provide program oversight for the Operational Range Assessment Program IAW DoDI 4715.14, *Operational Range Assessments,* and DAFPAM 32-7003, *Operational Range Assessment Program Management*.

2.5.2. Provide environmental policy consultation and support to SAF/IE for Office of the Secretary of Defense meetings related to Air Force ranges that require environmental representation.

2.5.3. Oversee implementation of policy and execution for applicable EIAP issues and documents concerning Air Force-operated ranges IAW 32 CFR Part 989. The NGB oversees the EIAP for ANG range actions.

2.5.4. Provide environmental consultation in support of Military Munitions Rule compliance under Title 42 U.S.C. Sections 6901-6992, *Resource Conservation and Recovery Act*.

2.5.5. Implement range-related environmental policy to ensure compliance with DoD policy and regulatory requirements.

2.6. Headquarters Air Force Test and Evaluation Policy, Programs, and Resources Division (AF/TEP) will:

2.6.1. Perform all PEM responsibilities for the following test and evaluation infrastructure and support areas: Air Force Operational Test and Evaluation Center O&M (PE 0207426F); Management HQ – Test and Evaluation (PE 0606398F); Threat Simulator Development (PE 0604256F); Major Test and Evaluation Investment (PE 0604759F); Initial Operational Test and Evaluation (PE 0605712F); Test and Evaluation Support (PE 0605807F); Facility Sustainment – Test and Evaluation Support (PE 0605978F); and Facility Restoration and Modernization – Test and Evaluation Support (PE 0605976F).

2.7. NGB. The NGB is the channel of communication between the Air Force and states for all matters pertaining to ranges held for the purpose of the ANG. The NGB will assume the role of the Headquarters Air Force under the oversight of the Secretary of the Air Force as described in DoDD 5105.77, *National Guard Bureau (NGB).*

2.8. MAJCOM Commanders. Unless a MAJCOM is specified in parentheses at the beginning of the sub-paragraph, all references to MAJCOMs in this manual include the ANG and Commander, Air Force Forces in Combatant Commands. MAJCOM Commanders, through appropriate level staff (including the Air Force Installation and Mission Support Center), will

2.8.1. **(ACC)** Review and forward requests for non-Tiered and T-0 waivers to this manual to AF/A3TI. Serve as the T-1 waiver authority or delegate this authority to the appropriate MAJCOM Director.

2.8.2. Review urban training facility proposals from ranges and submit these proposals to AF/A3TI for coordination with the Urban Training Facilities Review Group.

2.8.3. Develop policy, advocate for resources, define requirements, and manage the oversight of MAJCOM ranges.

2.8.4. Contract for services to support execution of range missions IAW AFI 63-138, *Acquisition of Services*, and AFI 63-101/20-101, *Integrated Life Cycle Management.* When practical, partner with other MAJCOMs to achieve economies of scale.

2.8.5. **(ACC)** Serve as the lead MAJCOM for the overall program management of the Center Scheduling Enterprise (CSE), to include development of utilization guidance, system implementation, and sustainment. Collect and adjudicate requirements from all MAJCOMs and stakeholders. Ensure range scheduling and utilization data are standardized across the Air Force, and are recorded, reported, and archived in the CSE.

2.8.6. **(ACC)** Serve as the lead MAJCOM for Weapon Danger Zones (WDZ).

2.8.7. Ensure ranges use the most current training Surface Danger Zones (SDZ)/WDZs concurrent with the introduction of new aircraft, weapons, and tactics or training requirements.

2.8.8. Establish procedures and training requirements for electronic warfare operations personnel performing duties associated with **paragraph 4.13**.

2.8.9. Support efforts of ranges to conserve and restore the natural and man-made infrastructure at their ranges.

2.8.10. Coordinate with other public and private interests and agencies as required to support MAJCOM range requirements.

2.8.11. Fund laser and directed energy weapon range certifications.

2.8.12. **(Air Force Materiel Command [AFMC])** Through the 711th Human Performance Wing, maintain the institutional capability to conduct range certifications for lasers and directed energy weapons.

2.8.13. **(ACC, Air Force Global Strike Command [AFGSC], Pacific Air Forces [PACAF], Air Education and Training Command [AETC], Air Force Special Operations Command [AFSOC], United States Air Forces in Europe-Air Forces Africa [USAFE-AFAFRICA])** Provide O&M inputs for Readiness Training Ranges (PE 27604F). ACC will serve as lead MAJCOM for representing PE 27604F to the Air Force Operational Training and Readiness (OT&R) Panel, which is responsible for representing programmatic details within PE 27604F as part of the Air Force corporate process. Each MAJCOM will represent their portion of PE 27604F through their MAJCOM's corporate process up to the OT&R Panel.

2.8.14. **(ACC)** Support the PEM for the Nevada Test and Training Range and the Utah Test and Training Range (PE 27428F). ACC will serve as lead MAJCOM for representing PE 27428F to the Operational Training and Readiness (OT&R) Panel, which is responsible for representing programmatic details within PE 27428F as part of the corporate process.

2.8.15. **(ACC, PACAF, AETC, AFSOC, USAFE-AFAFRICA, ANG, Air Force Reserve Command [AFRC], AFGSC, Air Mobility Command [AMC])** Provide input for training range investment and equipment, including threat system modernization (PEs 27429F/64735F). ACC will serve as lead MAJCOM for representing PEs 27429F/64735F to the Operational Training and Readiness (OT&R) Panel, which is responsible for representing programmatic details within these two PEs as part of the corporate process. ACC will develop and publish a process for soliciting MAJCOM input.

2.8.16. **(ANG)** Support the PEM for ANG Ranges, O&M, and personnel (PE 52634F). The ANG will represent their portion of PE 52634F, through the NGB corporate process, up to the Operational Training and Readiness (OT&R) Panel, which is responsible for representing programmatic details within PE 52634F as part of the Air Force corporate process.

2.8.17. **(AFMC)** Support the PEM for AFMC Ranges, Operations and Sustainment, and Personnel (PE 65807F) IAW the Air Force and MAJCOM Corporate process.

2.8.18. **(AFMC)** Support the PEM for the Test and Evaluation investment PEs.

2.8.19. **(ACC)** Host a biennial Air Force Range and Airspace Working Group (RAWG).

2.8.20. Provide oversight and funding to ensure that range systems meet cybersecurity risk management framework requirements IAW AFI 17-101, *Risk Management Framework (RMF) for Air Force Information Technology (IT)*.

2.8.21. **(AFMC)** Through, the Air Force Installation and Mission Support Center:

2.8.21.1. Perform all PEM responsibilities for the following range support areas: Military Construction; Range Facility Sustainment, Restoration, Modernization; Facilities Operation for Range Utilities; and operational range environmental support such as operational range assessments and responses, and natural and cultural resources program execution.

2.8.21.2. Identify environmental requirements eligible for funding in environmental PEs. Ensure AFCEC programs projects IAW DAFI 32-7001, *Environmental Management*, DAFI 32-7020, *Environmental Restoration Program*, or AFI 32-7091, as appropriate in Automated Civil Engineers System, or successor system.

2.8.21.3. AFCEC Technical Support Division, AFCEC/CZT, will:

2.8.21.3.1. Serve as Office of Primary Responsibility for execution of the Operational Range Assessment Program.

2.8.21.3.1.1. Maintain permanent records, documenting the technical data gathered and the analysis to determine if there is a release or substantial threat of a release of munitions constituents of concern from an operational range to an off- range area that creates an unacceptable risk to human health or the environment.

2.8.21.3.1.2. Maintain an inventory of Air Force operational ranges.

2.8.21.4. AFCEC National Environmental Policy Act Division, AFCEC/CZN, will:

2.8.21.4.1. Initiate planning for comprehensive range planning activities and land acquisition consisting of more than 1,000 acres.

2.8.21.4.2. Support range proponents in land acquisition efforts IAW AFI 32-1015, *Integrated Installation Planning*, AFI 32-9001, *Acquisition of Real Property*, and DoDI 4165.71, *Real Property Acquisition*. Ensure all National Environmental Policy Act compliance requirements in support of land acquisition are met.

2.8.22. **Headquarters, U.S. Air Force, The Judge Advocate General (AF/JA)** through the AF/JA-Operations and International Law, Environmental Law and Litigation Division (AF/JAOE). AF/JAOE:

2.8.22.1. Provides legal expertise to Headquarters Air Force (HAF), MAJCOMs, reserve components, AFIMSC, AFCEC and installations on all applicable laws, regulations, and Environmental Oversight requirements impacting DAF hazardous material and environmental policy implementation.

2.8.22.2. Provides legal expertise on applicability of this manual to locations outside the U.S. to ensure there is no conflict with applicable country-specific Foreign Guidance Supplement or, in countries where no Foreign Guidance Supplement exists, the Overseas Environmental Baseline Guidance Document, and Uniform Commercial Code directives.

2.8.22.3. Ensures coordination with Department of Justice, the Deputy Assistant Secretary, Environment, Safety, and Infrastructure (SAF/IEE), and the Air Force General Counsel, Installations, Energy, and Environmental Law (SAF/GCN) regarding the payment of Clean Air Act or other applicable penalties. Provides approval before settlement of any administrative action where the terms of the settlement include provision for the payment of fines or supplemental environmental projects. The appropriate Regional Counsel Office (East or West), established under AF/JAOE, will assist in resolving enforcement actions processed against DAF installations.

2.8.22.4. Assists in the review and validation of all proposed air quality National Emissions Standards for Hazardous Air Pollutants and coordinates with SAF/GCN, SAF/IEE, and AF/A4C as warranted.

2.9. Range Operating Authority. The wing or Combat Readiness Training Center commander responsible for operating the range will serve as the range operating authority. Range operating authority responsibilities may be delegated, in writing, to a unit commander or designated representative, however, contractors cannot serve as range operating authorities. For ANG-operated ranges and Nevada Test and Training Range, the range commander is designated as the range operating authority. ANG range operating authorities will be confirmed by the United States Property and Fiscal Officer prior to assumption of duties. Range operating authorities will:

2.9.1. Ensure compliance with these manual and other directives applicable to range programs. **(T-1)** Range operating authorities may delegate the daily scheduling, operation, maintenance, and management of the range to a subordinate unit.

2.9.1. **(EDWARDSAFB) PIRA Standard Operating Hours.** The PIRA standard operating hours are Monday through Friday, 0700 to 1700 local time. **NOTE**: The PIRA is **closed** during all other days and times outside of these standard operating hours due to hazardous conditions that exist in the range areas. **EXCEPTION**: If the PIRA is required outside of standard operating hours, then the unit scheduler will send an After-Hours Support Request (AHSR) within the 412 TW Center Scheduling Enterprise (CSE) tool to initiate the AHSR coordination process. The AHSR is described in the Edwards AFB Aircraft and Air/Ground Support Scheduling Guide (hereafter as the EAFB Scheduling Guide).

2.9.1.1. **(Added-EDWARDSAFB) PIRA Surface Vehicle Procedures During Standard Operating Hours.** All 412 TW organizations to include tenant units, contractors, military and civilian law enforcement units, fire department units, and safety offices shall contact the 412 RANS Range Operations Officer at DSN 525-6148 (Commercial 661-275-6148) or the 412 RANS DOWNFALL Tower Range Control Officer at DSN 525-5603 (Commercial 661-275-5603) to receive specific training before approved PIRA access is granted to execute duty responsibilities during PIRA standard operating hours. The DOWNFALL Tower Range Control Officer is the control point for all surface activities. All surface vehicles shall obtain approval from DOWNFALL Tower Range Control Officer before entering the PIRA during standard operating hours. The DOWNFALL Tower Range Control Officer maintains a log of surface vehicles and

personnel that enter or exit the PIRA, provides travel instructions to surface vehicles and personnel that are approved entry in the PIRA, and provides surface vehicles and personnel with required two-way radio equipment to communicate with DOWNFALL Tower. During approved after-hours support, surface vehicles follow the same DOWNFALL Tower procedures to request entry into the PIRA.

2.9.1.1.1. **(Added-EDWARDSAFB) PIRA Surface Vehicle Procedures Outside of Standard Operating Hours.** When access to the PIRA outside of standard operating hours is required for mission support or to execute duty responsibilities, an approved AHSR is required. Before approved entry is granted to the PIRA, the user will contact the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer to receive specific surface vehicle procedures training to mitigate hazardous condition risks. Without specific training, user access to the PIRA outside of standard operating hours is not permitted by any surface vehicle and person.

2.9.1.1.2. **(Added-EDWARDSAFB) Edwards Command Post Contact Procedures for Users Requiring PIRA Access Outside of Standard Operating Hours.** All users completing the specific training from the 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer to access the PIRA outside of standard operating hours will contact the Edwards Command Post at 661-277-3040 or UHF 304.0 MHz. All trained users will provide the Edwards Command Post with their point of entry, time of entry, expected time on the PIRA, and destination. Users will notify the Edwards Command Post when PIRA access is completed and their departure location.

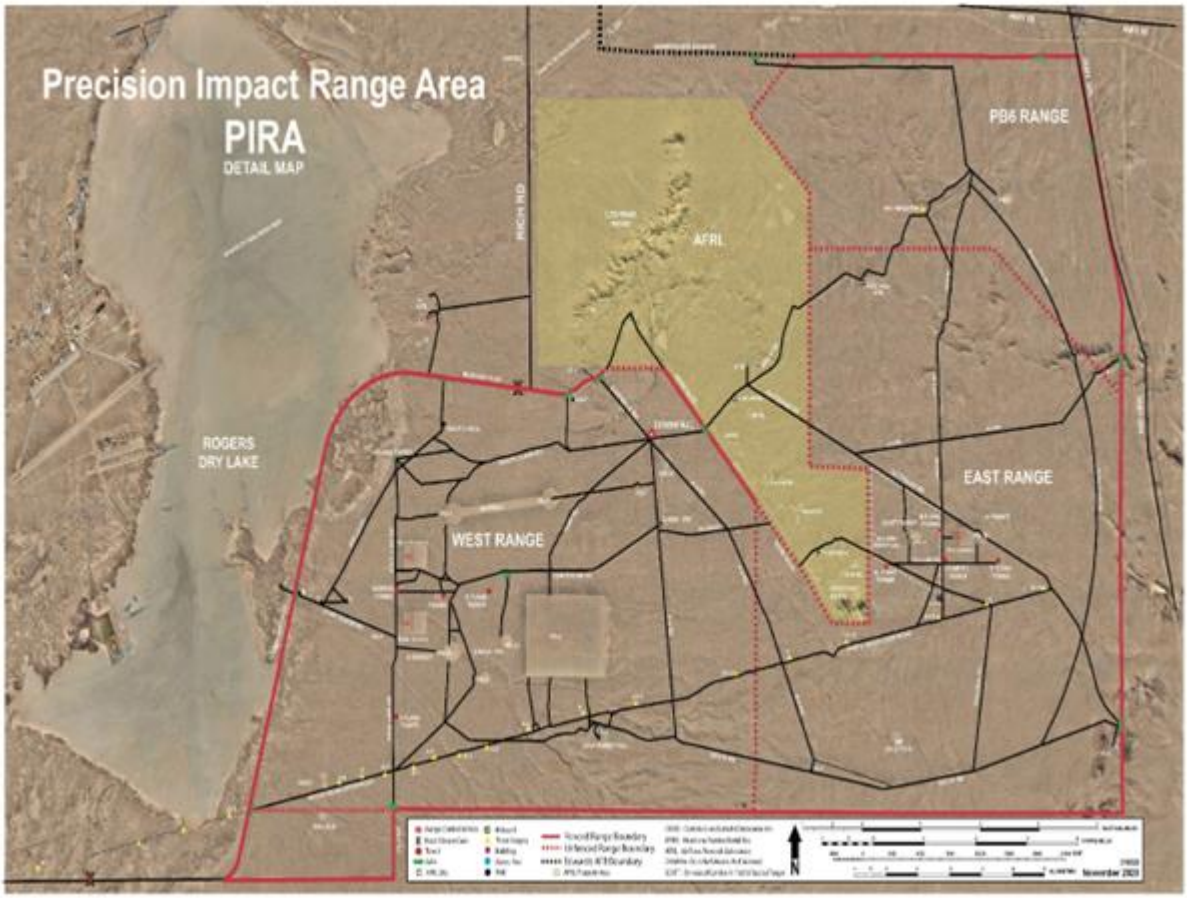
2.9.1.2. **(Added-EDWARDSAFB) Range Scheduling Procedures.** Unit schedulers will submit 412 RANS, PIRA, or 412 TW resource requests to the 412 TW CSE tool or to the 412 TW Scheduling Office at DSN 527-4110, commercial 661-277-4110, or e-mail at **412OSS.OSOS.ROC@us.af.mil**. Requesting schedulers or users without access to the 412 TW CSE tool will submit range resource requests to the appropriate 412 TW unit scheduler. Unit schedulers will coordinate with requesting users and the assigned program 412 RANS Range Control Specialists to ensure resources are correctly scheduled. The 412 RANS Operations Duty Officer at DSN 527-2091 or commercial 661-277-2091 is the focal point to plan and implement range resources in the weekly 412 TW CSE tool. The 412 RANS Operations Duty Officer will coordinate with the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officers, or the program 412 RANS Range Control Specialists to confirm scheduled range resources for mission support.

2.9.1.2.1. **(Added-EDWARDSAFB) Non-DoD or Off-Base User Scheduling Requests.** A non-DoD or off-base user requesting 412 RANS, PIRA, or 412 TW resources will contact the 412 TW Initial Point of Contact (412 TW/IPOC) office at Edwards AFB link **http://www.edwards.af.mil/contact-us/**. Go to the "Want to test at Edwards? Send us an email here" link. The 412 TW/IPOC coordinates with the 412th Test Management Group to determine if the requested support can be provided.

2.9.1.2.2. **(Added-EDWARDSAFB) 412 RANS Mission Support Cancellation Charge Procedures.** A test customer with a scheduled 412 RANS test resource has until 1230 local the workday prior to cancel the use of that resource without any cancellation charge. After 1230 local the workday prior, a test customer will be charged for the direct cost incurred of the expended test resource support up to the time that the official cancellation, abort, or resource release is received. A test customer will be charged for the direct cost incurred for resource set-up, teardown, or support expended to a scheduled mission. On execution day and after the support start time, when a test customer subsequently delays mission support and then deems that the 412 RANS test resource or support is no longer needed based on a change to mission execution, the test customer will be charged from the original start time to the official resource release or cancellation time of the test resource or support, and any remaining unused time will not be charged.

2.9.1.3. **(Added-EDWARDSAFB) General Range Description.** The designated range area for Edwards AFB is the PIRA. The PIRA satellite view diagram, see **Figure 2.1**, below, provides key identifiers for visual reference. The PIRA is located on the eastern portion of the Edwards AFB reserved area, covers approximately 75 square miles, and is subdivided into the West Range, the East Range, and the Precision Bombing (PB) 6 Range. The PIRA is used for air-to-ground gunnery, photo and infrared resolution tests, spin tests, aerial decelerator tests, precision instrumentation tests, precision bombing tests, air-to-ground laser tests, weapon or munitions separation tests, object separation tests, cargo tests, payload tests, chute tests, and ground-to-ground laser tests.

Figure 2.1. (Added-EDWARDSAFB) Precision Impact Range Area.



2.9.1.3.1. **(Added-EDWARDSAFB) West Range.** The West Range is a conventional Class A or Class B dual air-to-ground gun or bomb and limited rocket range with associated airspace. The range contains five PB circles (PB-1, PB-2, PB-9, PB-10, and PB-13), a PB-3 with a vehicle array for imagery, a PB-8 recovery target area, a saturation bombing target (Barbell Target which is located between PB-1 and PB-2) area for missions requiring multiple weapons releases, and the Dual Air-To-Ground Range (DAGRAG) area. PB-1 and PB-10 have scoring instrumentation. PB-13 is approved for limited release of live ordnance. The "B" Target is a permanent reflector and a hot barrel stand for infrared testing. The "B" Target is a National Geospatial-Intelligence Agency surveyed target.

2.9.1.3.2. **(Added-EDWARDSAFB) East Range.** The East Range is a conventional Class A or Class B, single air-to- ground gunnery, bombing and limited rocket range with associated airspace. The East Range has two PB circles (PB-6 and PB-12), a single air-to-ground gunnery or rocket range, and sensor only Mobile Target System Simulated Combat Airfield and Tactical Target range. PB-6 is part of the East Range but is not approved for bombing. PB-6 is primarily used for laser or sensor testing. PB-12 has video bomb scoring capability and is the only target available for bombing. No Southwest bombing run-ins are allowed on PB-12 when COWBELL Tower is manned due to the location of the tower. The gunnery range has a left-hand pattern for strafing

with 10 strafe targets. Strafing lanes one and two have the acoustic scoring capability. The “A” Target is a permanent reflector and a hot barrel stand for infrared testing. The “A” Target is a National Geospatial-Intelligence Agency surveyed target.

Table 2.1. (Added-EDWARDSAFB) PIRA Range Classifications.

Class A	A manned, ground-scoring capable range with an RCO present on range and controlling surface activities and operations
Class B	A manned or unmanned, ground-scoring capable range where no RCO is required for controlling surface activities and operations (commonly known as Flight Lead Control)
Class C	An unmanned range with no scoring and no RCO control of ground activities or operations

2.9.1.3.3. **(Added-EDWARDSAFB) PIRA Classifications.** The PIRA classifications are defined at **Table 2.1.**, below. The range classification (Class A, Class B, or Class C) service may change due to the type of activity being conducted on that range at that time.

2.9.1.3.4. **(Added-EDWARDSAFB) 412 RANS DOWNFALL Tower Range Control Officer.** The 412 RANS DOWNFALL Tower Range Control Officer is responsible for the on-site, real-time PIRA safety and is the final safety authority while conducting the on-site, real-time PIRA range operations during each class of range service. Class A range operations requires a qualified DOWNFALL Tower Range Control Officer to control air-to-ground operations. The DOWNFALL Tower Range Control Officer is the final authority for safe airborne range operations and is required during all Class A range operations, strafe operations, and mission profiles with flight below 300 feet above ground level. Class B (Flight Lead Control) range operations requires a qualified DOWNFALL Tower Range Control Officer to monitor air-to-ground operations, either on-site or a remote site, and is the final authority for safe airborne range operations. The DOWNFALL Tower Range Control Officer will control aircraft operations and will ensure that all safety restrictions identified by the Test Safety Review process and applicable hazard footprints are employed prior to range activities.

2.9.1.3.5. **(Added-EDWARDSAFB) 412 RANS PIRA Services Available.** The PIRA provides range services such as PB targets, single and dual air-to-ground gunnery targets, a photo and infrared resolution range, a tactical imaging and sensor range, a Simulated Combat Airfield and Tactical Target range, tactical mobile targets, range instrumentation for real-time and near real-time weapons scoring, and laser event measurements.

2.9.1.3.6. **(Added-EDWARDSAFB) PIRA Precision Bombing Targets.** The PIRA has six bombing targets and five auxiliary targets that are indicated at **Table 2.2** below. The West Range has five primary PB targets. PB-1 and PB-10 are instrumented bombing targets scored by triangulation from range towers equipped with video camera scoring. Real-time scoring information is accurate to approximately 2 feet and can be voice transmitted from the DOWNFALL Tower Range Control Officer to the aircraft pilot in less than 1 minute after impact. The PB-8 target is a 1.2 square mile cleared area, 6,500 feet by 8,500 feet, used primarily to drop parachutes, capsules, tip-tanks, drone vehicles, and other items, as well as to recover these items when required. Three other low-level bombing targets (PB-2, PB-9, and PB-13) can be used when a specific altitude, speed, and method of release dictate use of these targets. Scoring is accomplished by post-flight surveys. The East Range consists of the PB-12 target circle and is used for instrumented bombing like the PB-1 and PB-10 target circles on the West Range. PB-6 target circle is not approved for bombing. The PB-6 target circle is primarily used for laser or sensor testing.

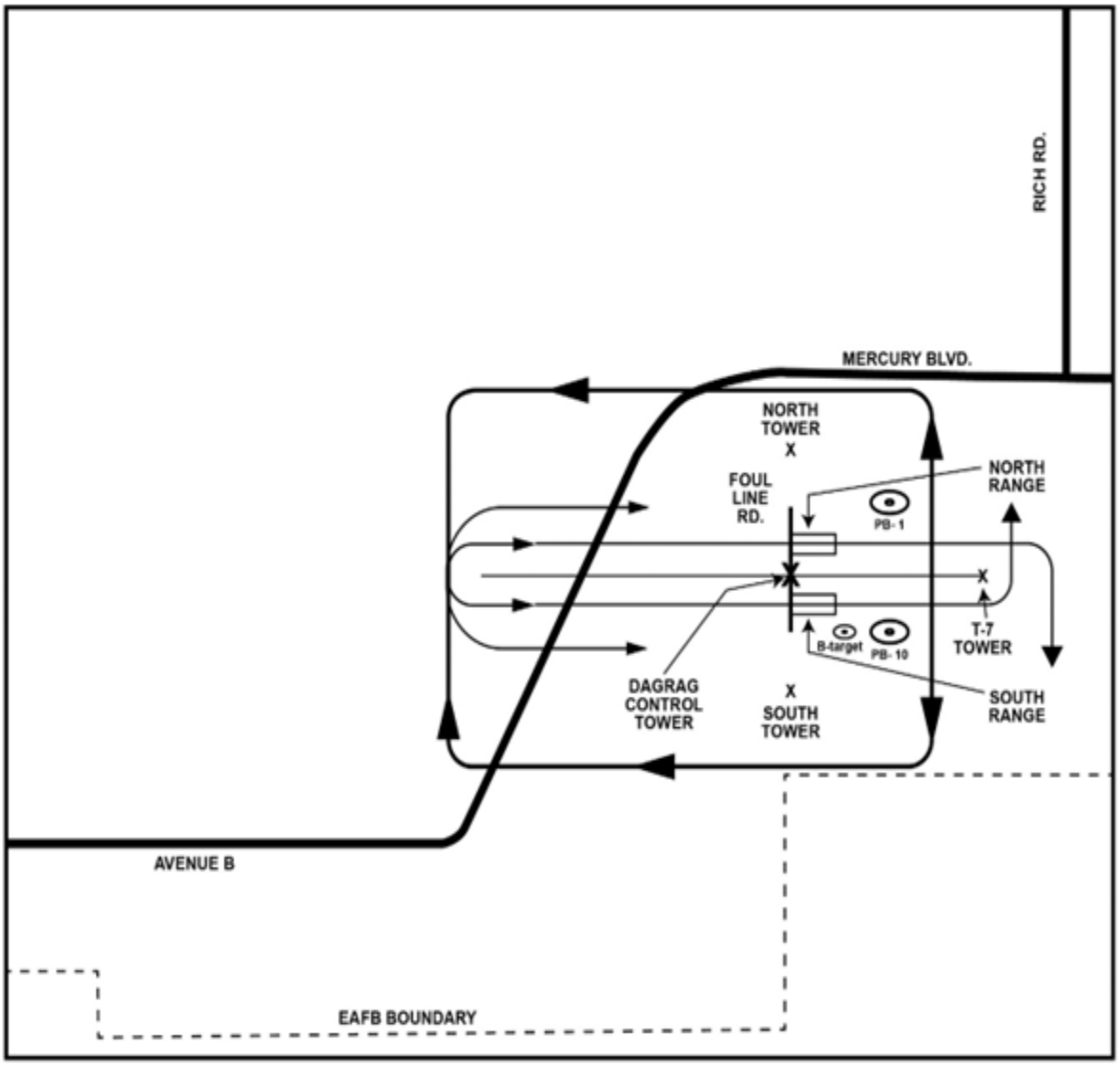
Table 2.2. (Added-EDWARDSAFB) Precision Bombing Targets.

PIRA PRECISION BOMBING TARGETS				
Target Area	Primary Function	Instrumented	Range	Size
PB-1	Bombing	Yes	West	1600’ DIA
PB-2	Bombing	Yes (Post-flight)	West	1600’ DIA
PB-3	Radar/IR Resolution	No	West	600’ DIA
A Target	Radar/IR Resolution	No	East	200’ DIA
PB-6	Laser	No	East	1000’ DIA
B Target	Radar/IR Resolution	No	West	200’ DIA
PB-8	Air-launched ordnance/ Recovery Area	No	West	8500’ x 6500’
PB-9	Bombing	Yes (Post-flight)	West	1000’ DIA
PB-10	Bombing	Yes	West	1600’ DIA
PB-12	Bombing	Yes	East	1000’ DIA
PB-13	Bombing	Yes (Post-flight)	West	1000’ DIA

2.9.1.3.7. **(Added-EDWARDSAFB) . Dual Air to Ground Range (DAGRAG).** The DAGRAG (**Figure 2.3. and Figure 2.4.**) is a conventional low altitude, Class A or Class B, air-to-ground gunnery, bombing and limited rocket range with associated airspace. The DAGRAG has a North Range (right traffic) and a South Range (left traffic) shown in the expanded view at **Figure 4.1**., DAGRAG Traffic Patterns, (see page 38). Each range has one bomb or rocket circle, five strafe targets, three flank instrumentation towers and one common range control tower. South Range has the acoustic scoring capability on lanes six and seven. A conventional “3 x 4” nautical mile flight pattern supports the ranges. Patterns are limited to Mercury

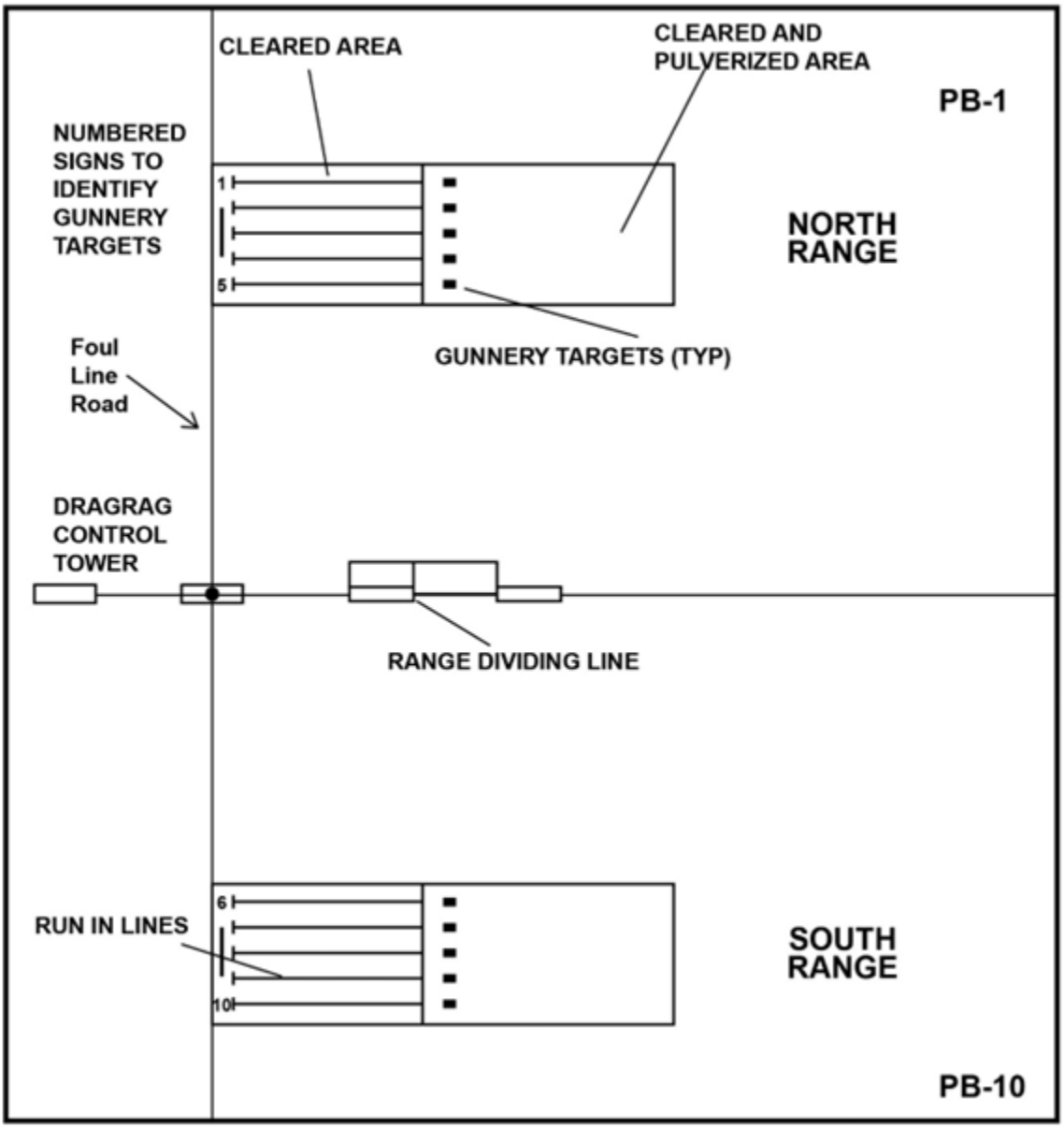
Boulevard to the north, Avenue B to the south, and a west base leg 1 nautical mile east of the north and south lakebed runway. Normally, the eastern crosswind legs of the patterns are 1 nautical mile beyond the target for a single aircraft. When two or more aircraft are in the same pattern, each aircraft will adjust the eastern crosswind leg to maintain adequate separation. There is no minimum altitude restriction on the north range pattern. Restrict south range pattern base leg to 4,800 feet mean sea level minimum until passing the south base runway centerline extension. The airspace extends vertically to 14,000 feet mean sea level. Refer also to **Figure 4.1**., DAGRAG Traffic Patterns to view the expanded map.

Figure 2.3. (Added-EDWARDSAFB) DAGRAG North and South Range.



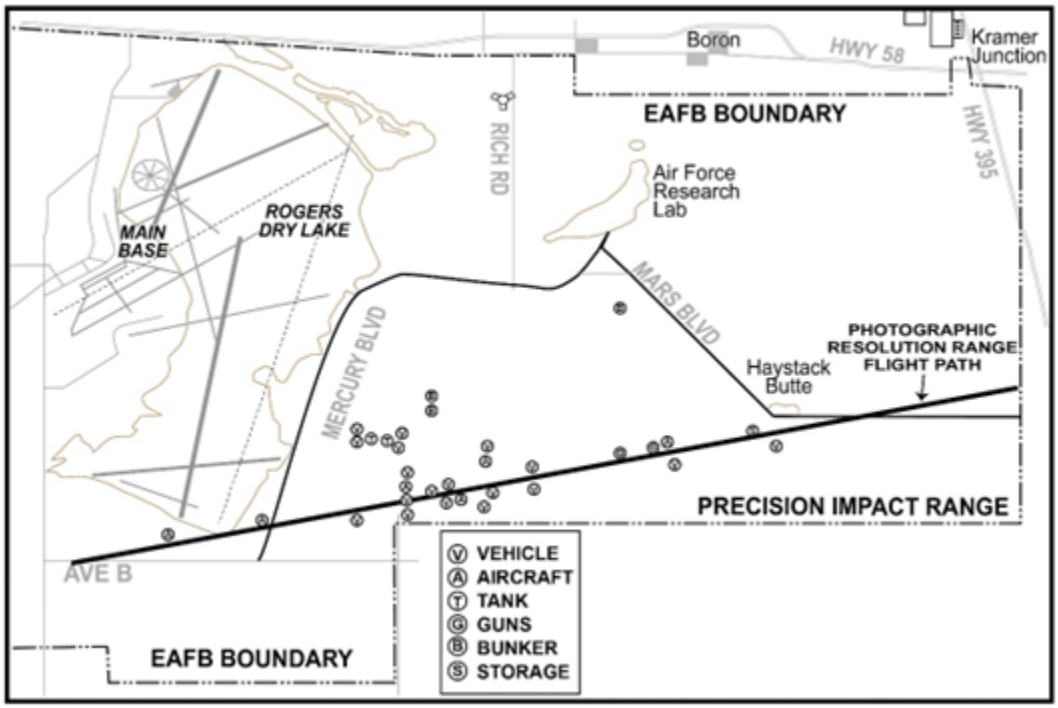
2.9.1.3.8. **(Added-EDWARDSAFB) DAGRAG Foul Line Road.** A hard surface roadway extending north and south from North Range to South Range towers immediately west of the DAGRAG on the West Range, see **Figure 2.3**, and **Figure 2.4** below. Strafe targets are located 2,000 feet from Foul Line Road. On the East Range, the road runs east to west immediately south of the strafe target. Strafing passes on DAGRAG are authorized on a west to east heading and a south to north heading on the East Guns area. Strafing will terminate at or prior to crossing Foul Line Road. All strafe missions, live or dry, below 300 feet above ground level on the PIRA require Class A range service.

Figure 2.4. (Added-EDWARDSAFB) DAGRAG Foul Line Road.



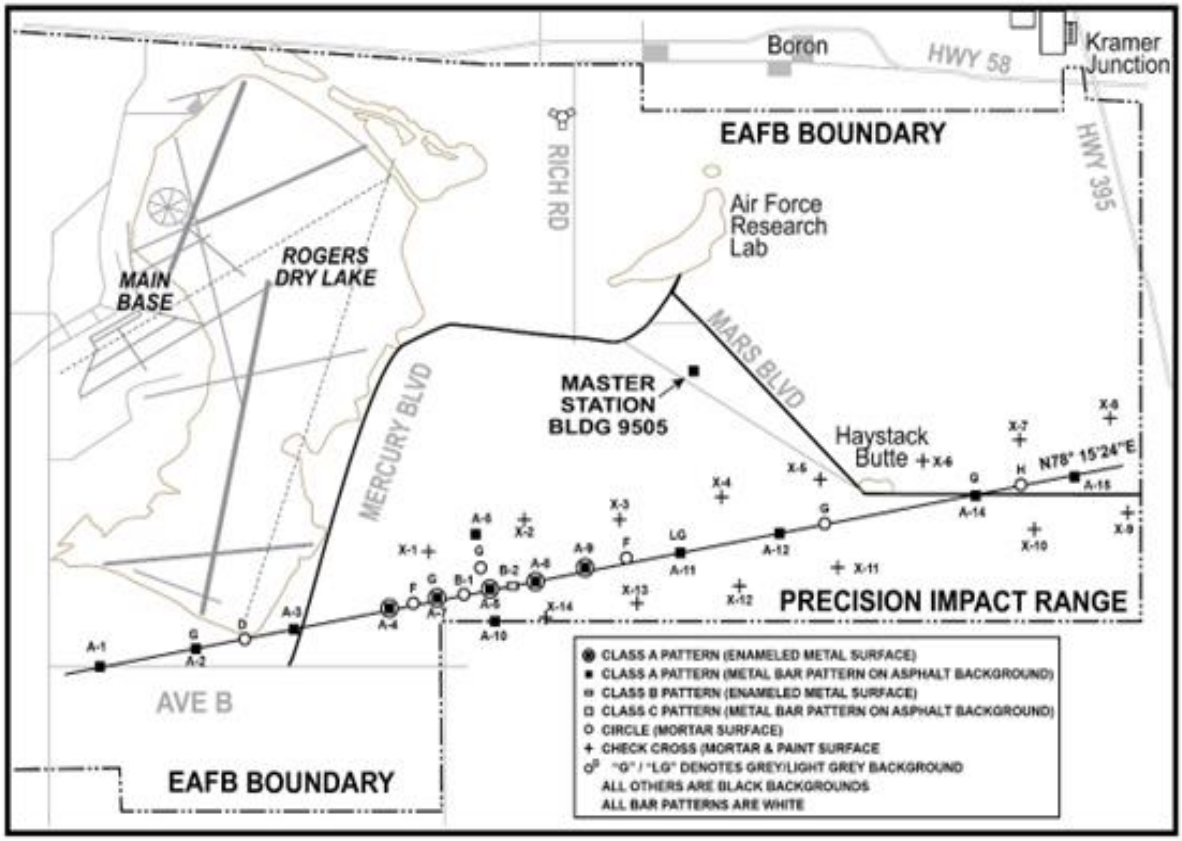
2.9.1.3.9. **(Added-EDWARDSAFB) Photo Infrared Tactical and Photo Infrared Resolution Ranges.** The Photo Infrared Tactical Range (**Figure 2.5**) and the Photo Infrared Resolution Range (**Figure 2.6**) are in the Alpha Corridor and PIRA. Both ranges consist of a variety of targets at sited locations along the resolution range and the PB-7 Strip Range.

Figure 2.5. (Added-EDWARDSAFB) Photo Infrared Tactical Range.



2.9.1.3.9.1. **(Added-EDWARDSAFB) Photo Infrared Resolution Range.** The Photo Infrared Resolution Range is in the southern part of the PIRA (**Figure 2.6.**) and provides various targets to test the resolution of airborne photographic equipment and film, and to test airborne infrared tracking systems. The range area is 2 nautical miles wide by 21 nautical miles long with 18 three-bar type resolution targets of various sizes, one tri-density target, five circle targets, one oblique target, and 14 check-cross patterns. The photo resolution patterns are built according to DoD requirements.

Figure 2.6. (Added-EDWARDSAFB) Photo Infrared Resolution Range.



2.9.1.3.9.1.1. **(Added-EDWARDSAFB)** Before entering the PIRA Photo Infrared Resolution Range, aircrews shall contact the SPORT Military Radar Unit (MRU) on mission frequency or on UHF 343.7 MHz to confirm range time and entry clearance.

2.9.1.3.9.1.2. **(Added-EDWARDSAFB) PIRA Entry Communication Procedures when SPORT MRU is Closed.** When SPORT MRU is closed, aircrews will contact Edwards AFB Control Tower for takeoff or departure before entering the PIRA. The DOWNFALL Tower Range Control Officer will notify the Edwards AFB Control Tower and the FAA Joshua Control Facility that the PIRA is active. Aircrews will contact the FAA Joshua Control Facility before entering the PIRA to request R-2515 work area information. The DOWNFALL Tower Range Control Officer will maintain contact with Edwards AFB Control Tower and advise when the mission is complete.

2.9.1.3.10. **(Added-EDWARDSAFB) Tactical Imaging and Sensor Range.** The Tactical Imaging and Sensor Range consists of a variety of targets used to determine the accuracy and fidelity of airborne imaging and sensor systems. Located within the Alpha Corridor and the PIRA, the Tactical Imaging and Sensor Range is comprised of tactical targets to include aircraft, land vehicles, tanks, missiles, guns, bunkers and infrared target boards. The Mobile Target System is part of the Tactical Imaging and Sensor Range.

2.9.1.3.10.1. **(Added-EDWARDSAFB) Mobile Target System.** The Mobile Target System consists of mobile targets which are military, commercial, or other type vehicles such as armored personnel carriers, jeeps, fast attack vehicles, heavy trucks, and specialty vehicles. The mobile targets are requisitioned from the Defense Logistics Agency Disposition Services or from other DoD methods and used on the PIRA to support test programs or projects requiring this type of target. Mobile targets are maintained, and safety inspected according to a maintenance schedule to support operator safety and test safety requirements. Mobile targets are used as static targets, moving target arrays, or automated mobile target arrays or configurations. Most mobile targets can be configured with the Time-Space-Position Information System equipment for post-flight analysis. The Mobile Target System is used for sensor testing only.

2.9.1.3.10.1.1. **(Added-EDWARDSAFB) Mobile Target System Management.** The 412 RANS Range Operations Officer executes the following procedures to manage the mobile target system:

2.9.1.3.10.1.1.1. **(Added-EDWARDSAFB)** Coordinate with the 412 RANS Logistics Section to request mobile target allocations and to turn-in mobile targets according to 412 RANS, 412 LRS/LGRDL, and Defense Logistics Agency Disposition Services processes.

2.9.1.3.10.1.1.2. **(Added-EDWARDSAFB)** Coordinate with the PIRA Equipment Custodian to account for all mobile targets on the Custodian Authorization and Custody Receipt Listing equipment account.

2.9.1.3.10.1.1.3. **(Added-EDWARDSAFB)** Ensure mobile targets have "**Mobile Target – Range Use Only**" stenciled onto the target surface.

2.9.1.3.10.1.1.4. **(Added-EDWARDSAFB)** Assign a 412 RANS registration number to each mobile target for accounting and maintenance tracking purposes.

2.9.1.3.10.1.1.5. **(Added-EDWARDSAFB)** Identify mobile targets that are disabled beyond repair, have mechanical discrepancies, or have safety concerns and coordinate with 412 RANS Logistics Section to initiate turn in actions. Identify mobile targets that can be repurposed to become permanent static targets.

2.9.1.3.10.1.1.6. **(Added-EDWARDSAFB)** Ensure the 412 RANS Operations and Maintenance contractor maintains the mobile target system according to contractual provisions.

2.9.1.3.10.1.2. **(Added-EDWARDSAFB) Mobile Target System Operations.** The 412 RANS Range Operations Officer executes the following procedures to safely train and operate the mobile target system:

2.9.1.3.10.1.2.1. **(Added-EDWARDSAFB)** Ensure mobile target system training consists of lesson plans developed from applicable technical data for the targets being utilized.

2.9.1.3.10.1.2.2. **(Added-EDWARDSAFB)** Ensure operators only operate mobile targets for which they have completed training and certification actions.

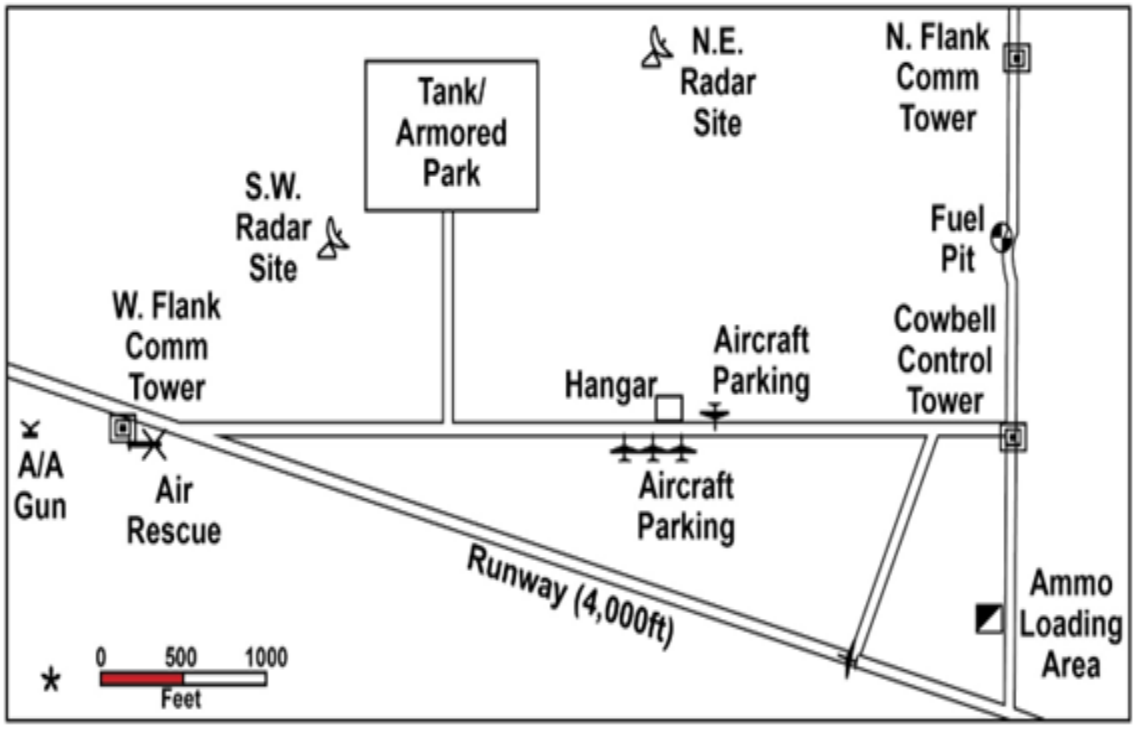
2.9.1.3.10.1.2.3. **(Added-EDWARDSAFB)** Ensure operator certification is signed off in the operator's training records to annotate the completion of the mobile target lesson plan by the supervisor or the trainer who performed the training.

2.9.1.3.10.1.2.4. **(Added-EDWARDSAFB)** Ensure operators do not operate mobile targets with mechanical discrepancies that could compromise operator safety or place an operator in an unsafe condition.

2.9.1.3.10.1.2.5. **(Added-EDWARDSAFB)** Ensure operators utilize mobile targets for intended purposes on the PIRA or on designated 412 TW areas in support of flight or ground test projects. **Note:** Mobile targets are not used in the same function as normal government transportation vehicles.

2.9.1.3.11. **(Added-EDWARDSAFB) Simulated Combat Airfield and Tactical Target Range.** The Simulated Combat Airfield and Tactical Target Range (**Figure 2.7**) is used for imaging or sensor test and training. It is configured to look like a combat airfield with aircraft, runway, hangar, fuel depot, storage area, and tank or an armored park.

Figure 2.7. (Added-EDWARDSAFB) The Simulated Combat Airfield and Tactical Target Range.



2.9.1.3.12. **(Added-EDWARDSAFB) Radiometric Analysis and Measurement System.** The Radiometric Analysis and Measurement System is a mobile van that provides instrumentation support for projects that develop, test, and evaluate ground and airborne electro-optical systems or sensors. The Radiometric Analysis and Measurement System support accounts for and quantifies the radiation across the optical spectrum from visible to the far infrared.

2.9.2. Appoint a range operations officer, in writing, to supervise range operations, management, planning, and maintenance. **(T-1)** The range operations officer will be a military officer or DoD civilian employee. **(T-1) Note:** Range operating authorities must ensure requests for waivers forwarded to the MAJCOM are reviewed by the Staff Judge Advocate and Manpower offices for compliance with applicable law and DoDI 1100.22, *Policy and Procedures for Determining Workforce Mix*. **(T-1)**

2.9.2. **(EDWARDSAFB)** The 412 TW/CC appoints the 412 RANS Range Operations Officer to supervise PIRA operations, management, planning, and maintenance. The 412 RANS Director of Operations supports the 412 RANS Range Operations Officer in the execution of assigned functions and duties appointed by the 412 TW/CC.

2.9.2.1. The range operating authority will designate the responsibilities of the range operations officer. Normally the range operations officer has authority over all range activities that support mission readiness training and serves as the range operating authority's primary point of contact for range issues.

2.9.2.1. **(EDWARDSAFB) 412 RANS Range Operations Officer responsibilities are to:**

2.9.2.1.1. **(Added-EDWARDSAFB)** Ensure DOWNFALL Tower Range Control Officers conduct their responsibilities and functions in compliance with AFMAN 13-212V1 requirements.

2.9.2.1.2. **(Added-EDWARDSAFB)** Notify the 412 TW Command Post when the PIRA and DOWNFALL Master Control Tower is open or closed. The on-duty DOWNFALL Tower Range Control Officer may also notify the 412 TW Command Post when the PIRA is open or closed according to Range Operations Officer delegation.

2.9.2.1.3. **(Added-EDWARDSAFB)** Control personnel access to and operations on the PIRA. Authority to control access to the PIRA is delegated to the DOWNFALL Tower Range Control Officer who is performing radio operations duties in the DOWNFALL Master Control Tower.

2.9.2.1.4. **(Added-EDWARDSAFB)** Ensure operational controls are effective for personnel safety on the PIRA.

2.9.2.1.5. **(Added-EDWARDSAFB)** Report PIRA operations incidents, deviations, or hazards to the 412 RANS Operations Duty Officer, 412 RANS Director of Operations, and the 412 TW Range Safety Office. Reports should be sent as quickly as possible. The DOWNFALL Tower Range Control Officer will make log entries to reflect the circumstances of any occurrence requiring such reporting.

2.9.2.1.6. **(Added-EDWARDSAFB)** Coordinate with 412 RANS personnel to initiate appropriate action by 412 TW organizations responsible for servicing and maintaining the PIRA.

2.9.2.1.7. **(Added-EDWARDSAFB)** Ensure all PIRA support schedule changes, add-ons, and cancellations are sent to the PIRA flight and sections. The DOWNFALL Tower Range Control Officer is the focal point of contact for the 412 RANS Operations Duty Officers and the program 412 RANS Range Control Specialists for changes to the daily PIRA support schedule requirements.

2.9.2.1.8. **(Added-EDWARDSAFB)** Monitor progress or status of missions requiring road closures to ensure that the PIRA flight establishes appropriate roadblocks. The 412 RANS Range Operations Officer will delegate authority to the DOWNFALL Tower Range Control Officer(s) to perform the monitoring function as required for mission support.

2.9.2.1.9. **(Added-EDWARDSAFB)** Ensure all 412 RANS personnel are aware of the Bird Aircraft Strike Hazard procedures as briefed during the annual range training. Personnel conducting missions on the PIRA will alert the DOWNFALL Tower Range Control Officer of bird hazards so that aircraft can be advised.

2.9.2.1.10. **(Added-EDWARDSAFB)** Delegate 412 RANS Range Operations Officer duties to the DOWNFALL Tower Range Control Officer(s) when the 412 RANS Range Operations Officer is not on duty.

2.9.2.1.11. **(Added-EDWARDSAFB)** Monitor the number of live munitions releases on PB-13 and notify the 412 RANS Director of Operations of any mission restrictions resulting from the environmental compliance requirement.

2.9.2.1.12. **(Added-EDWARDSAFB)** Coordinate all live munitions operations on PB-13 no later than 3 days prior to mission execution with the 812th Civil Engineering (812 CES/CED) Explosive Ordnance Disposal (EOD) unit.

2.9.2.2. The range operations officer develops range procedures, interfaces with support agencies and flying units related to range operations, maintenance, planning, and compliance.

2.9.2.2. **(EDWARDSAFB)** The 412 RANS Range Operations Officer will develop the PIRA procedures and will proactively develop coordination interfaces with 412 TW support agencies and flying units related to range operations, maintenance, planning, and compliance.

2.9.3. Appoint a range safety officer. **(T-1)** The range safety officer will be a military officer or DoD civilian employee. **(T-1) Note:** Range operating authorities must ensure requests for waivers forwarded to the MAJCOM are reviewed by the Staff Judge Advocate and Manpower offices for compliance with applicable law and DoDI 1100.22**. (T-1)**

2.9.3. **(EDWARDSAFB)** The 412 TW/CC appoints the 412 TW Range Safety Officer (412 TW/SETR). The 412 TW Safety Office (412 TW/SE) supports the 412 TW Range Safety Officer in the execution of assigned functions and duties appointed by the 412 TW/CC.

2.9.4. Appoint or obtain the support of a military officer or civilian as a Flight Safety Officer IAW AFI 91-202, *The U.S. Air Force Mishap Prevention Program,* if the range accommodates flight test or training. **(T-3)**

2.9.5. Appoint a Laser Safety Officer IAW AFI 48-139, *Laser and Optical Radiation Protection Program,* if the range accommodates laser operations. **(T-3)**

2.9.5. **(EDWARDSAFB)** The 412 TW/CC appoints the 412 TW Laser Safety Officer and the 412 RANS Laser Safety Officer. The 412 RANS Director of Operations and the 412 RANS Range Operations Flight support the 412 RANS Laser Safety Officer in the execution of assigned functions and duties appointed by the 412 TW/CC. The 412 TW Safety Office supports the 412 TW Laser Safety Officer in the execution of assigned functions and duties appointed by the 412 TW/CC. The 412 TW Safety Office supports the 412 TW Test Laser Safety Officer program to ensure the safe use of lasers and laser systems, and to coordinate the requirements for 412 TW laser operations under a test operations environment. The 412 TW Laser Safety Officer, 412 TW Test Laser Safety Officer, and 412 RANS Laser Safety Officer work together to support 412 TW laser missions. Refer to Air Force Test Center Instruction 91-202_412th Test Wing Supplement, 412th Test Wing Safety Review Policy, for 412 TW Laser Safety Officer and 412 Test Laser Safety Officer responsibilities.

2.9.6. Appoint a Directed Energy Weapon Safety Officer IAW AFI 91-401, *Directed Energy System Safety,* or obtain Directed Energy Weapon Safety Officer support from a parent or associated unit if the range has been certified to conduct directed energy weapon operations. **(T-1)**

2.9.7. Certify range control officers and Test Area Control Officers in writing. **(T-2)** The range operating authority will establish the education, training, experience, and affiliation (contract or government) requirements for these positions. **(T-1)**

2.9.7.1. For ranges providing Class A service, the range control officer at a minimum will have experience as a rated aircrew member, an Air Traffic Controller, an Air Battle Manager, a Terminal Attack Controller, a Weapons Director, a previous range control officer or equivalent test experience. **(T-1)**

2.9.7.2. For test sites, identify actions, requirements, and responsibilities of the designated range control officers and test area control officers. **(T-1)**

2.9.7.3. For air-to-air and air-to-surface training, range control officer duties and responsibilities follow AFI 11-214, *Air Operations Rules and Procedures* guidance.

2.9.8. Certify Range Training Officers in writing. **(T-2)** For air-to-air training on ranges and other airspace providing Class D service, the primary Range Training Officer responsibilities in addition to flight safety are to facilitate training, provide real-time kill removal, and assist flight leads in mission reconstruction during debriefs. During autonomous air-to-air training (conducted without Ground-Control Interception or Airborne Warning and Control System), the Range Training Officer may assist aircrews in maintaining aircraft inside airspace boundaries and expedite rejoins upon request. The Range Training Officer may be military, or contractor furnished, but as a minimum, Range Training Officers must possess "rated mission expertise" in the activity being performed or have Air Battle Manager/Weapons Director experience. **(T-2)**

2.9.9. Appoint a data steward to create, maintain, and manage the range functional geospatial data and/or mission datasets. **(T-3)** The range data steward coordinates with the Installation Geospatial Integration Office on all mapping and geospatial integration requirements IAW paragraphs **3.2.2** and **3.2.10**. of AFI 32-10112, *Installation Geospatial Information and Services (Installation GI&S)*.

2.9.10. Ensure all personnel appointed or assigned to the range are qualified and trained for their positions IAW **Attachment 5**, Range Personnel Training, and MAJCOM supplements before assuming duties. **(T-1)** All training must be documented. **(T-2)** The method of documentation is up to the supervisor, if not otherwise prescribed by the applicable MAJCOM or related chain of authority.

2.9.11. Ensure that authorized visitors and personnel who infrequently visit the range are appropriately briefed on range safety. **(T-1)**

2.9.12. Participate in DoDI 5000.89_DAFI 99-103, *Capabilities-Based Test and Evaluation*, safety review process by assessing test project safety plans, ensuring that they identify and mitigate all health and safety risks. **(T-1)**

2.9.13. As requested, provide information and assistance to test planners regarding range costs, availability, and how the range establishes priorities for training and testing requirements.

2.9.14. Prior to authorizing test events on a primary training range (PTR), obtain MAJCOM approval of test footprints or WDZ surrogate footprints and risk assessments IAW paragraphs **4.11.3.1.2** and **4.11.3.1.3**.

2.9.15. Enter into written agreements in coordination with the MAJCOM and IAW **paragraph 4.2** and AFI 25-201, *Intra-Service, Intra-Agency, and Inter-Agency Support Agreements Procedures*. **(T-1)**

2.9.16. Develop and maintain a range supplement to this manual, which is approved and published at wing level. **(T-1) Exception:** Supplements for ANG ranges and Nevada Test and Training Range may be approved and published by the range operating authority.

2.9.16.1. Ranges may use the most suitable format for the supplement (i.e., integrated or standalone) in accordance with DAFMAN 90-161.

2.9.16.2. Range operating authorities must submit the range supplement to the parent MAJCOM for review and coordination prior to publication. **(T-1)** When supplements require update due to a significant change in range operations, submit an updated supplement to the MAJCOM within 120 days.

2.9.16.3. At a minimum the supplement must include: general range description; services available; hours of operation; range diagrams; range scheduling procedures; operations; joint operating procedures; laser and attack restrictions, target and weapons authorizations and restrictions (to include composition of target and authorized ordnance); electronic warfare operating procedures; safety; emergency and jettison procedures; authorized frequency clearances; range control officer procedures; chaff and flare operations; night lighting diagrams; night operations; range control officer night vision device (NVD) operations; ground laser procedures; support agency procedures and agreements; range clearance operations; pyrotechnic employment procedures and restrictions; and fire suppression responsibilities. **(T-2)**

2.9.17. Ensure proposals to construct or upgrade urban training facilities are properly evaluated and/or certified IAW DoDI 1322.27 and this manual. **(T-0)**

2.9.18. Maintain scheduling control over all range space and equipment using CSE and document scheduling procedures in a letter of agreement or the range supplement. **(T-1)**

2.9.19. Post on CSE applicable range information, points of contact, updated procedures, access to the local supplement to this manual, scheduling information, and links to other large reference sources, such as target and range imagery, stored outside of CSE. **(T-1)**

2.9.20. Although Civil Engineering (CE) manages the EOD program and provides personnel for range clearance mission, the range operating authority funds all aspects of range clearance support. **(T-1)** All aspects will include the following: lodging, PPE, equipment, vehicles, fuel, and required maintenance.

2.9.20. **(EDWARDSAFB)** Although the 412th Civil Engineering Group manages the 812 CES/CED EOD program and provides personnel for the PIRA clearance mission, the 412 TW/CC is the Range Operating Authority and funds all aspects of PIRA clearance support. All aspects will include the following: lodging, PPE, equipment, vehicles, fuel, and required maintenance.

2.9.20.1. **(Added-EDWARDSAFB) Scheduling EOD Open Burn/Open Detonation Operations.** The 812 CES/CED EOD unit will initiate coordination with 412 RANS Operations Duty Officer (661-277-2091) to deconflict and propose a schedule of Open Burn/Open Detonation (OB/OD) operations. The resources are scheduled in the 412 TW CSE with the selection of primary and back-up dates and times for the OB/OD area, the PIRA West Range airspace, the altitude or the overflight restriction, and other range resources, as required. When coordination of these resources with the 412 RANS Operations Duty Officer are completed, the 812 CES/CED EOD unit will schedule OB/OD operations in the 412 TW CSE tool according to the EAFB Scheduling Guide.

2.9.21. Work with the supporting EOD unit and environmental office to develop the annual range clearance plan 6 months in advance and ensure the clearance schedule is readily available. **(T-3)** Work with the supporting LRS unit to provide the supporting EOD flight proper vehicle and trailer authorizations to best support their specific mission needs.

2.9.22. Maintain a record of clearance activities that accurately captures the geographic scope of the clearance activity. **(T-1)** Copies of EOD range clearance reports suffice provided the reports include this required information.

2.9.23. Develop and implement training, inspection, and maintenance procedures for range NVDs utilized by range control officers and other range personnel. **(T-2)**

2.9.24. Establish procedures for range, occupational and explosive safety. **(T-1)** Range operating authorities must ensure safety of landing zones and drop zones are in accordance with DAFMAN 13-217 and that emergency procedures are readily available to the range control officer and other ground personnel. Range operating authorities will contact the supporting installation Weapons Safety Manager to assist in licensing or siting munitions storage locations. **(T-1)**

2.9.25. Implement a risk management process for the range in accordance with AFI 90-802, *Risk Management*. **(T-1)**

2.9.25.1. For changes to range operations or procedures where the risk increases to high (or above), the range operating authority will notify the parent MAJCOM prior to implementation in accordance with AFI 91-202, *The US Air Force Mishap Prevention Program*. **(T-2)**

2.9.25.2. For planned or implemented changes, the range operating authority will perform and maintain records of a WDZ analysis for all targets, manned facilities/sites, and equipment on the range. **(T-1)**

2.9.26. Make reasonable effort to make information concerning hazard areas associated with ranges under the range operating authority's control available to the public.

2.9.27. Ensure range boundary signs are posted IAW **paragraph 4.5.1**. **(T-1)** On ranges that fall within the boundaries and direct control of another agency responsible for the safety and security of the range, such as a shared-use Army range, the range operating authority shall coordinate with the owning agency to determine appropriate signage consistent with the intent of **paragraph 4.5.1**. **(T-1)**

2.9.28. Establish an outreach program to educate installation personnel and the public about the dangers of trespassing and UXO hazards. **(T-1)** Using appropriate forums, address range issues that have the potential to influence the surrounding community. **(T-3)**

2.9.29. Establish procedures for basic first aid and medical evacuation of any personnel injured during range operations. **(T-1)** Ensure EOD range clearance activities receive medical support IAW DESR 6055.09_AFMAN 91-201 and AFMAN 32-3001.

2.9.30. Establish range security procedures to include physical safeguards for all range equipment and facilities. **(T-1)**

2.9.31. Control access to hazardous areas, as described in **Chapter 4**, using physical safeguards or procedures to protect all personnel and property. **(T-1)**

2.9.32. Conduct a risk assessment to establish published procedures and restrictions for the expenditure of all munitions (weapons, flares, threat simulators, etc.). **(T-1)** Review the risk assessment annually. **(T-2)**

2.9.33. Restrict the use of depleted uranium (DU) to sole use impact areas that have been permitted by Air Force Medical Readiness Agency Radioisotope Committee. **(T-0)**

2.9.34. Prepare requests for disposal or recycling of DU penetrators and fragments IAW AFMAN40-201, *Radioactive Materials (RAM) Management*, and coordinate with the supporting Installation Radiation Safety Officer. **(T-0)**

2.9.35. Prohibit training that employs improved conventional munitions or munitions that contain submunitions (e.g., cluster bombs) unless specifically authorized by AF/A3TI. **(T-0)**

2.9.35.1. When authorized, establish sole-use target or impact areas where feasible for training activities involving improved conventional munitions or submunitions. **(T-0)**

2.9.35.2. Make all reasonable attempts to employ inert and live improved conventional munitions on different targets. **(T-0)**

2.9.36. For research, development, test and evaluation activities that involve improved conventional munitions or submunitions, establish sole-use target or impact areas. **(T-0)**

2.9.37. To the maximum extent practicable, designate separate impact areas for live and inert ordnance training. **(T-1)**

2.9.38. Ensure no target or target material is placed on a range until all hazardous material, including potentially hazardous waste regulated by the Resource Conservation and Recovery Act, 42 U.S.C. §§ 6901-6992, has been removed from the target or target material, except to the extent that the presence of such hazardous material is essential to the operation or effectiveness of the target or target material. **(T-0)**

2.9.39. Implement a preventive and corrective maintenance program to renovate, remove, and overhaul range targets, systems, and debris. **(T-1)**

2.9.40. Whenever possible, design, locate, and maintain targets to minimize future debris clearance costs to include use of innovative range clearance technologies and best management practices (target material substitution, recycling, etc.).

2.9.41. Immediately report if a release of ordnance results in off-range impacts IAW AFI 91-202.

2.9.42. Immediately report off-range releases in the United States or the substantial threat of an off-range release of munitions constituents of concern in the United States, when such release or threat of release poses an imminent and substantial threat to human health or the environment IAW DoD Response authorities under 10 U.S.C. § 2701, *Environmental Restoration Program*, and 42 U.S.C. § 9604, *Response Authorities*. **(T-0)** Support EOD and environmental offices that are responsible for response actions under the Air Force environmental restoration program.

2.9.43. Implement, where appropriate or recommended as a result of an operational range assessment, proactive management practices to prevent or mitigate actual or potential migration of munitions constituents beyond the range boundary. **(T-0)**

2.9.44. Lead efforts to sustain, restore, and modernize the natural and man-made infrastructure on the range. **(T-3)** This includes identifying the range natural infrastructure requirements and regularly evaluating the health of the natural infrastructure.

2.9.45. Review, coordinate or approve all plans, proposals, and documents that potentially could affect range operations to ensure compatibility. **(T-1)**

2.9.45.1. Within 90 days of taking over such duties and at least annually thereafter, the range operating authority will coordinate with the supporting installation CE environmental office and ensure that range operations are in compliance with applicable environmental requirements and within the scope of all relevant environmental analyses, including any required management actions or mitigation. **(T-1)**

2.9.45.2. As required, identify and resolve in consultation with the officials responsible for the existing agreements (to include renegotiation of existing agreements if necessary) any management or mitigation actions that are having or could have a significant adverse impact on range operations.

2.9.46. Participate in the T/TSNS process and coordination IAW DAFMAN 13-201 and this manual.

2.9.47. Be the approval authority for coordination with the delegated lease execution authority on any enhanced use leases, out grants, leases, or easements on Air Force range property. **(T-1)** See also **paragraph 4.4.2.3**.

2.9.48. Identify, as early as possible, future range activities/actions which may require environmental analysis under EIAP and provide this information to the servicing CE environmental office for inclusion in the appropriate activity management plan which allows for documentation of any required funding needed to perform the analysis. **(T-2)**

2.9.49. Implement climate change mitigation and adaptation strategies documented in the Integrated Natural Resources Management Plan and/or Enterprise Range Plan as necessary and appropriate. **(T-0)**

2.9.50. Review, as early as possible, proposed activities (e.g., exercises) which may involve foreign nations employing weapons at the range. **(T-2)** Collect data for validation of WDZ footprints and identify range limitations which may restrict or prevent foreign participation. **(T-2)** Communicate findings and recommendations to the responsible MAJCOM office. **(T- 2)**

2.9.51. Locations where air-to-air gunnery training is performed using towed targets and drones, ensure that the aircraft performance capabilities and the appropriate munitions ballistics tables are used to determine the airspace and surface area required for safe operation. **(T-1)**

2.9.52. When PTRs and non-Major Range and Test Facility Base (MRTFB) ranges, have been declared in excess to the range requirements by the range operating authority, the MAJCOM and AF/A3TI, notify the supporting CE unit with real property responsibility of the excess status. **(T-1)** Obtain MAJCOM approval for any government-owned range equipment or facilities to be deactivated, transferred, or decommissioned. **(T-2)**

2.9.53. Submit requests to modify existing range Combat Training Systems (CTS) to ACC/A3A using Air Force Form 1067, *Modification Proposal*. **(T-1)**

2.9.54. Ensure all range radio frequency spectrum and communication requirements are met IAW **Chapter 4**.

2.9.55. Designate in writing, instructor personnel to conduct electronic warfare range/electronic warfare site training for government personnel.

2.9.56. Authorize laser/directed energy weapon operations on the range only after the laser/directed energy weapon and the range have been certified for use by the 711 Human Performance Wing, Bioeffects Division (711 HPW/RHD). Ensure the range laser/directed energy weapon certification is re-accomplished before it expires. Coordinate the use of lasers/directed energy weapons by any non-Air Force unit on Air Force ranges with 711 HPW/RHD. **Note:** Research and Test and Evaluation activities involving laser devices/directed energy weapon do not require range certification if conducted in accordance with DoDI 5000.89_DAFI 99-103 and applicable research and test management instructions.

2.9.57. All test ranges will upload their contact information and test capabilities in the Test Resource Management Center's Test Capabilities Directory found at: **https://tcd.dtic.mil/tcd/**. **(T-1)** The Test Capabilities Directory will be updated annually by the first month of the fiscal year. **(T-1)**

2.9. (EDWARDSAFB) Range Operating Authority. The 412 TW/CC is responsible for operating the Edwards AFB PIRA and will serve as the Range Operating Authority.

2.10. Installation/Center/Wing Chief of Safety will:

2.10.1. In coordination with the range operating authority, develop and implement required programs, practices, and procedures to reduce risk to the public, personnel, and resources, and maximize mission accomplishment. **(T-1)**

2.10.2. Advise commanders on range safety requirements for all operations. **(T-1)**

2.10.3. Provide trained and certified experts for range safety oversight and guidance on operational and test and evaluation issues. **(T-1).**

2.10.4. Provide mission safety requirements for operational and test and evaluation procedures. **(T-1)**

2.10.5. Develop standardized range safety requirements. **(T-1)**

2.10.5. **(EDWARDSAFB) Range Safety Requirements for PIRA Operations.** All 412 TW Range users must comply with the Test Safety Review process in accordance with USAF and Air Force Test Center guidance. Users seeking to employ new systems or methodologies must undergo an independent Test Safety Review process to allow an independent team to formally review the test unit's safety planning and to ensure such employment may be safely executed on the PIRA. The Test Safety Review process will review all hazards, will determine if the PIRA can support the activity, and will recommend sufficient mitigation procedures on items such as a footprint, containment area, additional radio calls as needed for test safety, a deviation, or an access restriction. Range users must be prepared to submit technical documents and any previous hazard analysis. Range users must proactively coordinate with the 412 TW Safety Office, must be aware that early insertion into the Test Safety Review process is essential to determining if a new system is ready to be scheduled for test on the PIRA, and must be aware that test scheduling delays can occur if documentation or coordination is not completed. For any class of PIRA range operations, the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer will be the Range Operations and Maintenance representatives to provide additional expertise or optional reviewer coordination as required by the Test Safety Review process.

2.10.6. Manage the safety mishap reporting program. **(T-1)**

2.10.7. Investigate, report, and identify corrective actions for range safety deficiencies, high accident potentials, and mishaps. **(T-1)**

2.10.8. Conduct analysis and advise commanders of in-flight impact, explosive, toxic, laser, directed energy, radiological, and acoustic hazards. **(T-1)**

2.10.9. Review and coordinate all range operations and training documentation. **(T-3)**

2.10.10. Assist in the development of a Wildland Fire Management Plan. **(T-3)**

2.10.11. Before exposing people, equipment, or the environment to hazards during test and evaluation, obtain and use safety releases from program offices. **(T-1)**

2.11. Range Safety Personnel will:

2.11.1. **(All range safety officers and Flight Safety Officers)** Advise the range operating authority on the safe conduct of range operations.

2.11.2. Test safety personnel have additional safety responsibilities and reflect those of the person conducting on-site or direct, real-time monitoring of range operations. Duties may vary by the type of operation: For Class T services, the range safety officer duties are typically performed by the range control officer/test area control officer. For tests requiring the monitoring of a system or operation from a mission control facility (e.g., a weapon flight test with a Flight Termination System installed) the safety function is usually performed by a range safety officer on a console. For small test operations (e.g., a munitions/explosives test) the on-site safety function may be performed by the Test Director/Manager. Regardless of the personnel or event, the range safety officer will:

2.11.2. **(EDWARDSAFB) Performing the On-Site Range Safety Officer Duty.** For each class of PIRA range operations, the on-site Range Safety Officer duty will be performed as follows: when a PIRA Class A range operation supports a test or a non-test mission activity and the mission requires no real-time monitoring of a system or operation, then the Range Safety Officer duty will be performed on-site at DOWNFALL Tower by the 412 RANS Range Control Officer; when a PIRA Class A range operation supports a specific test mission activity, such as use of a Flight Termination System, and the mission requires real-time monitoring of a system or operation from a 412 RANS mission control room, then the Range Safety Officer duty to monitor a system or operation will be performed on-site from a 412 RANS mission control room console by the 412 TW Range Safety Officer. Concurrently, the 412 RANS Range Control Officer at DOWNFALL Tower continues to perform existing safety responsibilities to control flight or ground range operations during this specific test mission activity occurring on the PIRA. Safety procedures shall be established and coordinated with the 412 RANS Range Operations Officer for this specific test mission activity to ensure clear delineation of safety responsibilities between the 412 RANS Range Control Officer and the 412 TW Range Safety Officer; when a PIRA Class B range operation supports a Flight Lead Control mission activity, then the Range Safety Officer duty will be performed on-site at DOWNFALL Tower or at a designated PIRA Tower site by the 412 RANS Range Control Officer; when a PIRA Class C range operation supports a mission activity, then the Range

Safety Officer duty will be performed on-site at DOWNFALL Tower or at a designated PIRA Tower site by the 412 RANS Range Control Officer; there is no PIRA Class D range operation that is performed by the 412 RANS; there is no PIRA Class T range operation that is performed by the 412 RANS; and there is no PIRA small munition test or small explosive test range operation mission activity that is performed by the 412 RANS.

2.11.2.1. Be on-site at the operation or in a position to have real-time situational awareness of events, personnel, and assets. **(T-1)**

2.11.2.2. Receive a briefing from (coordinate with) range control/safety on hazards and mitigating measures associated with the test activity. **(T-1)** This may vary from the range safety standard operating procedures with respect to the restrictions and mitigating measures established by the DoDI 5000.89_DAFI 99-103 mandated safety review.

2.11.2.2. **(EDWARDSAFB)** The 412 TW Range Safety Officer must brief the aircrews on the concerns and mitigation procedures on an item such as a footprint, containment area, a delegated radio call as necessary, a deviation, or an access restriction in order to ensure that aircrews are aware of a known risk and can evaluate this risk against acceptable parameters for test mission execution. The developed brief will include discussion of weapons release authority, safety oversight criteria for a weapon or object release, arm and release clearance calls, flight path conditions, ordnance hazard footprint, monitoring of flight conditions, and keeping positive communications with the aircraft for safe ordnance release within the designated impact area. This developed briefing must be given to the aircrews no later than 3 days prior to mission execution. After coordination with the program test or safety engineers, and analysis of the test or training program activity requirements, the 412 TW Range Safety Officer will provide the Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and the mission aircrew with mission information to include approved release profiles and footprint data no later than 3 days prior to mission execution. The Range Operations Officer and the DOWNFALL Tower Range Control Officer will review the release profile(s) and footprint data and coordinate with the 412 TW Range Safety Officer to resolve any mission support execution concerns.

2.11.2.2.1. **(Added-EDWARDSAFB)** If deviations to the mission release profile or footprint need to occur, then the 412 TW Range Safety Officer must complete coordination with the Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and the Detachment 7, Air Force Research Laboratory (Det 7, AFRL) representative (as applicable) no later than 3 days prior to mission execution. If a safety concern for the deviation is not resolved before mission execution, the Range Operations Officer in coordination with the DOWNFALL Tower Range Control Officer will notify the 412 RANS Operations Duty Office to cancel the PIRA mission support.

2.11.2.3. Ensure an applicable footprint/hazard area has been applied to the activity. **(T- 1)**

2.11.2.3. **(EDWARDSAFB)** The 412 TW Range Safety Office will obtain and verify the appropriate test weapon data or training weapon data from the program engineer in order to calculate an impact footprint and provide the footprint data to the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and the Det 7, AFRL representative (as applicable) to review.

2.11.2.3.1. **(Added-EDWARDSAFB)** The 412 TW Range Safety Officer will provide a range safety Surface Danger Zone or Weapon Danger Zone test footprint or test hazard area for a test weapon used during a test activity no later than 7 days prior to mission execution to the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and the Det 7, AFRL representative (as applicable) for review.

2.11.2.3.2. **(Added-EDWARDSAFB)** The 412 TW Range Safety Officer will provide a range safety Surface Danger Zone or Weapon Danger Zone training footprint or training hazard area for an inventory weapon used during a training activity no later than 7 days prior to mission execution to the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and Det 7, AFRL representative (as applicable) for review.

2.11.2.3.3. **(Added-EDWARDSAFB)** The 412 TW Range Safety Officer will provide the reviewed and approved procedure for a mitigation(s), a restriction(s), and a master arm point for a test mission involving a weapon release, or a master arm point when an aircrew is planning to drop, to fire, or to launch ordnance or an object with an undetermined ballistic characteristics no later than 7 days prior to mission execution to the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, the SPORT MRU Supervisor, and the Det 7, AFRL representative (as applicable) for review. The aircrew must review the procedure to ensure the mitigation(s) or restriction risk(s) are acceptable to execute the test mission.

2.11.2.3.4. **(Added-EDWARDSAFB)** All test or training programs planning to conduct live or inert ordnance delivery will contact the 412 TW Range Safety Office at **412TW.SETR.Workflow@us.af.mil**, DSN 527-5297, or Commercial 661-277-5297, a minimum of 15 days in advance of a scheduled mission to provide data on ordnance type, flight profile, and planned release conditions.

2.11.2.3.5. **(Added-EDWARDSAFB) PIRA Pyrotechnic Employment Procedures and Restrictions.**

2.11.2.3.5.1. **(Added-EDWARDSAFB)** Pyrotechnics may only be expended within approved and scheduled PIRA ground profiles as determined by the 412 TW Range Safety Office. The use of pyrotechnics within 100 meters of the PIRA reservation boundary is prohibited.

2.11.2.3.5.2. **(Added-EDWARDSAFB)** 412 TW units will coordinate with the 412 TW Safety Office and 412 RANS Range Operations Officer to determine if pyrotechnics can safely be used on the PIRA to support test or training requirements. 412 TW units will operate pyrotechnics in accordance with guidance as established in DoD service directives.

2.11.2.3.5.3. **(Added-EDWARDSAFB)** Smokey Surface-to-Air Missile Simulators have a 1,500-foot radius hazard area from the launch point and extends to 2,000 feet above ground level. With a hardtop shelter to protect from falling Smokey Surface-to-Air Missile debris, PIRA personnel are permitted no closer than 100 feet from the launch point.

2.11.2.4. Obtain status of the range and authorization to conduct activity/fire from range control. **(T-1)**

2.11.2.5. Ensure the hazard area is clear. **(T-1)**

2.11.2.6. Ensure proper Personal Protective Equipment is worn and applicable safety measures are in place. **(T- 1)**

2.11.2.7. Monitor flight path and status of weapon/asset, range boundaries and areas of concern; execute Flight Termination System as required. **(T-1)**

2.11.2.8. Suspend activity in event of mishap or significant unforeseen hazard. **(T-1)** Range safety personnel will contact Wing/Center Safety as resumption of activity may require Test Execution Authority authorization and a mishap investigation. **(T-1)**

2.11.3. Obtain training and certification as required by the range operating authority. **(T-3)**

2.12. Installation Civil Engineer will:

2.12.1. Provide, sustain, restore, and modernize the range infrastructure, facilities, and environment necessary to support range operations. **(T-1)** The supporting CE unit augments these functions if organic or contracted CE teams assigned to the range are not sufficient.

2.12.2. Maintain an accurate inventory of Air Force-controlled range real property in the Real Property Accountable Property System of Record (APSR), NexGen IT, or legacy system if not transitioned. **(T-1)**

2.12.3. Establish standards for the comprehensive planning, design, construction, operation, revitalization, and maintenance of range real property facilities to sustain their value to the Air Force, consistent with Air Force policies. **(T-1)**

2.12.4. Assist in the development and submission of funding requests for range infrastructure and facility requirements. **(T-1)** For environmental requirements, assist in the development of and coordination of funding requests with the AFCEC. **(T-1)**

2.12.5. Deliver services and products normally associated with facility management and operations to include real property transactions; construction, maintenance, and revitalization; utilities; and support of real property installed equipment. **(T-1)**

2.12.6. Provide EOD support for range clearance operations IAW AFMAN 32-3001. **(T-1)**

2.12.7. Review, coordinate, or approve all range-related capability documents to ensure compatibility with community planning, EOD, UXO, environmental regulatory requirements, and range clearance operations. **(T-1)**

2.12.8. Coordinate range-related environmental programs, plans or actions with the range operating authority or designated representative (e.g., range operations officer) prior to final signature or implementation. **(T-1)**

2.12.9. Act as the liaison office or delegate this responsibility, for compliance issues with environmental regulatory agencies. **(T-1)**

2.12.10. Establish local procedures and provide subject matter expertise with regard to environmental compliance and environmental program requirements. **(T-1)** To facilitate this, the supporting CE unit will:

2.12.10.1. Provide education and training, to include shop-level training, to meet environmental compliance requirements, as needed. **(T-1)**

2.12.10.2. Review all environmental permits and permitting requirements to ensure that ranges are in compliance. **(T-1)**

2.12.10.3. Prepare and implement Federal, DoD, and DAF-required environmental plans (e.g., Integrated Natural Resources Management Plan, Wildland Fire Management Plans, and Integrated Cultural Resources Management Plans) IAW associated AFIs and applicable guidance. See the applicable laws and guidance publications, listed in **Attachment 1**, for detailed guidance.

2.12.10.4. Develop procedures to address conflicts between military mission objectives and environmental and cultural resource constraints. **(T-1)**

2.12.10.5. Serve as the installation-level environmental planning function for applicable EIAP issues and documents concerning Air Force-operated ranges IAW 32 CFR Part 989. **(T-0)**

2.12.10.6. Provide environmental consultation and support for compliance with the Military Munitions Rule under Resource Conservation and Recovery Act. **(T-1)**

2.12.10.7. Track and report toxic and hazardous material information IAW Title 42 U.S.C. Sections 11001-11050, *Emergency Planning and Community Right-to-Know*, including annual Toxic Release Inventory reporting.

2.12.11. Implement, manage, and provide support for the range-related portion of Installation Geospatial Information and Services Program IAW AFI 32-10112. **(T-3)** Maintain consistent range geospatial data visualization, analysis, and integration capabilities IAW Air Force data strategies and associated standards.

2.12.11.1. Support range data stewards to assure quality and protection of range geospatial data, to include recommendations for safeguarding Geographic Information System data stored within the range organization.

2.12.11.2. Facilitate the identification of (and access to) those geospatial data that are maintained through the GeoBase Service.

Chapter 3

COMPREHENSIVE RANGE PLANNING

3.1. Comprehensive Range Planning. Comprehensive range planning is accomplished to ensure training and testing requirements are met. This process evaluates range requirements against current and projected capabilities to identify shortfalls and to guide sustainable range development to mitigate shortfalls. This manual and a hierarchy of plans provide the policy and guidance that documents the current sustainment and future range development.

3.1.1. Operational Test and Training Infrastructure Flight Plan. AF/A3TI drafts and publishes the Operational Test and Training Infrastructure Flight Plan. Since ranges are part of the Air Force's operational training infrastructure, the Operational Test and Training Infrastructure Flight Plan provides the long-term strategic vision and direction for Air Force ranges. The plan establishes a vision of a realistic, integrated training environment that allows forces to train mission relevant employment schemes to achieve and sustain full-spectrum readiness. It also describes the long-term operational test and training infrastructure requirements to achieve the vision set forth in the 2015 Strategic Master Plan and Air Force Future Operating Concept.

3.1.2. Enterprise Range Plan. AF/A3TI will produce a Range Capabilities Development Plan in coordination with the MAJCOMs. The plan should have a 10-year planning horizon and support achievement of the vision established in the Operational Training Infrastructure Flight Plan.

3.1.2.1. AF/A3TI will ensure the Range Capabilities Development Plan identifies the current mission of each range and the range requirements. This includes detailing necessary test and training capabilities as well as specifying training requirements for assigned users. It should compare current and known future requirements against the current range capabilities to identify shortfalls and describe investment actions.

3.1.2.2. Range Capabilities Development Plan development provides an opportunity for all MAJCOM organizations and staff to provide input, coordinate actions within the command, and across MAJCOMs.

3.1.2.3. The Range Capabilities Development Plan shall take into consideration climate change-related risks and opportunities (e.g., sea level rise, shifts in precipitation, increases in droughts or floods, increases in severe weather events and temperature). Authoritative climate data will be obtained from the 14th Weather Squadron (**https://climate.af.mil**).

3.1.3. Air Force Training Requirements. Range training support requirements are based on the Mission Design Series (MDS), Designed Operational Capability statement missions, and training events of the assigned units. The Ready Aircrew Program (RAP), Major Weapon System Volume 1, training publications and syllabi normally provide the specific events that require range support. Attachments **3** and **4**, "Range Supported Mission Design Series Training Events" and "Assigned Range Users and Training Events," are a compilation of the applicable events, required range capability and infrastructure for each specific training event and are based on current practices. The MDS-specific figures in **Attachment 4** list the units, equipped with that particular MDS, that normally utilize specific ranges and the various training events that they may conduct on those specific ranges. A unit being listed in one or more of the figures in **Attachment 4** does not imply priority for access to, or scheduling on, those specific ranges;

nor does a unit not being listed in those figures imply that the unit does not have access to those specific ranges. Access to a specific range should be coordinated between the unit leadership and the respective range operating authority. As MDS, mission types or training events, and/or tactics, techniques, and procedures change to meet emerging threats and technologies, the MAJCOM Realistic Training Review Boards are instituted to maximize the quality of unit training and convene (usually annually) to update training events, frequency, and standards.

3.1.3.1. At a minimum, MAJCOMs shall biennially review applicable portions of the Combat Air Forces, Mobility Air Forces, and Special Operations Forces Realistic Training Review Board recommendations; range user assignments; the range-supported MDS events; and other applicable training requirements documents. Following the review, the MAJCOMs, through ACC as the lead MAJCOM, will provide AF/A3TI with inputs on recommended changes to Attachments **3 and 4**.

3.1.3.1.1. The biennial review should ensure a specific range has been tasked to support each specific training event and that it has the required capability to do so. MAJCOMs must support shortfalls necessitating a change to assigned users, or capability and infrastructure investments based on the published data.

3.1.3.1.2. User-event assignments should consider geographic proximity as well as the scope and attributes required for the assigned event. Units will not be assigned to more than one range for the same training event.

3.1.3.1.3. When a MAJCOM does not have an adequate or close-proximity range to support one of its units, the MAJCOM will engage other range-owning MAJCOMs and services for support and assignment of their units. If the shortfall cannot be serviced, the proponent MAJCOM will submit a T/TSNS to AF/A3TI for resolution.

3.1.3.1.4. During the biennial review, MAJCOMs should review test and training requirements to determine the live and synthetic mix and if changes to range requirements are needed. This drives a continuous modernization process that meets test objectives, increases combat realism, aids tactics development, enhances day-to-day training, and moves toward a balanced and appropriate live and synthetic mix.

3.1.3.2. Non-Air Force Training Requirements at PTRs. Support of non-Air Force unit training events on Air Force ranges is often based on historical agreements and/or related to basing decisions. This support may establish additional range financial obligations and requirements and range operating authorities must document this with a written agreement as outlined in **paragraph 4.2**. **(T-1)**

3.1.3.2.1. The range operating authority, in coordination with the MAJCOM, will identify the specific events, training priority, and related range capabilities and infrastructure needed to support the training **(T-2)**; and will determine if the range has the capacity to support the event(s) while continuing to meet its obligations to the assigned Air Force users. **(T-2)**

3.1.3.2.2. If additional operating hours, capabilities, equipment or infrastructure are necessary to support the training, the range operating authority will document those costs in the written agreement and coordinate for reimbursement as authorized IAW **paragraph 4.2**. **(T-1)** Ranges will not assume training that incurs additional costs to support users that have not been assigned to them in this manual, unless documented in a written agreement which has been approved by the range-operating MAJCOM and AF/A3TI. **(T-1)**

3.1.3.3. Training Priority at PTRs. Priority is normally determined by the range- operating Wing. The range-operating Wing and individual ranges should schedule effectively to adequately support each assigned user in the assigned event as well as non- Air Force users who have entered into a written agreement for support. Ranges are authorized to provide range access and training support on a space-available basis to non- assigned units, but those units should not normally have priority above assigned users.

3.1.3.4. Training-Support Requirements at MRTFB Ranges. IAW DoDD 3200.11, *Major Range and Test Facility Base (MRTFB),* the MRTFB may be used by DoD users and by users outside the Department such as United States Government Agencies, State and local governments, allied foreign governments, and commercial entities for training events. Users must clearly articulate training requirements to determine the level of support needed and the priority of the activity.

3.1.3.5. Training Priority at MRTFB Ranges. IAW DoDD 3200.11, scheduling of the MRTFB is based upon a priority system that gives equitable consideration to all DoD Components and accommodates DoD acquisition program priorities. The Test Resource Management Center implements the composition, sizing, and usage for the MRTFB. Priority for training operations on MRTFB ranges is coordinated at the local level through a robust scheduling and prioritization process that takes into account both test and training requirements needed to support all users.

3.1.3.6. Range Test-Support Requirements. Test requirements are based on the characteristics and attributes of the specific test and are governed by departmental and service-level directives and instructions. MAJCOMs, acquisition programs, SAF/AQ, and AF/TE collaboratively define the test requirements and the resources necessary to support them.

3.1.4. Test and Training Resource Programs. Test and training requirements are funded through a combination of institutional funds and reimbursements and/or customer fees at MRTFB ranges. Test and training resource funding is managed through Air Force and other DoD unique PEs. Execution is accomplished by various Air Force field activities. Headquarters Air Force Test and Evaluation Policy, Programs, and Resources Division (AF/TEP) provides test and evaluation resource oversight and guidance to the MRTFB ranges through the parent MAJCOM. AF/A3TI provides training resource oversight and guidance to MRTFB ranges and together with the ANG, Air Space and Ranges Division (NGB A2/3/6/10YR) to PTRs.

3.1.4.1. Aligning and Deconflicting Test and Training Requirements. Within the Air Force, the investments needed for test requirements may also be needed for training, and vice versa. Infrastructure investments at MRTFB ranges are planned and allocated through different processes, PEs and special access programs. It is imperative to avoid redundancy and gain synergy by closely aligning and deconflicting test and training requirements.

3.1.4.1.1. Training O&M. PE 27604 and PE 52634 fund PTR O&M, and are programmed by ACC and NGB A2/3/6/10YR (respectively) through the Air Force Operational Training and Readiness (OT&R) Panel. Individual MAJCOMs execute funding in these PEs for their assigned ranges. The program funds range contracts, certain range personnel and O&M of range capabilities. It also funds technical and service contracts to maintain operating hours and the basic infrastructure for scoring activities, threat and debrief systems, data link capabilities and other range services such as road maintenance and range clean up.

3.1.4.1.2. Nevada Test and Training Range and Utah Test and Training Range O&M. PE 27428 funds Nevada Test and Training Range and Utah Test and Training Range O&M, and is programmed by ACC through the Air Force Operational Training and Readiness (OT&R) Panel. ACC is also the executing agent for this program. The program funds range contracts, certain range personnel, and O&M of range capabilities, technical and service contracts to maintain operating hours and the basic infrastructure for scoring activities, telemetry, threat, and debrief systems, data link capabilities and other range services such as road maintenance and range clean up.

3.1.4.1.3. Foreign military use of air-to-air/air-to-ground training ranges/airspace must be coordinated with appropriate country offices within SAF/IA, AF/A3TI, Air Force Security Assistance Training, MAJCOM offices supporting the foreign partner (MAJCOM/IA for Foreign Military Sales, A4 for Acquisition and Cross-Service Agreements, etc.), and MAJCOM staff Range and Airspace Operations authorities. As part of the coordination and approval process, MAJCOM offices supporting the foreign partner will coordinate with MAJCOM respective finance and Range and Airspace authorities (contract program managers) to determine the range/airspace usage fees which will include all direct and indirect costs to include, but not limited to: range contracts, certain range personnel performing O&M of range capabilities, technical and service contracts to maintain operating hours and the basic infrastructure for targets/scoring activities, telemetry, threat and debrief systems, data link capabilities, and other range services such as range clean-up. This level of coordination is done with the intent to satisfy the Arms Export Control Act provision of no cost to the U.S. government.

3.1.4.2. Improvements/Investments.

3.1.4.2.1. Test and Training Space. Test and training space attributes accommodate missions based on tactics, techniques, and procedures and event requirements. When range operating authorities/MAJCOMs project a unit readiness decline due to a deficiency or gap in range capabilities, MAJCOMs should first attempt to shift the unit assignment to a range with the required capability, or submit a T/TSNS to meet the shortfall.

3.1.4.2.1.1. Actions involving the fielding of new capabilities, changes in land use or the acquisition of new land or airspace typically generate long-term sustainment costs, environmental impacts, and public interest. Therefore, any action to establish, modify, or delete test/training space except those purely administrative in nature (i.e., changes that do not impact the capabilities or capacity of the range), must be reviewed by units, MAJCOMs, and Headquarters Air Force via the T/TSNS process detailed in AFI 13-201. **(T-1)** Examples of actions requiring a T/TSNS include: acquiring new land or airspace; lowering the minimum altitude of a Military Operations Area; fielding a new capability with a recurring sustainment cost; moving or establishing a live munitions area. Equipment replacement governed by a program office does not require a T/TSNS. Actions that only impact ANG equipment purchased with National Guard and Reserve Equipment Account funds do not require a T/TSNS.

3.1.4.2.1.1.1. Requests from non-Air Force entities to move onto an Air Force range must be vetted through the Strategic Basing Process as defined in AFI 10-503, *Strategic Basing*. Additionally, approved non-Air Force tenants desiring to increase their manpower by 35 or more positions must also receive approval from the Air Force Strategic Basing Process.

3.1.4.2.1.1.2. Range operating authorities and MAJCOMs must follow the guidelines and approval process established in DoDI 3200.18, *Management and Operation of the Major Range and Test Facility Base (MRTFB)*, for any changes/closures to MRTFB ranges that affect the MRTFB capabilities or capacity. **(T-0)**

3.1.4.2.1.1.3. For proposals involving expansion of range boundaries, whether permanent or otherwise, range operating authorities and MAJCOMs must also coordinate the T/TSNS with the AFCEC Military Munitions Response Program office to identify conflicts and impacts related to current or projected Air Force Military Munitions Response Program projects. **(T-1)**

3.1.4.2.1.2. Airspace and Land Actions.

3.1.4.2.1.2.1. Airspace. When establishing a need for new airspace, proponents must ensure they comply with AFI 13-201 and applicable FAA Orders and Directives. **(T-0)**

3.1.4.2.1.2.2. Land Acquisition. Land acquisitions require lead times as long as 7 years. There are a variety of mechanisms for acquiring land. Examples include traditional buy/lease transactions, or withdrawal under the Federal Land Policy and Management Act (see also “Land Withdrawal” in terms section of **Attachment 1**). Once the T/TSNS is approved, AF/A3TI will work with the proponent and the Deputy Assistant Secretary, Installations (SAF/IEI) to pursue the correct mechanism.

3.1.4.2.1.2.3. Closing or Transferring Ranges. MRTFB ranges must follow Office of the Secretary of Defense processes before reducing or eliminating capability IAW DoDI 3200.18. **(T-0)** For PTRs and non-MRTFB ranges, when declared in excess to the range requirements by the range operating authority, the MAJCOM and AF/A3TI, the land will be surface cleared IAW Air Force standards. **(T-1)** The range operating authority will notify the supporting CE unit with real property responsibility of the excess status for subsequent action to determine a new use and accomplish required clearance for the new use. **(T-1)** It is Air Force policy to clear to a level deemed safe for the proposed future land use and issue Certificates of Clearance for excess ranges or ranges converted to other uses but remaining on the installation property inventory. The supporting CE unit will ensure clearance IAW DoDI 3200.16 *Operational Range Clearance (ORC)*, excess declaration IAW AFI 32-9004, *Disposal of Real Property*, and National Environmental Policy Act actions IAW 32 CFR Part 989. **(T-0)** Return, or partial return, of overseas ranges operated or maintained by United States personnel or forces to host nation control is conducted IAW DoDI 4165.69, *Realignment of DoD Sites Overseas*. **(T-0)** Airspace transfer is IAW AFI 13-201, and range operating authorities must ensure disposition of government-owned range equipment and facilities is IAW MAJCOM guidance. **(T-2)**

3.1.4.2.2. Range Equipment and Infrastructure. Range equipment and infrastructure provide instrumentation for numerous functions, including test and evaluation, readiness training, tactics development and evaluation, command and control, safety, and real-time flight monitoring. Ideally, these systems provide precise monitoring and reconstruction, and facilitate post-mission debriefing of range users for various mission sets (i.e., single or multiple aircraft, ground training, and test missions), to include reactions to air defense threats, live or simulated ordnance exchanges, and realistic simulations of the projected threat environment. The requirement for continuous modernization of these systems drives investments in this area.

3.1.4.2.2.1. Test Equipment and Infrastructure. The Test Investment Planning and Programming Process provides the venue for investing in test infrastructure. Headquarters AFMC/A3 manages the Air Force Test Investment Planning and Programming Process to identify test resource investments needed to support military systems testing. Those investments that have possible multi-service applicability may be referred to the Central Test and Evaluation Investment Program for funding.

3.1.4.2.2.2. Training Equipment and Infrastructure. When a range projects that equipment or infrastructure deficiencies will prohibit them from supporting an assigned training event or an emerging requirement, the range operating authority should coordinate with the MAJCOM to have the mission reassigned, arrange for O&M funding, or coordinate acquisition of the required equipment or infrastructure with the lead MAJCOM (ACC). AFSOC may seek funding that supports Special

Operations Forces-unique training and has Special Operations Forces sustainment. In these cases, AFSOC will coordinate with the lead MAJCOM if it affects other Air Force range users. **(T-1)** All equipment or materials purchased for ANG range operations with state or National Guard and Reserve Equipment Account funds will remain with the State when and if the range real property is reassigned. **(T-0)**

3.1.4.2.2.2.1. The Air Force Life Cycle Management Center / Range Systems Division (AFLCMC/HBZ) and the Air Combat Command, Ranges Branch (ACC/A3AR) shall address improvement of existing CTS using feedback from equipment users, jointly establishing priorities for future efforts, developing solutions to reliability and maintainability problems, and providing updates concerning on-going efforts.

3.1.4.2.2.2.2. Headquarters ACC is the lead agency for procurement and life-cycle fleet management of instrumentation, threat systems, and test/training range engineering development supporting Combat Air Forces requirements.

3.1.4.2.2.2.3. The Air Force Operational Test and Evaluation Center will coordinate on Combat Air Forces requirements that need operational test and evaluation input or may affect Operational Test and Evaluation as well as Developmental Test and Evaluation, and Expeditionary Force Experiments. The Air Force Joint Test Program Office will coordinate on directed Combat Air Forces requirements that need Joint Test and Evaluation IAW DAFI 99-106, *Joint Test and Evaluation Program*. **(T-1)**

3.1.4.2.2.2.4. Headquarters ACC/A3A will develop and maintain an equipment inventory and configuration control system, as part of the Enterprise Range Plan. It should accurately inventory, track equipment, and document the configuration and threat lay-down plan of threat emitters and simulators used for training.

3.1.4.2.2.2.5. The reallocation of threat systems will be coordinated by Headquarters ACC/A3A with the affected MAJCOM(s) and AF/A3TI, with AF/A3 arbitrating any disputes. **Note:** This paragraph is not intended to apply to maintenance-related movements, such as transfers to/from depot.

3.1.4.2.2.3. Requests for Disposition of Equipment. Range operating authorities must obtain MAJCOM approval to deactivate, transfer or decommission any range Combat Training System. **(T-2)** All CTS deactivation/decommission and final disposition requests will be approved by the parent MAJCOM and Headquarters ACC with AF/A3TI arbitrating any disputes. This does not apply to Test and Evaluation equipment on MRTFB ranges which is managed IAW DoDI 3200.18.

3.1.4.2.2.4. Urban Training Facilities. Constructing or upgrading semi-permanent and permanent urban training facilities, as defined and covered in DoDI 1322.27, requires advance coordination. The parent MAJCOM will evaluate proposals. Proposals which are funded via Military Construction (MILCON) funding also require certification by the Under Secretary of Defense for Personnel and Readiness (USD (P&R)). Range operating authorities shall submit proposal information (to include the items listed in Enclosure 3 of DoDI 1322.27) through the parent MAJCOM and ACC to AF/A3TI prior to initiating development or procurement of these capabilities. **(T-0)** AFSOC may submit proposals directly to AF/A3TI for projects that support Special Operations Forces-unique training and have Special Operations Forces sustainment. AF/A3TI will consult with Office of the Secretary of Defense, as appropriate, to identify evaluation and certification requirements for the proposal.

Chapter 4

RANGE OPERATIONS AND SAFETY

4.1. Range Operations. Range operating authorities will ensure that range operations are conducted IAW this manual, DoDD 4715.11, *Environmental and Explosives Safety Management on Operational Ranges Within the United States*, AFI 13-201, AFI 11-214, DESR 6055.09_AFMAN 91-201, *Explosives Safety Standards*, Technical Order (TO) 1-1M-34, *Aircrew Weapons Delivery Manual*, TO 1-1M-34-1, *Aircrew Weapons Delivery Manual (classified)*, aircraft specific weapons delivery TOs, aircraft specific AFI 11-2 Mission Design Series, and United States Air Force Weapons School instructional texts. For ranges outside of the United States, range operating authorities must also ensure operations are conducted IAW DoDD 4715.12, DoDI 4715.22, and AFI 32-7091, whichever apply. For operations involving ground weapons, range operators should consult AFI 36-2654 (as applicable); weapons system TOs; test-specific procedures, and/or specific SDZ profiles for the types of operations and weapons used.

4.1.1. **(Added-EDWARDSAFB) SPORT MRU.** SPORT MRU is authorized to provide Command and Control service functions to participating aircrews in R-2515 airspace. Additional information on SPORT MRU Command and Control services is found in Edwards AFB Instruction 13-204, *Flying and Airfield Operations*. SPORT MRU provides the following:

4.1.1.1. **(Added-EDWARDSAFB)** Controls airborne access to the PIRA and the Alpha Corridor.

4.1.1.2. **(Added-EDWARDSAFB)** Provides hot or cold status advisories for the Alpha Corridor.

4.1.1.3. **(Added-EDWARDSAFB)** Uses surveillance radars to provide precise positioning, airspace surveillance, traffic advisories, and boundary calls to aircraft.

4.1.1.4. **(Added-EDWARDSAFB)** Coordinates directly with DOWNFALL Tower Range Control Officers on all air operations that involve overflight of the PIRA at or below 4,000 feet mean sea level.

4.1.1.5. **(Added-EDWARDSAFB)** Provides transfer of communications to DOWNFALL Tower Range Control Officers for aircraft operations within the confines of the PIRA boundary which require Class A, Class B, or Class C service operations, for Flight Lead Control (FLC) operations, for UAV operations, or for lasing operations.

4.1.1.6. **(Added-EDWARDSAFB)** Deconflicts aircraft using the PIRA through altitude or lateral separation.

4.1.1.7. **(Added-EDWARDSAFB)** Advises the DOWNFALL Tower Range Control Officers of the deconfliction altitude.

4.1.1.8. **(Added-EDWARDSAFB)** Has the authority to suspend or terminate all PIRA air operations and recall airspace as required.

4.1.2. **(Added-EDWARDSAFB) General PIRA Procedures.** The DOWNFALL Tower Range Control Officer will provide a trained escort(s) for mission support requirements. The DOWNFALL Tower Range Control Officers, DAGRAG Tower Range Control Officers, and COWBELL Tower Range Control Officers will:

4.1.2.1. **(Added-EDWARDSAFB)** Notify SPORT MRU when PIRA conditions change (for example, when Mercury Boulevard is closed or mission complete for a vehicle or personnel movement operation).

4.1.2.2. **(Added-EDWARDSAFB)** Direct Hot Run abort when PIRA circumstances warrant this action, and will transmit "ABORT, ABORT, ABORT" on mission frequency.

4.1.2.3. **(Added-EDWARDSAFB)** Advise SPORT MRU and mission aircrew when a violation of PIRA safety rules occurs and will determine the need to suspend PIRA operations.

4.1.2.4. **(Added-EDWARDSAFB)** Notify Det 7, AFRL Site Operations Control Center of scheduled supersonic flights through the Alpha Corridor and PIRA, through Haystack Butte, operations on the East Range, and operations on PB-6.

4.1.2.5. **(Added-EDWARDSAFB)** Monitor the assigned mission frequency as required for mission support during PIRA operations and will monitor land mobile radios.

4.1.2.6. **(Added-EDWARDSAFB)** Ensure hazard areas are clear of personnel.

4.1.3. **(Added-EDWARDSAFB) PB-13 Operations and Procedures.** The DOWNFALL Tower Range Control Officer will:

4.1.3.1. **(Added-EDWARDSAFB)** Maintain direct communication with the mission aircraft and SPORT MRU.

4.1.3.2. **(Added-EDWARDSAFB)** Permit the release of a live munition only on the PB-13 target and restrict the release to one live munition per pass on the PB-13 target.

4.1.3.3. **(Added-EDWARDSAFB)** Limit release of a live munition on PB-13 to a Net Explosive Weight (NEW) of 500 pounds or less. NOTE: The NEW limit of 500 pounds or less is based on the May 2005 Environmental Assessment of Armed Munitions Integration Testing on the PIRA.

4.1.3.4. **(Added-EDWARDSAFB)** Prohibit the release of a weapon with live sub-munitions or a release of a munition containing depleted uranium on the PIRA.

4.1.3.5. **(Added-EDWARDSAFB)** Permit the release of only precision guided munitions that use laser, GPS, and infrared precision targeting methods.

4.1.3.6. **(Added-EDWARDSAFB)** Coordinate with the 412 TW Range Safety Office to provide safety procedures and the approval on a case-by-case basis for the use of a non-precision targeting method no later than 3 days prior to mission execution.

4.1.3.7. **(Added-EDWARDSAFB)** Terminate the mission and close PB-13 in the event a munition fails to detonate.

4.1.3.8. **(Added-EDWARDSAFB)** Coordinate with the 812 CES/CED EOD unit to perform an evaluation of craters to check for uncovered weapons, inspect those munitions for condition or hazard risk, and conduct a surface sweep of the target area.

4.1.3.8.1. **(Added-EDWARDSAFB)** If the munition is located on the surface, the 812 CES/CED EOD unit will dispose of the weapon according to operational procedures. The DOWNFALL Tower Range Control Officer will resume live munition release once 812 CES/CED EOD clearance is completed.

4.1.3.8.2. **(Added-EDWARDSAFB)** If the munition is determined by 812 CES/CED EOD to be below the surface, then the DOWNFALL Tower Range Control Officer will record the event in the Expended Munitions Tracking Report. The DOWNFALL Tower Range Control Officer will resume live munition release once 812 CES/CED EOD clearance is completed.

4.1.3.9. **(Added-EDWARDSAFB)** Prohibit the release of inert munitions on the PB-13 target.

4.1.3.10. **(Added-EDWARDSAFB)** Limit no more than 100 live munition releases on PB-13 target in any calendar year to comply with environmental requirements.

4.1.3.11. **(Added-EDWARDSAFB)** Visually monitor the PB-13 target area from a location outside the Surface Danger Zone or Weapons Danger Zone footprint area to provide visual confirmation of the live munition detonation.

4.1.4. **(Added-EDWARDSAFB) AFRL 1-36 Site Procedures.** AFRL 1-36 Pad D is within the East Range boundary. When the Det 7, AFRL Motor Behavior Complex Experimental Area 1-36 Pad D is active, the Det 7, AFRL personnel who are assigned to 1-36 Pad D are not required to obtain permission from DOWNFALL Tower Range Control Officer to enter the East Range as long as these personnel travel the specified route to and remain within the assigned 1-36 Pad D area boundary. Det 7, AFRL personnel are required to contact DOWNFALL Tower Range Control Officer if their travel requires them to travel off the specified route to or from 1-36 Pad D or extend beyond the assigned 1-36 Pad D area boundary.

4.1.4.1. **(Added-EDWARDSAFB)** When 1-36 Pad D area is not active, the overflight altitude restriction is established as the same overflight altitude as the rest of the PIRA. NOTE: Refer to the R-2515 Users Handbook, Airspace and Flying Procedures, for the "No Fly Areas" avoidance restrictions and the altitude restriction to not overfly the AFRL below 5,300 feet mean sea level.

4.1.4.2. **(Added-EDWARDSAFB)** The Det 7, AFRL Safety Office will notify the DOWNFALL Tower Range Control Officer when 1-36 Pad D area is active as scheduled and the established overflight altitude restriction. Subsequently, the DOWNFALL Tower Range Control Officer will notify the 412 RANS Operations Duty Officer when 1-36 Pad D area is active as scheduled and the established overflight altitude restriction.

4.1.4.3. **(Added-EDWARDSAFB)** The Det 7, AFRL Safety Office will notify the DOWNFALL Tower Range Control Officer when 1-36 Pad D area is mission complete. Subsequently, the DOWNFALL Tower Range Control Officer will notify the 412 RANS Operations Duty Officer when 1-36 Pad D area is mission complete.

4.1.5. **(Added-EDWARDSAFB) Aircrew Alpha Corridor and PIRA Procedures.** All aircrews utilizing the Alpha Corridor and PIRA will:

4.1.5.1. **(Added-EDWARDSAFB)** Contact SPORT MRU and the DOWNFALL Tower Range Control Officer prior to departure to confirm the scheduled mission event for the Alpha Corridor and PIRA, to provide ordnance release parameters if dropping ordnance, and to provide parameters for combat laser, non-eye safe laser, training laser, or eye-safe laser operations.

4.1.5.2. **(Added-EDWARDSAFB)** Coordinate entry point for the Alpha Corridor or the PIRA with SPORT MRU. SPORT MRU will coordinate and approve specific missions to be released to the DOWNFALL Tower Range Control Officer. SPORT MRU will transfer control of the aircraft to DOWNFALL Tower, COWBELL Tower, or DAGRAG Tower, as coordinated, for mission control within the PIRA.

4.1.5.2.1. **(Added-EDWARDSAFB)** Request Alpha Corridor and PIRA clearance from SPORT MRU on mission frequency or on UHF 343.7 MHz (Channel 5) prior to entry.

4.1.5.2.2. **(Added-EDWARDSAFB)** Call SPORT MRU for Alpha Corridor clearance off the PIRA when the mission event is complete. The DOWNFALL Tower Range Control Officer will notify SPORT MRU via the hot line, "CALL SIGN, MISSION COMPLETE, RETURNING TO SPORT CONTROL".

4.1.5.3. **(Added-EDWARDSAFB)** Be familiar that entry to the East Range is generally from north of Highway 58 in the vicinity of Kramer Junction.

4.1.5.4. **(Added-EDWARDSAFB)** Remain at or above 5,300 feet mean sea level when overflying Det 7, AFRL, unless otherwise coordinated with Det 7, AFRL Site Operations Control Center representative, and SPORT MRU.

4.1.5.4.1. **(Added-EDWARDSAFB)** Enter PB-6 from the north or west, remain north of Det 7, AFRL, if possible, for the scheduled mission event.

4.1.5.5. **(Added-EDWARDSAFB)** Confirm the weapons release and firing information with SPORT MRU before the mission event's first run:

4.1.5.5.1. **(Added-EDWARDSAFB)** Type and quantity of weapons to release or fire.

4.1.5.5.2. **(Added-EDWARDSAFB)** Weapons are live, inert, or training.

4.1.5.5.3. **(Added-EDWARDSAFB)** Aircraft maneuvers (for example, level, dive, turning, etc.)

4.1.5.5.4. **(Added-EDWARDSAFB)** Target scheduled.

4.1.5.6. **(Added-EDWARDSAFB)** Acknowledge position advisories and PIRA constraints.

4.1.5.7. **(Added-EDWARDSAFB)** Restrict supersonic approach activity to the Alpha Corridor and PIRA Supersonic Corridor Areas. Refer to the R-2515 Users Handbook for PIRA Supersonic Corridor procedures.

4.1.5.8. **(Added-EDWARDSAFB)** Forward mission plan change(s) to SPORT MRU as soon as they are known to allow for sufficient time to coordinate the changes with the DOWNFALL Tower Range Control Officer.

4.1.5.9. **(Added-EDWARDSAFB)** Coordinate the following actions for an Alpha Corridor only mission, for a PIRA with the Alpha Corridor mission, or for a PIRA mission when under SPORT MRU control:

4.1.5.9.1. **(Added-EDWARDSAFB)** Provide the DOWNFALL Tower Range Control Officer with pre-mission planning and briefing information.

4.1.5.9.2. **(Added-EDWARDSAFB)** Use cold spotting charges for all air-to-ground weapon (for example, BDU-33) deliveries to reduce the risk of fire from spotting charges during April through September months.

4.1.5.9.3. **(Added-EDWARDSAFB)** Use of hot spots for test plans, test preparation, test build-up sorties, or for test are permissible, if required.

4.1.5.9.3.1. **(Added-EDWARDSAFB)** Use more accurate delivery methods for hot spot activity.

4.1.5.9.3.2. **(Added-EDWARDSAFB)** F-16 aircraft may use “CCIP”, “CCRP”, or “DTOS” delivery methods.

4.1.5.9.3.3. **(Added-EDWARDSAFB)** Other Mission Design Series aircraft will plan for the most accurate combination of weapon type, release parameters, and targeting method for proficient weapon deliveries.

4.1.5.9.3.4. **(Added-EDWARDSAFB)** Contact the 412 RANS Range Operations Officer DSN 525-6148 (661-275-6148) or the 412 RANS Director of Operations DSN 527-2727 (661-277-2727) to request on a case-by-case basis an exception to the delivery method policy for hot spot use.

4.1.5.10. **(Added-EDWARDSAFB)** Provide SPORT MRU and the DOWNFALL Tower Range Control Officer at DSN 525-5603 (661-275-5603) with the following range information when planning to drop, fire, and shoot an object onto the PIRA, regardless of the intent to use instrumentation radar or other range assets, no later than 1 hour prior to departure.

4.1.5.10.1. **(Added-EDWARDSAFB)** ORDNANCE:

4.1.5.10.1.1. **(Added-EDWARDSAFB)** Type, quantity of weapon(s), devices to be released, fired, and jettisoned.

4.1.5.10.1.2. **(Added-EDWARDSAFB)** Size and weight.

4.1.5.10.1.3. **(Added-EDWARDSAFB)** High drag or low drag.

4.1.5.10.2. **(Added-EDWARDSAFB)** PARAMETERS:

4.1.5.10.2.1. **(Added-EDWARDSAFB)** Type of release: single, ripple, salvo, automatic, manual.

4.1.5.10.2.2. **(Added-EDWARDSAFB)** Aircraft maneuver: dive angle, level, dive toss, turn.

4.1.5.10.2.3. **(Added-EDWARDSAFB)** Weapon is inert or live (live release is restricted to PB-13)

4.1.5.10.3. **(Added-EDWARDSAFB)** AREA:

4.1.5.10.3.1. **(Added-EDWARDSAFB)** Range: West, East, or Alpha Corridor.

4.1.5.10.3.2. **(Added-EDWARDSAFB)** Target, DAGRAG, or PB number.

4.1.6. **(Added-EDWARDSAFB) Target and Weapons Authorizations and Restrictions.** Target and weapon authorizations are provided by the 412 TW Range Safety Office. Minimum release, recovery altitudes, weapons delivery minimum altitudes, weapons employment minimum recovery altitudes, and restrictions are according to the applicable AFMAN 11-2 mission design and volume series, AFMAN 11-214 guidance, or publications.

4.1.7. **(Added-EDWARDSAFB) Joint Operating Procedures (Simultaneous Mission Operations).** Aircrews will be familiar with these procedures for conducting simultaneous mission operations. Refer to Edwards AFB Instruction 13-204 for use of the SPORT MRU Pre-Brief Sheet to provide detailed profile information.

4.1.7.1. **(Added-EDWARDSAFB)** The lead aircrew commences final coordination to de-conflict their respective operation after preliminary agreement to conduct simultaneous operations between missions or between test squadrons. Aircrews should exchange significant details of each operation to prevent conflicts. Do not place the burden of de-conflicting missions on a single user. Do not require visual separation from a drop zone at extreme altitude or when restrictions to flight visibility exist.

4.1.7.2. **(Added-EDWARDSAFB)** Aircrews from conflicting missions agree to conduct simultaneous operations. Each aircrew will contact SPORT MRU Supervisor and the DOWNFALL Tower Range Control Officer to brief their respective mission, confirm each aircrew has coordinated with the other aircrew, and confirm agreement with other aircrew to conduct simultaneous operations.

4.1.7.3. **(Added-EDWARDSAFB)** For mission changes, the aircrews of conflicting missions will contact SPORT MRU prior to departure to brief the mission changes, to verify coordination with the other aircrew of the changes and verify agreement with the other aircrew to conduct simultaneous operations with those changes.

4.1.7.4. **(Added-EDWARDSAFB)** The Survival School, Erickson and Enad drop zones are within the Alpha Corridor. PIRA missions and paradrop operations in the Alpha Corridor may operate concurrently under SPORT MRU control. Aircrews will advise SPORT MRU of specific flight paths, altitudes, and expected proximity to a drop zone when simultaneous operations involve paradrop missions. Refer to R-2515 Users Handbook Airspace and Flying Procedures for specific drop zone information.

4.1.7.5. **(Added-EDWARDSAFB)** SPORT MRU is the final authority for simultaneous or continued simultaneous operations.

4.1. (EDWARDSAFB) Range Operations for PIRA and Alpha Corridor. Range Operations for the PIRA and Alpha Corridor are scheduled according to the EAFB Scheduling Guide. The PIRA and Alpha Corridor procedures and Range Operations within the range areas are according to this AFMAN 13-212V1 EAFB Supplement. The 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officers, and SPORT MRU work cooperatively to ensure operational coordination is completed to support daily 412 TW missions.

4.2. Written Agreements. AFI 25-201 details the required procedures for entering into written agreements. Range operating authorities should closely coordinate with the MAJCOM any time the range is agreeing to provide support to an unassigned range user and a written agreement is required. DoDD 3200.11 and DoDI 5000.89_DAFI 99-103 address the written agreements and documentation for test support. Range operating authorities will ensure that all written agreements are maintained IAW AFI 33-322 and are disposed IAW the Air Force Records Disposition Schedule. **(T-1)** All agreements pertinent to their range should be readily available to range personnel and users.

4.2.1. Types of Written Agreements.

4.2.1.1. Support Agreement (DD Form 1144). In accordance with AFI 25-201, range operating authorities must use the DD Form 1144, *Support Agreement,* to document recurring reimbursable support when the Air Force is the supplier. **(T-0)** The range operating authority, in coordination with the MAJCOM, shall complete an Inter-Service Support Agreement when providing training support to any other DoD component. Specifically, an Inter-Service Support Agreement is required when a non-Air Force user from another DoD Component requests more than 12 days of range use per calendar year. **(T-1)**

4.2.1.1. **(EDWARDSAFB) Support Agreement (DD Form 1144).** According to AFI 25-201, a DD Form 1144 will be used to document recurring reimbursable support where the Air Force is the supplier. The Edwards AFB Range Operating Authority, in coordination with AFMC, shall complete an Inter-Service Support Agreement when providing training support to any other DoD component, to any non-Air Force user requesting more than 12 days of training support on the PIRA per calendar year, or to any non-Air Force user requesting a significant portion of recurring PIRA utilization for training support. The 412 RANS Support Agreement Manager, in coordination with the 412 RANS Range Operations Officer, will author or review any DD Form 1144, AF Form 7600A, or similar formatted agreement with the requesting organization that requires 412 RANS or PIRA resources, services, or capabilities, and continue coordination of the DD Form 1144, AF Form 7600A, or similar formatted agreement through existing processes used by the 412 TW, and through AFMC if necessary.

4.2.1.1.1. Ranges may entertain requests and discuss user requirements with any non-Air Force user, but the range operating authority will coordinate and receive approval from the MAJCOM prior to entering into any agreement for support of non-Air Force training events that require commitment of additional resources. **(T-2)**

4.2.1.1.1. **(EDWARDSAFB)** The 412 RANS Range Operations Officer will consider requests and discuss user requirements with any other DoD component or with any non-Air Force user. However, the Edwards AFB Range Operating Authority will coordinate and receive approval from AFMC prior to entering into any agreement for support of non-Air Force training events that require commitment of additional resources.

4.2.1.1.2. Written support agreements are not required for occasional/limited use or during Air Force-sponsored exercises, deployments, evaluations, or inspections.

4.2.1.2. Memorandum of Understanding. A memorandum of understanding defines general areas of understanding between two or more parties, when reimbursement is not required. It neither includes commitment of resources nor binds a party to a specific action. Individual units normally author memorandums of understanding and coordinate them with the parent MAJCOM. Ensure memorandums of understanding involving other nations are reviewed by the MAJCOM Judge Advocate and the State Department Mission in the host country for legal implications. See DoDI 4000.19, *Support Agreements*, Section 4, for memorandum of understanding content and format requirements.

4.2.1.2. **(EDWARDSAFB) Memorandum of Understanding (MOU).** An MOU defines general areas of understanding between two or more parties, when reimbursement is not required. It neither includes commitment of resources nor binds a party to a specific action. The 412 RANS Support Agreement Manager will author or review any MOU with the requesting organization that requires 412 RANS or PIRA resources, services, or capabilities, and continue coordination of the MOU through existing processes used by the 412 TW, and through AFMC if necessary.

4.2.1.3. Memorandum of Agreement. Memorandums of agreement document the specific terms and responsibilities that two or more parties agree to in writing, especially those that involve reimbursement or financial obligations (but without the recurring requirement which drives an AF Form 7600A). It includes either a commitment of resources or binds a party to a specific action. Individual units normally author memorandums of agreement and coordinate them with the parent MAJCOM. Ensure memorandums of agreement involving other nations are reviewed by the MAJCOM Judge Advocate and Financial Management, and the State Department Mission in the host country for legal and financial implications. See DoDI 4000.19, Section 4, for memorandum of agreement content and format requirements.

4.2.1.3. **(EDWARDSAFB) Memorandum of Agreement (MOA).** An MOA documents the specific terms and responsibilities that two or more parties agree to in writing, especially those that involve reimbursement or financial obligations (but without the recurring requirement which drives an AF Form 7600A or similar formatted agreement). It includes either a commitment of resources or binds a party to a specific action. The 412 RANS Support Agreement Manager will author or review any MOA with the requesting organization that requires 412 RANS or PIRA resources, services, or capabilities, and continue coordination of the MOA through existing processes used by the 412 TW, and through AFMC if necessary.

4.2.2. Written Agreements for Test and Evaluation Activities. DoDI 5000.89_DAFI 99-103 details the extensive documentation processes required for test and evaluation activities. MRTFB ranges may enter into written agreements via formal documentation and coordination at the range operating authority level.

4.2.2. **(EDWARDSAFB) Written Agreements for Test and Evaluation Activities.** DAFI 99-103 details the extensive documentation processes required for test and evaluation activities. The 412 TW or the 412 RANS may enter into written agreements for test and evaluation activities. The 412 TW or the 412 RANS may enter into written agreements via formal documentation and coordination at the Edwards AFB Range Operating Authority level.

4.2.2.1. **(Added-EDWARDSAFB)** The 412 RANS may enter into written agreements for range user support for a non-test and evaluation activity, for a test support activity, for a non-test support activity, or for a training activity which requires use of 412 RANS or PIRA resources, services, capabilities, personnel, or the transport of data to 412 TW or external units. The 412 RANS will evaluate the range user support request and provide a rough order of magnitude or capability and resource estimate which outlines costs. The 412 TW or the 412 RANS may enter into written support agreements such as MOAs, MOUs, DD1144s, AF Form 7600A, Statements of Capability, Letters of Agreement, or other approved DoD or 412 TW methods between a range user and the Edwards AFB Range Operating Authority or subordinate level authority for required support. The 412 RANS Support Agreement Manager will author or will review agreements to verify 412 RANS support and complete coordination with the 412 TW Support Agreement Office.

4.2.3. Foreign User Training Support Agreements. Foreign users will submit written requests for range support through appropriate country offices within SAF/IA to AF/A3TI. AF/A3TI will coordinate with the appropriate MAJCOM to determine the level of support and to integrate the support requirement into the range mission. The MAJCOM will then task the range operating authority to develop the agreement resulting in a memorandum of understanding or memorandum of agreement between the range and appropriate foreign authority. These agreements may also include specific letters of agreement, a Host-Tenant Support Agreements, other support agreements detailed on an AF Form 7600A and/or Foreign Military Sales Letter of Offer and Acceptance. Each signatory ensures that these agreements meet the needs of the organization without compromising the mission and obligating the organization beyond its intent or authority and are consistent with any applicable international agreements with the host nation. Range operating authorities will coordinate the draft agreement with the MAJCOM Office of the Staff Judge Advocate. **(T-1)** If a conflict arises regarding one of these agreements, the range operating authority should resolve the issue at the appropriate level.

4.2.3. **(EDWARDSAFB) Foreign User Training Support Agreements.** Foreign users will submit written requests for PIRA support through appropriate country offices within SAF/IA to AF/A3OJR. AF/A3OJR will coordinate with AFMC to determine the level of support and to integrate the support requirement into the 412 TW's range mission. AFMC will then task the Edwards AFB Range Operating Authority to develop the agreement resulting in an MOU or MOA between the 412 TW and appropriate foreign authority for PIRA utilization. These agreements may also include specific Letters of Agreement, a Host-Tenant Support Agreement, an AF Form 7600A, a Foreign Military Sales Letter of Offer and Acceptance, or other details of PIRA support. Each signatory ensures that these agreements meet the needs of the foreign user organization without compromising the 412 TW mission and without obligating the 412 TW beyond its intent or authority. Edwards AFB Range Operating Authority will coordinate the draft agreement with the 412 TW Office of the Staff Judge Advocate. If a conflict arises regarding one of these agreements, the Edwards AFB Range Operating Authority should resolve the issue at the appropriate level.

4.2.4. **(Added-EDWARDSAFB) Other Agency Coordination.** The Edwards AFB Range Operating Authority will ensure liaison and coordination with appropriate agency officials of federal, state, and local governments requiring use of 412 TW or 412 RANS resources. Agreements with federal, state, and local government agencies should be finalized and signed during the acquisition phase. It is important that any issues affecting 412 TW, or 412 RANS resources, services, capabilities, personnel, or safety be settled or be clearly coordinated with other agency government officials before the required activation or need date.

4.2. (EDWARDSAFB) Written Agreements and Support Agency Procedures. AFI 25-201 details the required procedures for entering into written agreements. The Edwards AFB Range Operating Authority should closely coordinate with AFMC any time the 412 RANS is agreeing to provide support to an unassigned range user. DoD Directive 3200.11 and AFI 99-103 address the written agreements and documentation for test support. The Edwards AFB Range Operating Authority will ensure that all 412 RANS written agreements are maintained according to AFMAN 33-363. All agreements pertaining to the 412 RANS resources, services, capabilities, and the PIRA shall be coordinated through the 412 RANS Support Agreement Manager for review and processing with the 412 TW Agreements Manager. The agreements pertaining to 412 RANS support will be available to 412 RANS personnel and range users.

4.3. Joint Use of Ranges. Joint use of ranges will be established by the appropriate written agreement(s), license, contract or other written test documentation between the range operating authority or test authority (in coordination with the MAJCOM) and the joint user IAW AFI 25-201 or test instructions. **(T-1)** All written agreements should ensure the range operating authority can terminate user activities detrimental to the range's natural, cultural, or physical infrastructure and obtain compensation or redress.

4.3.1. Use of Air Force Ranges for Non-Air Force Contract Air Support. PTRs can only be used by contract air support providers when it is in support of a DoD contract. MRTFB ranges may be used by contract air support providers when supporting a non-DoD contract (e.g., contracts supporting other United States Government Agencies, State, or local governments, allied foreign governments, or commercial entities). When use of an Air Force range by a contract air support provider in support of another service's contract is requested, the other service's Technical Airworthiness Authority must provide confirmation that an airworthiness assessment has been conducted and a flight release has been issued.

4.3.1.1. When another service requests use of an Air Force range for their contracted air support, the range parent MAJCOM must verify the aircraft status (civil aircraft operations or public aircraft operations) with the other service's Technical Airworthiness Authority prior to allowing the contractor to operate on an Air Force range. **Note:** An aircraft must be in public aircraft operations status in order to drop ordnance on a range and have an approved WDZ footprint, WDZ surrogate footprint, or waiver.

4.3.1.2. If a Combatant Command has contracted air support for another service's training (e.g., if USSOCOM contracts air support for Navy Seal training) and requests use of an Air Force range, the range parent MAJCOM must verify the aircraft status (civil aircraft operations/public aircraft operations) with the supported service's (the service that is receiving the training) Technical Airworthiness Authority prior to allowing the contractor to operate on an Air Force range. Combatant Commands do not have a Technical Airworthiness Authority; therefore, the service receiving the support must act as the Technical Airworthiness Authority. **Note:** For assistance with verifying airworthiness, contact the Air Force Technical Airworthiness Authority, AFLCMC/EN-EZZ, at **USAF.Airworthiness.Office@us.af.mil**. For the Army, contact USARMY CCDC AVMC, and for the Navy contact USN NAWCAD.

4.4. Shared Use of Ranges.

4.4.1. Shared Use for Test and Training Activities. Range operating authorities will address user requirements and scheduling to develop a priority policy. **(T-3)** In many instances, a written agreement between the range scheduling office and assigned range users may be appropriate.

4.4.1.1. 10 U.S.C. § 2681 authorizes the DoD to contract for the use of MRTFB resources by commercial entities. DoDD 3200.11 and DoDI 3200.18 provide guidance for these activities. Commercial activities on any range cannot compromise public safety, detract from mission accomplishment, nor impair range operations or scheduling in any way. Prioritization of MRTFB assets for commercial use is generally made available only on a non-interference basis.

4.4.1.2. PTRs are not designed or structured to accommodate test and evaluation activities as defined in DoDI 5000.89_DAFI 99-103. Should use of a PTR be necessary for test and evaluation activities, that cannot be accommodated by an MRTFB range, all applicable test and evaluation governing directives, instructions, safety and risk management procedures, necessary range enhancements, training, environmental impact analysis, additional personnel or upgrades required for the activity will be accomplished and complied with at the cost to the proponent. **(T-1)** Proponents must obtain approval from the range's parent MAJCOM. **(T-1)**

4.4.1.2.1. Requests for performing test and evaluation activities on PTRs must be submitted to the parent MAJCOM and verify the following information:

4.4.1.2.1.1. That there are no scheduling impacts to the training mission of the range.

4.4.1.2.1.2. That there are no costs incurred by the range or the range O&M contractor, to include overtime.

4.4.1.2.1.3. That the test team(s) planned to use dedicated test facilities per DoDI 5000.89_DAFI 99-103 but were deferred by the MRTFB due to capacity or other reasons.

4.4.1.2.1.4. That there are no planned permanent impacts to the range, i.e., the range will be returned to the same state as prior to the test.

4.4.1.2.1.5. That test activities are IAW the approved National Environmental Policy Act process developed for the range.

4.4.1.2.1.6. That the test team(s) is aware of the PTR's capabilities (or lack of) and potential issues such as limited instrumentation, and that any associated impacts will not adversely affect the test.

4.4.1.2.1.7. That risk management is accomplished prior to conducting any portion of the test on the PTR. The tester must provide test-specific assessment to the range operating authority for consideration, and the range operating authority must determine whether the test activity can be safely conducted on their range.

4.4.1.3. IAW DoDD 3200.11, the MRTFB may be used by other DoD users, including DoD training users.

4.4.2. Shared Use for Activities other than Test and Training. Shared use of range land with non-DoD users is encouraged when it does not compromise public safety, detract from mission accomplishment, or impair range operations. IAW DESR 6055.09_AFMAN 91-201, *Explosives Safety Standards*, public access is prohibited in areas known or suspected to contain UXO or other munitions that have experienced abnormal environments. **(T-0)** Shared use is characterized as either concurrent or non-concurrent.

4.4.2.1. Concurrent Shared Use. Civil activities may occur concurrently with range operations provided human access is prohibited in the hazard area during operations. Agriculture, grazing and timber management are examples of activities that are conducive to concurrent shared use.

4.4.2.1.1. Grazing programs can be a complementary activity on Air Force ranges. Grazing programs are conducted IAW AFMAN 32-7003, *Environmental Conservation*. Grazing programs on Department of the Interior lands withdrawn for Air Force use are generally the responsibility of the Bureau of Land Management. IAW AFMAN 32-7003, the supporting CE unit must prepare an Integrated Natural Resources Management Plan that addresses all issues related to natural resources for all ranges.

4.4.2.1.2. Other Commercial Activities. Air Force ranges may also offer other commercial uses such as timber management, agricultural out-leasing, and mining. Again, many of these programs are generally the responsibility of the Bureau of Land Management. The Bureau of Ocean Energy Management of the Department of Interior manages the oil and gas exploration on the outer continental shelf applicable to Air Force over-water ranges. For ranges located outside the United States, Combatant Command and in-theater Headquarters Air Force Component Command (e.g., USAFE-AFAFRICA, PACAF, Air Forces Central Command), issuances may prohibit or restrict commercial uses.

4.4.2.2. Non-Concurrent Shared Use. Civil use of range land may be allowed within the hazard area (exclusive of areas known or suspected to contain UXO) IAW **paragraph 4.5.2**. Recreation and education programs are often compatible because many ranges encompass large airspace, land, or water areas and may contain significant natural resources, wild areas, historical sites, or archaeological sites. As custodian of this public property, the Air Force has the responsibility to maintain it in the best manner possible consistent with the military mission. IAW AFI 32-7064, the supporting CE unit will ensure that range Integrated Natural Resources Management Plans address public access to natural resources. This plan is prepared in cooperation with the United States Fish and Wildlife Service and the state fish and game agency. Additionally, the supporting CE unit will ensure that Integrated Cultural Resources Management Plans, IAW AFMAN 32-7003, address public restrictions to range areas containing, archaeological and cultural resources, including sacred sites. The range operating authority is the approval authority for all non-concurrent shared use of range land. **(T-2)**

4.4.2.3. Enhanced Use Leases, out grants, leases, and easements. Normally Air Force ranges are not used for enhanced use leases, outgrants, leases, or easements, as they can encroach on or otherwise adversely impacts range activities and capabilities. However, if one of the real estate actions is deemed beneficial to the Air Force, the appropriate authority, as delegated by HAFMD 1-18, *Assistant Secretary of the Air Force (Installations, Environment, and Energy)*, may enter into an agreement only after the installation commander, the range operating authority, the MAJCOM, and AF/A3TI coordinate on approval of the transaction. The range operating authority may deem an enhanced use lease compatible with range activity on a case-by-case basis, provided it does not adversely impact range capability (to include MRTFB capabilities). Approval of enhanced use lease consideration should run from the range operating authority to the installation commander to the MAJCOM to AF/A3TI to SAF/IEI, in coordination with the Air Force Civil Engineering Center/Installations Directorate (AFCEC/CI). Range sustainment, repair, modernization, and other range facility investments shall be the primary objective for application of enhanced use lease proceeds derived from ranges. **(T-1)**

4.5. Range Access. MAJCOMs and range operating authorities will develop procedures for permitting and deconflicting ground party and non-military activities on the range. **(T-1) Note:** Where the terms "essential personnel," and "mission essential personnel" are used below, reference the "Terms" section in **Attachment 1** of this manual.

4.5.1. Warning Signs. In addition to the installation warning signs required by DAFI 31-101, *Integrated Defense (CUI)*, range operating authorities will post safety warning signs (multi-lingual where appropriate) on the range boundaries. **(T-1)** Refer to DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards,* for sizing and general design criteria. Where practical, signs should include both text and pictograms/symbols describing the hazard. The type of sign (e.g., "Danger" or "Warning") is selected based on the degree of hazard associated with range activities. Ranges conducting laser operations shall also post laser

warning signs utilizing the signage guidance in Military Handbook 828C, *Range Laser Safety*. **(T-1)** For ranges located on another service's/host nation's property, and where another agency is responsible for range access, security and signage, the range operating authority may apply that service/nation's rules, specified text, graphics, or procedures. Range operating authorities will ensure signs required by this section are placed along the boundaries: **(T-1)**

4.5.1. **(EDWARDSAFB) Warning Signs.** The PIRA perimeter, where crossed by roads, paths, trails, streams, streambeds or railways, will be posted with an adequate number of signs reading: "Danger – Bombing / Gunnery Range - Active Seven Days/Week". The 412 RANS Range Operations Officer will ensure all PIRA signs are in place and maintained according to current guidance.

4.5.1.1. Where crossed by roads, paths, trails, streams, streambeds, or railways.

4.5.1.1. **(EDWARDSAFB)** All access roads and trails into the PIRA, with the exception of the paved road leading to DOWNFALL Tower, will have a sign posted reading, "Danger - Bombing Range - No Entry". The road called "DOWNFALL Road" is the only authorized road leading to DOWNFALL Tower, Building 9505. Access to "DOWNFALL Road" and Building 9505 is controlled through the main gate. Visitors will contact DOWNFALL Tower at DSN 525-5603 (661-275-5603). Building 9505 is the normal daily entrance point for access and entry on the PIRA. All other PIRA access and entry routes require specific radio contact and clearance from the DOWNFALL Tower Range Control Officer to proceed. A sign reading, "Stop—Sign In Before Proceeding", is posted at DOWNFALL Tower, Building 9505, adjacent to the PIRA access entry gate. A red "Bravo Flag" is flown from a pole adjacent to the PIRA access entry gate whenever DOWNFALL Tower is manned. The red flag indicates the PIRA is active.

4.5.1.2. At 200-meter intervals where roads, paths, trails, or railways parallel the range boundary within 500 meters.

4.5.1.3. At 1000-meter intervals in other readily accessible areas not included above. **Note:** Signs are not required in areas that are not readily accessible on foot or with wheeled vehicles (e.g., mountainous terrain, swamps, lakes).

4.5.2. Hazard Areas. The hazard area is a composite of all WDZs, SDZs, Laser SDZs, and Directed Energy WDZs for all authorized weapon delivery events and represents operational hazards as well as residual hazards following munitions deliveries. Hazard areas may be segmented based upon the targets, weapons, tactics, maintenance, or clearance being utilized or conducted at a given time. If hazard areas are segmented, range operating authorities will ensure segment borders are readily identifiable by ground personnel. **(T-3)** Access during operations into the hazard area is limited to mission essential personnel, unless specifically authorized by the range operating authority. During intentional detonations, personnel shall be afforded protection standards per DESR 6055.09_AFMAN 91-201, *Explosives Safety Standards*. The range operating authority shall ensure that personnel accessing a Laser SDZ or Directed Energy WDZ have received applicable laser/directed energy weapon safety training and wear the appropriate personal protective equipment. **(T-2)** Routine access to hazard areas not in use should be limited to essential personnel. Public access to hazard areas is prohibited

until all ordnance employed in the applicable portion of the hazard area is accounted for and appropriate remediation actions (i.e., decontamination and risk management assessment) have been conducted. Access to hazard areas is determined locally based on a risk management assessment IAW **paragraph 4.18**. On ranges where another agency controls hazard area access, the range operating authority will establish procedures to verify access control and inform that agency of potential hazards. **(T-2)**

4.5.2. **(EDWARDSAFB) Mission Essential Personnel Training of PIRA Hazard and Impact Areas.** Mission essential personnel who require frequent access to the PIRA for a limited time period to perform test program or non-test program support duty may be granted unescorted entry. Before granting unescorted entry, the 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will provide unescorted access and entry training to test program or non-test program support personnel with specific PIRA access and operations focus to ensure safe conduct and awareness of hazard or impact areas. If less than 1 year access, the 412 RANS Range Operations Officer will determine the effective period of the unescorted access training for test program support personnel.

4.5.2.1. Impact areas. The impact area is that area on a range immediately surrounding the target(s) or desired point(s) of impact approved for actual ordnance delivery. Public access to impact areas is prohibited at all times. Range operating authorities will ensure when range impact area(s) or specific desired points of impact are not in use that access is limited to essential personnel. **(T-1). Exception:** Impact area access restrictions may not apply when range land use is governed the provisions of a real property permit (e.g., joint-land use permit). The impact area demarcation is determined locally using risk management assessment IAW **paragraph 4.18** but should normally be no less than either (a) 500 feet from the center of a target or desired point of impact approved for live ordnance, or (b) 300 feet from the center of a target or desired point of impact used solely for inert or practice ordnance.

4.5.2.1. **(EDWARDSAFB)** The 412 RANS Range Operations Officer and the DOWNFALL Tower Range Control Officers maintain a list of personnel assigned to the PIRA, maintains a list of personnel who routinely conduct range maintenance, or personnel who require access to maintain water wells located on the PIRA. DOWNFALL Tower Range Control Officers maintain a list of call signs for those personnel who have to perform range maintenance or require access to water wells. All other personnel who require access to the PIRA must sign in at DOWNFALL Tower, building 9505, prior to receiving approval into the PIRA, and must sign out at DOWNFALL Tower when departing. The AF Form 1109 will be used for sign in and sign out requirement. Personnel will be required to identify their name, organization, purpose of visit, and specific site to be visited on the PIRA. Personnel will receive training on use of their assigned Personnel Equipment Tracking System unit and assigned land mobile radio prior to be allowed access onto the PIRA. The Personnel Equipment Tracking System unit is a Global Positioning System transmitter that DOWNFALL Tower Range Control Officers use to monitor personnel actual location while on the PIRA. All personnel who access the PIRA will be issued a Personnel Equipment Tracking System transmitter prior to PIRA entry. DOWNFALL Tower displays situational awareness position data on personnel location to ensure personnel are out of any surface danger zone, weapon danger zone, or hazard zone during PIRA access.

4.5.2.1.1. **(Added-EDWARDSAFB) PIRA Escort Privileges Training.** The 412 RANS Range Operations Officer will grant escort privileges to mission essential personnel who have a validated need for PIRA access and entry. Escort privileges are not granted for "convenience or a desired request" of conducting test program or operational support. The 412 RANS Range Operations Officer or designated DOWNFALL Tower Range Control Officer will provide specific escort training. NOTE: Trained escorts are responsible for the escorted personnel's actions and safety on the PIRA. Failure to properly monitor and control escorted personnel can result in removal of the escort's access to the PIRA.

4.5.2.1.2. **(Added-EDWARDSAFB) Test Program Personnel Requiring Access on the PIRA.** Programs with test personnel requiring access to the PIRA will submit a written or an e-mail request to the 412 RANS Range Operations Officer a minimum of seven workdays prior to entry with the name(s), organization, mission status (essential/non-essential), function, and frequency of visits for everyone. The 412 RANS Range Operations Officer may approve access for mission essential test program personnel requiring frequent entry to the PIRA for one year. Mission essential test program personnel who are granted one year access must contact the 412 RANS Range Operations Officer to schedule annual safety or refresher training.

4.5.2.1.3. **(Added-EDWARDSAFB) Non-mission Essential Personnel on the PIRA.** Non-mission essential personnel are defined as "personnel deemed by the 412 RANS Range Operations Officer, the 412 RANS Director of Operations, or the 412 TW Safety Officer to not be mission essential or test program support essential". For example, personnel categorized as the general public, visitors, program management personnel not directly supporting test execution, the media, public affairs, and any personnel who can be excluded with no effect on PIRA operations. All non-mission essential personnel or personnel requiring infrequent access to the PIRA will always be escorted. Access to the PIRA during operations by non-mission essential personnel is solely at the discretion of the 412 RANS Range Operations Officer following a formal hazard analysis. The 412 RANS Range Operations Officer will coordinate with 412 TW Safety Office for a hazard analysis safety review. Non-mission essential personnel will receive a face-to-face PIRA safety briefing from the 412 RANS Range Operation Officer or designated DOWNFALL Tower Range Control Officer before each entry or access to the PIRA. Non-mission essential personnel will not be permitted within PIRA safety clearance zones, hazardous areas, surface danger zones, and weapon danger zones.

4.5.2.1.4. **(Added-EDWARDSAFB) Apprehension of Unauthorized Personnel on the PIRA.** The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officers will contact the 412 TW Security Forces Squadron (Law Enforcement Desk or AFRL) at DSN 527-3340 (661-277-3340) if unauthorized personnel are identified on the PIRA. The 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer can contact the AFRL Security Office at DSN 525-5550 (661-275-5550) if their assistance is needed.

4.5.2.2. WDZ. The WDZ encompasses the ground and airspace for lateral and vertical containment of a user-determined percentage of projectiles, fragments, debris, and components resulting from the firing, launching, and/or detonation of aviation delivered ordnance. This three-dimensional zone accounts for weapon accuracy, failures, ricochets, and broaches (resurfacing) of a specific weapon or munitions type delivered by a specific aircraft type under planned specific delivery release parameters.

4.5.2.3. SDZ. The SDZ encompasses the ground and airspace designated for vertical and lateral containment of a percentage of projectiles, fragments, debris, and components resulting from the firing, launching, or detonation of weapon systems to include explosives and demolitions. See the United States Army's Range Manager's Toolkit Software for specific SDZ information.

4.5.2.4. Laser SDZ and Directed Energy WDZ. Laser SDZs and Directed Energy WDZs encompass the ground area where laser radiation or directed energy levels may exceed Maximum Permissible Exposure levels thereby requiring controls. In addition, a Directed Energy WDZ may include proximity restrictions to the testing or employment of directed energy weapons with or in vicinity of conventional munitions to prevent inadvertent functioning or damage. Active Laser SDZs or Directed Energy WDZs become part of the hazard area while such equipment is in use.

4.5.3. Observation Posts. Notwithstanding paragraphs **4.5.2 and 4.11.5**, observation posts within a hazard area may be used by Tactical Air Control Parties, Joint Terminal Attack Controllers (JTAC), or other briefed personnel involved in Close Air Support (CAS)/air-to-ground training, consistent with the following criteria:

4.5.3.1. Range operating authorities will conduct a risk assessment of the proposed observation post or maneuver area prior to approving their use. **(T-1)**

4.5.3.2. Mission Essential Personnel operating inside an active hazard area must remain outside the minimum safe distance for Ground Parties (Training Use Only: Live Fire) published in Air Force Tactics, Techniques, and Procedures (AFTTP) 3-2.6(I), *JFIRE Multi-Service Tactics, Techniques and Procedures for the Application of Firepower*. **(T-1)** IAW AFI 11-214, if the planned operations or deliveries are inconsistent with the assumptions used to derive the minimum safe distance values, personnel must remain outside the WDZ.

4.5.3.3. Weapons danger zones (WDZ) are the required separation distance from the target for ground personnel. Minimum Safe Distances (MSD) for the weapon or aircraft are to be used for quick reference checks and are not to be used to circumvent range restrictions derived from the WDZs. Submit requests to the participating MAJCOM/A3, Service equivalent organization, and the range operating authority for review and approval. Personnel shall not be allowed inside the WDZ unless such approval is granted. **(T-1)**

4.5.3.4. While inside the hazard area, all personnel will wear protective gear IAW AFI 11-214 and AFI 13-112V1, *Joint Terminal Attack Controller (JTAC) Training Program.*

4.5. (EDWARDSAFB) Range Access, and Control of Personnel Access and Operations on the PIRA. The 412 RANS Range Operations Officer and DOWNFALL Tower Range Control Officers control access and entry to the PIRA. Operational control is performed from DOWNFALL Tower, which is the PIRA's Master Control Tower, call sign "DOWNFALL". The safety of personnel and property on the PIRA is of the utmost importance and shall take precedence over all other considerations. Non-EOD personnel will receive appropriate training from EOD personnel before assisting in PIRA clean-up. Authorized visitors or personnel who infrequently visit the PIRA must be briefed on range safety or must be escorted and kept under continuous surveillance by properly briefed and escort trained personnel. Visitors will be signed into the PIRA using the AF Form 1109 (Visitor Registration Log). The AF Form 1109 will be kept on file for one month. DOWNFALL Tower Range Control Officers will keep an Events Log to note daily operations and keep on file for one month.

4.6. Radio Frequency Spectrum Issues. Range operating authorities will identify the radio frequency spectrum required for range operation and coordinate requirements with the appropriate Installation Spectrum Manager. **(T-1)** The range operating authority will ensure that a frequency assignment has been obtained by the user (or appropriate Spectrum Management Office supporting the user's radio frequency requirements) prior to using any radio frequency emitting equipment. **(T-1)**

4.6.1. The range operating authority will ensure AFI 91-208, *Hazards of Electromagnetic Radiation to Ordnance (HERO) Certification and Management*, requirements are met. Seek assistance with Hazards of Electromagnetic Radiation to Ordnance requirements from the supporting installation Weapons Safety Manager prior to the use or presence of ordnance containing electrically initiated devices during range operations.

4.6.1. **(EDWARDSAFB)** The 412 TW Spectrum Management Office is responsible for resolving all harmful spectrum interferences issues on the PIRA according to AFI 17-221, *Spectrum Interference Resolution Program*, DAFI 17-220, and Allied Communications Publication 190. Mission customers impacted by interference will contact the 412 TW Spectrum Management Office to determine if interference is potentially from other scheduled customer activity. The 412 TW or 412th Operations Group will issue a "cease operations" order to unauthorized emitters or to those emitters causing harmful interference from these missions and coordinate resolution with the 412 TW Spectrum Management Office. In the event that a frequency interference source is determined to be from a non-DoD or non-Federal entity, the 412 TW Spectrum Management Office is required to immediately contact the Area Frequency Coordinator to take corrective actions to resolve the interference.

4.6. (EDWARDSAFB) 412 TW Radio Frequency Spectrum Issues. Authorized frequency clearance is issued from the 412 TW Spectrum Management Office. The 412 TW Scheduling Office is the focal point for the deconfliction and proper scheduling of all mission spectrum requirements according to the radio frequency authorizations issued from the 412 TW Spectrum Management Office. The 412 TW Spectrum Management Office is the spectrum focal point and is responsible for all range and mission spectrum requirements. The 412 TW test organizations are required to contact the 412 TW Spectrum Management Office for new or changing spectrum requirements, and to ensure spectrum requirements, to include voice communications, electronic attack, chaff, telemetry, Global Positioning System test events, and other situations are addressed according to DAFI 17-220, *Spectrum Management,* and Allied Communications Publication 190, *Guide to Spectrum Management in Military Operations*. All new or changing requirements will

be coordinated with the 412 TW Scheduling Office through the Radio Frequency Authorizations process from the 412 TW Spectrum Management Office. Site licenses or Radio Frequency Authorizations will be processed and issued by the 412 TW Spectrum Management Office to all using activities for spectrum use on the installation and the associated range. The 412 TW Spectrum Management Office will coordinate all spectrum activities with the DoD Area Frequency Coordinator through appropriate procedures as referenced in DAFI 17-220 and Allied Communications Publication 190.

4.7. Communications Requirements. The range operating authority will submit all normal communication capability requirements to the supporting wing communications function. **(T-2)** Test-specific systems used on MRTFB ranges may be obtained by local acquisition methods.

4.7.1. Ranges providing Class A, B and C services will have reliable point-to-point communication with the parent or using base and manned sites on the range. **(T-2)** During Class A service, the range control tower will have redundant communication capability with the flank tower(s) and operations building(s). **(T-2)** All parties or individuals must maintain two-way voice contact with the main tower, range office, and/or parent base while in the hazard area. **(T-1)** Portable or mobile radios may be used for backup communication and to maintain contact with personnel in the hazard area.

4.7.1. **(EDWARDSAFB) PIRA Communication Capability.** The "DOWNFALL", "PIRA", "DAGRAG" and "RANGE" communication nets used on Edwards AFB land mobile radios are monitored by the DOWNFALL Tower Range Control Officers at DOWNFALL Tower, or at the designated COWBELL Tower and DAGRAG Tower. The DOWNFALL Tower telephone number is DSN 525-5603 or commercial 661-275-5603. The DOWNFALL Tower Range Control Officers maintain an events log to record activity on the PIRA as well as mission support status or mission changes. The PIRA uses UHF and VHF ground-to-air radios to remain in contact with aircraft on the range. All clearance requests are recorded and include call signs, times, and PIRA destinations.

4.7.2. Ranges providing Class A, B and C services will have ground-to-air radios to communicate with aircraft on the range. **(T-2)** Primary systems are normally ultra-high frequency radios, but very-high frequency, or frequency modulation radios are authorized at ranges where users are similarly equipped. Provision of dual, multi- band radios should be considered to maximize mission flexibility and safety monitoring capability.

4.7.2.1. When a range provides Class A service, the range control officer is responsible for all range operations and air/occupational safety, except in cases where weapons release clearance is delegated. The range control officer must maintain continuous radio communication with all aircraft and ground personnel on the range and have a backup, independent radio. **(T-2)** The range control officer must remain in either the main or flank tower (or another MAJCOM approved location) while providing Class A service. **(T-2)**

4.7.2.2. During Class B service, ranges must have the capability to communicate with the aircraft to provide scores. **(T-2)**

4.7.2.3. During Class A and B service, ranges will ensure all ultra-high frequency, very-high frequency, frequency modulation and Land Mobile Radio voice frequencies being used on range are recorded at all times. **(T-2)**

4.7.2.4. The range control officer will ensure that all recorders are working properly. **(T- 3).** The range control officer will ensure each recording contains the range name, date, and time. **(T-3)** The range operating authority or range operations officer will establish procedures to ensure recordings are maintained for a minimum of 7 days. **(T-3)** In addition, the range control officer will keep and secure recordings containing records of any range incident until released by the investigating agency. **(T-3)**

4.7.2.5. All aircraft utilizing the range will monitor Guard frequency. **(T-0)**

4.7.2.6. Aircraft using ultra-high frequency, very-high frequency, or frequency modulation radios on the range will utilize range assigned ultra-high frequency, very-high frequency, or frequency modulation frequencies. **(T-1)**

4.7.3. Ranges providing Class D service where operations are monitored by a Range Training Officer should have ground-to-air radios to communicate with aircraft on the range and network connectivity to ground instrumentation receivers when live-monitoring of the mission is mandatory.

4.7.4. Communication Operations. Range personnel normally operate all communication equipment used on range. Communications personnel will be assigned as radio operators only when special conditions warrant (e.g., when sufficient range personnel are unavailable due to unplanned or unforeseen deployments).

4.7.4. **(EDWARDSAFB) PIRA and 412 RANS Communications Operations.** Radio communications for the PIRA to execute 412 RANS mission support, such as ground or vehicle operations, are critical to maintaining positive control during real-time mission activities. 412 RANS personnel normally operate radio equipment used on the PIRA and will observe proper radio discipline to ensure radio transmissions are brief, clearly spoken, and contain the information pertinent to the situation. When operating on the PIRA, radio operators will monitor radio communications. Radio operators will repeat the instruction received verbatim to validate the understanding of the instruction. This is especially important for PIRA hold short instructions given during vehicle operations. Do not proceed into the PIRA prior to reading back instructions.

4.7.4.1. **(Added-EDWARDSAFB) Radio Communication Techniques.** 412 RANS radio operators will use a professional, courteous voice when conducting radio transmissions on the PIRA. Personnel will not use slang words, non-standard calls, unnecessary verbiage, or abbreviations. Do not shorten the name of locations or fixes, unless it is common knowledge and published.

4.7.4.1.1. **(Added-EDWARDSAFB)** 412 RANS radio operators will avoid non-mission related communication. During aircraft operations such as when aircraft are taxiing, operating at the end-of-runway, or airborne, DO NOT transmit questions on future missions or administrative issues since this is distracting non-mission related communication to the aircrew and may lead to a potential unsafe situation. Non-mission related questions can be raised with the aircrew after the mission is completed.

4.7.4.1.2. **(Added-EDWARDSAFB)** 412 RANS radio operators will listen to the PIRA communications between other radio operators and DOWNFALL Tower Range Control Officers before transmitting on the radio. Operators should formulate phraseology before keying the radio transmitter. Speech should be short, precise, and to the point in a conversational tone. Allow a pause between keying the transmitter and speaking the first word to ensure a clear transmission. Be aware that when a radio operator is talking, keying the radio transmitter will be ineffective and will likely override or block out the receiver. Maintain situational awareness while waiting for a response from the DOWNFALL Tower Range Control Officer before retransmitting due to concurrent real-time operational tasks. Be alert for sounds, or lack of sounds, on your radio. Check the volume and channel to ensure the microphone is not stuck in the transmit position. If the radio microphone is stuck in the transmit position, then this is a “hot mic” resulting in the radio receiver being disabled and DOWNFALL Tower Range Control Officer transmissions being blocked.

4.7.4.1.3. **(Added-EDWARDSAFB) Use of Phraseology.** This EAFB Supplement describes phraseology terms and definitions that PIRA vehicle operators and ground personnel will use to communicate with the DOWNFALL Tower Range Control Officers. Using proper phraseology is key to ensure the DOWNFALL Tower Range Control Officers understand ground personnel or vehicle operator requests before issuing control instructions, as well as ground personnel or vehicle operators understanding the phraseology and control instructions issued by DOWNFALL Tower Range Control Officers. The examples of terms, definitions, common use phrases, and meanings are described in the subsequent paragraphs.

4.7.4.1.3.1. **(Added-EDWARDSAFB)** During PIRA ground operations, supporting ground personnel or vehicle operators shall not use the words "clear,” “cleared,” or “clearance” while communicating with DOWNFALL Tower Range Control Officers to avoid the risk of miscommunication errors with aircrews during weapons release operations in the range areas. An exception to use these words for ground or vehicle operations is not permitted.

4.7.4.1.3.2. **(Added-EDWARDSAFB) PIRA Vehicle Operator Procedures.** Vehicle operators shall contact the DOWNFALL Tower Range Control Officers each and every time they enter or exit the PIRA. When entering the PIRA, vehicle operators shall provide DOWNFALL Tower Range Control Officers with three actions: (1) WHO you are, (2) WHERE you are, and (3) WHAT your intentions are. Vehicle operators must always acknowledge communications with DOWNFALL Tower to ensure Range Control Officers know that your message was received and understood.

4.7.4.1.3.2.1. **(Added-EDWARDSAFB)** Vehicle operators will always read back instructions verbatim before taking any action and allow time for correction, if misinterpreted. If vehicle operators do not understand the DOWNFALL Tower Range Control Officer instructions, then vehicle operators will request DOWNFALL Tower Range Control Officer to repeat the message or instruction by stating “Say Again”.

Table 4.1. (Added-EDWARDSAFB) Read Back Instructions or Say Again.

Vehicle Operator. "PIRA One understands approved to enter PIRA at 395 Gate" "PIRA One, Holding Short of PB-10" "DOWNFALL, PIRA One at JACKRABBIT" "DOWNFALL, PIRA One, Say Again" **DOWNFALL:** "DOWNFALL, Roger"

4.7.4.1.3.2.2. **(Added-EDWARDSAFB)** If the vehicle operators are issued hold short instructions, then the operators are required to provide a read back to DOWNFALL Tower. Example "Hold Short" instructions:

Table 4.2. (Added-EDWARDSAFB) Hold Shot Instructions.

DOWNFALL. "PIRA One, proceed K-2 Road, Hold Short at South Road" **Vehicle Operator:** "DOWNFALL, Roger" **DOWNFALL:** "PIRA One, Read Back Hold Short Instructions" **Vehicle Operator:** "PIRA One, proceed K-2 Road, Hold Short at South Road"

4.7.4.1.3.2.3. **(Added-EDWARDSAFB)** When the vehicle is out of or off the PIRA, vehicle operators must inform DOWNFALL Tower of the exit. Example Exit Complete instructions:

Table 4.3. (Added-EDWARDSAFB) Exit Complete Instructions.

Vehicle Operator: "DOWNFALL, RAMS One" **DOWNFALL:** "RAMS One, DOWNFALL"

Vehicle Operator:

"RAMS One, Off the Range at 200 Gate"

DOWNFALL:

"RAMS One, Roger"

4.7.4.1.3.2.4. **(Added-EDWARDSAFB) Uncontrolled PIRA Entry or Exit Procedures.** When the PIRA is closed, vehicle operators must contact CONFORM (Command Post) for PIRA Entry and Exit on UHF 304.0 MHz or call DSN 527-3040 or (661) 277-3040.

4.7.4.1.3.2.5. **(Added-EDWARDSAFB) Common Use Phrases and Meanings.**

Table 4.4. (Added-EDWARDSAFB) Common Use Phrases and Meanings.

Word or Phrase Said:	Word or Phrase Meaning:
Acknowledge	Confirms the operator received and understood the message sent.
Advise Intentions	Request made to the operator to inform of the action to be taken.
Affirmative	Yes
Correction	An error has been made in the transmission, and the correct transmission follows.
Go Ahead	Request made to the operator to speak the intended message. **NOTE:** "Go Ahead" is not authorization for an operator to proceed with a control action.
Hold / Hold Short	Directive to an operator to keep vehicle or personnel in a specified area or at a specified point while awaiting further instruction.

How do you hear me? or How Copy?

Question to an operator asking about the quality of the transmission or question about receiving the last spoken transmission.

Immediately or Expedite

Directive to an operator when compliant action is required to avoid an imminent situation.

Negative

Directive to an operator to mean “No”, “Permission Not Granted”, or “Not Correct”.

Read Back

Directive to an operator to repeat the previous message or instruction.

Roger

Radio transmission received by an operator.

NOTE: Does not indicate compliance or reaction by an operator.

Say Again

Repeat your message or instruction.

Stand By

To wait or pause for a period of time while another priority action takes place.

NOTE: When the wait or pause is too lengthy, re-establishing contact should occur.

Unable

Cannot comply as requested or directed with a specific instruction or message.

Verify

Directive to investigate or confirm the action, information, or message.

Wilco

Will comply with received instruction, directive, or message.

4.7.4.1.3.3. **(Added-EDWARDSAFB) Phonetic Alphabet.** To provide clarity in transmissions, the International Civil Aviation Organization uses the following phonetic alphabet to reduce pronunciation errors with similar sounds. As an example, the phrase “Taxiway B” is referred as Taxiway Bravo when transmitting

on the radio. In **Table 4.5**, below the phonetic alphabet is used by the DOWNFALL Tower Range Control Officers to communicate with vehicle operators and ground personnel during PIRA operations.

Table 4.5. (Added-EDWARDSAFB) International Civil Aviation Organization Phonetic Alphabet.

Character	Word	Pronunciation
0	Zero	**ZE**-**RO**
1	One	**WUN**
2	Two	**TOO**
3	Three	**TREE**
4	Four	**FOW**-**ER**
5	Five	**FIFE**
6	Six	**SIX**
7	Seven	**SEV**-**EN**
8	Eight	**AIT**
9	Nine	**NIN**-**ER**
20	Twenty	**TWIN TEE**
30	Thirty	**THUR TEE**
40	Forty	**FORE TEE**
50	Fifty	**FIF TEE**
60	Sixty	**SIX TEE**
70	Seventy	**SEVEN TEE**
80	Eighty	**AIT TEE**
90	Ninety	**NIN TEE**
X00	Hundred	**HUN DRED**
X,000	Thousand	**TAW SUND**
A	Alpha	**AL**-FAH
B	Bravo	**BRAH**-**VOH**
C	Charlie	**CHAR**-LEE
D	Delta	**DELL**-TAH
E	Echo	**ECK**-OH
F	Foxtrot	**FOKS**-TROT
G	Golf	GOLF
H	Hotel	HOH-**TELL**
I	India	**IN**-DEE-AH
J	Juliet	**JEW**-LEE-**ETT**
K	Kilo	**KEY**-LOH

L	Lima	**LEE**-MAH
M	Mike	**MIKE**
N	November	NO-**VEM**-BER
O	Oscar	**OSS**-CAR
P	Papa	PAH-**PAH**
Q	Quebec	KEH-**BECK**
R	Romeo	**ROW**-ME-OH
S	Sierra	SEE-AIR-RAH
T	Tango	**TANG**-GO
U	Uniform	**YOU**-NEE-FORM
V	Victor	**VIK**-TOR
W	Whiskey	**WISS**-KEY
X	X-ray	**ECKS**-RAY
Y	Yankee	**YANG**-KEE
Z	Zulu	**ZOO**-LOO
NOTE: Emphasize syllables in **bold face** when transmitting.		

4.7. (EDWARDSAFB) Range Operations Communication Requirements. The 412 RANS operates and manages communication capabilities that include ultra-high frequency or very-high frequency ground-to-air radios, land mobile radios, telephone networks, telephone lines, and voice recording equipment to support operations information and communication requirements for 412 TW test missions. The 412 RANS uses voice logger equipment to record ground-to-air radios, telephone networks, and telephone lines that are used by test support personnel to execute test missions, to coordinate use of resources, to schedule mission requests, and to coordinate operational support actions. The 412 RANS will ensure recordings are maintained for a minimum of 7 days and up to a maximum of 60 days for normal (i.e., non-incident or non-mishap reporting) operational support actions. To support incident or mishap reporting preservation of evidence protocols required by the investigating agency (i.e., Interim Safety Board), the 412 RANS will ensure recordings are maintained for a minimum of 7 days and held until the recordings are released to the investigating agency representative.

4.8. Armament Safety Procedures. For ranges capable of aircraft ordnance delivery, prior to first release when carrying expendable ordnance (live, inert, or practice), final switch configuration for weapon release is not accomplished until the aircraft is in such a position that any accidental release will be contained within the range. **(T-1)** MAJCOMs will develop specific guidance for armament system configurations for multiple passes. After completing final weapons delivery, each flight member will perform a weapons system safety check. **(T-1)** Refer to AFI 11-214, aircraft specific Air Force 11-2MDS series operating procedures, and individual range supplements for additional guidance.

4.8.1. **(Added-EDWARDSAFB)** Verify cleared to arm and release points.

4.8.2. **(Added-EDWARDSAFB)** Verify receiving expected weapon, device, or target impact area information.

4.8.3. **(Added-EDWARDSAFB)** Verify range status information.

4.8.4. **(Added-EDWARDSAFB)** Verify the appropriate weapon or ordnance footprint data from the 412 TW Range Safety Office no later than 7 days before dropping or firing any object or ordnance with an undetermined ballistic characteristic or a distinctive ballistic characteristic.

4.8.5. **(Added-EDWARDSAFB)** Not authorize an aircraft to arm if an inadvertent release of the device would impact outside the PIRA boundary.

4.8.6. **(Added-EDWARDSAFB)** Give a clearance to arm that ensures the aircraft is positioned within one nautical mile of the PIRA boundary.

4.8.7. **(Added-EDWARDSAFB)** Notify the 412 RANS Operations Duty Officer if a weapon or device is inadvertently fired, released, or impacts off the PIRA.

4.8.7.1. **(Added-EDWARDSAFB)** The 412 RANS Operations Duty Officer will contact the 412 TW Range Safety Office to notify of the inadvertent off PIRA release or impact. The 412 RANS Operations Duty Officer will contact the SPORT MRU Supervisor if assistance is needed.

4.8.8. **(Added-EDWARDSAFB) Armament / Arming Safety Procedures.**

4.8.8.1. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer will give specific clearance for the aircrew to select or enable the arming switch. The DOWNFALL Tower Range Control Officer will give: “CLEARED HOT”. The phrase “CLEARED HOT” in subsequent paragraphs is a control procedure that is used to unlock weapons and grant weapons release clearance.

4.8.8.2. **(Added-EDWARDSAFB)** When conducting Flight Lead Control Operations, aircrews will follow the permissible control procedures to select or enable the arming switch.

4.8.8.3. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer, the SPORT MRU controller, or the Flight Lead Control aircrew will not give clearance to a test mission or a training operation to select or enable the arming switch without assurance that:

4.8.8.3.1. **(Added-EDWARDSAFB)** The aircraft has crossed into the PIRA airspace, or

4.8.8.3.2. **(Added-EDWARDSAFB)** The gun fire, gravity bomb, or other devices would impact within the PIRA if an inadvertent release would occur at the moment of arming, or

4.8.8.3.3. **(Added-EDWARDSAFB)** The device being deployed would impact on Rogers Dry Lakebed within the confines of the Alpha Corridor or PIRA if Mercury Boulevard is closed and an inadvertent release occurs. **NOTE**: If an arming sequence requires additional time, then a test program will request a specific target and weapon authorization process review from 412 TW Range Safety Office. Refer to EAFB Scheduling Guide for authorized times when Mercury Boulevard may be scheduled closed.

4.8.8.4. **(Added-EDWARDSAFB) Radar Position Information Procedures.** The following is required if SPORT MRU controller is using radar to obtain position information on the test aircraft. The SPORT MRU controller will not issue "OPERATION APPROVED AS REQUESTED" when:

4.8.8.4.1. **(Added-EDWARDSAFB)** Position or identity of test aircraft is uncertain or questionable.

4.8.8.4.2. **(Added-EDWARDSAFB)** The controller is dissatisfied with the radar presentation.

4.8.8.4.3. **(Added-EDWARDSAFB)** Unable to issue clearance.

4.8.8.4.3.1. **(Added-EDWARDSAFB)** When unable to issue clearance, the SPORT MRU controller will instruct the aircraft to "CONTINUE" the approach. The word "CONTINUE" is a control term that means to continue the present maneuver and does not imply a change in clearance to engage or expend ordnance. This call lets the aircrew know to continue the approach and await the next instruction.

4.8.8.4.3.2. **(Added-EDWARDSAFB)** When the airspace along the intended flight path is clear the SPORT MRU controller will call: "(AIRCRAFT CALL SIGN), (ALPHA-CORRIDOR/ WEST RANGE/ EAST RANGE/ PIRA), ENTRY APPROVED, REMAIN AT OR BELOW (altitude), STANDBY FOR DOWNFALL." This call lets the aircrew know that the flight path ahead is clear of aircraft.

4.8.8.4.3.3. **(Added-EDWARDSAFB)** The next call would come from the DOWNFALL Tower Range Control Officer to call: "(AIRCRAFT CALLSIGN) CLEARED HOT." This call allows the pilot to select or enable the arming switch and the aircrew must wait to release until on conditions are met in the approved and briefed footprint.

4.8.8.5. **(Added-EDWARDSAFB) Cleared to Arm Points Procedures.** Aircrews will ensure armament switches are safe until the DOWNFALL Tower Range Control Officer issues the "CLEARED HOT" call. **NOTE**: If an arming sequence requires additional time, then a test program will request a specific target and weapon authorization process review from 412 TW Range Safety Office. The "CLEARED HOT" call will be given to the aircrews after:

4.8.8.5.1. **(Added-EDWARDSAFB)** Crossing Mercury Boulevard on west to east runs into West Range.

4.8.8.5.2. **(Added-EDWARDSAFB)** Crossing Lancaster Boulevard when Mercury Boulevard is closed. **NOTE**: The Test Safety Review process may require additional mitigation procedures for test execution.

4.8.8.5.3. **(Added-EDWARDSAFB)** Crossing Highway 395 on east to west runs into West Range.

4.8.8.5.4. **(Added-EDWARDSAFB)** Established on the range south of Haystack Butte on south to north runs into East Range.

4.8.8.5.5. **(Added-EDWARDSAFB)** Crossing Highway 58 on north to south runs into the East Range.

4.8.8.6. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer will abort a hot run any time circumstances warrant. Aircrews will acknowledge and confirm that all arming switches are safe when the DOWNFALL Tower Range Control Officer directs ("ABORT, ABORT, ABORT") to abort the run.

4.8.8.7. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer will notify SPORT MRU of suspected malfunctions or anomalies.

4.8.8.8. **(Added-EDWARDSAFB)** Automatic delivery and separation by onboard radar or other methods is authorized.

4.8.8.9. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer will make the “CONTINUE DRY” and “CLEARED HOT” calls.

4.8.8.10. **(Added-EDWARDSAFB)** The aircrews first run will be a dry run unless pre-coordinated with the DOWNFALL Tower Range Control Officer and the SPORT MRU controller; otherwise, a first hot pass requires Class A Range Service procedures (reference AFMAN 11-214). The aircrew will determine hot pass or dry run status before beginning each new run.

4.8.8.11. **(Added-EDWARDSAFB)** All missions involving the release of live ordnance are required to conduct a dry run pass without exception.

4.8.8.12. **(Added-EDWARDSAFB)** The SPORT MRU controller will:

4.8.8.12.1. **(Added-EDWARDSAFB)** Provide a qualified controller to direct and monitor PIRA missions.

4.8.8.12.2. **(Added-EDWARDSAFB)** Brief the DOWNFALL Tower Range Control Officer on pending supersonic flights in the Alpha Corridor to include the number of runs and duration.

4.8.8.12.3. **(Added-EDWARDSAFB)** Verify range status with the DOWNFALL Tower Range Control Officer and brief expected weapon, device impact areas, and targets prior to aircraft range entry.

4.8.8.12.4. **(Added-EDWARDSAFB)** Confirm the following aircraft and weapons data with the aircrew before the first run:

4.8.8.12.4.1. **(Added-EDWARDSAFB)** Type and quantity of weapons(s) to be release and fired.

4.8.8.12.4.2. **(Added-EDWARDSAFB)** Aircraft maneuver (for example, level, dive turn, etc.).

4.8.8.12.4.3. **(Added-EDWARDSAFB)** Planned target.

4.8.8.12.4.4. **(Added-EDWARDSAFB)** Weapons status (for example, live or inert).

4.8.8.12.5. **(Added-EDWARDSAFB)** Transmit, "ABORT, ABORT, ABORT" to terminate a hot run or a hot pass when circumstances warrant.

4.8.8.12.6. **(Added-EDWARDSAFB)** Confirm weapon(s) switches are "OFF HOT/DRY SWITCHES SAFE" after actual, attempted release, or fire.

4.8.8.12.7. **(Added-EDWARDSAFB)** Verify with the DOWNFALL Tower Range Control Officer that the PIRA is clear prior to conducting a weapon, device drop, separation, or laser operations.

4.8.8.12.8. **(Added-EDWARDSAFB)** Provide traffic advisories and boundary calls as required.

4.8.8.12.9. **(Added-EDWARDSAFB)** Terminate range activity when safety is in question.

4.8.8.12.10. **(Added-EDWARDSAFB)** Confirm no restrictions are implied for maneuvers associated with either simulated evasive maneuver by the aircrew or when unusual maneuvers are required to satisfy test objectives.

4.8.9. **(Added-EDWARDSAFB) Monitor Progress and Status of Missions Requiring Road Closure.** The 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer will:

4.8.9.1. **(Added-EDWARDSAFB)** Ensure 412 RANS personnel are sufficiently trained to perform road closure duties.

4.8.9.2. **(Added-EDWARDSAFB)** Review the 412 TW CSE for the closure of Mercury Boulevard. Plan to monitor the progress and status of missions requiring road closure.

4.8.9.3. **(Added-EDWARDSAFB)** Ensure Mercury Boulevard is closed when:

4.8.9.3.1. **(Added-EDWARDSAFB)** The mission profile as determined by the Test Director, Test Conductor, the program Range Control Specialist, or the 412 TW Range Safety Officer indicates a hazard to personnel or equipment using Mercury Boulevard.

4.8.9.3.2. **(Added-EDWARDSAFB)** As part of a test scenario, a clearance to arm is required west of Mercury Boulevard and an inadvertent release of the device would result in impact west of the PIRA, on the lakebed, or west of Mercury Boulevard.

4.8.9.3.3. **(Added-EDWARDSAFB)** Doubt exists as to the flight characteristics of the device being dropped (for example, fuel cells or devices with sequential chute deployments) and impact west of the PIRA is a possibility.

4.8.9.3.4. **(Added-EDWARDSAFB)** A laser hazard would exist on Mercury Boulevard caused by the use of a non-eye safe laser in other than a point track mode.

4.8.9.3.5. **(Added-EDWARDSAFB)** Forward firing projectiles or gunfire cross Mercury Boulevard.

4.8.9.3.6. **(Added-EDWARDSAFB)** Any munitions or stores separations are conducted east to west on the West Range.

4.8. (EDWARDSAFB) Pre-Weapons Release Safety Procedures. Prior to the aircraft entering the PIRA and before weapon release, the DOWNFALL Tower Range Control Officer will:

4.9. Weapons Release Authority. Acceptance of weapons release authority incurs sole responsibility for the safe release of ordnance.

4.9.1. Terminology Conventions. To prevent miscommunication, when authorizing weapons release personnel will only use the term "cleared." **(T-1)** Personnel will use the term "continue" in the manner described in Joint Publication 3-09.3, *Close Air Support*, to acknowledge requests or indicate approval to proceed with an attack profile or action without providing clearance to release any ordnance yet. **(T-1)**

4.9.1. **(EDWARDSAFB)** When unable to issue clearance, the SPORT MRU controller will instruct the aircraft to "CONTINUE" the approach. The word "CONTINUE" is a control term that means to continue the present maneuver and does not imply a change in clearance to engage or expend ordnance. This call lets the aircrew know to continue the approach and await the next instruction.

4.9.1.1. **(Added-EDWARDSAFB)** When the airspace along the intended flight path is clear then the SPORT MRU controller will call: "(AIRCRAFT CALL SIGN), (ALPHA-CORRIDOR / WEST RANGE / EAST RANGE / PIRA), ENTRY APPROVED, REMAIN AT OR BELOW (altitude). STANDBY FOR DOWNFALL." This call lets the aircrew know that the flight path ahead is clear of aircraft.

4.9.1.2. **(Added-EDWARDSAFB)** The next call would come from the DOWNFALL Tower Range Control Officer to call: "(AIRCRAFT CALLSIGN) CLEARED HOT." This call allows the aircrew to select or enable switches and the aircrew must wait to release until on conditions and in the approved briefed footprint.

4.9.2. On ranges providing Class A service, weapons release authority resides inherently with the range control officer. For each pass, the range control officer will authorize weapons release with a "cleared hot" radio call unless release authority has been delegated. **(T-1)** The range control officer may delegate weapons release authority to a qualified flight lead, individual pilot, Forward Air Controller (Airborne), JTAC or other briefed person by issuing an "Authorized (flight lead, individual pilot, Forward Air Controller (Airborne), JTAC, etc.) Control" radio call. The range control officer will remain in either the main or flank tower (or another MAJCOM approved location), will retain overall range authority at all times and can withdraw release clearance or abort a release at any time. **(T-2)**

4.9.2. **(EDWARDSAFB) DOWNFALL Tower Range Control Officer Procedures for Class A Range Service.** A qualified DOWNFALL Tower Range Control Officer is required in DAGRAG Tower or COWBELL Tower when weapons delivery altitudes, missions, or aircrew require Class A Range Service operations. Exceptions for maneuvering deliveries or laser-designated deliveries, which create a hazard for personnel in the DAGRAG or COWBELL Towers, must be approved in the applicable safety plans.

4.9.2.1. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer performs a DAGRAG Tower or a COWBELL Tower tour of duty which is supervised by the 412 RANS Range Operations Officer and is part of the DOWNFALL Tower Range Control Officer training program. The DAGRAG Tower is located between the north and south gun targets and the COWBELL Tower is located between the SCATT Range and the PB-12 target.

4.9.2.2. **(Added-EDWARDSAFB) Responsibilities for DOWNFALL Tower Range Control Officers Providing Class A Range Service:**

4.9.2.2.1. **(Added-EDWARDSAFB)** Controls DAGRAG Tower or COWBELL Tower missions under the provisions of AFMAN 13-212V1, this Supplement, and applicable guidance governing tactical fighter or mission design series weapons delivery procedures.

4.9.2.2.2. **(Added-EDWARDSAFB)** Monitors safety of all DAGRAG Tower or COWBELL Tower operations during tour of duty.

4.9.2.2.3. **(Added-EDWARDSAFB)** Updates PIRA personnel on scheduled mission(s) as needed.

4.9.2.2.4. **(Added-EDWARDSAFB)** Monitors aircraft in the pattern with primary concern for aircraft on final.

4.9.2.2.5. **(Added-EDWARDSAFB)** Monitors passes for dive angle, minimum altitude, fouls and dangerous passes.

4.9.2.2.6. **(Added-EDWARDSAFB)** Monitors strafe passes closely for flat passes, firing past foul line, lazy pull offs and improper pattern procedures.

4.9.2.3. **(Added-EDWARDSAFB) DAGRAG Tower and COWBELL Tower Procedures.** The DOWNFALL Tower Range Control Officers will:

4.9.2.3.1. **(Added-EDWARDSAFB)** Contact DOWNFALL Tower DSN 525-5603 (661-275-5603) for ground range entry approval.

4.9.2.3.2. **(Added-EDWARDSAFB)** Inspect the target areas to be used prior to mission start for safety hazards and ricochet potentials.

4.9.2.3.3. **(Added-EDWARDSAFB)** Provide the aircrew with the following advisory information: wind direction, wind speed in miles per hour, target condition (i.e., container in center), and BASH status.

4.9.2.3.4. **(Added-EDWARDSAFB)** Be responsible for ground personnel within the DAGRAG or the COWBELL range areas.

4.9.2.3.5. **(Added-EDWARDSAFB)** Maintain a complete mission or PIRA range log.

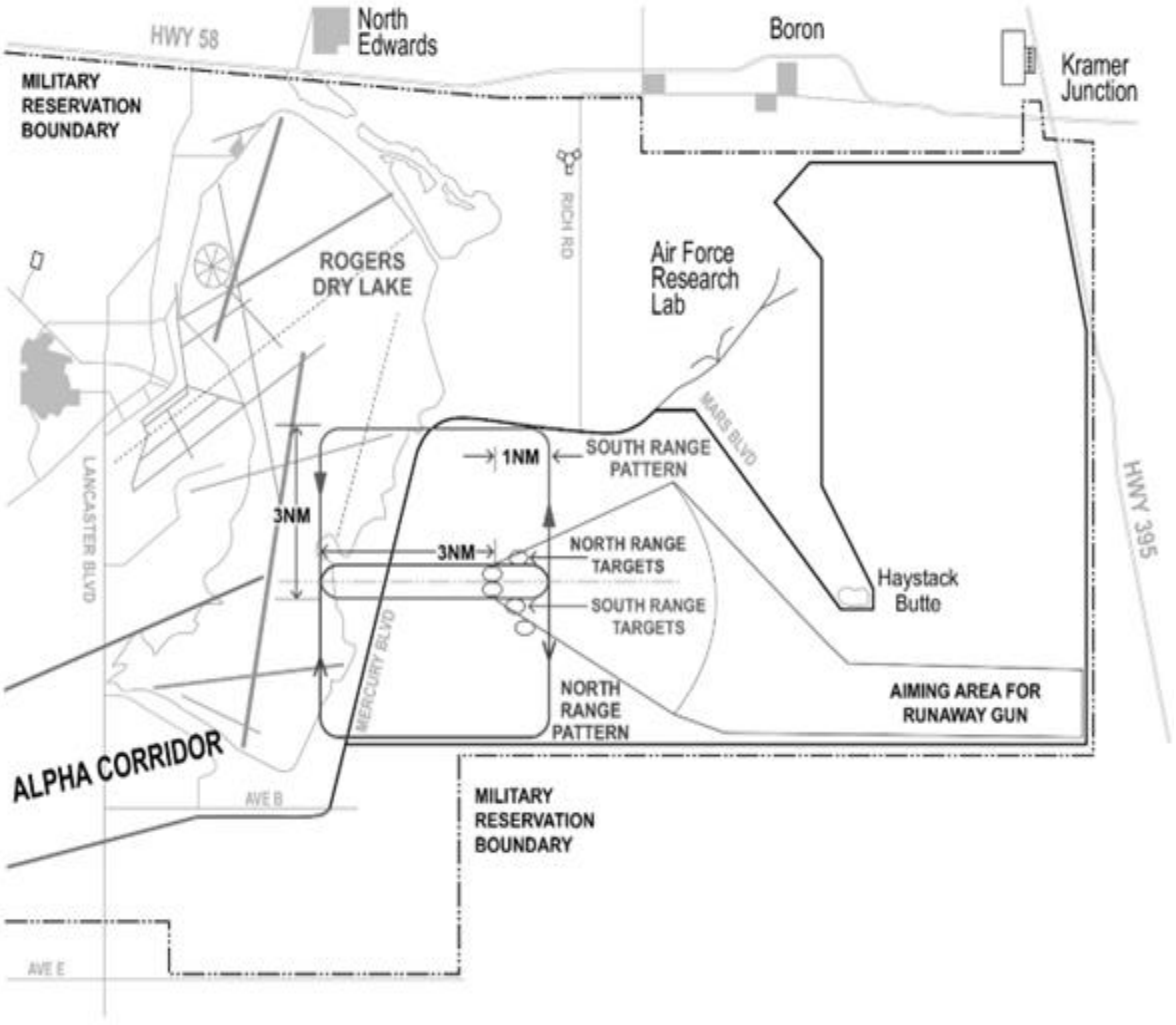
4.9.2.3.6. **(Added-EDWARDSAFB)** Ensure the minimum recovery altitude is flown according to the applicable AFMAN 11-2 mission design and volume series, AFMAN 11-214 guidance, or publications. If a lower minimum recovery altitude is required, then this altitude must be included in the safety plan approved through the Test Safety Review process.

4.9.2.4. **(Added-EDWARDSAFB) Aircrew Procedures for Class A Range Service.** Aircrews will brief the DOWNFALL Tower Range Control Officer on the mission including ordnance and type(s) of delivery, minimum altitudes, type patterns, and foul criteria for Class A Range Service operations and review pattern communication procedures.

4.9.2.4.1. **(Added-EDWARDSAFB) Aircrew Procedures for DAGRAG.** Aircrews will contact SPORT MRU on mission or common frequency for range entry. Hold as directed by the SPORT MRU controller while awaiting DAGRAG Tower approval.

4.9.2.4.2. **(Added-EDWARDSAFB)** After receiving approval from SPORT MRU, aircrews will contact DAGRAG Control on the assigned mission frequency. DAGRAG Control and the DAGRAG Range Control Officer assumes control of the mission in the DAGRAG Traffic Pattern, see **Figure 4.1**.

Figure 4.1. (Added-EDWARDSAFB) DAGRAG Traffic Patterns.



4.9.2.4.3. **(Added-EDWARDSAFB)** Aircrews will fly at least one dry pass on each target complex (for example, PB-10, strafe, PB-12, etc.).

4.9.2.4.4. **(Added-EDWARDSAFB)** Aircrews will obtain DAGRAG Control clearance for each sortie or pass.

4.9.2.4.5. **(Added-EDWARDSAFB)** Aircrews will comply with mandatory radio calls in the DAGRAG Traffic Pattern as follows:

4.9.2.4.5.1. **(Added-EDWARDSAFB)** Before rolling in on final, aircrews will transmit "(AIRCRAFT CALL SIGN) IN HOT" or "(AIRCRAFT CALL SIGN) IN DRY." Aircrews will not switch on armament master switches until after rollout on final and cleared by the DAGRAG Range Control Officer. Aircrews will "safe" the switches as soon as practicable after pullout.

4.9.2.4.5.2. **(Added-EDWARDSAFB)** After release, aircrews will "safe" the armament master switch and call: "(AIRCRAFT CALL SIGN), OFF HOT SWITCHES SAFE" or "(AIRCRAFT CALL SIGN), OFF DRY SWITCHES SAFE."

4.9.2.4.6. **(Added-EDWARDSAFB)** Aircrews will follow SPORT MRU position advisories, if applicable.

4.9.2.4.7. **(Added-EDWARDSAFB)** Aircrews will maintain separation from other aircraft while flying DAGRAG missions.

4.9.2.4.7.1. **(Added-EDWARDSAFB)** When more than one aircraft is in the same DAGRAG Traffic Pattern, each aircrew adjusts position in the pattern to maintain proper spacing. Aircrews will extend the pattern as far east as the western boundary of the East Range, or as far as required within the PIRA to maintain separation when the East Range is not in use.

4.9.2.4.7.2. **(Added-EDWARDSAFB)** To optimize range use time and to provide time between each sortie for weapon scoring, the DAGRAG Range Control Officer may request the aircrew to extend the pattern three miles east of Foul Line Road before turning crosswind. This minimizes the potential ricochet hazard and ensures when one aircraft is turning final, the second aircraft is turning downwind. Further, both aircraft will intercept Foul Line Road simultaneously.

4.9.2.4.8. **(Added-EDWARDSAFB)** Aircrews will notify the DAGRAG Range Control Officer and SPORT MRU of a suspected ordnance malfunction or an anomaly.

4.9.2.4.9. **(Added-EDWARDSAFB)** Aircrews will notify the DAGRAG Range Control Officer and SPORT MRU of mission completion.

4.9.2.5. **(Added-EDWARDSAFB) PIRA Flight Lead Control Operations.** Flight Lead Control Operations allow the Flight Lead (FL) or Aircraft Commander (AC) to assume air / range safety responsibilities and eliminates the requirement for the DOWNFALL Tower Range Control Officer to clear the aircrew to arm and release on every pass. **NOTE**: The DOWNFALL Tower Range Control Officer will continue to monitor the overall safety of the PIRA.

4.9.2.5.1. **(Added-EDWARDSAFB)** Flight Lead Control Operations are authorized for fighter and trainer aircraft on test or training missions using training munitions with a known hazard footprint. FL or AC is not authorized for missions using live ordnance.

4.9.2.5.2. **(Added-EDWARDSAFB)** Under Flight Lead Control, SPORT MRU and the DOWNFALL Tower Range Control Officer monitor mission frequency and will call the aircrew for clarification (for example, “Call Sign, Confirm OFF HOT, SWITCHES SAFE”) as required for safety. SPORT MRU controls airborne access to the PIRA.

4.9.2.5.3. **(Added-EDWARDSAFB) Flight Lead Control Procedures:**

4.9.2.5.3.1. **(Added-EDWARDSAFB)** The FL or AC will contact DOWNFALL Tower Range Control Officer and SPORT MRU to pre-brief the mission prior to stepping. Briefings will cover type and amount of training munitions, planned events, pattern ground tracks, and run-in headings.

4.9.2.5.3.2. **(Added-EDWARDSAFB)** DOWNFALL Tower Range Control Officer will provide the aircrew run-in restrictions, pattern restrictions, and location of manned sites on the range.

4.9.2.5.3.3. **(Added-EDWARDSAFB)** The FL or AC must fully understand restrictions imposed and the location of manned sites. The FL or AC will advise SPORT MRU and DOWNFALL Tower Range Control Officer of plans to operate within the PIRA under flight lead control. It is critical the FL or AC clearly communicate expectations, mission requirements and planned profile to the DOWNFALL Tower Range Control Officer and SPORT MRU.

4.9.2.5.3.4. **(Added-EDWARDSAFB)** After takeoff the FL or AC will check-in with SPORT MRU and DAGRAG Range Control Officer or COWBELL Range Control Officer on mission frequency for restrictions or planned events update. SPORT MRU will clear the FL or AC for Flight Lead Control Operations.

4.9.2.5.3.5. **(Added-EDWARDSAFB)** The FL or AC assumes range safety responsibilities once cleared for Flight Lead Control Operations. **NOTE**: The DOWNFALL Tower Range Control Officer will continue to monitor the overall safety of the PIRA.

4.9.2.5.4. **(Added-EDWARDSAFB)** SPORT MRU or DOWNFALL Tower Range Control Officer has the authority to terminate Flight Lead Control Operations or procedures.

4.9.2.5.5. **(Added-EDWARDSAFB)** Due to the small size of the PIRA, the weapon's footprint for an unintentional release may not remain within the range boundary. Therefore, the FL or AC will “safe” the delivery system and call "OFF HOT, SWITCHES SAFE or OFF DRY, SWITCHES SAFE" after each pass.

4.9.2.5.6. **(Added-EDWARDSAFB)** FL or AC may arm the delivery system when established on base leg within the range boundary (for example, the conventional pattern).

4.9.2.5.7. **(Added-EDWARDSAFB)** When the release pattern either exits or starts outside the range boundary, the FL or AC may arm the delivery system when entering the range and the weapon’s footprint for an inadvertent release is within the range boundary (for example, on final for a radar release pattern after crossing the range boundary) and does not endanger a manned site.

4.9.3. On ranges providing Class B and C service, weapons release authority follows a hierarchy from the JTAC, to the Forward Air Controller (Airborne), to the flight lead, to an individual pilot or other briefed person who may then delegate the authority as desired.

4.9. (EDWARDSAFB) Weapons Release Authority. The DOWNFALL Tower Range Control Officer will authorize weapons or ordnance release on the PIRA during each class of range service. An ordnance hot run or ordnance hot pass is a maneuver where the aircrew intends to release or fire a weapon, an object, or a projectile from the aircraft. For missions within the confines of the PIRA, the following procedures apply:

4.10. Simulated Weapons Release Authority. Except during flight-lead or individual control, to emulate the clearance procedures contained in Joint Publication 3-09.3, the weapons release authority should issue a “continue dry” radio call to signify authority to simulate weapons release.

4.10.1. **(Added-EDWARDSAFB)** When unable to issue clearance, SPORT MRU will instruct the aircraft to "CONTINUE" the approach. The word “CONTINUE” is a control term that means to continue the present maneuver and does not imply a change in clearance to engage or expend ordnance. This call lets the aircrew know to continue approach and await next instruction.

4.10.2. **(Added-EDWARDSAFB)** When the airspace along the intended flight path is clear SPORT MRU will call: “(AIRCRAFT CALL SIGN), (ALPHA-CORRIDOR / WEST RANGE / EAST RANGE / PIRA), ENTRY APPROVED, REMAIN AT OR BELOW (altitude). STANDBY FOR DOWNFALL.” This call lets the aircrew know that the flight path ahead is clear of aircraft.

4.10.3. **(Added-EDWARDSAFB)** The next call would come from the DOWNFALL Tower Range Control Officer to make the cold or dry run continue call: “(AIRCRAFT CALL SIGN) CONTINUE DRY”. This call allows the aircrew to continue a cold or dry employment run on the condition parameters and the briefed footprint.

4.10. (EDWARDSAFB) Simulated Ordnance Cold / Dry Run Release Authority. A cold run or dry run is a maneuver where the aircrew simulates weapon releasing or firing an object, a projectile, or a laser from the aircraft. For missions within the confines of the PIRA, the following procedures apply:

4.11. Weapons Employment and Airdrop Operations.

4.11.1. Personnel will not employ weapons outside approved DoD Special Use Airspace. **(T-0) Note:** Research and Test and Evaluation activities do not require Special Use Airspace if conducted IAW DoDI 5000.89_DAFI 99-103, provided appropriate coordination and approvals are obtained from FAA/spectrum authorities as required.

4.11.2. Range operating authorities must ensure the land or sea underlying the hazard area or failure footprint used for actual weapon employment are protected by purchase, lease, or other means to ensure the safety of personnel, structures, and the public from expended weapons, laser and electromagnetic emissions, and target debris. **(T-0)**

4.11.2. **(EDWARDSAFB)** For all classes of range operations, and prior to conducting a weapon firing or release, an airdrop or separation operation, the DOWNFALL Tower Range Control Officer also performs the on-site safety function responsibility and ensures the applicable hazard footprint, the associated airspace, and the underlying PIRA land is clear prior to authorizing weapons or airdrop employment operations. Prior to laser employment operations, the DOWNFALL Tower Range Control Officer receives notification from the on-site Laser Safety Officer that the ground hazard footprint area is clear. The DOWNFALL Tower Range Control Officer coordinates with SPORT MRU to confirm the area under altitude of the aircraft remains clear.

4.11.2.1. For events where the weapons footprint falls outside the lateral confines of the established range boundaries, the range operating authority will notify the public and conduct a risk assessment and take actions to minimize the hazard to the public and non-DoD property. **(T-1)**

4.11.2.2. Range operating authorities at test ranges will ensure a Flight Termination System or other containment measures are applied to aerial test items (weapons, missiles, unmanned aerial systems, etc.) having the potential to exceed range boundaries and endanger public or DoD assets and personnel. **(T-1)** Range operating authorities will establish a method of determining the position of the item in relation to its footprint and the range boundary and assess the need for redundancy of position information, telemetry, and Flight Termination System signal transmission based on the potential hazard. **(T-1)**

4.11.2.2. **(EDWARDSAFB)** Ordnance with a maximum energy footprint that exceeds the boundaries of the PIRA may be dropped provided that the footprint and risk hazard analysis is performed by the 412 TW Range Safety Office with reference to AFMAN 13-212V1, from **paragraph 4.11.2.1** through **paragraph 4.11.3.2**, guidance. If the footprint and risk hazard analysis is not acceptable per AFMAN 13-212V1, then the device must be equipped with a 412 TW Range Safety Office approved Flight Termination System capable of containing the weapon within the PIRA. The Flight Termination System approval and certification process can be lengthy, so test project coordinators shall contact the 412 TW Range Safety Office as early as possible to determine the Flight Termination System requirements.

4.11.3. WDZ Tool Methodology. The WDZ tool application methodology allows the range operating authority to identify possible target locations, modify allowable delivery ground tracks to eliminate or reduce hazards, identify the best location for range improvements, and design a new impact area or range. The methodology is based on WDZ footprints developed from a combination of actual weapon impact data and simulation results. A WDZ defines the minimum land required to employ a given munition safely, using a certain aircraft and delivery tactic, over a specific soil density and target type. Each WDZ footprint incorporates a probability distribution function, which provides the information necessary to perform a quantitative risk assessment and evaluate the relative risk of an identified hazard.

4.11.3.1. Except as noted below, range operating authorities will prohibit weapons employment unless an applicable SDZ/WDZ footprint, WDZ surrogate footprint or test footprint has been applied to the target for the specific delivery platform, weapon and employment parameters. **(T-1)** Range operating authorities may expedite analysis by grouping targets together or combining aircraft type, weapon type, and tactics, as appropriate. Test footprints are not authorized for training missions, nor are applicable training footprints, by themselves, sufficient for test missions, but training footprints may be used as part of the range operating authority's overall risk assessment.

4.11.3.1. **(EDWARDSAFB)** In the case of unguided weapon integration tests, including weapon separation flight test, the 412 TW Range Safety Office will enlist the help of the Air Force Seek Eagle Office at Eglin AFB or the contractor (in the case of Total System Performance Responsibility programs), to obtain as much data as possible to assist the 412 TW Range Safety Office in footprint development. To account for ballistic parameter uncertainties, the 412 TW Range Safety Office will use several techniques and tools to ensure the footprint can be contained safely on the PIRA. Techniques may include limiting the release parameters, limiting the size and location of the shot box, increasing required containment levels, and applying wind limits. Tools may include the weapon danger zone footprint, the use of "FOOTWIN" legacy footprint analysis tool, other generic footprint determination tools, and historical data.

4.11.3.1.1. Where containment is software selectable, range operating authorities will apply WDZs and SDZs that provide the following containment of projectiles, fragments, debris, and components as a minimum: **(T-1)**

4.11.3.1.1.1. Surface fires: 99.9999% (Expressed as 1:1,000,000 escapement for SDZs).

4.11.3.1.1.2. Aviation-delivered gun ammunition: 99.999%.

4.11.3.1.1.3. All other aviation-delivered ordnance: 99.99%.

4.11.3.1.2. WDZ surrogate footprints may be used and do not require ACC/A3A approval when:

4.11.3.1.2.1. Test events are conducted on MRTFB ranges or test sites.

4.11.3.1.2.2. AFMC-sponsored test events are conducted on a PTR and a test footprint or WDZ surrogate footprint has been approved by AFMC/SE, the range operating authority, and the range-owning MAJCOM.

4.11.3.1.3. WDZ surrogate footprints must be used and require ACC/A3A approval when:

4.11.3.1.3.1. Training operations involve specific delivery platforms, weapon and employment parameter combinations, including foreign aircraft and weapons, that do not exist in the WDZ tool.

4.11.3.1.3.2. Test events are conducted on a PTR except as noted in 4.11.3.1.2.2.

4.11.3.1.3.3. Training operations involve specific delivery platforms, weapon and employment parameter combinations that are classified and cannot be generated in the WDZ tool (it is an unclassified system).

4.11.3.1.4. The use of WDZ surrogate footprints may be approved by completing the following actions:

4.11.3.1.4.1. The event-sponsoring MAJCOM requests a permanent WDZ footprint from ACC/A3A as the lead Air Force agent for procuring and modifying WDZ footprints as necessary.

4.11.3.1.4.2. Range operating authority performs a risk assessment using a WDZ surrogate footprint and 100% containment of projectiles, fragments, debris and components and submits it to the range-owning MAJCOM.

4.11.3.1.4.3. The range-owning MAJCOM reviews and provides concurrence with the range operating authority's WDZ surrogate footprint and risk assessment to the event-sponsoring MAJCOM.

4.11.3.1.4.3.1. When use of a WDZ surrogate footprint is requested for contracted air support for another service or MAJCOM:

4.11.3.1.4.3.1.1. The service or MAJCOM requesting use of the surrogate footprint is responsible for performing a risk assessment and generating surrogate footprint(s) using 100% containment of projectiles, fragments, debris, and components and submitting them to the range-owning MAJCOM.

4.11.3.1.4.3.1.2. The range-owning MAJCOM must verify that the contractor meets airworthiness requirements IAW that Service's Technical Airworthiness Authority and that the aircraft will be in public aircraft operations status (i.e., has an Interim Flight Clearance for the Navy or an Airworthiness Release for the Army).

4.11.3.1.4.3.1.3. Contact the Air Force Technical Airworthiness Authority, AFLCMC/EN-EZZ, at **USAF.Airworthiness.Office@us.af.mil** for assistance with verifying airworthiness.

4.11.3.1.4.3.1.4. The range-owning MAJCOM must include verification of airworthiness with the WDZ surrogate footprint request that is submitted to ACC/A3A.

4.11.3.1.4.4. The event-sponsoring MAJCOM submits WDZ surrogate footprint and risk assessment to ACC/A3A for approval.

4.11.3.1.5. For foreign countries using same type United States aircraft and United States weapons, range operating authorities may use a United States aircraft WDZ in their risk management assessment. Notify the MAJCOM and AF/A3TI of the use of this applicable WDZ prior to actual employment. Range operating authorities may also apply a larger buffer, for risk mitigation purposes, if the foreign country is not a routine participant and unfamiliar with the range and exercise/training scenario.

4.11.3.1.6. MRTFB and test range operating authorities are responsible for ensuring test footprints are developed and applied for test missions. If the application of test footprints would preclude the accomplishment of test objectives, the range operating authority, in conjunction with the DoDI 5000.89_DAFI 99-103-mandated safety review, may authorize the use of a risk management policy. The range operating authority will perform a risk assessment, which is validated by the safety review, to identify and minimize hazards consistent with test objectives (see DoDI 5000.89_DAFI 99-103). **(T-1)**

4.11.3.1.6. **(EDWARDSAFB)** In the special case of the Joint Direct Attack Munition class smart weapons, an operational footprint will be generated following the AFMAN 13-212V1 approved method of triple checking that the aircraft has provided the proper target coordinates to the weapon. If the weapon telemetry is available to verify target coordinates prior to release, the triple check is not required.

4.11.3.1.6.1. **(Added-EDWARDSAFB)** In the special case of a weapon, ordnance, or an object with an undetermined or a specifically distinctive ballistic characteristic, the program engineer shall provide the 412 TW Range Safety Office with sufficient data to calculate an impact footprint when planning test missions that require a drop, fire, or launch activity.

4.11.3.2. Procuring WDZ Footprints. Headquarters ACC/A3A is the lead Air Force agent for procuring and modifying WDZ footprints for Combat Air Forces requirements. MAJCOMs requiring unique or fast-track WDZ footprints will coordinate with ACC/A3A at least 6 months prior to the need date. Foreign countries requiring WDZ support should be directed to SAF/IAPX.

4.11.4. Precautions with Expendable Munitions. Aircraft with free-falling and/or forward-firing ordnance will not over fly or point their aircraft or guns at manned sites with intent to expend or employ weapons. **(T-1)** Aircraft with moveable guns, such as helicopters and AC-130 gunships, will not point their guns at any manned site. **(T-1)** Range operating authorities must ensure that facilities where the risk of damage is deemed unacceptable are not within any footprint. **(T-1)**

4.11.5. Weapons Employment near Manned Equipment, Facilities and Sites.

4.11.5.1. Training Operations. In general, manned equipment, facilities, and sites, such as range towers, simulated threat emitter systems, and observation posts, may be located within the hazard area during range operations, provided the associated personnel are designated Mission Essential, inert ordnance is used, and a risk assessment is conducted and approved by the range operating authority IAW **paragraph 4.18**. Also see **paragraph 4.5.3** for additional criteria associated with observation posts.

4.11.5.2. Test Operations. Range operating authorities must ensure the presence of manned equipment, facilities and sites are addressed in the risk assessment process described in **paragraph 4.11.3.1.4**. **(T-1)** Authorized test activities employing live munitions may locate personnel or equipment (e.g., telemetry or recording devices) within the hazard area only if essential for mission accomplishment, and only with documented risk management application and approval. Permanently locating or constructing non-essential manned facilities within a hazard area with continuous exposure requires the application of AFMAN 91-201 processes and approval authorities following range operating authority coordination.

4.11.6. Improved Container Delivery System and Joint Precision Airdrop System Deliveries.

4.11.6.1. Range operating authorities will approve Improved Container Delivery System (ICDS) and Joint Precision Airdrop System (JPADS) deliveries only if the Precision Airdrop System- Mission Planner derived footprints for success and chute failure overlay only government-owned, leased or otherwise controlled land with no non-mission essential personnel present and an Airdrop Damage Estimation acceptable to the range operating authority has been accomplished. **(T-1)** Additional ICDS and JPADS information and restrictions are published in DAFMAN 13-217.

4.11.6.2. If the load is projected to, or may transit non-restricted airspace during its fall (e.g., footprints are not wholly contained within restricted airspace), range operating authorities will also ensure the user complies with applicable rules contained in Title 14 CFR Subpart B, Section 105.25, *Information required, and notice of cancellation or postponement of jump* (Federal Aviation Regulation 105.25). **(T-0)**

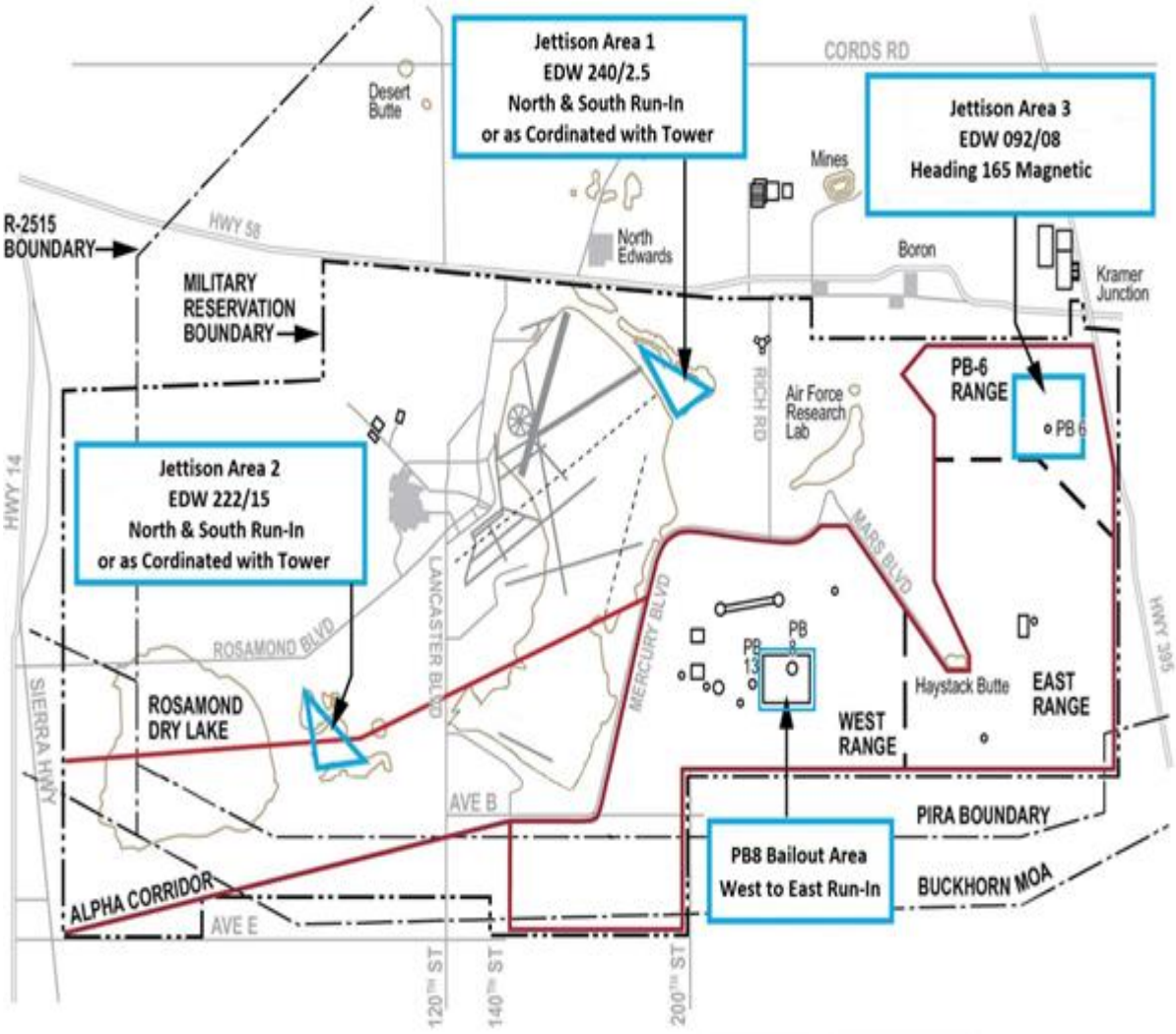
4.11.6.3. Equipment, facilities, and mission essential personnel are permitted within the success, chute failure and the guidance failure footprints if the range operating authority has conducted and approved a risk assessment IAW **paragraph 4.18**.

4.11.6.4. Because weather observations and dropsonde (an expendable, aircraft-launched weather measurement device) wind updates are used to revise footprint analyses up until the time of delivery, range operating authorities should ensure the user is aware of the range operating authority's exclusion requirements and be provided with the location of airspace/range boundaries and any personnel or equipment the range operating authority cannot or chooses not to allow in the footprint as part of his/her risk analysis.

4.12. Jettison Areas. Range operating authorities will ensure that all ranges with aerial delivery capability have an area within the hazard area designated for jettison of ordnance and stores. **(T-1)** The location of jettison areas and jettison procedures are designed to minimize the hazard to the public, ground personnel and range structures as well as aircraft and aircrew.

4.12.1. **(Added-EDWARDSAFB)** Non-explosive and explosive ordnance jettison areas are described in **Figure 4.2**.

Figure 4.2. (Added-EDWARDSAFB) Non-Explosive/Explosive Jettison Areas and Bailout Area.



4.12.2. **(Added-EDWARDSAFB)** Aircrews will use the PIRA non-explosive jettison Area 3 located on the East Range (**Figure 4.2**) when the aircraft flight path after jettison is in doubt. The jettison altitude is between 5,000 feet to 6000 feet mean sea level. If possible, contact SPORT MRU or Edwards AFB Control Tower before jettison. Continue on heading 165 magnetic, reduce throttles to idle, and eject, if necessary.

4.12.3. **(Added-EDWARDSAFB)** Aircrews will inform SPORT MRU or Edwards AFB Control Tower of their intent to jettison. SPORT MRU will, upon request, provide radar vectors to the area, traffic advisories, and advise when entering and leaving the area. SPORT MRU will not inform the crew when to jettison.

4.12.4. **(Added-EDWARDSAFB)** Aircrews will overfly the area, if weather or time permits, before jettisoning to ensure it is clear of personnel and vehicles.

4.12.5. **(Added-EDWARDSAFB)** If SPORT MRU assistance is not available, Edwards AFB Control Tower may give only a general description of the area and may advise other aircraft to avoid the area.

4.12. (EDWARDSAFB) Jettison Area Procedures. The PIRA is the primary jettison area for non-explosive and explosive ordnance. Normally, jettison non-explosive or explosive ordnance on the range being worked. Non-explosive stores can be jettisoned in Area 1, Area 2, or in PIRA Area 3, or as coordinated. The aircrew chooses the jettison area. Both Edwards AFB Control Tower and SPORT MRU provide advisory service to the aircrew. As a last resort, jettison explosive ordnance on PB-13 under SPORT MRU or DOWNFALL Tower Range Control Officer control. Refer to Edwards AFB Instruction 13-204 for Stores Jettison Area and Bailout Area Procedures.

4.13. Electronic Warfare Training Operations. Electronic warfare training facilities provide a realistic electronic threat environment through the use of surface-to-air missile and anti-aircraft artillery threat emitter simulators. These facilities provide electronic warfare support for composite force training, unit exercises, unit gunnery competitions, normal training missions, and higher Headquarters exercises or inspections.

4.13.1. Electronic Warfare Personnel Training. To provide a realistic threat environment, electronic warfare threat operators require a working knowledge of electronic warfare doctrine and employment concepts.

4.13.1.1. Training Programs. Management criteria, administrative practices, and training policies required to conduct a comprehensive electronic warfare training program include academics and practical application in live and simulated environments. Electronic warfare threat operators should be familiar with applicable MAJCOM training events and requirements defined in Air Force 11-2MDSv1s and RAP Tasking Memos.

4.13.1.2. Responsibilities. Range operating authorities will select and designate in writing instructor personnel responsible for conducting site training for government personnel. The unit training officer/site manager will develop the unit training programs, training schedules, quarterly and annual unit training plans, and lesson plans, and is the unit approving authority for all unit-developed training courses or programs. For contractor run operations, the range operating authority must ensure the contract contains the minimum training requirements listed in **Attachment 5**. **(T-1)**

4.13.2. Range operating authorities should make every effort to provide scores and feedback for requested electronic warfare activity. These are documented and delivered in a mutually agreed upon format.

4.13. (EDWARDSAFB) Electronic Warfare Training Operations. No information added due to the 412 TW not having a test mission requirement for this specific range capability on the PIRA.

4.14. Flare and Chaff Employment. Unless further restricted by MAJCOM or local supplements, flares will be employed in accordance with AFI 11-214, while chaff will be employed in accordance with the latest version of Chairman of the Joint Chiefs of Staff Manual (CJCSM) 3212.02E, *Performing Electronic Attack in the United States and Canada for Tests, Training and Exercises*, and Air Force 11-2MDS series publications. This guidance assumes use of current inventory flares and chaff, a frequency authorization for chaff, and an adequate environmental analysis of the use of flare and chaff. The use of flare or chaff cartridges having significantly different characteristics (pyrophoric flares or double squibbed chaff) is restricted to authorized test and evaluation activities until their use is authorized by the appropriate MAJCOM.

4.14.1. **(Added-EDWARDSAFB)** Chaff is authorized for release on the PIRA. While using the PIRA, chaff will be deployed according to the annual USAF Spectrum Management Office authorization memorandum. **EXCEPTION:** The use of chaff is prohibited while over the PIRA during EOD operations.

4.14.2. **(Added-EDWARDSAFB)** Flares are authorized for release above 5,000 feet above ground level while using the PIRA. **EXCEPTION:** The use of flares is prohibited while over the PIRA during EOD operations.

4.14.3. **(Added-EDWARDSAFB)** The use of flares below 5,000 feet above ground level while using the PIRA requires a low fire danger condition as determined by the 412 TW Range Safety Office or the applicable 412 TW organization.

4.14.4. **(Added-EDWARDSAFB)** The use of flares requires a burn out or consumption at 1,000 feet above ground level within the PIRA.

4.14.5. **(Added-EDWARDSAFB)** Flares are not authorized over inhabited areas or manned sites. This includes MJU-7/10 and M-206 flares.

4.14.6. **(Added-EDWARDSAFB)** Aircrews shall pre-brief SPORT MRU prior to flight on the intended flare or chaff operations. Refer to R-2515 Users Handbook, *Airspace and Flying Procedures*, on aircraft use of chaff or flares.

4.14. (EDWARDSAFB) Chaff and Flare Employment Operations.

4.15. Global Positioning System (GPS) Electronic Attack. GPS electronic attack in DoD tests, training, and exercises are employed IAW the latest version of CJCSM 3212.03A, *Performing Tests, Training, and Exercises Impacting the Global Positioning System*, and classified supplement. These activities require approval from the Commander, United States Strategic Command.

4.15.1. GPS Electronic Attack Requests and Approvals. Units requesting GPS electronic attack must have official sanction from a DoD organization or authorized contractor (under contract with a DoD organization) and be conducted to ensure forces can operate in an intense electronic warfare environment. **(T-0)** Units must develop and submit requests IAW the procedures established in CJCSM 3212.03A. **(T-0)**

4.15.2. GPS Electronic Attack Employment. Teams conducting these operations will maintain positive control of GPS jammers. **(T-0) Note:** Operating jammers via remoting equipment is considered to be positive control. For range safety, the range operating authority will provide the teams operating GPS jamming equipment with a primary and alternate means of communication. **(T-1)**

4.15.2.1. Ranges must provide dedicated Cease Buzzer telephone numbers for the duration of the GPS electronic attack employment. **(T-0)** The range or proponent must provide these telephone numbers with the initial request package. **(T-1)** Any change to Cease Buzzer numbers must be provided to the Air Force Spectrum Management Office by either the range or the proponent in an updated request to be forwarded to United States Strategic Command and received by FAA Headquarters not later than 7 days prior to event execution. **(T-0)**

4.15.2.2. Ranges must provide the GPS jamming teams with the capability to monitor Guard and range operations channels. **(T-0)**

4.15.2.3. Ranges must make notifications prior to and post GPS electronic attack operations IAW the GPS electronic attack approval message from the Commander, United States Strategic Command (and FAA or the United States Coast Guard concurrence messages, if applicable). **(T-0)**

4.15.2.4. Ranges must make Cease Buzzer notifications IAW GPS electronic attack approval message from the Commander, United States Strategic Command and the procedures defined in CJCSM 3212.03A. **(T-0)**

4.16. Laser and Directed Energy Weapon Operations.

4.16.1. Laser Range Operations. The range operating authority will not authorize laser operations unless the range is certified and the laser either complies with the Federal Performance Standard for Laser Products issued by the Food and Drug Administration, Center for Devices and Radiological Health, in Title 21, CFR, Subchapter J, Parts 1040.10 and 1040.11 or has a current and valid exemption issued by the Directed Energy Safety Board. **(T-1)** Examples of lasers include designators, rangefinders, and illuminators/pointers.

4.16.1.1. Laser operators shall be trained annually on safe use procedures by their Unit Laser Safety Officer. Unit Laser Safety Officers may contact the supporting Bioenvironmental Engineering Flight for laser safety training information.

4.16.1.2. Military-Specific Laser Operations. Prior to commencement of laser operations on Air Force ranges:

4.16.1.2.1. All Air Force military-specific lasers shall have an exemption from the Air Force Directed Energy Safety Board and be certified for use on Air Force ranges by the 711 HPW/RHD.

4.16.1.2.2. Other DoD components using non-Air Force military-specific lasers on laser-certified Air Force ranges shall provide the range operating authority with their service's laser use approval. Additionally, non-Air Force military-specific laser systems must have their corresponding safety evaluation reviewed by the 711 HPW/RHD, and the laser must be included on the list provided by the 711 HPW/RHD as a reviewed non-Air Force laser system. As long as these requirements have been met, non-Air Force personnel may be authorized to operate non-Air Force military-specific lasers on an Air Force range. Air Force personnel may participate in joint training, but Air Force personnel may not operate non-Air Force military-specific lasers. For any person within the laser hazard zone, proper personal protection equipment (PPE), such as laser eye protection (LEP), is required.

4.16.1.2.3. Foreign nations using non-U.S. Air Force military-specific lasers on U.S. Air Force ranges shall provide the range operating authority with an independent (i.e., not conducted by the manufacturer) laser safety evaluation, if available. The range operating authority will coordinate laser use with the 711 HPW/RHD. U.S Air Force personnel may participate, but U.S. Air Force personnel may not operate non-U.S. Air Force lasers. For any person within the laser hazard zone, proper personal protection equipment (PPE), such as laser eye protection (LEP), is required. The foreign nation must comply with North Atlantic Treaty Organization (NATO) Allied Range Safety Publication (ARSP)-4, Edition B, Version 2, *Laser Safety for Military Use*, or, for non-North Atlantic Treaty Organization nations, an equivalent standard to be approved by the range operating authority on a case-by-case basis.

4.16.2. Directed Energy Weapon Range Operations. The range operating authority will not authorize directed energy weapon operations (to include, but not limited to, high-energy lasers, weaponized microwave and millimeter wave beams, explosive-driven electromagnetic pulse devices, acoustic weapons, laser induced plasma channel systems, non-lethal directed energy devices, and atomic-scale and subatomic particle beam weapons) unless the weapon has been certified (or approved for testing) IAW AFI 91-401, and the range has been certified for the weapon. **(T-1)**

4.16.3. Research and Test and Evaluation Activities.

4.16.3.1. Research and Test and Evaluation activities involving lasers or directed energy weapons do not require range certification if conducted IAW DoDI 5000.89_DAFI 99-103 and applicable research and test management instructions. However, lasers used for research, development, test and evaluation events or demonstrations on a range must be reviewed by the 711 HPW/RHD, and they also must be approved by the Directed Energy Safety Board if the risk level cannot be mitigated to "Low." The range operating authority will include the Installation Laser Safety Officer and Bioenvironmental Engineering Office, as applicable, on the Safety Review Team cited in DoDI 5000.89_DAFI 99-103. **(T-1)** The safety review must show control of the hazards from the beginning of operations until termination. **(T-1)**

4.16.3.2. The safety review assessment team must show that hazards extending beyond restricted airspace or controlled government land have the appropriate levels of federal and local coordination in place. **(T-1)** The Installation Commander or range operating authority is the final approval authority in lieu of formal range certification. Additional expertise regarding Test and Evaluation can be obtained by contacting the 711th Human Performance Wing (711 HPW) or other Air Force Research Laboratory unit with applicable expertise.

4.16.4. Range Certification. The range operating authority will request laser and/or directed energy weapon range certification through the MAJCOM to the 711 HPW/RHD. **(T-2)** The 711 HPW/RHD will report certification results to the range operating authority, the base Bioenvironmental Engineering office, the parent MAJCOM, and AF/A3TI. **(T-2)**

4.16.4.1. Laser range evaluations are conducted by 711 HPW/RHD in collaboration with other agencies as needed to enable safe mission accomplishment. 711 HPW/RHD will include in the evaluation an on-site survey, a review of proposed activities (weapons, targets, and tactics) and provide range-specific laser safety footprints, safety recommendations, flight profiles, and systems approved for use on that range. **(T-1)**

4.16.4.2. Directed energy weapon evaluations will be a collaborative effort between 711 HPW/RHD, the Air Force Research Laboratory / Directed Energy Directorate (AFRL/RD), and any other agency needed to ensure foreseeable health and material safety concerns are considered. **(T-1)**

4.16.5. Certification Annual Reviews. The Laser Safety Officer and/or Directed Energy Weapon Safety Officer will review applicable operations with the range operations officer or range operating authority annually (within 45 days of the certification report's anniversary) to ensure activities are within the scope of the current range certification. **(T-2)** If any activities or physical changes to the range are found to be outside the scope, the range operations officer/range operating authority will terminate the affected operations. The range operations officer/range operating authority will then contact the 711 HPW/RHD to evaluate the changes and incorporate them into the range certification. **(T-1)** Contact the 711 HPW/RHD for assistance. **Note:** The previous two sentences do not apply to research or Test and Evaluation activities, provided the conditions in **paragraph 4.16.3.1** have been met. Range operating authorities will report findings from the annual review to the parent MAJCOM and 711 HPW/RHD. **(T-2)**

4.16.6. Certification Validity. Certifications expire at the end of the third fiscal year following issuance (e.g., a certification dated 3 December 2017 expires on 30 September 2021). Range operating authorities must ensure the certification is re-accomplished IAW **paragraph 4.16.4** before covered activities may resume. **(T-2)** Prior to expiration, the certification is valid for the equipment, aircraft, targets, tactics and weapons evaluated as long as the report's recommendations are complied with, land and airspace boundaries remain unchanged, annual reviews are submitted to the MAJCOM and 711 HPW/RHD within 45 days of the initial report's anniversary, and laser specular hazards are periodically removed (as specified in **paragraph 4.16.8.4**.). The reinstatement or extension of a certification is at the discretion of the 711 HPW/RHD based on the scope of any reported changes.

4.16.7. Commander, United States Strategic Command Coordination. Use of Category II or III lasers or laser systems that have the potential to propagate above the horizon requires range operating authority notification to United States Strategic Command Commander in accordance with DoDI 3100.11, *Management of Laser Illumination of Objects in Space*, and Chairman of the Joints Chiefs of Staff Instruction (CJCSI) 3225.01A, *Procedures for Management* of *Illumination of Objects in Space*. **(T-0)**

4.16.8. General Requirements for Lasers and Laser Directed Energy Weapons.

4.16.8. **(EDWARDSAFB) General PIRA Requirements for Lasers and Laser Energy Operations.**

4.16.8.1. Lasers will not be operated without the approval of the Laser Safety Officer and the range operations officer (range control officer during Class A service). **(T-2)**

4.16.8.1. **(EDWARDSAFB) Combat or Non-Eye Safe Laser Operations.** Lasers will not be operated without the approval of the 412 RANS Laser Safety Officer and 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer during Class A range service.

4.16.8.1.1. **(Added-EDWARDSAFB) Roles and Responsibilities.** For the purposes of this Supplement, the terms "area track" and "point track" are interchangeable and indicate a targeting pod mode that will continue to track a specific geographic area without aircrew inputs.

4.16.8.1.2. **(Added-EDWARDSAFB)** The program Range Control Specialist, Test Conductor, Test Director, or aircrew will perform pre-mission planning and coordination with the SPORT MRU and the DOWNFALL Tower Range Control Officer prior to conducting laser operations.

4.16.8.1.3. **(Added-EDWARDSAFB)** Pilot pre-flight briefing with SPORT MRU and DOWNFALL Tower Range Control Officer will include combat laser attack plans.

4.16.8.1.4. **(Added-EDWARDSAFB)** Once airborne, SPORT MRU will confirm combat laser planned attack.

4.16.8.1.5. **(Added-EDWARDSAFB) Laser Clearance.** SPORT MRU issues laser clearances (for example, "(AIRCRAFT CALL SIGN), CLEARED TO LASE").

4.16.8.1.6. **(Added-EDWARDSAFB) Airborne Combat Laser Operations.** The aircrew will ensure laser switches are off or safe until cleared by SPORT MRU. Prior to the combat laser pass, the aircrew will inform SPORT MRU of the target, planned altitude, run-in direction, and any ordnance to be dropped, (for example, "(AIRCRAFT CALL SIGN), NEXT ATTACK DRY WITH COMBAT LASER AGAINST PB-1, WEST-TO-EAST RUN-IN, FL200").

4.16.8.1.7. **(Added-EDWARDSAFB) Prior To and During Airborne Laser Operations.** Combat laser communications prior to and during airborne laser operations on the PIRA are described using the following scenario examples.

4.16.8.1.7.1. **(Added-EDWARDSAFB)** The aircrew will achieve area track or point track on the target and will inform SPORT MRU and DOWNFALL Tower Range Control Officer once this is accomplished (for example, "ZOOM 01, POINT TRACK ON PB-1, REQUEST PERMISSION TO LASE" or "ZOOM 01, AREA TRACK ON PB-1, REQUEST PERMISSION TO LASE").

4.16.8.1.7.2. **(Added-EDWARDSAFB)** SPORT MRU will then check for traffic between the lasing aircraft and the target, receive clearance from DOWNFALL Tower Range Control Officer for laser operations and clear the aircraft to lase (for example, "ZOOM 01, CLEARED TO LASE PB-1").

4.16.8.1.7.3. **(Added-EDWARDSAFB)** If there is a conflict, SPORT MRU will call "ZOOM 01, ABORT", until the conflict is resolved.

4.16.8.1.7.4. **(Added-EDWARDSAFB)** The aircrew will acknowledge clearance to lase (for example, "ZOOM 01, CLEARED TO LASE, PB-1")

4.16.8.1.7.5. **(Added-EDWARDSAFB)** The aircrew will inform SPORT MRU the laser is armed (for example, "ZOOM 01, LASER – ARM")

4.16.8.1.7.6. **(Added-EDWARDSAFB)** The aircrew will call, if the time permits, when lasing begins (for example, "ZOOM 01, LASING")

4.16.8.1.7.7. **(Added-EDWARDSAFB)** The aircrew completes the lasing run and will inform SPORT MRU they are complete, and the laser is off, or the laser is safe (for example, "ZOOM 01, LASER SAFE").

4.16.8.1.7.8. **(Added-EDWARDSAFB)** When complete with an actual weapons delivery in conjunction with laser operations, the aircrew will call "ZOOM 01, OFF HOT, SWITCHES SAFE, LASER SAFE". The SPORT MRU and DOWNFALL Tower Range Control Officer will acknowledge this call (for example, "ZOOM 01, ROGER").

4.16.8.1.7.9. **(Added-EDWARDSAFB)** DOWNFALL Tower Range Control Officer will notify SPORT MRU that Mercury Boulevard is closed.

4.16.8.1.7.10. **(Added-EDWARDSAFB) Dragging Laser Passes.** For passes where the laser will be dragged across the ground or other than area track and point track on the targeting pod, pilots will use plain terminology to communicate requirements to SPORT MRU and DOWNFALL Tower Range Control Officer.

4.16.8.1.7.11. **(Added-EDWARDSAFB) Lasing Contingencies.** Lasing aircraft will maintain vigilant lookout in the targeting pod for an unstable pod track and for air or ground traffic. If traffic is encountered or if the pod stops tracking the target, the lasing aircraft will immediately stop lasing, break track and switch laser to safe. The aircrew will call "KNOCK-IT-OFF" over the radio.

4.16.8.1.7.12. **(Added-EDWARDSAFB)** SPORT MRU will control and issue all lasing clearances.

4.16.8.1.7.13. **(Added-EDWARDSAFB)** SPORT MRU may transfer control of the aircraft and lasing clearances to DOWNFALL Tower Range Control Officer during Flight Lead Control and other mission specific operations.

4.16.8.1.7.14. **(Added-EDWARDSAFB)** For eye-safe, non-eye safe, and combat laser operations conducted from an aircraft or airborne platform, a DOWNFALL Tower Range Control Officer or a 412 RANS Laser Safety Officer will conduct real-time monitoring of laser operations at DOWNFALL Tower or at a 412 RANS test site to ensure sufficient situational awareness and safe laser operations are conducted on the PIRA.

4.16.8.2. The user will notify the range operations officer or range control officer upon termination of laser activity. **(T-3)** The range operations officer or range control officer will acknowledge termination and record the start and stop time of range periods when laser operations take place and will also record the laser(s) used during each period. **(T-3)**

4.16.8.2. **(EDWARDSAFB)** The aircrew or range test program customer will notify the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer upon termination of laser activity. The 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer will acknowledge termination and will record the start and stop time of range periods when laser operations take place.

4.16.8.3. Range users will not direct ground lasers at targets or hazard areas where the laser beam ocular hazard will extend beyond the range boundary unless the hazardous laser energy exclusively transits restricted airspace (or equivalent) up to an altitude of 60,000 feet (Flight Level 600) and terminates in space without affecting non-participating satellites. **(T-0) Note:** A laser does not need to be an ocular hazard to humans for it to be a hazard to satellites.

4.16.8.3. **(EDWARDSAFB) PIRA Ground Laser Procedures.** Ground lasers will not be directed at targets or hazard areas where the laser beam will terminate with an ocular hazard beyond the PIRA boundary unless the hazardous laser energy will exclusively transit restricted airspace (or equivalent) up to an altitude of 60,000 feet (Flight Level 600) and terminate in space without affecting non-participating satellites.

4.16.8.4. Range personnel shall periodically inspect the targets, impact area, and Laser SDZ for specular reflectors and remove or cover as needed IAW MIL-HDBK-828. **(T-2)**

4.16.8.4. **(EDWARDSAFB)** The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will ensure specular hazards are periodically removed by PIRA personnel from the Laser Surface Danger Zone surrounding the PIRA targets. Reference guidance in MIL-HDBK-828B, *Range Laser Safety*, as necessary.

4.16.8.5. IAW AFI 48-139, the supporting Aerospace Medicine function will accomplish medical surveillance examination requirements for range personnel (as determined by the Installation Occupational and Environmental Health Working Group) before working in a laser hazard environment. **(T-2)**

4.16.8.5. **(EDWARDSAFB)** As referenced in AFI 48-139, the director of Edwards AFB Medical Services will be notified of the specific types of laser activities that occur on the PIRA and will accomplish medical surveillance examination requirements for PIRA personnel, as determined by the Edwards AFB Occupational and Environmental Health Working Group or a qualified medical provider, before working in a laser hazard environment.

4.16.8.6. IAW AFI 48-139, all range personnel within the Laser SDZ during laser operations will wear laser eye protection of appropriate optical density for the wavelength of the laser being utilized. **(T-2)**

4.16.8.6. **(EDWARDSAFB)** As referenced in AFI 48-139, all PIRA personnel within the Laser Surface Danger Zone during laser operations will wear laser eye protection of appropriate optical density.

4.16.8.7. **(Added-EDWARDSAFB)** For eye-safe, non-eye safe, and combat laser operations conducted on the ground or a surface platform, a DOWNFALL Tower Range Control Officer, a 412 RANS Laser Safety Officer, or a designated program Range Control Specialist will conduct real-time monitoring of laser operations at DOWNFALL Tower or at a 412 RANS test site to ensure sufficient situational awareness and safe laser operations are conducted on the PIRA.

4.16.9. General Requirements for Non-Laser Directed Energy Weapons.

4.16.9.1. Range operating authorities will ensure non-laser directed energy weapon systems are operated only with approval from the Directed Energy Weapon Safety**. (T-2)**

4.16.9.2. The user will notify the range operations officer or range control officer upon termination of directed energy weapon activity. **(T-3)** The range operations officer or range control officer will acknowledge termination and record the start and stop time of range periods when directed energy weapon operations take place. **(T-3)**

4.16.10. Accident/Incident Injury Reporting. Report all actual or suspected laser accidents, incidents, and injuries that impact DoD personnel and coordinate immediate expert medical advice by calling the Tri-Service Laser Injury Hotline (1-800-473-3549) and the Unit Laser Safety Officer. **(T-0)**

4.16.11. Laser and Directed Energy Weapon Certification on Foreign and Other United States Service Ranges. When Air Force aircraft operate on another nation's/service's range, MAJCOMs will ensure safe use procedures have been developed and are promulgated by the host nation/service prior to employing lasers and/or directed energy weapons. In the event procedures have not been developed, range users will comply with this manual and NATO ARSP-4, Edition B, Version 2. All military-specific lasers used by Air Force personnel, in any location, shall have a safety approval from the Air Force Directed Energy Safety Board. **(T-1)**

4.16. (EDWARDSAFB) Laser Energy Operations and Attack Restrictions. Laser operations guidelines and attack restrictions are provided in the latest 711 HPW/RHDO Special Report: Edwards PIRA Certification for the Safe Use of Lasers. Contact the 412 RANS Range Operations Officer, the 412 RANS Director of Operations, the 412 RANS Laser Safety Officer, or the 412 TW Range Safety Office for a copy of the latest document. Any laser system that is not identified in the latest 711 HPW/RHDO report must be reviewed and approved by 412 TW Range Safety Office. The use of lasers must be coordinated with the 412 RANS Laser Safety Officer, 412 RANS Range Operations Officer, and the 412 RANS Director of Operations prior to laser energy operations on the PIRA.

4.17. Range Safety.

4.17.1. Visual Identification of Manned Equipment, Facilities and Sites. To the maximum extent possible, equipment and facilities, manned or unmanned, that are not targets should include visual identification systems, such as high contrast paint (white or orange), infrared radiation strobes, etc. Range personnel will not use white or orange paint on any "bombable" targets with the exception of specific test targets where white supports test objectives. **(T-1)**

4.17.1. **(EDWARDSAFB) Visual Identification of Manned Equipment, Facilities and Sites.** To the maximum extent possible, equipment and facilities, manned or unmanned, that are not PIRA targets should include visual identification systems, such as high contrast paint (white or orange), or infrared radiation strobes. White or orange paint shall not be used on any "bombable" targets except for strafe "rags" or specific test targets that use white paint to support test objectives.

4.17.2. Range Visitor Procedures. Each range operating authority must publish or reference procedures in their local range supplement to ensure positive control of all spectators. **(T-3)** Range operating authorities must analyze visitor locations using the appropriate hazard area(s) to ensure that spectators are not within these areas during operations. **(T-2)** The range operating authority will have control in those limited cases where DoD personnel or contractors are within the hazard area during operations to accomplish a specific mission-related task. **(T-2)**

4.17.2. **(EDWARDSAFB) Range Demonstrations and Visitor Procedures.** The 412 RANS Range Operations Officer will publish or reference procedures in this Supplement to ensure positive control of all spectators in accordance with Edwards AFB Range Operating Authority. Visitor locations must be analyzed using the appropriate weapons footprint or hazard area program to ensure that spectators are not within the hazard area during PIRA operations. The Edwards AFB Range Operating Authority will have control in those limited cases where DoD personnel or contractors must be within the hazard area during operations to accomplish a specific mission-related task.

4.17.2.1. **(Added-EDWARDSAFB) Positive control procedures for visitors requiring access to the PIRA.** Visitors will perform the following:

4.17.2.1.1. **(Added-EDWARDSAFB)** Sign in and out at DOWNFALL Tower, building 9505.

4.17.2.1.2. **(Added-EDWARDSAFB)** Sign out a PIRA assigned radio.

4.17.2.1.3. **(Added-EDWARDSAFB)** Receive a brief by the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer on the PIRA rules, hazards, communication protocols, and mission schedule for that day.

4.17.2.1.4. **(Added-EDWARDSAFB)** Follow the communication procedures as briefed by the 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer.

4.17.2.2. **(Added-EDWARDSAFB) PIRA Escort Requirements.** 412 RANS PIRA personnel will escort visitors as necessary. Visitors requiring escorted access to the PIRA will sign in and out at DOWNFALL Tower, building 9505.

4.17.2.2.1. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will validate the need for visitors to enter the PIRA.

4.17.2.2.2. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, or designated escorts will confirm the visitor's identity via picture identification or personal recognition.

4.17.2.2.3. **(Added-EDWARDSAFB)** 412 RANS escorts will ensure visitors are continuously escorted while on the PIRA.

4.17.3. Flight Safety.

4.17.3.1. Airspace Activation. The range operating authority will ensure range airspace is activated with the local air traffic control facility or air traffic representative IAW local agreements between the range operating authority and the controlling agency. **(T-0)** The request for activation should be timed (before a scheduled mission) to allow shared or joint users to clear the area and for the controlling agency to make internal adjustments.

4.17.3.2. Weather Observation. The range control officer will monitor weather conditions to include altimeter setting, temperature, ceiling, visibility, and winds. **(T-2)** Observations may be obtained from the local base weather station, command post, Automated Weather Observing System, or air traffic control agency at least hourly, if the capability is not available at the range. The supporting weather detachment or command post should advise the range control officer of any sudden adverse weather changes that might affect range operations or safety.

4.17.3.2. **(EDWARDSAFB) Edwards Weather Observation.** The DOWNFALL Tower Range Control Officer will monitor weather conditions to include altimeter setting, temperature, ceiling, visibility, and winds. Observations may be obtained from the Edwards AFB Weather Office, Automated Weather Observing System (AWOS), or SPORT MRU at least hourly. The Edwards AFB Weather Office or specified 412 TW agency should advise DOWNFALL Tower of any sudden adverse weather changes that might affect PIRA operations or safety.

4.17.3.2.1. **(Added-EDWARDSAFB) Adverse Weather Impacts to 412 RANS Mission Support.** 412 RANS mission support equipment and facilities will not be used during periods of adverse weather where there is a probability of damage to equipment, degradation of data beyond acceptable limits, or a hazard to personnel and property.

4.17.3.2.1.1. **(Added-EDWARDSAFB)** Any 412 RANS personnel who suspect that equipment or facility limits of operation are being exceeded due to adverse weather conditions or may affect real-time mission support will immediately notify the supervisor or leader-in-charge.

4.17.3.2.1.2. **(Added-EDWARDSAFB)** The supervisor or leader-in-charge will assess the situation to determine if adverse weather conditions are exceeding the established equipment or facility limits and will notify the 412 RANS Operations Duty Officer that mission support equipment be placed on a "weather hold" or support terminated. The adverse weather limits are:

4.17.3.2.1.2.1. **(Added-EDWARDSAFB)** The wind limit for the 8-foot and the 5-meter fixed telemetry antennas are 60 miles per hour (52 knots).

4.17.3.2.1.2.2. **(Added-EDWARDSAFB)** The wind limit for the gunnery strafe snow fence target is 20 knots.

4.17.3.2.1.2.3. **(Added-EDWARDSAFB)** The wind limit for the optical tracker is 20 knots with gusts no greater than 30 knots, the rain limit is the first evidence of precipitation, and a zero-degree Fahrenheit temperature chill factor.

4.17.3.2.1.2.4. **(Added-EDWARDSAFB)** The wind limit for the Radiometric Analysis and Measurement System with the antenna mast extended is 30 miles per hour (26 knots) and there is no wind limit when the mast is not extended.

4.17.3.2.1.2.5. **(Added-EDWARDSAFB)** The wind limit for the Solar Active Edge Target hemispheres and tarps is 15 knots.

4.17.3.2.1.2.6. **(Added-EDWARDSAFB)** The wind limit for the telemetry ground station 6-foot roof mounted antenna is 55 miles per hour (48 knots) and the 8-foot towed antenna is 60 miles per hour (52 knots).

4.17.3.2.1.3. **(Added-EDWARDSAFB)** The 412 RANS Operations Duty Officer will notify the program Range Control Specialist, the DOWNFALL Tower Range Control Officer, and the 412 RANS Director of Operations of the action taken to secure the equipment or facility and reduce the hazards to personnel during adverse weather conditions.

4.17.3.3. Avian Hazard Advisory System. Where internet connectivity permits, range control officers will utilize Avian Hazard Advisory System (**http://www.usahas.com**) to monitor bird movement and bird strike risk levels for the airspace associated with the range. **(T-2)**

4.17.3.3. **(EDWARDSAFB) Avian Hazard Advisory System (AHAS).** Where internet connectivity permits, the DOWNFALL Tower Range Control Officers will utilize AHAS (**http://www.usahas.com**) to monitor bird movement and bird strike risk levels for the airspace associated with the PIRA.

4.17.3.4. Aircraft Accident Procedures. In case of an aircraft crash during Class A service, the range control officer will initiate all necessary emergency actions. **(T-1)** These include appropriate notifications, closing of the range, and assuming responsibility as the interim on-scene commander until the crash response team can be organized. During Class B, Class C, and Class D service, the flight lead, individual pilot, Forward Air Controller (Airborne), JTAC or other briefed person will start the emergency actions, and if feasible, act as the interim on-scene commander. **(T-2)** Investigations are conducted IAW DAFI 91-204, *Safety Investigations and Reports.*

4.17.3.4. **(EDWARDSAFB) Aircraft Accident Emergency Procedures.**

4.17.3.4.1. **(Added-EDWARDSAFB) Lost Communications.** An aircrew will abort the run if radio contact is lost with SPORT MRU, DOWNFALL Tower, DAGRAG Control, or COWBELL Control at or prior to countdown to a release or a fire.

4.17.3.4.1.1. **(Added-EDWARDSAFB)** Aircrews will attempt to contact SPORT MRU on common frequency. If unable, contact Edwards AFB Control Tower and return to base. Do not use the PIRA without clearance.

4.17.3.4.1.2. **(Added-EDWARDSAFB)** SPORT MRU will immediately notify Edwards AFB Control Tower.

4.17.3.4.2. **(Added-EDWARDSAFB) Hung Ordnance.** For hung ordnance, an aircrew will:

4.17.3.4.2.1. **(Added-EDWARDSAFB)** Notify SPORT MRU and DOWNFALL Tower.

4.17.3.4.2.2. **(Added-EDWARDSAFB)** Remain with SPORT MRU or DOWNFALL Tower.

4.17.3.4.2.3. **(Added-EDWARDSAFB)** Follow instructions from SPORT MRU or DOWNFALL Tower Range Control Officer to jettison ordnance on the PIRA.

4.17.3.4.2.4. **(Added-EDWARDSAFB)** Notify SPORT MRU and DOWNFALL Tower if jettison is unsuccessful.

4.17.3.4.3. **(Added-EDWARDSAFB) Runaway Gun.** For a runaway gun, an aircrew will:

4.17.3.4.3.1. **(Added-EDWARDSAFB)** Direct fire into the targets as long as possible and then toward an uninhabited area, preferably a point approximately 1 nautical mile south of Haystack Butte if on DAGRAG or West Range.

4.17.3.4.3.2. **(Added-EDWARDSAFB)** Continue direct fire to the target area for an aircraft with a rear firing gun.

4.17.3.4.3.3. **(Added-EDWARDSAFB)** Fly a course parallel to Foul Line Road then circle the target area directing fire into the target area for an aircraft with a side-firing gun.

4.17.3.4.4. **(Added-EDWARDSAFB) Aircraft Crash Procedures.** If an aircraft crash occurs, the 412 RANS Range Operations Officer and DOWNFALL Tower Range Control Officer will:

4.17.3.4.4.1. **(Added-EDWARDSAFB)** Immediately notify SPORT MRU, who in turn notifies Edwards AFB Control Tower to activate the Primary Crash Phone.

4.17.3.4.4.2. **(Added-EDWARDSAFB)** Provide SPORT MRU with aircraft emergency information to include aircraft type and call sign, crash location, condition of crew (if known, use the "DIM" code as outlined in the 412 TW Installation Emergency Management Plan 10-2), and the aircraft's unit or organization, if known.

4.17.3.4.4.3. **(Added-EDWARDSAFB)** Terminate all air activity on the PIRA except that associated with search and rescue.

4.17.3.4.4.4. **(Added-EDWARDSAFB)** Advise 412 TW Command Post (UHF 304.0 MHz), the 412 RANS Director of Operations, and the 412 RANS Operations Duty Officer.

4.17.3.4.4.5. **(Added-EDWARDSAFB)** Provide Incident Commander status and direct rescue efforts until relieved by the Edwards AFB Fire Department.

4.17.3.4.4.6. **(Added-EDWARDSAFB)** Maintain communications and provide assistance.

4.17.4. Occupational and Explosive Safety.

4.17.4.1. Fire Prevention. Identify range activities likely to cause range fires and develop and implement procedures to mitigate them. Identify a single agency or office responsible for determining the fire danger hazard level, typically in cooperation with the land management agency. Develop a decision matrix and checklist to standardize fire prevention actions and ensure appropriate responses as fire conditions change. Range operating authorities will ensure fire prevention is included in the Wildland Fire Management Plan and range supplement. **(T-3)**

4.17.4.1. **(EDWARDSAFB) PIRA Fire Prevention.**

4.17.4.1.1. **(Added-EDWARDSAFB)** When a mission involves flares and tracer munitions, the DOWNFALL Tower Range Control Officer will notify the Edwards AFB Fire Department 24 hours prior to the scheduled mission start time. If the mission cancels, then notify the Fire Department.

4.17.4.1.2. **(Added-EDWARDSAFB)** The DOWNFALL Tower Range Control Officer will confirm the Edwards AFB Fire Department coordination (661-275-5232 / 5181 or DSN 525-5232 / 525-5181) 30 minutes before the scheduled mission start time. The DOWNFALL Tower Range Control Officer will terminate Edwards AFB Fire Department standby action when the mission concludes.

4.17.4.2. Fire Suppression Equipment. On ranges providing Class A or manned Class B or T service, appropriate fire suppression equipment and personnel must be available to deal with local fire hazards. **(T-2)** This capability may be provided through Air Force, host military organization for tenant units, landowner, in-service civilian, independent contractor assets, or through agreement with another government agency such as the Bureau of Land Management or United States Forest Service. The range operating authority, in coordination with the agency or organization providing fire support, shall identify the fire suppression equipment requirements and will ensure that fire suppression equipment sufficient to protect the lives of personnel on the range. **(T-2)**

4.17.4.2. **(EDWARDSAFB) Fire Suppression Responsibilities.**

4.17.4.2.1. **(Added-EDWARDSAFB)** In the event that any fire is observed or reported within the PIRA Master Control Tower compound, regardless of the fire size, the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer will:

4.17.4.2.1.1. **(Added-EDWARDSAFB)** Call 911 or Call the AFRL to report the building number, location of fire, and nature of fire.

4.17.4.2.1.2. **(Added-EDWARDSAFB)** Direct 412 RANS personnel and range users within the compound to take specific actions.

4.17.4.2.1.3. **(Added-EDWARDSAFB)** Assign 412 RANS personnel tasks to assist the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer in fire suppression or operations actions.

4.17.4.2.1.4. **(Added-EDWARDSAFB)** Use telephone or radio to notify SPORT MRU and the 412 RANS Operations Duty Officer of the compound fire.

4.17.4.2.2. **(Added-EDWARDSAFB)** In the event that any fire is observed or reported within or adjacent to the West Range or East Range areas, the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, 412 RANS personnel, or range users will:

4.17.4.2.2.1. **(Added-EDWARDSAFB)** Call 911 or Call the AFRL to report the location of the fire and the nature of the fire.

4.17.4.2.2.2. **(Added-EDWARDSAFB)** Use telephone or radio to notify SPORT MRU and the 412 RANS Operations Duty Officer of the fire.

4.17.4.2.2.3. **(Added-EDWARDSAFB)** Announce on the DOWNFALL Tower Radio Net the location of the fire.

4.17.4.2.2.4. **(Added-EDWARDSAFB)** Halt all PIRA missions.

4.17.4.3. Emergency Procedures. Range operating authorities must ensure emergency procedures are readily available to the range control officer and other ground personnel, such as JTACs, for use during range operations. **(T-2)**

4.17.4.3. **(EDWARDSAFB) Emergency Procedures.** DOWNFALL Tower Range Control Officers will have emergency procedures readily available during PIRA operations.

4.17.4.4. Safety Briefing Requirements. The range operating authority or contractor will provide and document initial and yearly occupational and explosive safety briefings, IAW AFMAN 91-201, to personnel assigned to operate and maintain Air Force ranges. **(T-2)** The range operating authority will ensure all personnel participating in range clearance operations receive additional training IAW **paragraph 5.4.2** before assisting in range clearance. **(T-2)** The range operating authority must also ensure that authorized visitors and personnel who infrequently visit the range are appropriately briefed on range safety. **(T-1)**

4.17.4.4. **(EDWARDSAFB) Safety Briefing Requirements.** The 412 RANS Range Operations Officer will provide and document initial and yearly ground and explosive safety briefings, as referenced in AFI 91-202, to 412 RANS personnel assigned to operate and maintain the PIRA.

4.17.4.4.1. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure all personnel participating in range maintenance and clearance operations receive ground and explosive safety briefings by the 812 CES/CED EOD before assisting in range clearance.

4.17.4.4.2. **(Added-EDWARDSAFB)** The safety of personnel and property on the PIRA is of the utmost importance and shall take precedence over all other considerations. Non-EOD personnel will receive appropriate safety briefings and range clearance training from the 812 CES/CED EOD office before assisting in range maintenance and clearance operations.

4.17.4.4.3. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will brief authorized visitors and infrequent visitors on PIRA safety. The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will provide escort and keep these visitors under continuous surveillance by properly briefed and trained PIRA personnel.

4.17.4.5. Hazard Condition Watch. The range control officer and all range personnel must continually watch for hazardous conditions such as trespassers, fires, bird activity, etc. **(T-3)** Range users will be notified immediately of any hazardous conditions on the range. **(T-3)** If safety is in question, the range control officer or other appropriate authority will stop range operations until the situation is remedied. **(T-3)**

4.17.4.5. **(EDWARDSAFB) Hazard Condition Watch.** The DOWNFALL Tower Range Control Officer and 412 RANS personnel must continually watch for hazardous conditions such as trespassers, fires, and bird activity conditions. Program or project users will be notified immediately of any hazardous conditions on the PIRA. If safety is in question, the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, 412 RANS authority, or 412 TW authority will stop range operations until the situation is remedied.

4.17.4.5.1. **(Added-EDWARDSAFB)** The timely reporting of incidents, deviations, or hazards during PIRA operations shall be made to the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officer, the 412 RANS Operations Duty Officer, or the 412 RANS Director of Operations. The 412 TW Range Safety Office will be notified, as required, of ground or flight incidences, deviations or hazards. The DOWNFALL Tower Range Control Officer will make log entries and describe the circumstances of any occurrence requiring such reporting.

4.17.4.5.2. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure all DOWNFALL Tower Range Control Officers and PIRA personnel are trained on applicable BASH procedures as they relate to PIRA operations.

4.17.4.5.3. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure DOWNFALL Tower Range Control Officers notify Airfield Management and the Supervisor of Flying of the bird hazard on the PIRA if other than low conditions while the PIRA is hot. Aircrews are required to notify DOWNFALL Tower Range Control Officer of adverse bird conditions on specific ranges.

4.17.4.6. Performance during Critical Tasks. During the performance of critical situational tasks such as controlling or monitoring the movement of aircraft or ground parties using the range, personnel must not conduct unrelated activities which divert their attention, degrade situational awareness or prevent timely communication. **(T-3)** Such distracting prohibitions include composing and sending emails and text messages unrelated to the mission at hand, browsing web pages unrelated to the mission and making and receiving unofficial phone calls.

4.17.4.6. **(EDWARDSAFB) Performance During Critical Tasks.** During the performance of critical situational tasks such as controlling or monitoring the movement of aircraft or ground parties using the PIRA, the 412 RANS Range Operations Officer, the DOWNFALL Tower Range Control Officers, or designated 412 RANS personnel must not conduct unrelated activities which divert their attention, degrade situational awareness, or prevent timely communications. Such distracting prohibitions include composing and sending e-mails or text messages unrelated to the mission at hand, browsing web pages unrelated to the mission, and making or receiving unofficial phone calls.

4.17.4.6.1. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officers performing duties in DOWNFALL Tower must be vigilant of all PIRA activities and operations.

4.17. (EDWARDSAFB) Range Safety During PIRA Operations. Timely reporting of incidents, deviations, or hazards during PIRA operations shall be made to the 412 RANS Range Operations Officer, the 412 RANS Operations Duty Officer, the 412 RANS Director of Operations, or to the 412 TW Range Safety Officer, as appropriate. DOWNFALL Tower Range Control Officer provides log entry to reflect the circumstances of any occurrence requiring such notification.

4.18. Risk Management. Range operations are complex, and the risk management process requires a dedicated team representing multiple functional areas such as operations, CE (to include EOD), legal, public affairs, and safety. Utilize the risk management process prior to any change to range operations, boundaries, or procedures with the potential to increase flight, occupational or explosive safety risks. The risk management assessment examines type of ordnance authorized, employment tactics, weapons footprints, type of proposed joint or shared use, geographic features, frequency of EOD sweeps, etc. Coordinate assessments through the wing (or equivalent) POC for risk management matters. The AF Form 4437, *Deliberate Risk Assessment Worksheet*, and the ACC Form 167, *ACC Risk Management (R M) Worksheet*, are two tools that may be used to assist with documenting risk management assessments.

4.19. Night Operations.

4.19.1. Covert or blacked-out lighting operations will not be conducted during Class A service unless the range control officer is equipped with binocular, generation-III NVDs. **(T- 2)**. NVDs will be tested, adjusted, and focused by the range control officer before use with (in order of preference) the Hoffman ANV-20/20 Tester, a unit eye lane, or equivalent tester. **(T- 2).** Range control officers will perform NVD testing, adjustment and focusing procedures IAW appropriate TOs, manufacturer instructions or MAJCOM guidance. **(T-2)**

4.19.1. **(EDWARDSAFB)** Aircraft reduced, covert and blacked-out lighting operations will not be conducted during Class A Range Service unless a DOWNFALL Tower Range Control Officer is on the PIRA and is equipped with binocular, generation-III Night Vision Devices (NVDs). NVDs will be tested, adjusted, and focused by the DOWNFALL Tower Range Control Officer before use with (in order of preference) a DoD approved unit eye lane or equivalent tester. NVD testing will be performed using adjustment and focusing procedures according to appropriate technical orders and MAJCOM guidance.

4.19.1.1. **(Added-EDWARDSAFB) DOWNFALL Tower Range Control Officer NVD Operations.** The DOWNFALL Tower Range Control Officers must be day-qualified before upgrading to night operations. Night Range Control Officer training will consist of ground training and observation of at least two night flights under the control of a night-qualified DOWNFALL Range Control Officer. To qualify for NVD duties, a DOWNFALL Tower Range Control Officer must be night qualified.

4.19.2. Range Lighting.

4.19.2. **(EDWARDSAFB) Range Night Lighting Diagrams.** There is no range night lighting diagram or pattern information contained in this Supplement due to Edwards AFB not having a test mission requirement for this specific range capability.

4.19.2.1. Class A Service Identification Lighting. Ranges providing Class A services that support night operations must have available a distinctive pattern of lights visible by aircrews, with and without NVDs, to ensure positive orientation and identification of the range and target area. **(T-2)** These lights should be readily identifiable but not so as to distract aircrews during weapons delivery or wash out target locations. Range operating authorities will ensure that no similar pattern of lights exists near the range that could be misidentified as targets or the impact area, but cultural lighting inside or outside the range boundary may serve as a portion of the distinctive pattern of lights. **(T-2)**

4.19.2.2. Class B Service Identification Lighting. Ranges providing Class B services that support night operations will have lighting as described above unless the range operating authority has determined that the lighting is not required based on a documented, risk management assessment (based on range size, remoteness, etc.) **(T-2)**

4.19.2.3. Target Lighting. Lighting for the specific purpose of illuminating the target is only required on ranges that support users accomplishing unaided visual deliveries. If required (or desired by other users), target lighting may be accomplished using incandescent lights, lanterns, or flares. Lights and lanterns should be NVD compatible or reduced in intensity to produce the minimum halo around each light when viewed through NVDs. Every effort should be made to prevent the target from appearing as a light with a halo to allow for positive target identification.

4.19.2.4. Range Facility Identification.

4.19.2.4.1. Range Tower and Routinely-Manned Facility Lighting. Lighting is required on all routinely-manned range facilities and range towers to facilitate positive identification by aircrew. **(T-2)** Proper management of tower lighting is important to maximize performance of NVDs used by range personnel. Tower lighting (interior and exterior) should be minimized to those necessary to conduct operations and all unnecessary lights should be extinguished or taped over. NVD compatible lighting is highly recommended. Consider moving tower, routinely-manned facilities and obstruction strobe lights to locations that can best support night operations.

4.19.2.4.1. **(EDWARDSAFB) DOWNFALL Tower and Routinely Manned Facility Lighting.** Lighting is required on all routinely manned PIRA facilities and PIRA towers to facilitate positive identification by the aircrew. Proper management of PIRA tower lighting is important to maximize performance of NVDs used by 412 RANS personnel. PIRA tower interior and exterior lighting should be minimized to those necessary to conduct operations and all unnecessary lights should be extinguished or taped over. NVD compatible lighting is highly recommended.

4.19.2.4.2. Other Manned Locations. Other manned locations (such as observation posts, visitor locations, vehicles, etc.) will, as a minimum, mark their location by any means described in Joint Publication 3-09.3 for "friendly marking" that facilitates positive identification by aircrew. **(T-0)**

4.19.3. Infrared Radiation Pointers and/or Lasers. Range users will not use a single, steady infrared radiation point as the sole means of target identification. **(T-1)** If using an infrared radiation pointer to mark targets, range users must ensure either the target or friendly terminus of the pointer is positively verified by another means (snaking, friendly location marking, etc.). **(T-1)** When ground-based lasers/designators are used to designate targets, range users must ensure attack headings within ±10° of the laser designator to target line (safety zone) are prohibited and run-in headings are restricted to a 50° cone on either side of the safety zone. **(T-1)** Per AFI 48-139, personnel will not be exposed to laser radiation in excess of the Maximum Permissible Exposure levels as defined in American National Standards Institute Z136.1, *Safe Use of Lasers*, and avoid unnecessary exposures to laser radiation below the Maximum Permissible Exposure. **(T-2)**

4.20. Unmanned Aerial Systems/Remotely Piloted Aircraft. Range operating authorities shall conduct a risk management assessment of Unmanned Aerial System/Remotely Piloted Aircraft operations and apply appropriate mitigation measures prior to employment on the range. **(T-0)** At a minimum consider:

4.20.1. Kinetic energy potential for lethality and property damage.

4.20.2. Flight paths to avoid endangering concentrations of DoD and civil assets/personnel.

4.20.3. System maturity and employment methodology.

4.20.4. Mid-air collision avoidance through see, sense and avoid, spotter, or chase aircraft.

4.20.5. Airspace and Air Traffic Control coordination. See FAA publications for flight requirements in the National Airspace System.

4.20.6. Robustness and redundancy of vehicle control or telemetry.

4.20.7. Containment measures such as a Flight Termination System, lost link software, or parachute.

Chapter 5

RANGE MAINTENANCE AND CLEARANCE OPERATIONS

5.1. Range Clearance Guidelines. Each MAJCOM/range operating authority is responsible for the clearance of operational ranges under its control IAW DoDI 3200.16, DoDI 4140.62, *Material Potentially Presenting an Explosive Hazard (MPPEH)*, DoDM 4140.72, *Management of Material Potentially Presenting an Explosive Hazard*, and Title 40 CFR, Parts 260-270. Furthermore, each MAJCOM/range operating authority must dispose of radioactive wastes IAW the Air Force Radioactive Recycling and Disposal Office and AFI 40-201. **(T-0)**

5.1.1. Funding and Scheduling Range Clearance. The range operating authority is responsible for funding and scheduling all Material Potentially Presenting an Explosive Hazard (MPPEH)/range clearance. The range operating authority should publish a readily accessible clearance schedule. The range operating authority or range operations officer may temporarily postpone range clearance on a case-by-case basis for severe weather or other unforeseen circumstances that warrant delays.

5.1.1. **(EDWARDSAFB) Funding and Scheduling PIRA Clearance.** The 412 TW/CC is responsible for funding and scheduling all MPPEH range clearance. Unless otherwise amended by the 412 TW/CC, the PIRA clearance for the East Range shall be scheduled for incremental cleanup each April. The PIRA clearance for the West Range shall be scheduled for incremental cleanup each October. The 412 TW/CC or 412 RANS Range Operations Officer may temporarily postpone range clearance on a case-by-case basis for severe weather or other unforeseen circumstances that warrant delays. The exact dates for range clearance to include scheduling interior range areas shall be established by the 412 RANS Range Operations Officer in coordination and cooperation with the 812 CES/CED EOD unit.

5.1.1.1. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer will evaluate the PIRA East and West Range areas to determine targets or designated sites to be scheduled for cleanup whenever they become saturated with ordnance or debris, regardless of the number of weapons dropped, or when residue could impact test results. These conditions or other off-cycle cleanup activities shall be coordinated with 812 CES/CED EOD unit and scheduled as required by the 412 RANS Range Operations Officer.

5.1.1.2. **(Added-EDWARDSAFB) Open Burn/Open Detonation.** The 812 CES/CED EOD unit will evaluate the OB/OD Range to determine the designated areas to be scheduled for cleanup whenever they become saturated with ordnance or debris. The 812 CES/CED will initiate coordination with the 412 RANS ODO (661-277-2091) for an OB/OD area cleanup time and then schedule this time in CSE. The OB/OD Range is located in the southwest corner of the PIRA. The OB/OD Range is a 700-foot by 1400-foot rectangular area surrounded by an 8-foot chain-linked fence. Access to the OB/OD Range is restricted to and controlled by 812 CES/CED EOD personnel. 812 CES/CED EOD unit will coordinate with the 412 RANS Range Operations Officer or the DOWNFALL Tower Range Control Officer to request access to the PIRA and then follow a specific route to enter the OB/OD Range.

5.1.2. EOD Support for Range Clearance. Development of the annual range clearance plan will be a collaborative effort between the range operating authority and the supporting EOD unit (or MAJCOM) and be finalized at least 6 months prior to the first clearance activity. If there is no organic EOD capability, range operating authorities will establish a written agreement for range clearance support with the nearest Air Force EOD unit, if practicable; or if not with the nearest non-Air Force EOD unit via a memorandum of agreement and/or Inter-Service Support Agreement. **(T-2)** Range operating authorities will attempt to synchronize clearances with EOD deployment cycles or periods of high EOD operations tempo.

5.1.2.1. When the requirement for range clearance support exceeds the capability of the associated EOD unit, shortfalls are managed within the EOD Information Management System Range Support Tasking Program. The supporting EOD unit (or MAJCOM) will enter Range Support Tasking requests into EOD Information Management System as required to meet clearance requirements. **(T-2)**

5.1.2.2. Range operating authorities will prepare a plan for range clearance operations that at a minimum considers the number of personnel involved, types of ordnance anticipated to be encountered and/or recovered, expected levels of contamination, and support requirements. **(T-2)**

5.1.2.2.1. The range operating authority will ensure the plan provides sufficient time to complete the planned and anticipated range clearance operations safely. **(T-2)**

5.1.2.2.2. Consider requesting an advanced echelon element from the supporting EOD unit to conduct a GPS mapping survey of UXO-contaminated areas and debris-densities/distances around the targets at least 2 weeks prior to arrival of the main body for workload estimates and to finalize clearance planning.

5.1.2.3. Range operating authorities must finalize all range scheduling and EOD Render Safe Procedure support at least 45 days before the start of the clearance operation. **(T-3)**

5.1.3. To identify, safeguard, and dispose of UXO, range operating authorities will ensure the following are accomplished:

5.1.3.1. Safely clear UXO from ranges consistent with the stated mission of the range and for the continuing viability of the range. **(T-1)** Resolve all conflicts between explosive safety and other requirements with the objective of minimizing explosive hazards. Ensure controlled burning is not used to accomplish UXO clearance, but may be used to support UXO clearance if environmental and safety requirements are met. **(T-1)**

5.1.3.2. Promptly report any UXO or aircraft incident that falls outside of range or installation boundaries to proper authorities to protect personnel and property IAW AFI 32-3001. **(T-0)**

5.1.3.3. Notify installation personnel and the public, as appropriate, if any range operation presents a potential explosive hazard off the range. **(T-2)** This includes informing the public of any mishap that could impact the local community and may require additional precautions and/or restrictions. For range operations that occur outside the United States, comply with DoDD 4715.12 and relevant Combatant Command and in-theater Headquarters Air Force Component Command issuances, if any.

5.1.3.4. On operational ranges, the procedures for evaluating and responding to explosives safety, human health, and environmental risks are IAW DESR 6055.09_AFMAN 91-201 and Title 40 CFR Parts 260-270.

5.1.3.5. Maintain permanent records of:

5.1.3.5.1. All expenditures (types, quantities, locations, using organization, and estimated dud rates) per target of ammunition and explosives IAW DoDD 4715.11 and DoDD 4715.12. **(T-0)**

5.1.3.5.2. All mishaps attributed to UXO that occur on or off the range/installation IAW DoDI 6055.07, *Mishap Notification, Investigation, Reporting, and Record Keeping*. **(T-0)**

5.1.3.5.3. All EOD incidents and range clearance operations conducted on ranges. **(T-0)**

5.1.3.5.4. All areas known or suspected to contain UXO using a geographic information system, range maps and/or installation master planning maps in compliance with the Spatial Data Standards for Facilities, Infrastructure, and Environment. Conduct historical research as necessary. **(T-0)**

5.1.3.6. Remove all munitions debris from targets awaiting disposal. **(T-0)**

5.1. (EDWARDSAFB) Range Clearance Guidelines. The 412 RANS Range Operations Officer is responsible for the clearance of the PIRA to ensure that all processed target residue material released to recycling markets is demilitarized according to DoDI 4140.62, *Material Potentially Presenting an Explosive Hazard (MPPEH).* The 412 RANS Range Operation Officer will make a conscious effort to improve the target residue handling process, to provide sound stewardship of natural and man-made resources, and to enable a safe recycling strategy which ensures that trash and hazardous waste is properly disposed of according to applicable environmental statues.

5.2. Range Clearance Operations. Range clearance includes the removal or disposal of all ordnance, inert ordnance debris, training projectile ammunition, and other range debris reasonably possible to detect (normally down to 4 inches in size). Non-explosive debris should, at a minimum, be marked clear but is not required to be removed from targets by EOD personnel.

5.2.1. **WARNING:** Ranges will not conduct maintenance and clearance operations when snow covers the ground. **(T-1)** However, if specifically authorized in the MAJCOM supplement, activities such as clearance, target removal, or target replacement may be performed during these periods.

5.2.2. Ordnance Removal and Disposal. EOD Personnel or UXO-Qualified Personnel, as defined in the current version of the DoD Explosives Safety Board Technical Paper 18, *Minimum Qualifications for Personnel Conducting Munitions and Explosives of Concern-Related Activities,* will inspect munitions, munitions components, and targets for the presence of explosive hazards and mark inspected items that are “safe to move”. Only EOD Personnel will perform explosive operations to detonate/destroy UXO and energetic material. **(T-1)** With an approved waiver, UXO-Qualified Personnel may perform limited explosives operations to support specific range clearance objectives, limited to venting practice bombs or destroying spotting charges. UXO-Qualified Personnel contracted to perform target maintenance or

manage the installations MPPEH Program on operational ranges, will mark all UXO or munitions that contain explosives and report the item and location to the appropriate agency to coordinate with the supporting EOD unit to schedule a disposal/render safe procedures. Special attention should be provided during target inspection to ensure UXO are not embedded or contained within voids. "Safe to move" may be indicated by any readily distinguishable means that is mutually agreeable to EOD and the range operating authority.

5.2.2. **(EDWARDSAFB) Ordnance Removal and Disposal.** 812 CES/CED EOD personnel will inspect munitions, munitions components, range target residue, targets, and trash for the presence of explosive hazards or ignitable residue and will mark inspected items that are "safe to move". Only 812 CES/CED EOD personnel will perform explosive operations to detonate or destroy UXO and energetic material and will ensure that all explosive hazards are removed. Special attention will be provided during target inspection to ensure UXOs are not embedded or contained within voids. "Safe to move" may be indicated by any readily distinguishable means that is mutually agreeable to EOD and the Range Operations Officer.

5.2.2.1. **(Added-EDWARDSAFB) Personnel Limitations.** There will be at least two (2) EOD technicians serving as munitions residue inspectors. One (1) of the two EOD technicians will be a Staff Sergeant or above with a 7-skill level acting as the supervisor.

5.2.2.2. **(Added-EDWARDSAFB) Range Target Residue Operational Areas.** Range target residue is found in the PIRA, in the PIRA Munitions Residue Storage Area (MRSA), or in specified areas that could generate target munitions residue.

5.2.2.3. **(Added-EDWARDSAFB) Target Vehicle Inspection Procedures.** All expended range vehicle target material will be inspected for imbedded munitions residue and Low-Level Radioactive Waste (LLRW) residue by an EOD munitions inspector.

5.2.2.3.1. **(Added-EDWARDSAFB)** Any large driveline components with visual evidence of munitions residue impact will be drained, cut open, and visually checked for munitions.

5.2.2.3.2. **(Added-EDWARDSAFB)** All automotive fluids will be containerized and taken to Building 9508 or the approved government waste accumulation point that supports or services PIRA operations.

5.2.3. Personnel assisting in range clearances will only handle items that have been inspected and marked "safe to move" by EOD Personnel or UXO-Qualified Personnel. **(T-1)** Personnel assisting will be supervised by either EOD Personnel or UXO-Qualified Personnel to ensure hazardous items are not inadvertently moved or removed. **(T-1)**

5.2.3. **(EDWARDSAFB)** Personnel assisting in PIRA clearances will only handle items that have been inspected and marked "safe to move" by 812 CES/CED EOD personnel. Personnel assisting will be supervised by these EOD personnel to ensure hazardous items are not inadvertently moved or removed.

5.2.4. MPPEH Management. All munitions, munitions debris, and target debris removed from a range are considered MPPEH and are managed IAW explosive safety standards. Processing of MPPEH results in a classification of the material as either Material Documented as an Explosive Hazard (MDEH) or Material Documented as Safe (MDAS). Ranges must institutionalize chain of custody processes for MPPEH from collection to final disposition of the hazardous and safe material. **(T-0)** This is particularly necessary when two or more agencies (e.g., military and contracted) are involved in the MPPEH process. Refer to DAFMAN 21- 201, *Munitions Management*, for additional guidance on MPPEH management and disposition.

5.2.4. **(EDWARDSAFB) PIRA MPPEH Management.** All munitions, munitions debris, and target debris removed from the PIRA is considered MPPEH and shall be managed in accordance with explosive safety standards. Processing of MPPEH results in either Material Documented as an Explosive Hazard (MDEH) or Material Documented as Safe (MDAS). The Range Operations Officer ensures an institutionalized chain of custody process for MPPEH from collection to the final disposition of the hazardous and safe material. This is particularly necessary when two or more agencies (for example, military and contracted) are involved in the MPPEH process. Refer to AFI 21-201, *Conventional Munitions Management*, for additional guidance on MPPEH management and disposition.

5.2.4.1. MPPEH processing (also known as "Range Residue Removal") is normally accomplished by UXO-Qualified Contractors where demilitarization complements the inspection process to enable a more thorough examination.

5.2.4.1. **(EDWARDSAFB)** MPPEH processing (also known as "Range Residue Removal") is normally accomplished by the 812 CES/CED EOD where demilitarization complements the inspection process to enable a more thorough examination.

5.2.4.2. Range operating authorities will ensure that MPPEH awaiting processing and inspection is in a properly sited residue holding area. **(T-0)** When non-EOD personnel handle and transport material marked "safe to move," EOD Personnel or UXO-Qualified Personnel will supervise handling and loading for consolidation at the residue holding area. **(T-1)** Only EOD Personnel or UXO-Qualified Personnel will handle material after placement in the residue holding area. **(T-1)** Ranges may accumulate MPPEH until a cost-effective processing quantity is reached but shall not delay processing in order to maximize the value of recyclable materials. **(T-0)**

5.2.4.2. **(EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure that MPPEH awaiting processing and inspection is secured in a properly sited Residue Holding Area (RHA). When non-EOD personnel handle and transport material marked "safe to move", EOD personnel will supervise handling and loading for consolidation at the RHA. Only 812 CES/CED EOD personnel are authorized to handle material after placement in the RHA. The 412 RANS may accumulate MPPEH until a cost-effective processing quantity is reached but shall not delay processing in order to maximize the value of recyclable materials. The 412 RANS Range Operations Officer will:

5.2.4.2.1. **(Added-EDWARDSAFB)** Coordinate with EOD munitions inspectors to ensure procedures are followed in approved technical order data documents.

5.2.4.2.2. **(Added-EDWARDSAFB)** Stop any unsafe practice related to PIRA operations and coordinate with EOD munitions inspectors to enforce compliance with safety standards.

5.2.4.2.3. **(Added-EDWARDSAFB)** Ensure that an EOD munitions inspector inspects the range target residue for any hazards or explosives contamination upon completion of a disposal or demilitarization operation. Target residue will then be placed in a clearly defined area, separated to ensure any un-cleared residue does not contaminate the cleared residue.

5.2.4.2.4. **(Added-EDWARDSAFB)** Provide, as requested, PIRA manpower and heavy equipment to support EOD munitions inspectors to thoroughly inspect vehicles, munitions, and other range target residue.

5.2.4.2.5. **(Added-EDWARDSAFB)** Separate all Resource Conservation and Recovery Act regulated materials such as batteries, antifreeze, oil, fuel, and other materials that are encountered and deliver to the PIRA hazardous waste accumulation point located at Building 9508 for eventual disposal by the government.

5.2.4.3. EOD Personnel or UXO-Qualified Personnel must double-inspect MPPEH IAW DoDI 4140.62 to identify and remove Material Documented as an Explosive Hazard before it can be declared Material Documented as Safe. **(T-0)** To prevent co-mingling, once Material Documented as an Explosive Hazard and Material Documented as Safe are identified and documented as such, EOD Personnel or UXO-Qualified Personnel must segregate them from one another and from MPPEH awaiting inspection. **(T-0)** Segregation may be accomplished by any means that ensures physical separation and controlled access (e.g., bins, buildings, or fenced compounds under lock and key).

5.2.4.3. **(EDWARDSAFB)** After all range target munitions residue completes inspection the first time, the range target munitions residue will be inspected a second time by a different EOD munitions inspector to confirm it is explosive free.

5.2.4.3.1. **(Added-EDWARDSAFB)** After all target munitions residue completes inspection for a second time, all target munitions residue shall be taken to the PIRA MRSA and shall be placed inside lockable hoppers.

5.2.4.3.2. **(Added-EDWARDSAFB)** EOD munitions inspectors shall control access of these hoppers by locking each hopper.

5.2.4.3.3. **(Added-EDWARDSAFB)** EOD munitions inspectors shall keep the hopper combinations in their control to ensure no un-cleared target munitions residue is inadvertently mixed with cleared target munitions residue.

5.2.4.4. Disposal of Material Documented as Safe may be accomplished by any means consistent with DoDI 4140.62 and may be accomplished by the Defense Reutilization and Marketing Service under a memorandum of agreement or through direct commercial sales or contracts. Range operating authorities must ensure only properly inspected and documented Material Documented as Safe are released to the public. **(T-0)** For Material Documented as Safe at ranges located outside the United States, ensure compliance with DoDD 4715.12 and relevant Combatant Command and in-theater Headquarters Air Force Component Command issuances, if any.

5.2.4.4. **(EDWARDSAFB) Disposal of Material Documented as Safe Procedures.** The 412 RANS Range Operations Officer will ensure only properly inspected and documented MDAS are released to the public. Range target residue is primarily composed of tires, plastic vehicle components, retarder parachutes, wood, expended munitions and parts thereof, such as large and small practice bombs, trash and debris, tires, case fragments of live bombs, inert cannon projectiles, rockets, and expended target materials such as salvaged vehicles, aircraft, and other materials used to fabricate targets. Much of the range target residue has value on the recycling market due to its content of commercially viable metals such as steel, aluminum, and brass. The 412 RANS Range Operations Officer will:

5.2.4.4.1. **(Added-EDWARDSAFB)** Manage procedures and actions to safely process to approved landfills any range target residue and other debris which are commonly scavenged from PIRA targets.

5.2.4.4.2. **(Added-EDWARDSAFB)** Avoid contamination of range target residue with unexploded ammunition or explosive components, ignitable residue from spotting charges with phosphoric fillers, low-level radiological contaminates, or waste automotive fluids.

5.2.4.4.3. **(Added-EDWARDSAFB)** Direct Un-recyclable Trash Removal Procedures.

5.2.4.4.3.1. **(Added-EDWARDSAFB)** Sort range target residue and un-recyclable trash, inspect for imbedded UXOs, inspect for LLRW, and demilitarize, if necessary, prior to transportation to a landfill.

5.2.4.4.3.2. **(Added-EDWARDSAFB)** Sort tires and inspect for imbedded UXOs, inspect for LLRW, and demilitarize, if necessary, prior to transportation to a processing center or landfill authorized to receive tires.

5.2.4.4.3.3. **(Added-EDWARDSAFB)** Segregate concrete and wood, inspect for imbedded UXOs, inspect for LLRW, and demilitarize, if necessary, prior to transportation to a processing center or landfill authorized to receive construction debris.

5.2.4.4.4. **(Added-EDWARDSAFB) Transportation of Processed Range Residue Procedures.** Before any range target residue or munitions target residue is shipped or leaves the PIRA for recycling, the Range Operations Officer, DOWNFALL Tower Range Control Officer, or PIRA Lead shall ensure a DD Form 1348-1A, *Issue Release/Receipt Document*, certification document is accomplished and signed by both EOD munition inspectors in accordance with Air Force Technical Order (AFTO) 11A-1-60, *General Instruction - Inspection of Reusable Munitions Containers and Scrap*.

5.2.4.4.4.1. **(Added-EDWARDSAFB)** All certified munitions target residue shall have a legible certificate of clearance as the turn-in or shipping document. This clearance certificate document will be kept with the driver during transportation.

5.2.4.4.4.2. **(Added-EDWARDSAFB) Truck Scales Procedures.** Truck scales at the Air Force Research Laboratory (adjacent to the PIRA) will be used to account for range target residue or scrap leaving the PIRA, to ensure the truck's payload is within highway weight limits, to positively control all range target residue material leaving the PIRA, and to comply with Department of Transportation requirements.

5.2.4.5. MPPEH disposal should not be viewed as a funds generating activity. Management and disposal of MPPEH should consider safety first and ensure compliance with the Resource Conservation and Recovery Act Military Munitions Rule and (where applicable) hazardous waste management requirements. Additionally, IAW DoDI 4715.23, *Integrated Recycling and Solid Waste Management*, the following categories are excluded from being sold through a qualified recycling program: items that require demilitarization; ships, planes or weapons that require demilitarization or mutilation prior to sale; Munitions List items or Strategic List items. Range operating authorities must ensure the proceeds from their sale are NOT returned to a qualified recycling program. **(T-0)**

5.2.4.5. **(EDWARDSAFB)** To assist 812 CES/CED EOD munitions inspectors with removal of explosive or ignitable residue hazards, the 412 RANS Range Operations Officer will:

5.2.4.5.1. **(Added-EDWARDSAFB)** Separate range target materials into defined categories. Examples of categories are Steel, Sheet Metal, Aluminum, and Brass.

5.2.4.5.2. **(Added-EDWARDSAFB)** Optimize the value of recyclable materials by removing non-value added material, when economically feasible.

5.2.4.5.3. **(Added-EDWARDSAFB)** Direct LLRW Removal Procedures.

5.2.4.5.3.1. **(Added-EDWARDSAFB)** Search for LLRW which usually consists of luminous dials on range target vehicles.

5.2.4.5.3.2. **(Added-EDWARDSAFB)** Package LLRW for shipment and place it in an open-top steel drum at the PIRA site storage area for eventual disposal.

5.2.4.5.3.3. **(Added-EDWARDSAFB)** Mark the steel drum by describing the material, quantity, and any instrument readings encountered during LLRW monitoring.

5.2.4.5.3.4. **(Added-EDWARDSAFB)** Prepare LLRW for transportation or turn in.

5.2.4.5.4. **(Added-EDWARDSAFB)** Remove Resource Conservation and Recovery Act regulated material. Removal includes the characterization of the material, the preparation of the material for transport, and the transportation of the material to the initial accumulation point.

5.2.4.5.5. **(Added-EDWARDSAFB)** Halt processing activities and notify EOD munitions inspectors immediately if any UXO or suspected UXOs are identified.

5.2. (EDWARDSAFB) PIRA Clearance Operations. PIRA clearance includes the removal or disposal of all ordnance, inert ordnance debris, training projectile ammunition, and other range debris, which is reasonably possible to detect, normally down to four inches in size.

5.3. Range Clearance Types and Requirements. On active ranges programmed for continued use, clear the surface of all UXO, MPPEH, and inert debris as described below. Ranges located outside the United States will comply with paragraphs **5.3.1- 5.3.4.4** to the extent allowed by applicable international agreements. When seeking Air Force Safety Center / Weapons Safety Division (AFSEC/SEW) concurrence for operations, review requirements in DESR 6055.09, Vol. 7.

5.3.1. The following general rules apply to all clearance activities:

5.3.1.1. Mechanized gathering procedures coordinated with the supporting EOD unit and approved by the Air Force Safety Center may be used to clear sub-scale practice munitions (e.g., BDU-33) as long as un-probed munitions are only handled by EOD Personnel or UXO-Qualified Personnel.

5.3.1.2. EOD Personnel or UXO-Qualified Personnel will only perform subsurface UXO and subsurface MPPEH removal as required in current or former hazard areas as required for construction, cable burial, etc. **(T-1)** EOD Personnel or UXO-Qualified Personnel will report subsurface MPPEH removal and range clearance separately. **(T-1)**

5.3.1.3. Range operating authorities will ensure that hazard areas used specifically for live munitions containing extremely hazardous fuzing (e.g., influence, random-delay) are identified as "Extremely Hazardous Contaminated hazard areas." **(T-1)** Range clearance requirements of “Extremely Hazardous Contaminated hazard areas” are based on an Environmental, Safety, and Occupational Health risk assessment, and if warranted, range operating authorities must accomplish a waiver request IAW **paragraph 1.3** and staff it through the appropriate Air Force Environmental, Safety, and Occupational Health agencies (i.e., CE Environmental, Safety, and/or Bioenvironmental Engineering). **(T-1)**

5.3.1.4. Ranges adjacent to the land of another service that prohibits clearance activities are exempt from clearance requirements on those portions when the other service provides written acknowledgement of the ramifications. These areas will be considered “Extremely Hazardous Contaminated impact areas” and managed accordingly by the range operating authority. **(T-1)**

5.3.1.5. Range operating authorities will receive authorization from adjacent facilities prior to decontaminating/clearing adjacent impact areas. **(T-1)** Range operating authorities will document all access denials for clearance activities and report occurrences to the parent MAJCOM and AF/A3TI. **(T-2)**

5.3.2. Low-Angle Strafe Targets. If used for low-angle strafe by aircraft that fly past or over the target within 500 feet of the target (slant range), range operating authorities will ensure that the target is hand-policed daily when used and cleared after 15,000 scored rounds (or 12 use days for non-scored targets). **(T-2)**

5.3.3. Improved conventional munition targets (e.g., Cluster Bomb Unit grids) are normally cleared after accumulating a specific number of weapons. Range operating authorities will specify the maximum accumulation amount for improved conventional munition targets in the range supplement for approval by the MAJCOM with the concurrence of AF/A4C. **(T-1)** After the maximum accumulation amount is achieved, EOD Personnel will clear improved conventional munition targets to the radius where the debris density factor is less than five items per acre or at least 500 feet from the target. **(T-1)**

5.3.4. Range Clearance Requirements. EOD Personnel will clear active ranges used for munitions expenditures IAW the minimum requirements defined in the following sub-paragraphs. **(T-1)** Conduct clearance more frequently or to a greater distance as required to maintain targets or realistic appearance, or where an uneven distribution of weapons results from restricted run-in headings. EOD Personnel will:

5.3.4.1. Clear target access roads and the area 50 feet on either side during clearance operations or target maintenance activities (only those roads noted on range maps and those used for sole access to targets/arrays). **(T-1)** Consider clearing to a greater side distance to accommodate anticipated vehicle traffic (e.g., towed load), vehicle characteristics (e.g., turn radius), and potential obstacles (e.g., road damage, UXO).

5.3.4.2. Semiannually or after 150 use days (whichever comes later), clear the area around each desired point of impact associated with a high-use target to a radius of 150 feet. **Note:** This criterion applies to individual desired points of impact (not multi-desired points of impact target arrays) that receive a high volume of munitions, where deferring clearance to the biennial interval would create a significant ricochet hazard or significantly increase the range clearance work effort due to the accumulation and/or density of UXO. **(T-1)** Range operating authorities may determine these high-use targets based on range-specific conditions, in consultation with the MAJCOM.

5.3.4.3. Biennially, no later than 2 years since last accomplished, clear the area around a target used for weapons expenditure as follows: **(T-1)**

5.3.4.3.1. Clear targets used for practice, inert, pyrotechnic, illumination or smoke munitions to either a) a radius of 300 feet, or b) the shorter radius where the debris density factor is less than five items per acre.

5.3.4.3.2. Clear targets used for high explosive munitions (including White Phosphorous) to either a) a radius of 500 feet, or b) the shorter radius where the debris density factor is less than five items per acre.

5.3.4.4. Decennially, no later than 10 years since last accomplished, clear the area around a target used for weapons expenditure to either a) a radius of 1,000 feet, or b) the shorter radius where the density of debris items is less than five per acre. **(T-1)** Range operating authorities may elect to clear all areas within 1,000 feet of targets every 10 years and may do so incrementally (e.g., 20% of the required area biennially for 10 years).

5.4. Range Maintenance and Clearance Safety.

5.4.1. **WARNING:** Non-EOD Personnel will not move or attempt to move munitions or targets unless they have been marked “safe to move” by EOD Personnel or UXO-Qualified Personnel. **(T-0)** This includes any object that appears to have been delivered on-range or had weapons fired against it. The sole exceptions are for training gun ammunition which has been expended on targets designated solely for that use, and expended Smokey Surface-to- Air Missile simulators.

5.4.1. **(EDWARDSAFB) WARNING:** Non-EOD personnel will not move or attempt to move munitions or targets unless they have been identified as "safe to move" by 812 CES/CED EOD personnel or UXO-qualified personnel. This includes any object that appears to have been delivered on-range or had weapons fired against it. The sole exceptions are for training gun ammunition which has been expended on targets designated solely for that use, and expended Smokey Surface-to-Air Missile Simulators.

5.4.1.1. **(Added-EDWARDSAFB)** Any items that a test project wants to recover from the PIRA will be cleared by the 812 CES/CED EOD. The 812 CES/CED EOD will ensure that the item is identified as "safe to move" before allowing it to be moved off the PIRA or being returned to the test project.

5.4.1.2. **(Added-EDWARDSAFB)** If a test project team member wants to locate a lost or unrecovered item on the PIRA, then such action will require either the DOWNFALL Tower Range Control Officers, 412 RANS PIRA personnel, or the 412 RANS Range Operations Officer to escort the test project team member to the general area to locate the item. If the test project team member notifies the 412 RANS that the item is classified, then the test project team leader has the responsible action to either clear in two of the 412 RANS personnel to locate the item or plan to clear in and use two 812 CES/CED EOD as escorts.

5.4.1.3. **(Added-EDWARDSAFB)** The test project leader will ensure that any items requiring removal from the PIRA are included in the Test Safety Review Process and prior coordination with 812 CES/CED EOD is planned to identify the item is "safe to move".

5.4.2. Safety Training Requirements.

5.4.2. **(EDWARDSAFB)** Safety Training Requirements.

5.4.2.1. Range operating authorities will ensure all range personnel engaged in range maintenance and clearance operations receive occupational and explosive safety briefings IAW **Attachment 5**. **(T-1)**

5.4.2.1. **(EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure all range personnel engaged in range maintenance and clearance operations receive occupational and explosive safety briefings according to AFMAN 13-212V1, **Attachment 5.**

5.4.2.2. Range operating authorities will ensure EOD personnel receive orientation on ordnance expected to be encountered, range vehicle operation, and emergency procedures. **(T-1)**

5.4.2.2. **(EDWARDSAFB)** The 412 RANS Range Operations Officer will ensure EOD personnel, DOWNFALL Tower Range Control Officers, and PIRA personnel receive an orientation brief on the ordnance expected to be encountered, range vehicle operation, and emergency procedures.

5.4.2.2.1. **(Added-EDWARDSAFB)** The 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will brief emergency procedures prior to conducting range maintenance and clearance operations.

5.4.2.2.1.1. **(Added-EDWARDSAFB)** Any individual detecting an unsafe act or condition will immediately take action to stop that act or address the situation to the 412 RANS Range Operations Officer, DOWNFALL Tower Range Control Officer, PIRA Supervisor Lead, EOD Team Chief, or EOD Inspector. All pertinent operations will cease until the situation is corrected.

5.4.2.2.1.2. **(Added-EDWARDSAFB)** If any UXO is discovered to be unsafe, such UXO will be disposed of by EOD munitions inspectors or designated EOD personnel.

5.4.2.2.1.3. **(Added-EDWARDSAFB)** Immediately notify the Edwards AFB Fire Department in the event of a fire and attempt to extinguish a fire, within reason. If explosives are involved, **immediately** evacuate the area to a safe distance.

5.4.2.2.1.4. **(Added-EDWARDSAFB)** Immediately notify the hospital in the event of injury. Render first aid and transport injured personnel if possible.

5.4.2.3. The supporting EOD unit will ensure non-EOD personnel who are assisting in range clearance operations are briefed on explosive safety hazards IAW AFMAN 91-201, the markings used for live, practice, and inert ordnance found in TO 11A-1-53, *General Instruction for Ammunitions Color Coding, Identification of Empty and Inert Loaded Ammunition Items and Components, and Assignment of Version Numbers to Training and Dummy Ammunition Items*, and the hazards they may encounter. **(T-1)**

5.4.2.3. **(EDWARDSAFB)** The supporting 812 CES/CED EOD will ensure non-EOD personnel who are assisting in range clearance operations are briefed on explosive safety hazards according to AFMAN 91-201, the markings used for live, practice, and inert ordnance found in TO 11A-1-53, *General Instruction for Ammunitions Color Coding, Identification of Empty and Inert Loaded Ammunition Items and Components, and Assignment of Version Numbers to Training and Dummy Ammunition Items*, and the hazards they may encounter.

5.4.3. In addition to communication requirements specified in **paragraph 4.7**, each working EOD team must have a signaling device for use in the event an uninformed flight crew attempts to use the range. **(T-3)**

5.4.3. **(EDWARDSAFB)** In addition to communication requirements specified in AFMAN 13-212V1, **paragraph 4.7.**, each working 812 CES/CED EOD team must have a signaling device for use in the event an uninformed flight crew attempts to use the range.

5.4.4. Requiring activities shall incorporate Occupational Safety and Health Administration standards and/or Department of Defense Manual 4145.26, *DoD Contractors' Safety Manual for Ammunition and Explosives*, into range support contracts, as applicable. **(T-0)** Requiring activities outside the United States may have additional, host nation requirements. Contact the servicing Safety Office for more information.

5.4. (EDWARDSAFB) PIRA Maintenance and Clearance Safety Procedures.

5.5. Range Operations during Range Maintenance and Clearance. Unless further restricted by MAJCOM or local range supplements, adhere to the following restrictions during range maintenance and clearance operations.

5.5.1. Only simulated weapons deliveries, IAW applicable directives, and training-mode laser operations are authorized for the duration of range maintenance and range clearance closures with the following exceptions. When the range is divided as described below all maintenance and clearance activities must be complete before air-to-ground activities can resume on that portion. **(T-2)** Range operating authorities will ensure that all restrictions are clearly identified in the range schedule and reinforced with range Notices to Airmen. **(T-2)**

5.5.1. **(EDWARDSAFB)** Only simulated weapons deliveries, in accordance with applicable directives, and training-mode laser operations are authorized for the duration of range maintenance and range clearance closures with the following exceptions. When the PIRA is divided as described below all maintenance and clearance activities must be complete before air-to-ground activities can resume on that portion. The 412 RANS Range Operations Officer will ensure that all restrictions are clearly identified in the range schedule and reinforced by an Edwards AFB Notice to Airmen for air-to-ground activities during range maintenance and clearance.

5.5.1.1. Class A Service Operations. Range maintenance and clearance operations can be conducted on one side of a dual Class A service range (except behind strafe targets) while the range control officer is controlling aircrew training missions on the other side of the range.

5.5.1.1. **(EDWARDSAFB) Class A Range Service Operations.** Range maintenance and clearance operations can be conducted on one of the PIRA ranges (i.e., West Range or East Range) for Class A Service Operations, except behind strafe targets, while a DOWNFALL Tower Range Control Officer is controlling aircrew training missions on the other range.

5.5.1.2. Class B and C Service Operations. Ranges providing Class B and C service are normally closed during maintenance and clearance operations, but if the range is of sufficient size the range operating authority may authorize continued operations on portions of the range (and adjacent ranges) with the approval of the ground party team chief. If approved, the range operating authority will brief aircrews concerning the location of ground personnel and emphasize the absolute need for positive target identification. **(T-2)**

5.5.1.2. **(EDWARDSAFB) Class B and Class C Range Service Operations.** Class B and Class C Service Operations are normally closed during maintenance and clearance operations; however, if the PIRA is of sufficient size for either class service operations, the 412 RANS Range Operations Officer may authorize continued operations on the PIRA with concurrence of the ground party team leader. If approved, the 412 RANS Range Operations Officer or DOWNFALL Tower Range Control Officer will brief aircrews concerning the location of all ground personnel on the PIRA and emphasize the absolute need for positive target identification.

5.5.2. Over Flight Procedures. Aircraft must obtain concurrence and approval from the ground party and range control officer/range operations officer for overflight over areas or portions of ranges during maintenance and clearance operations in conjunction with the following restrictions. **(T-1)** This ensures the protection of ground personnel and prevents aircraft damage by fragments from explosives operations. For the purpose of this manual, the overflight restrictions define a minimum distance in any direction, i.e., a "safety bubble." Strict adherence to the procedures outlined below is mandatory.

5.5.2. **(EDWARDSAFB) Over Flight Procedures.** Aircrew must obtain concurrence and approval from the ground party and the DOWNFALL Tower Range Control Officer or the 412 RANS Range Operations Officer for overflight over areas or portions of the PIRA West or East Ranges during maintenance and clearance operations in conjunction with the following restrictions. This ensures the protection of ground personnel and prevents aircraft damage by fragments from explosives operations. For the purpose of this Supplement, the over flight restrictions define a minimum distance in any direction, for example, a "safety bubble." Strict adherence to the procedures outlined below is mandatory.

5.5.2.1. When range maintenance or clearance personnel are present on the range and explosives operations are not planned, aircraft will remain above 3,000 feet above ground level unless a lower altitude and distance is specifically approved by the ground party and range control officer/range operations officer. **(T-1)**

5.5.2.1. **(EDWARDSAFB)** When range maintenance or clearance personnel are present on the PIRA and explosives operations are not planned, aircraft will remain above 3,000 feet above ground level unless a lower altitude and distance is specifically approved by the ground party and DOWNFALL Tower Range Control Officer or the 412 RANS Range Operations Officer.

5.5.2.2. When explosives operations are planned, aircraft will remain at or above 10,000 feet above ground level (accounts for potential rogue fragments IAW DESR 6055.09 AFMAN 91-201, Paragraph V5.E3.2.7.4.2). **(T-1)**

5.5.2.2. **(EDWARDSAFB)** When explosive operations are planned, aircraft will remain at or above 10,000 feet above ground level which accounts for potential rogue fragments according to the AFMAN 13-212V1 when over the PIRA.

5.5.2.3. Aircraft will not release chaff and flares over range maintenance and clearance operations. **(T-1)**

5.5.2.3. **(EDWARDSAFB)** Aircraft will not release chaff and flares over the PIRA during maintenance and clearance operations.

5.5.2.4. When no personnel are on the range aircraft operations may proceed with no additional altitude restrictions.

5.5.2.4. **(EDWARDSAFB)** When no personnel are on the PIRA, aircraft operations may proceed with no additional altitude restrictions.

5.5. (EDWARDSAFB) PIRA Operations during Range Maintenance and Clearance. Unless further restricted by AFMC, operations on the PIRA will adhere to the following restrictions during range maintenance and clearance operations.

Chapter 6

TEST AND TRAINING USE OF DEPLETED URANIUM (DU)

6.1. Policy and Procedures for Use of DU. Test and training operations involving DU are IAW this manual, AFI 40-201, and the applicable Air Force Radioactive Material Permit issued under the authority of Air Force Master Materials License by the Radioisotope Committee of the Air Force Medical Readiness Agency / Surgeon General Policy Branch (AFMRA/SG3PB).

6.2. Personnel Using or Handling DU will:

6.2.1. Comply with radiation safety procedures outlined in the license or permit authorizing the storage or use of DU, the referenced directives, local operating instructions or directives, and verbal instructions of their Radiation Safety Officer and supervisor. **(T-0)**

6.2.2. Halt any imminent danger situation immediately, place the operation in a safe configuration, and inform their Radiation Safety Officer or supervisor of unsafe or non-compliant radiological conditions and incidents or accidents involving DU. **(T-0)**

6.3. Authorized Locations. Range operating authorities must ensure the use of DU is restricted to sole use impact areas that have been permitted by the Radioisotope Committee. **(T-0)** Range operating authorities will ensure, except for combat mix (i.e., a mixture of armor piercing and high explosive ammunition), all other munitions, live or inert, practice or full-scale, are prohibited from use in DU impact areas. **(T-0)**

6.3.1. 30-millimeter Armor Piercing Incendiary DU Testing and Training. Testing of 30-millimeter armor piercing incendiary DU munitions and reliability certification flight tests are limited to the following locations:

6.3.1.1. Eglin Air Force Base, Florida, Area C-64. Used to conduct life cycle testing of 30-millimeter armor piercing incendiary (PGU-14/B) ammunition; rounds are fired into an enclosed gun butt. The range operating authority will ensure that only approved areas are used to handle, store, machine, and test research and development items within an enclosed test chamber. **(T-1)**

6.3.1.2. Nevada Test and Training Range, Nevada, Target 63-10. Used to conduct Test and Evaluation to verify ballistics, Operational Flight Program software changes, Low Altitude Safety and Targeting Enhancement upgrades, and to conduct United States Air Force Weapons School instructor pilot training and tactical employment evaluation using Combat Mix.

6.3.2. Stockpile Surveillance Flight Tests (Air Force-National Nuclear Security Administration Joint Flight Tests). National Nuclear Security Administration constructs Joint Test Assemblies for Joint Flight Tests based on National Nuclear Security Administration and Air Force test requirements. These Joint Test Assemblies may contain DU.

6.3.2.1. As stated in the Air Force- National Nuclear Security Administration memorandum of agreement addressing joint test and assessment activities, it is the joint policy of National Nuclear Security Administration and the Air Force to recover lost Joint Test Assemblies whenever possible, consistent with United States policy and law.

6.3.2.2. Air Force- National Nuclear Security Administration Joint Flight Tests are normally conducted at Utah Test and Training Range, Eglin Range, National Nuclear Security Administration's test range located within Tonopah Test Range, and Army's Kwajalein Atoll (Reagan Test Site).

6.3.2.2.1. Units must obtain advance coordination for use of any other Air Force range for employment of Joint Flight Test weapons containing DU with the range's parent MAJCOM, AF/A3TI and AF/TE. **(T-1)**

6.3.2.2.2. Joint Flight Test activities at Vandenberg Air Force Base, Nellis Air Force Base, Eglin Air Force Base, Whiteman Air Force Base, Barksdale Air Force Base, and Minot Air Force Base must limit operations to carrier/weapon interface and carrier launch only. **(T-1)**

6.3.2.3. Air Force ranges supporting Joint Flight Tests will work with the Radioisotope Committee to determine whether the impact area requires a radioactive material permit/license and obtain a permit/license when required. **(T-1)**

6.3.2.4. Use of DU for stockpile surveillance flight tests is authorized under Air Force auspices by Section 91(b) of the Atomic Energy Act of 1954 (Public Law 83-703, 42 U.S.C. § 2121(b), *Material for Department of Defense use*) and falls under the jurisdiction of AFSEC/SEW.

6.4. Disposal/Decommission Procedures.

6.4.1. Disposal of Expended DU Penetrators. DU authorized personnel (see the Terms section in **Attachment 1**) will collect, package and dispose of expended DU penetrators and visible fragments in coordination with the Installation Radiation Safety Officer and the Radioisotope Committee. **(T-0)** Before commencing clearance actions, an EOD Team will sweep the impact area and eliminate any UXO or explosive hazards. **(T-1)** Annually, DU authorized personnel will manually remove visible DU rounds and fragments, and package them for disposal. **(T-1)**

6.4.1.1. Disposal Packing Requirements. DU authorized personnel will package DU penetrators and fragments for disposal or recycling IAW instructions provided by Air Force Radioactive Recycling and Disposal Office. **(T-0)** The range operating authority will prepare requests for disposal or recycling IAW AFI 40-201 and coordinate with the Installation Radiation Safety Officer. **(T-0)**

6.4.1.2. Contaminated Targets. Vehicles and tanks that are no longer intended to be used as targets or are so damaged from use that they are in need of replacement will be identified and decontaminated IAW AFI 40-201 by DU authorized personnel or have arrangements made for proper disposal or recycling. **(T-1)**

6.4.2. Decommission Plan. IAW AFI 40-201, prior to expending DU at any authorized location, units will develop a decommissioning plan that identifies the cost to remediate DU contamination at the site based on current technology. The range operating authority will review and update the decommissioning plan biennially and provide a copy of the plan to the Radioisotope Committee. **(T-0)**

Chapter 7

ENVIRONMENTAL MANAGEMENT

7.1. Environmental Compliance. Air Force ranges will conduct their activities according to national environmental policy and will meet environmental compliance guidelines set forth in Air Force and DoD directive publications, including (but not limited to) DAFI 32-7001 and DoDI 4715.06, *Environmental Compliance at Installations Outside the United States*. The Air Force will comply with applicable Federal, State, and local environmental laws, standards, and reporting requirements. For ranges outside the United States, Air Force activities will comply with environmental criteria specified in applicable international agreements, and DoDI 4715.05, *Environmental Compliance at Installations Outside the U.S.*, and AFI 32-7091.

7.1.1. Military Munitions Rule. For ranges in the United States, the regulations governing cradle-to-grave management of hazardous waste are codified in Title 40 CFR Parts 260-270. The portion of those regulations that determines when military munitions become solid waste and provides standards for their management are codified in Title 40 CFR, Part 266, Subpart M, *Military Munitions* (40 CFR §§ 266.200 -- 266.206). MAJCOMs/range operating authorities will comply with DoDM 4715.26, *DoD Military Munitions Rule (MR) Implementation Procedures*, AFMAN 32-7002, *Environmental Compliance and Pollution Prevention*, and DAFMAN 21-201 to meet Environmental Protection Agency regulatory requirements on operational ranges (i.e., including both active and inactive ranges as defined by 40 CFR 266.201). **(T-0)** Range operating authorities must ensure the supporting CE unit/environmental planning function is involved in the process. **(T-3)**

7.1.2. MPPEH. MAJCOMs/range operating authorities will manage active and inactive ranges under their control to maintain long-term viability of DoD ranges while protecting human health and the environment and to minimize future clearance costs. **(T-1)** Ranges and munitions should be designed to minimize potential explosive hazards and harmful environmental impacts and to promote resource recovery and recycling. Each MAJCOM/range operating authority will comply with DoDI 4140.62, DoDD 4715.11, and DoDD 4715.12. **(T-0)**

7.1.3. Environmental Remediation outside the United States. To address environmental contamination on Air Force-operated ranges located in foreign countries, follow AFI 32-7091 to the extent that it does not conflict with the requirements of binding international agreements regarding use of ranges.

7.2. Integrated Natural and Cultural Resources Programs. Air Force ranges will conserve natural and cultural resources pursuant to the Sikes Act (16 U.S.C. § 670 et seq), Endangered Species Act (16 U.S.C. § 1531 et seq), and the National Historic Preservation Act (54 U.S.C. § 300101 et seq).

7.2.1. The management of range natural resources is governed by an approved Integrated Natural Resources Management Plan IAW AFMAN 32-7003.

7.2.2. The management of cultural resources present on ranges is governed by an approved Integrated Cultural Resources Management Plan IAW AFMAN 32-7003.

7.2.3. Each Integrated Natural Resources Management Plan and Integrated Cultural Resources Management Plan is written to support the current and future known mission requirements identified in the Range Capabilities Development Plan and shall be amended as mission requirements change.

7.3. Mission Sustainment. Range hazards must be identified and managed to ensure mission readiness, adequate natural infrastructure (air, land, and water) and frequency spectrum are available to test, train, and perform the Air Force's varied missions at its ranges. These resources can be degraded or denied in some locations due to environmental issues or encroachment. Encroachment (defined in AFI 90-2001, *Mission Sustainment*) comes in many forms, but often results in diminished operations or significant workarounds, inconveniences, and additional costs.

7.3.1. The range operating authority's Installation Mission Sustainment Team (IMST) will ensure the installation's Mission Sustainment Risk Reports (MSSRs) document all hazards which adversely affect, or have the potential to affect, mission operations within ranges or associated airspace. **(T-3)** Internal factors include, but are not limited to, wetlands, threatened and endangered species, hazardous waste sites, military frequency spectrum competition, and UXO. External factors include, but are not limited to, incompatible land use zoning, frequency spectrum restrictions, physical obstructions and renewable energy projects, and local and regional environmental constraints. The range operating authority will work with the IMST to minimize external and internal encroachment impacts to the range through the Mission Sustainment Risk Management Process. **(T-2)**

7.3.2. Environmentally-Sensitive Buffer Areas. Many range areas are bordered by environmentally-sensitive open space whose management is balanced to support both military training and natural ecosystems. The Readiness and Environmental Protection Integration (REPI) program purchases development easements from willing land owners managed by private conservation groups to help ensure the continued viability of training lands by removing the potential for incompatible development. These are often habitat areas for endangered species and may come with restrictions that affect training like avoiding nearby portions of the range during mating or migration seasons. Coordination with the partner organization managing the REPI easements is recommended to ensure that covenants are respected by all parties.

7.4. Pollution Prevention. Range operating authorities shall identify opportunities to optimize range processes or activities in terms of pollutant reduction, maximization of solid waste diversion, lower energy use, reduction in the use of natural resources, water conservation and improvements to health and safety IAW DAFI 32-7001.

7.5. Operational Range Assessment Program. Ranges shall work with the supporting CE environmental office and AFCEC to conduct operational range assessments IAW DoDD 4715.11, DoDD 4715.12, DoDI 4715.14, and DAFPAM 32-7003, *Operational Range Assessment Program Management*. **(T-0)**

7.6. Environmental Restoration Program. As necessary, range operating authorities will support environmental response actions (e.g., the identification and investigation of releases, removal actions or remedial actions) for a release or substantial threat of a release of hazardous substances, pollutants or contaminants on a range or migrating from a range to off-range areas in accordance with DAFI 32-7020. The Military Munitions Response Program is an Environmental Restoration Program, and removal of buried munitions on an operational range is a Military Munitions Response Program function.

7.7. Programming Guidance. AFCEC will identify environmental requirements eligible for funding in environmental PEs. **(T-1)** The range operating authorities shall identify environmental requirements when requested by AFCEC in conjunction with their servicing environmental office. **(T-1)** AFCEC and the servicing environmental office can assist in identifying projects eligible for environmental funding versus environmentally related operational requirements that may be funded with range O&M funds. AFCEC will program projects in accordance with DAFI 32-7001, DAFI 32-7020 or AFI 32-7091, as appropriate in the Automated Civil Engineers System or successor system.

7.8. Environmental Impact Analysis. As required by 32 C.F.R. **Part 989** and the National Environmental Policy Act (NEPA), the Air Force must analyze the potential impacts of its actions upon the environment prior to making decisions on range use and associated real property actions. Range operating authorities shall coordinate environmental impact analysis requirements through their supporting installation EIAP program manager. EIAP for comprehensive range planning, and for land acquisition comprising more than 1,000 acres, should be coordinated with AFCEC/ CZN. Comprehensive range planning and its associated EIAP should occur, at a minimum, every ten years to ensure environmental impact analysis remains current with range operations.

Chapter 8

TRACKING AND REPORTING REQUIREMENTS

8.1. Reporting Requirements. Ranges will utilize CSE to track range use such as scheduled and actual mission information, supporting range equipment, munitions used, etc. **(T-1) Note:** This requirement may be waived for ranges that provide Class-T service and do not manage or use airspace. Range operating authorities must ensure classified information, if any, is maintained outside of CSE. **(T-1)**

8.2. Expended Munitions Tracking. IAW DoDD 4715.11, and DoDD 4715.12, range operating authorities will maintain records of all expenditures (types, quantities, locations, using organization, and estimated dud rates) of ammunition and explosives for each target. **(T-0)** Range operating authorities and range personnel will record data in CSE when possible and by any other means when not. **(T-1) Note:** In addition to munitions employed by range users, this requirement also applies to munitions employed by the range operator in support of test or training activities (e.g., Smokey Surface-to-Air Missiles).

8.2. (EDWARDSAFB) Expended Munitions Tracking for the PIRA. The Edwards Range Operating Authority and the 412 RANS DOWNFALL Tower Range Control Officer will maintain a range log to record all ordnance expenditures to include the details of range sorties, types of ordnance expended, quantities of ordnance expended, location of expenditure, the unit expending the ordnance, estimated duds, ground safety incidents, air safety incidents, and other ordnance expended in support of test or training activities. The Edwards Range Operating Authority and 412 RANS will record data in CSE when possible and by any other means when not possible.

8.3. Range Clearance Report. This report is a detailed report of clearance events and range operating authorities will ensure it precisely identifies the areas that have been cleared and includes the number of people and the amount of money and material used to accomplish the work. **(T-0)** When closing or transferring ranges, range operating authorities will ensure the range clearance report is attached to the Certificate of Clearance which becomes an official document certifying that all dangerous and explosive materials that are reasonably possible to detect have been removed. **(T-0)**

8.4. Defense Readiness Reporting System. IAW AFI 10-403, *Deployment Planning and Execution*, report range readiness in the Defense Readiness Reporting System.

8.5. Threat Systems and Equipment Inventory. Based on the threat systems and equipment required for each range as outlined in **Attachment 3**, each range-owning MAJCOM shall produce and maintain an inventory of threat systems and equipment (types and quantities) each range possesses and an inventory of threat systems and equipment each range requires but does not possess.

8.5.1. Include Air CTS, threat, electronic attack, and feedback and scoring systems in the inventory.

8.5.2. Provide the inventory to ACC for inclusion in the Range Capabilities Development Plan.

JAMES C. SLIFE, Lt Gen, USAF
Deputy Chief of Staff for Operations

(EDWARDSAFB)

DOUGLAS P. WICKERT, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

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DAFMAN 13-217, *Drop Zone, Landing Zone and Helicopter Landing Zone Operations*, 22 April 2021

AFMAN 32-7002, *Environmental Compliance and Pollution Prevention*, 4 February 2020

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DAFPAM 32-7003, *Operational Range Assessment Program Management*, 9 September 2021

DESR 6055.09_AFMAN 91-201, *Explosives Safety Standards*, 28 May 2020

AFPD 13-2, *Air Traffic, Airfield, Airspace, and Range Management*, 3 January 2019

HAF Mission Directive 1-14, *General Counsel and the Judge Advocate General*, 29 December 2016

HAF Mission Directive 1-18, *Assistant Secretary of the Air Force (Installations, Environment, and Energy)*, 10 July 2014

HAF Mission Directive 1-54, *Deputy Chief of Staff, Operations*, 8 September 2015

(Added-EDWARDSAFB) R-2515 Users Handbook, *Airspace and Flying Procedures,* 9 January 2021

TO 11A-1-53, *General Instruction for Ammunitions Color Coding, Identification of Empty and Inert Loaded Ammunition Items and Components, and Assignment of Version Numbers to Training and Dummy Ammunition Items,* 21 January 2016

TO 1-1M-34, *Aircrew Weapons Delivery Manual,* 22 August 2013

TO 1-1M-34-1, *Aircrew Weapons Delivery Manual (Classified),* 25 March 2015

Other Publications

American National Standards Institute Z136.1, *Safe Use of Lasers,* 2014

DoD Explosives Safety Board Technical Paper 18, *Minimum Qualifications for Personnel Conducting Munitions and Explosives of Concern-Related Activities*, 1 September 2016

North Atlantic Treaty Organization Standard Allied Range Safety Publication (ARSP)-4, Edition B, Version 1, *Laser Safety for Military Use*, 4 October 2016

Prescribed Forms

None

Adopted Forms

ACC Form 167, *ACC Risk Management (RM) Worksheet*

(**Added-EDWARDSAFB**) AF 1109, *Visitor Register Log*

DAF Form 679, *Air Force Publication Compliance Item Waiver Request/Approval*

DAF Form 847, *Recommendation for Change of Publication*

AF Form 1067, *Modification Proposal*

AF Form 4437, *Deliberate Risk Assessment Worksheet*

DoD Form 1144, *Support Agreement*

Abbreviations and Acronyms

(Added-EDWARDSAFB) 412 TW/SCOTS—Frequency or Spectrum Management Office

ACC—Air Combat Command

ADE—Airdrop Damage Estimation

AETC—Air Education and Training Command

AFCEC—Air Force Civil Engineer Center

AFGSC—Air Force Global Strike Command

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFPD—Air Force Policy Directive

AFRC—Air Force Reserve Command

AFSOC—Air Force Special Operations Command

AMC—Air Mobility Command

ANG—Air National Guard

ARSP—Allied Range Safety Publication

CAS—Close Air Support

CE—Civil Engineering

CFR—Code of Federal Regulations

CJCSM—Chairman of the Joint Chiefs of Staff Manual

CSE—Center Scheduling Enterprise

CTS—Combat Training Systems

DAFI—Department of the Air Force Instruction

DoD—Department of Defense

DoDD—Department of Defense Directive

DoDI—Department of Defense Instruction

DU—Depleted Uranium

EIAP—Environmental Impact Analysis Process

EOD—Explosive Ordnance Disposal

ERP—Enterprise Range Plan

FAA—Federal Aviation Administration

GPS—Global Positioning System

HAF—Headquarters Air Force

IAW—In Accordance With

ICDS—Improved Container Delivery System

IMST—Installation Mission Sustainment Team

JPADS—Joint Precision Airdrop System

JTAC—Joint Terminal Attack Controller

LEP—Laser Eye Protection

MAJCOM—Major Command

MDAS—Material Documented as Safe

MDEH—Material Documented as an Explosives Hazard

MDS—Mission Design Series

MILCON—Military Construction

MPPEH—Material Potentially Presenting an Explosive Hazard

MRTFB—Major Range and Test Facility Base

(Added-EDWARDSAFB) MRU—Military Radar Unit

MSD—Minimum Safe Distance

NATO—North Atlantic Treaty Organization

NEPA—National Environmental Policy Act

NGB—National Guard Bureau

NVD—Night Vision Device

O&M—Operations and Maintenance

(Added-EDWARDSAFB) ODO—Operations Duty Officer

OT&R—Operational Training and Readiness

PACAF—Pacific Air Forces

PE—Program Element

PEM—Program Element Monitor

(Added-EDWARDSAFB) PIRA—Precision Impact Range Area

PPE—Personal Protection Equipment

PTR—Primary Training RangeRAP—Ready Aircrew Program

RAWG—Range and Airspace Working Group

REPI—Readiness and Environmental Protection Integration

SDZ—Surface Danger Zone

(Added-EDWARDSAFB) SM—Statute Mile

(Added-EDWARDSAFB) SOCC—Site Operations Control Center at Detachment 7, Air Force Research Laboratory

(Added-EDWARDSAFB) SPORT—Space Positioning Optics and Radar Tracking, and Call Sign for Edwards AFB Military Radar Unit

TO—Technical Order

T/TSNS—Test/Training Space Needs Statement

USAFE-AFAFRICA—United States Air Forces in Europe-Air Forces AFRICA

U.S.C.—United States Code

UXO—Unexploded Ordnance

WDZ—Weapon Danger Zone

Office Symbols

AF/A3—*Air Force Deputy Chief of Staff for Operations*

AF/A3T—*Air Force Directorate for Training and Readiness*

AF/A3TI—*Air Force Operational Training Infrastructure Division*

AF/A4C—*Air Force Directorate for Civil Engineering*

AF/A4CX—*Air Force Civil Engineering Readiness Division*

AF/JA—*Office of the Air Force Judge Advocate General*

AF/JAO—*Air Force Operational and International Law Directorate*

AF/JAOE—*Air Force Environmental Law and Litigation Division*

AF/TE—*Air Force Test and Evaluation Directorate*

AF/TEP—*Air Force Test and Evaluation Policy Division*

SAF/AA—*Administrative Assistant to the Secretary of the Air Force*

SAF/AQP—*Secretary of the Air Force, Acquisitions, Director for Global Power Programs*

SAF/GCN—*Air Force General Counsel, Installations, Energy, and Environmental Law*

SAF/IA—*Secretary of the Air Force, International Affairs*

SAF/IE—*Secretary of the Air Force, Installations and Environment*

SAF/IEE—*Deputy Assistant Secretary, Environment, Safety, and Infrastructure*

SAF/IEI—*Deputy Assistant Secretary, Installations*

ACC/A3A—*Air Combat Command, Airspace, Ranges, and Airfield Operations Division*

ACC/A3AR—*ir Combat Command, Ranges Branch*

AFCEC/CI—*Air Force Civil Engineering Center / Installations Directorate*

AFCEC/CZT—*Air Force Civil Engineering Center / Technical Support Division*

AFCEC/CZN—*Air Force Civil Engineering Center / National Environmental Policy Act Division*

AFLCMC/EN-EZZ—*Air Force Life Cycle Management Center / Safety Engineering Division*

AFLCMC/HBZ—*Air Force Life Cycle Management Center / Range Systems Division*

AFRL/RD—*Air Force Research Laboratory / Directed Energy Directorate*

AFSEC/SEW—*Air Force Safety Center / Weapons Safety Division*

AFMRA/SG3PB—*Air Force Medical Readiness Agency / Surgeon General Policy Branch*

711 HPW/RHD—*711 Human Performance Wing, Bioeffects Division*

USD (P&R)—*Under Secretary of Defense for Personnel and Readiness*

NGB A2/3/6/10YR—*National Guard Bureau/A2/3/6/10YR*

Terms

(Added-EDWARDSAFB) Alpha Corridor—A west to east air corridor into the West Range beginning at the southwest corner of Edwards AFB and extending east to the western boundary of the PIRA. The northern boundary runs across the north side of the settling ponds and intersects Mercury Boulevard near the point the road turns east towards Detachment 7, Air Force Research Laboratory.

(Added-EDWARDSAFB) Cold Run or Dry Run—A maneuver where the aircrew simulates weapon(s) release or simulates fire of an object, projectile, or laser from the aircraft.

(Added-EDWARDSAFB) COWBELL—Call sign "COWBELL" is used by the Range Control Officer when the East Range Tower operations are activated.

(Added-EDWARDSAFB) DAGRAG—Call sign "DAGRAG" is the Dual Air-To-Ground Range area used by the Range Control Officer when the West Range Tower operations are activated.

(Added-EDWARDSAFB) DOWNFALL Tower—Call sign "DOWNFALL" is the Master Control Tower which controls all ground activity within the PIRA. DOWNFALL Tower is located in the northeast corner of the West Range.

(Added-EDWARDSAFB) Foul Line Road—A hard surface roadway extending north and south from North Flank to South Flank Towers immediately west of the DAGRAG on the West Range.

(Added-EDWARDSAFB) Hot Run or Hot Pass—A maneuver where the aircrew intends weapon(s) release or fire of an object or projectile from the aircraft.

(Added-EDWARDSAFB) Hung Ordnance—Ordnance not released although a release was attempted.

(Added-EDWARDSAFB) Live Ordnance—Any munition(s) which contains energetic material in high enough concentration such that the material presents an explosive hazard and is capable of producing blast pressure, fragmentation, and incendiary or thermal effects. The term munition includes confined gaseous, liquid, and solid propellants, explosives, pyrotechnics, chemical and riot control agents, smokes and incendiaries, bulk explosives, chemical agents, chemical weapons, rockets, guided and ballistic missiles, bombs, warheads, mortar rounds, artillery ammunition, grenades, mines, torpedoes, depth charges, cluster munitions and dispensers, demolition charges and devices and components thereof.

Note: The purpose of this glossary is to help the reader understand the terms used in this publication. It does not encompass all pertinent terms.—The *DoD Dictionary of Military and Associated Terms*, and the *Air Force Glossary* contain standardized terms and definitions for DoD and Air Force use.

Air Combat Training Systems—A general term referring to aircraft instrumentation pods, ground receiving equipment, and interface/display equipment used for live monitoring, capture, and replay/debrief of training missions.

(Added-EDWARDSAFB) PIRA Definition—Originating at the intersection of Mercury Boulevard and Avenue B and running north along the east side of Mercury Boulevard; then east along the south side of Mercury Boulevard to Mars Boulevard; then southeast on the west side of Mars Boulevard to 34° 51' 45"N, 117° 37' 28"W; east to 34° 51' 45"N, 117° 36' 42"W; north to 34° 53' 45"N, 117° 36' 42"W; west to 34° 53' 45"N, 117° 37' 53"W north along the San Bernardino and Kern County line to 34° 57' 17"W, 117° 37' 53"W; northwest to 34° 58' 17" N, 117° 39' 05"W; northeast to 34° 59' 00"N, 117° 38' 12"W; east along northern reservation boundary to 34° 56' 08"N, 117° 31' 30"W; southeast to 34° 56' 08"N, 117° 31' 30"W; south along eastern reservation boundary to 34° 39' 17"N, 117° 31' 30"W; west along southern reservation boundary to 200th Street East; south along western side of 200th Street East to Avenue E; west to 140th Street East; north to Avenue B; east along south side of Avenue B to point of origin.

(Added-EDWARDSAFB) Precision Impact Range Area (PIRA) —The designated range area located at Edwards AFB and is established to support test activities such as air-to-ground gunnery, photo and infrared resolution, spin tests, aerial decelerator tests, tests requiring precision instrumentation, laser testing, ordnance release tests, and precision weapons or bombing tests.

(Added-EDWARDSAFB) Radio Mission Frequency—DOWNFALL Tower and SPORT Military Radar Unit monitors radio mission frequencies when aircraft operate on the PIRA.

(Added-EDWARDSAFB) Range Classifications—The range classifications are Class A, Class B, or Class C. The PIRA classification may change daily due to the type of activity or service being conducted on a specific range area at any given time.

(Added-EDWARDSAFB) Range Control Officer—The Range Control Officer is assigned to and supported by the 412 RANS. The 412 RANS Range Control Officer performs the DOWNFALL Tower assigned duties with the primary responsibility for PIRA safety during aircraft operations, aircraft emergencies, and ground activities during ordnance release or delivery, weapons fire, or any active events of this nature occurring on the PIRA. Weapons release clearance authority resides with the 412 RANS Range Control Officer. Safety procedures shall be established when the 412 RANS Range Control Officer has specifically delegated weapons release authority to a qualified flight lead, pilot, Forward Air Controller, or a designed briefed person.

(Added-EDWARDSAFB) Range Control Specialist—The Range Control Specialist is assigned to and supported by the 412 RANS. The 412 RANS Range Control Specialist supports the test project or test program and is the primary focal point to test organizations with responsibilities of evaluating range resources to meet test requirements, for coordinating inter- and intra-range support resources, for conducting mission control room assigned duties, and to ensure the appropriate range resources are scheduled with unit schedulers to satisfy real-time test mission execution requirements.

(Added-EDWARDSAFB) Range Operating Authority—The 412 TW/CC is responsible for operating and maintaining the 412 RANS PIRA and funds all aspects of PIRA clearance support.

(Added-EDWARDSAFB) Range Operations Officer—The Range Operations Officer is assigned to and supported by the 412 RANS. The 412 RANS Range Operations Officer supervises PIRA management and operations personnel, conducts planning of PIRA operations, ensure maintenance activities, performs day-to-day PIRA operations, and has overall responsibility and management of DOWNFALL Tower Range Control Officers.

(Added-EDWARDSAFB) Range Safety Officer—The Range Safety Officer is assigned to and supported by the 412 TW Safety Office. The 412 TW Range Safety Officer provides a range safety footprint and master arm point for test missions involving ordnance releases or when an aircrew is planning to drop, fire, or launch ordnance or objects with undetermined or specifically distinctive ballistic characteristics. The 412 TW Range Safety Officer provides a pyrotechnic ground profile to support test mission activity, determines if a low fire danger condition exists prior to using chaff or flares below 5,000 feet above ground level, and reviews any laser system that is not identified in the latest 711 HPW/RHDO report for approved use prior to laser energy operations on the PIRA.

(Added-EDWARDSAFB) South Lakebed—The area of Rogers Lakebed south of the extended centerline of South Base Runway 06/24.

(Added-EDWARDSAFB) SPORT—Call sign "SPORT" is the Military Radar Unit which controls airborne access to the PIRA, Alpha Corridor, and Restricted Area 2515.

Tethered Flight Instrumentation Range Systems—These systems provide Time Space Position Information and model weapons impact or flyout for aircrew kill notification using ground-based computers, towers, and communications relays and aircraft pods or internal avionics. These systems also provide real-time live monitor capability for Range Training Officer oversight.

Rangeless/Untethered Flight Instrumentation Range Systems—These systems use autonomous pods, aircraft avionics or recording devices of participating aircraft to provide capabilities similar to a tethered system. Pod-to-pod or aircraft data links can relay kill predictions to participating aircraft for near-real time kill notification. Rangeless/untethered systems do not provide a real-time live monitor data feed to a Range Training Officer.

Air Force Master Materials License—The Nuclear Regulatory Commission License issued to the United States Air Force Radioisotope Committee. The Master Materials License delegates to the Air Force regulatory authority over Byproduct, Source, and limited quantities of Special Nuclear Material.

Air Force Radioactive Material Permit—Written authorization from the United States Air Force Radioisotope Committee allowing Air Force organizations to receive, possess, distribute, use, transfer, or dispose of radioactive materials.

Air Force Radioisotope Committee—A committee established according to, and the named licensee on Air Force Master Materials License. The Radioisotope Committee coordinates the regulatory and administrative aspects of licensing, possessing, distributing, using, transferring, transporting, and disposing of all radioactive material in the Air Force.

Air-to-Air Test and Training—Air-to-Air test and training covers a wide range of mission requirements, such as air-to-air operations involving simulated and actual employment of missiles, air-to-air gunnery, aeronautical system testing, unmanned vehicles, and electronic warfare, which require a substantial amount of range space and a sophisticated range infrastructure. This infrastructure may include high fidelity simulators, visual simulators, end- game scoring capabilities, command and control systems, communication networks, data display/processing capabilities, instrumentation systems, flight termination systems, and flight hazard analysis/strike prediction capability.

Air-to-Surface Test and Training—Air-to-Surface test and training may require a substantial amount of range space and a sophisticated infrastructure to support complex, multi-aircraft operations; air-to-surface munition, cruise missile, and directed energy weapon employment; aeronautical system testing; unmanned vehicles; airdrop; and battlefield operations. This infrastructure may include high and/or low fidelity threat simulators, visual threat simulators, scoring capabilities, realistic target arrays, command and control systems, communication networks, data display/processing capabilities, instrumentation/debrief systems, flight termination systems, and flight hazard analysis/strike prediction capability.

Alert Area—Airspace designated to inform pilots of a high level of training activity or any unusual activity where prior knowledge would significantly enhance air safety. There are no restrictions placed on non-participating instrument flight rules or visual flight rules aircraft.

Armor Piercing Incendiary—A 30-millimeter round consisting of a 0.66-pound extruded DU penetrator, alloyed with 0.75 weight percent titanium, encased in a 0.8-millimeter-thick aluminum shell and windscreen.

Byproduct Material—Radioactive material (except Source or Special Nuclear Material) yielded in or made radioactive by exposure to radiation, incident to the process of producing or using Source or Special Nuclear Material.

Certificate of Clearance—An official document that certifies removal of all dangerous and explosive materials reasonably possible to detect. The certificate is dated, and a range clearance report is attached. The cleared areas are identified in red crosshatch on a map and are annotated in the map legend.

Combat Training Systems—The family of systems that includes range threat systems, Air CTS, and associated live training capabilities that may include connectivity to virtual and/or constructive systems.

Comprehensive Range Planning—An iterative process that examines current and projected operational needs to identify the best use for limited range development resources and the largest threats to the military value of ranges and airspace.

Controlling Agency—Air Traffic Control facility responsible for providing airborne control services in and around a designated airspace. With respect to a restricted area, the using agency may authorize transit through or flight within the restricted area according to a joint-use agreement.

Department of Defense Explosives Safety Board—DoD organization charged with recommending new and updated explosives and chemical agent safety regulations and standards and performing explosives safety management functions IAW DoDD 6055.09E.

Depleted Uranium—DU is the resultant “waste” of the Uranium-235 enrichment process. The radioisotope ratio in DU is 0.4% Uranium-235 (maximum) and 99.6% Uranium-238. DU is a dense, heavy metal and has two properties that make it ideal for military applications: extremely high density and the tendency to ignite on impact (especially with steel).

Depleted Uranium Authorized Personnel—Personnel trained in DU hazards and wearing the appropriate personal protective equipment, as determined by a qualified health physicist and the Installation Radiation Safety Officer.

Directed Energy Weapon—A weapon system that uses directed energy primarily as a means to deny, disrupt, degrade (damage), or destroy enemy equipment, facilities, or personnel.

Electronic Warfare—Locations supporting electronic warfare training must be capable of providing a simulated electronic threat environment for aircrew combat training. Electronic warfare ranges are located on a range (usually a range capable of receiving surface fires and actual ordnance delivery). Electronic warfare sites typically provide electronic warfare simulation and feedback, but are not located on a range. An electronic warfare site can be established at any location with proper security. Typically, they are located under or near a military training route or Military Operations Area.

Electronic Warfare Range—This term has traditionally taken on a narrow definition of referring only to those ranges with an electronic warfare capability that is associated with and located on a range capable of receiving surface fires and actual ordnance delivery. However, with the proliferation of electronic warfare capabilities across all Air Force ranges, the term is now generally understood to refer to any range with an electronic warfare capability.

Electronic Warfare Site—An electronic warfare site is a specific type of electronic warfare training area which is not located within the boundaries of another range. Typically, the land required for an electronic warfare site is minimal and they do not support actual ordnance delivery. They can be located at any location with proper security and are usually located under or near a Military Training Route or Military Operations Area. In previous versions of this publication, this type of training area was referred to as an Electronic Scoring Site, despite the fact that many of these sites did not have scoring capability.

Emitter/Simulator—Generic terms used to describe threat equipment operated at electronic warfare ranges and electronic warfare sites. However, the terms could refer to a variety of equipment such as, but not limited to, emitter only systems, emitter-receiver-processors, and replica type systems.

Encroachment—Any action by any governmental or non-governmental entity or individual that does, or is likely to inhibit, curtail, or impede current or future military activities within the installation complex and/or mission footprint; or any military activity that is, or is likely to be incompatible with a community's use of its resources.

Environmental Impact Analysis Process—The formal Air Force process for implementing the National Environmental Policy Act and the Council on Environmental Quality's National Environmental Policy Act Regulations, used to support Air Force decision making and assess environmental impacts resulting from a proposed action and its reasonable alternatives.

Environmental Planning Function—At every level of command, the environmental planning function is a key participant responsible for the EIAP. At the request of the proponent, the environmental planning function manages the preparation of environmental documents and assists the proponent in obtaining review of environmental documents.

Essential Personnel—Those personnel on a range, not required for ordnance employment, that participate and provide essential support for the range test, training or evaluation mission. Examples include maneuver elements, opposition forces, instructors, evaluators, range personnel conducting maintenance, etc.

Explosive Ordnance Disposal—The detection, identification, on-site evaluation, rendering safe, recovery, and final disposal of unexploded explosive ordnance. It may also include explosive ordnance that has become hazardous by damage or deterioration.

Explosive Ordnance Disposal Personnel/Technician—Military personnel who have graduated from the Naval School, EOD; are assigned to a military unit with a service-defined EOD mission; and meet service and assigned unit requirements to perform EOD duties. EOD Personnel have received specialized training to address explosive and chemical agent hazards during both peacetime and wartime. EOD Personnel are trained and equipped to perform render safe procedures on nuclear, biological, chemical, and conventional munitions, and on improvised explosive devices.

Final Governing Standard—A comprehensive set of country-specific substantive environmental provisions. A Final Governing Standard requirement is typically a technical limitation on effluent or discharge, or a specific management practice. Examples include technical limits on air emissions and wastewater discharges, as well as recordkeeping and reporting requirements.

Hazard Area—The composite area of all WDZs, SDZs, Laser SDZs, and Directed Energy WDZs for all authorized weapon delivery events.

Impact Area—The area on a range immediately surrounding a target or designated mean point of impact that is approved for the actual delivery of ordnance. The impact area demarcation should normally be no less than a) 500 feet from the center of a target or desired point of impact approved for live ordnance, or b) 300 feet from the center of a target or desired point of impact used solely for inert or practice ordnance.

Joint Terminal Attack Controller—An individual qualified IAW Joint Publication 3-09.3 to provide terminal control for the delivery of weapons by aircraft.

Joint Use—With respect to ranges, Joint Use means other MAJCOMs or services may use, as long as they conduct operations IAW this manual, as supplemented. With respect to range airspace, it means the use by civil or other military aviation when it is not active.

Land Withdrawal—The process used as a means to transfer administrative jurisdiction from one federal agency to another. A withdrawal creates a title encumbrance on the land restricting an agency's ability to manage its lands under multiple use management principles. The restrictions generally segregate the lands from some or all the public land laws and some or all of the mining and mineral leasing laws for a specific period of time (generally 20 years for post Federal Land Policy and Management Act withdrawals). The Secretary of Interior is authorized to make, modify, extend, or revoke withdrawals.

Laser Surface Danger Zone—Designated region or ground area where laser radiation levels may exceed maximum permissible exposure levels, thereby, requiring control during laser operation.

Lead Major Command—As defined in AFI 38-101, *Manpower and Organization*, a lead MAJCOM consolidates responsibilities for a particular function in a single MAJCOM, supporting the entire Air Force as applicable.

License—Nuclear Regulatory Commission written authorization delegating regulatory authority to receive, possess, use, or transfer Byproduct, Source, or Special Nuclear Material. The Nuclear

Regulatory Commission has issued a master materials license to the Air Force Radioisotope Committee, which in turn issues radioactive material permits to Air Force units.

Low-Angle Strafe Target—A target specifically designed for aircraft strafe events where only training projectile ammunition is authorized, i.e., strafe pit.

Major Range and Test Facility Base—The designated core set of DoD Test and Evaluation infrastructure and associated workforce that must be preserved as a national asset to provide Test and Evaluation capabilities to support the DoD acquisition system. (DoDD 3200.11)

Material Potentially Presenting an Explosive Hazard—Material owned or controlled by the DoD that, before determination of its explosives safety status, potentially contains explosives or munitions (e.g., munitions containers and packaging material; munitions debris remaining after munitions use, demilitarization, or disposal; and range-related debris) or potentially contains a high enough concentration of explosives that the material presents an explosive hazard (e.g., equipment, drainage systems, holding tanks, piping, or ventilation ducts that were associated with munitions production, demilitarization, or disposal operations). Excluded from MPPEH are: (1) Military munitions and military munitions-related materials, including wholly inert components (e.g., fins, launch tubes, containers, packaging material), that are to be used or reused for their intended purpose and are within a DoD Component-established munitions management system, (2) Non-munitions-related material (e.g., horseshoes, rebar, other solid objects) and munitions debris that are solid metal fragments that do not realistically present an explosive hazard, and (3) Other items (e.g., gasoline cans, compressed gas cylinders) that are not munitions or munitions-related material but may present an explosion hazard. (DoDI 4140.62)

Maximum Permissible Exposure—The level of laser radiation to which an unprotected person may be exposed without adverse biological changes in the eye or skin.

Military Operations Area—Special Use Airspace allocated to the military to separate/segregate certain military activities from Instrument Flight Rules traffic, and to identify the location of these military activities to Visual Flight Rules traffic. Visual Flight Rules aircraft are not restricted from transiting Military Operations Areas.

Military Training Route—A low-level, high-speed training route established in coordination with the FAA. Military Training Routes are used by DoD to conduct low altitude navigation and tactical training, in instrument and visual weather conditions, below an altitude of 10,000 feet Mean Sea Level and at airspeeds more than 250 Knots Indicated Airspeed. Routes are established as Instrument Flight Rules routes or Visual Flight Rules routes. The FAA has approval authority to implement Instrument Flight Rules routes and the appropriate MAJCOM approves Visual Flight Rules route implementation. Environmental documentation is required for implementation IAW 32 CFR Part 989. Visual Flight Rules routes are processed through the FAA via the Air Force Representative. Military Training Routes are published in the flight information publication AP/1B and charted on FAA Sectionals and DoD Low Instrument Flight Rules charts. Air Force Representatives assign all route numbers.

Mission Essential Personnel—Those personnel on a range directly required for the employment of ordnance (air/surface/electronic attack) in a test, training or evaluation mission. This may include JTACs, Tactical Air Control Parties, range control officers, scorers, and any other personnel identified as required by the range operating authority.

Ordnance—***Boosted Munitions (forward firing):*** Munitions such as the Air-to-Ground Missile-65 Maverick and the 2.75” folding fin rocket driven by propellant. These are also considered live munitions when they are equipped with an explosive or incendiary warhead.

Test—The act of generating empirical data during the research, development, or sustainment of systems, and the creation of information through analysis that is useful to technical personnel and decision maker for reducing design and acquisition risks. The process by which systems are measured against requirements and specifications, and the results analyzed so as to gauge progress and provide feedback.

Training—***Full-scale Inert:*** Concrete-filled or cast ductile iron bombs of the same size and weight of the live munition but containing no explosives, pyrotechnics, or chemical agents.

Practice Bombs—Practice bombs may be full-scale or sub-scale. Some practice bombs contain a small explosive charge or pyrotechnic that marks the point of impact with a small cloud of smoke or flash. For example, Bomb Dummy Unit-33 practice bombs contain a Mark 4 spotting charge, and Mark 82 practice bombs may contain 6.25 pounds of composition C-4 high explosive. British 1,000-pound class practice bombs may contain 50 pounds of Torpex. These bombs normally use a fuze to initiate the high explosive fillers.

Target Practice—Ball projectile gun ammunition that has no explosive in the projectile.

Live Munitions—Munitions containing a fuze and a high-explosive or incendiary warhead designed to detonate either prior to, upon, or shortly after impact. They can be bombs, missiles, rockets, bullets, etc.

Operational Range—A range that is under the jurisdiction, custody, or control of the Secretary of Defense and that is used for range activities; or although not currently being used for range activities that is still considered by the Secretary to be a range and has not been put to a new use that is incompatible with range activities. (10 U.S.C. 101(e)(3)(A) and (B)). Also includes

"military range," "active range," and "inactive range" as those terms are defined in 40 CFR § 266.201.

Overseas Environmental Baseline Guidance Document—A set of objective criteria and management practices designed to protect human health and the environment. It reflects generally accepted federal environmental standards that apply to DoD installations, facilities, and actions in the United States. It also incorporates requirements of United States law that have extraterritorial application (i.e., apply overseas). The DoD developed and maintains the Overseas Environmental Baseline Guidance Document.

Penetrator—Dense projectile component of ammunition round designed to pierce armor.

Permit (Noun)—***In General*:** an official document or certificate giving permission for something.

Radioactive Material Permit—United States Air Force or United States Navy Radioactive

Material Permit issued to a unit with the respective service, under the authority of that service's Master Materials License.

Primary Training Range—PTRs are established to accommodate training. They are not designed or structured to accommodate test and evaluation activities.

Prohibited Area—A specified area over the land of a state, or territorial waters adjacent thereto, within which the flight of aircraft is prohibited in the interest of national security and welfare.

Proponent—The office, unit, single manager, or activity at any level that initiates, or is responsible for an Air Force action.

Radiation—The emissions, either electromagnetic or particulate, resulting from the transformation of an unstable atom or nucleus.

Radiation Safety Officer—An individual with specific education, military training, and professional experience in radiation protection practice appointed by a Permittee or the United States Air Force Radioisotope Committee to manage radiation safety programs. The term "Radiation Safety Officer" is a functional title and does not denote a commissioned status or specialty code. The Radiation Safety Officer must have the education, military training, and professional experience needed for the job.

Radioactive Material—Material with unstable nuclei that decay by emission of ionizing radiation. The radiation emitted may be alpha or beta particles, gamma or X-rays, or neutrons.

Range—As set forth at 10 U.S.C. § 101(e)(1), the term “range”, when used in a geographic sense, means a designated land or water area that is set aside, managed, and used for range activities of the DoD. Such term includes the following:

(A)Firing lines and positions, maneuver areas, firing lanes, test pads, detonation pads, impact areas, electronic scoring sites, buffer zones with restricted access, and exclusionary areas.
(B)Airspace areas designated for military use IAW regulations and procedures prescribed bythe Administrator of the FAA.
As set forth at 10 U.S.C § 101(e)(2), the term “range activities” means—
(A)research, development, testing, and evaluation of military munitions, other ordnance, and weapons systems; and
(B)the training of members of the armed forces in the use and handling of military munitions, other ordnance, and weapons systems.

Active Range—For the purposes of this manual, an active range is an operational range that is being used for range activities.

ANG Range—For the purposes of this manual, ANG ranges include all buildings and property that is established by the lease, license, permit or other written agreement, for either exclusive or joint use by the ANG for weapons delivery operations.

Inactive Range—For the purposes of this manual, an inactive range is an operational range that is still considered to be a range and has not been put to a new use that is incompatible with range activities.

Range Clearance—Range clearance (previously known as “range residue clearance,” “range decontamination,” or “EOD clearance”) is the surface-removal or disposal of MPPEH from the targets and surrounding areas. MPPEH includes UXO, classified ordnance, inert ordnance debris, and any other range material fired on, or upon a military range.

Range Clearance Report—A narrative statement about the removal and disposal of UXO and MPPEH on a specific range and serves as a factual record of the debris clearance. Also known as a “Report of Clearance,” it is not a “Certificate of Clearance” (defined above).

Range Control Officer—The person with primary responsibility for matters of range safety during aircraft operations, aircraft emergencies, and ground weapons fire activities during active events of this nature occurring on the range. Weapons release clearance authority resides with the range control officer (except where the range control officer has specifically delegated this authority to a qualified flight lead, individual pilot, Forward Air Controller, or other briefed person).

Range Operating Authority—The wing commander, or designated commander responsible for operating and maintaining the range. For ANG-operated ranges, the range commander is designated as the range operating authority. The range operating authority may delegate the daily scheduling, management, and maintenance of the range to any appropriate subordinate unit.

Range Operations Officer—The person who supervises range management, planning, maintenance, and day-to-day operations. Specific responsibilities are designated by the range operating authority.

Range Service Levels—Ranges provide different levels of service dependent upon several factors, such as hours of manned operation, scheduled test and training activities, utilization, etc. For example, a MRTFB range may provide both Class A and Class T service on different parts of the range or at different times; a PTR may provide Class A service during daylight hours and Class C service during nighttime hours.

Class A Service—Class A service provides a manned, ground-scoring and/or electronic warfare capable range with a range control officer present on range and controlling surface activities and air-to-ground operations.

Class B Service—Class B service provides a manned or unmanned, ground scoring and/or electronic warfare capable range where no range control officer is present on range for controlling surface activities and operations. **Note:** Class B service includes ranges where a remotely-sited range operations center monitors air and ground operations and provides scoring feedback.

Class C Service—Class C service provides an unmanned range with no scoring, no electronic warfare services and no range control officer control of ground activities or operations.

Class D Service—Class D service provides an instrumented range supporting operations monitored by a Range Training Officer.

Class T Service—Class T service provides a manned or unmanned range or test site intended for test activities and explosives/weapons detonations, controlled by the range operating authority.

Range Test and Training Activities—The land or sea encompassed within the hazard area or underlying an air-to-air range used for actual weapon employment must be protected by adequate physical safeguards, legally sufficient real property acquisition documentation (e.g., fee interest purchase, lease) or other means to ensure the safety of personnel, structures, and the public from expended weapons, laser and electromagnetic emissions, and target debris.

Range Training Officer—The person responsible for monitoring Air CTS, passing kill removal, and providing debriefs. The Range Training Officer will establish communication with aircraft entering the range.

Readiness and Environmental Protection Integration—An OSD program to purchase development easements from willing land owners to help ensure the continued viability of training lands by removing the potential for incompatible development. REPI is authorized under 10 U.S.C. § 2684a. REPI enhances military readiness by preventing, mitigating, or reducing restrictions on the timing, frequency, and type of training activities caused by encroachment. REPI promotes compatible development and protects valuable habitat that supports unconstrained training, testing, and operations.

Restricted Area—An area (land, sea, or air) in which there are special restrictive measures employed to prevent or minimize interference between friendly forces or an area under military jurisdiction in which special security measures are employed to prevent unauthorized entry.

Airspace where the flight of aircraft, while not wholly prohibited, is subject to restriction. When not activated by the using agency, the controlling air traffic control facility may authorize Instrument Flight Rules or Visual Flight Rules operations in the area. If joint use is authorized, the name of the controlling air traffic control facility is annotated on the map.
An area that must contain all "Hazardous Activity" as defined by branch of service for specific type of aircraft using the range.

Shared Use—When the range operating authority authorizes non-DoD, private or government agencies or individuals to use range land or airspace concurrently or non-concurrently.

Special Use Airspace—Airspace of defined vertical and lateral dimensions wherein activities are confined. Special Use Airspace (SUA) is authorized and managed by the FAA as part of the National Airspace System (NAS). Use of SUA may need to be coordinated with FAA. Certain limitations or restrictions may be imposed on non-participating aircraft. Except for Controlled Firing Areas, Special Use Airspace is depicted on aeronautical charts. Types of Special Use Airspace include: alert areas, controlled firing areas, military operations areas, national security areas, prohibited areas, restricted areas, and warning areas.

Surface Danger Zone—The ground and airspace designated for vertical and lateral containment of projectiles, fragments, debris, and components resulting from the firing, launching, or detonation of weapon systems to include explosives and demolitions.

Surface-to-Air Test and Training—Surface-to-Air test and training covers a wide range of mission requirements. Ranges that support, for example, endo-atmospheric and exo- atmospheric missile intercepts, aeronautical system testing, and ballistic missiles require a substantial amount of range space and a sophisticated range infrastructure. This infrastructure may include high fidelity simulators, visual simulators, end-game scoring capabilities, command and control systems, communication networks, data display/processing capabilities, instrumentation systems, flight termination systems, and flight hazard analysis/strike prediction capability.

Surface-to-Surface Test and Training—Surface-to-Surface test and training covers a wide range of mission requirements, including survivability test, horizontal and vertical live-fire events, surface and subsurface tests in international waters, research and development, directed energy weapon, explosives operations, heavy weapons qualification, and EOD thermal treatment and disposal. Ranges that support Special Operations Forces and other ground test and training requirements may require substantial ground infrastructure, including shoot houses, urban villages, laser scoring, in-band imaging, complex communication and instrumentation, range surveillance radars and other facilities.

Test Areas Control Officer—The person responsible for range operations and safety at a test site when no air-to-ground weapons release authority is needed.

Test Sites—These sites are developed and used for specific developmental test and evaluation and operational test and evaluation requirements. They are considered ranges under this manual due to the scope, nature, and frequency of conducting many of the defined range activities.

Unexploded Ordnance—As set forth at 10 U.S.C. § 101(e)(5), the term "UXO" means military munitions that—

(A)have been primed, fuzed, armed, or otherwise prepared for action.
(B)have been fired, dropped, launched, projected, or placed in such a manner as to constitute a hazard to operations, installations, personnel, or material; and
(C)remain unexploded, whether by malfunction, design, or any other cause.

United States—The several States, District of Columbia, Commonwealths of Puerto Rico and Northern Mariana Islands, American Samoa, Guam, Midway and Wake Islands, United States Virgin Islands, any other territory or possession of the United States, and associated navigable waters, contiguous zones, and ocean waters of which the natural resources are under the exclusive management authority of the United States.

Urban Training Facility—Facilities constructed for the primary purpose of training in an urban environment. They include instrumentation, communications infrastructure, and the population needed to create a physical urban environment and capabilities to conduct training in that environment. Facilities can be composed of buildings, subterranean infrastructure, airfields, and representative building shells, bridges, tunnels, etc., that constitute target arrays for fixed and rotary wing aircraft. All urban training facilities will support live training, however, based on the training audience and their training objectives, may support virtual and constructive training as well. (DoDI 1322.27)

Unexploded Ordnance—**Qualified Personnel**—Personnel who have performed successfully in military EOD positions, or are qualified to perform in the following Department of Labor, Service Contract Act, Directory of Occupations, contractor positions: UXO Technician II, UXO Technician III, UXO Safety Officer, UXO Quality Control Specialist, or Senior UXO Supervisor. UXO-qualified personnel must also meet DoD Explosives Safety Board experience and training requirements. (See: DoD Explosives Safety Board Technical Paper 18.)

Warning Area—A specified area above, below, or within which there may be potential danger. Airspace of defined dimensions over international waters that contain activity that may be hazardous to non-participating aircraft. An area that must contain all "Hazardous Activity" as defined by branch of service for specific type of aircraft using the range.

Weapon Danger Zone—The ground and airspace for lateral and vertical containment of a user-determined percentage of projectiles, fragments, debris, and components resulting from the firing, launching, and/or detonation of aviation delivered ordnance.

Attachment 2

UNITED STATES AIR FORCE OPERATED OR OWNED RANGES GOVERNED BY THIS MANUAL

A2.1. Note: The "Owner" column in the tables contained in this attachment is intended to reflect the entity with real property accountability for the range land.

Table A2.1. Primary Training Ranges (PTR).

PTR Name	Range Location	Operator	Owner
Adirondack	Fort Drum, New York	ANG	United States Army
Airburst	Fort Carson, Colorado	ANG	United States Army
Avon Park	Florida	ACC	ACC
Barry M. Goldwater Range East	Arizona	AETC	AETC
Belle Fourche Electronic Warfare Site	Wyoming	AFGSC	AFGSC
Bollen Range	Pennsylvania	ANG	United States Army
Cannon	Fort Leonard Wood, Missouri	AFRC	United States Army
Claiborne Range	Louisiana	AFRC	AFRC
Dare County Ranges Air Force Dare County RangeNavy Dare County Range	North Carolina	ACC United States Navy	ACCACC
Draughon	Japan	PACAF	Japan
Falcon	Fort Sill, Oklahoma	AFRC	United States Army
PTR Name	**Range Location**	**Operator**	**Owner**
Grand Bay	Georgia	ACC	ACC
Grayling	Camp Grayling, Michigan	ANG	United States Army

Guam Electronic Warfare Site	Joint Region Marianas-Andersen, Guam	AFGSC	United States Navy
Hardwood	Volk Field, Wisconsin	ANG	ANG
Holloman Ranges Centennial Oscura Red Rio	New Mexico	AETC	United States Army
Indiana Air Range ComplexJefferson Atterbury	Indiana Camp Atterbury, Indiana	ANG	United States Army
Joint Okinawa Training Range Complex	Japan	PACAF	Japan
Joint Pacific Alaska Range Complex[1] Big Delta (R-2202) Blair Lakes (R-2211) Fairbanks (R-2205) Fort Greely (R-2201)	Alaska	PACAF	PACAF United States Army
McMullen Range	Texas	ANG	United States Navy
Melrose Range	New Mexico	AFSOC	AFSOC
PTR Name	**Range Location**	**Operator**	**Owner**
Mountain Home Range Complex Juniper Butte Saylor Creek Grasmere Electronic Warfare Site	Idaho	ACC	ACC
Pilsung Range	Korea	PACAF	Korea
Poinsett Range	South Carolina	ACC	ACC
Polygone Electronic Warfare Site	Germany/France	USAFE-AFAFRICA	United States/ Germany/ France
Razorback	Fort Chaffee, Arkansas	ANG	United States Army

Shelby	Mississippi	ANG	United States Army
Smoky Hill	Kansas	ANG	ANG
Snyder Electronic Warfare Site	Texas	AFGSC	AFGSC
Warren Grove	New Jersey	ANG	ANG
1. PACAF/A31 will designate the range operating authority for the Joint Pacific Alaska Range Complex.			

Table A2.2. Major Range and Test Facility Bases (MRTFB).

MRTFB Range Name	Range Location	Operator	Owner
412th Test Wing[1]	Edwards Air Force Base, California	AFMC	AFMC
96th Test Wing[1]	Eglin Air Force Base, Florida	AFMC	AFMC
Arnold Engineering Development Complex[1]	Holloman Air Force Base, New Mexico	AFMC	AFMC
Nevada Test & Training Range	Nevada	ACC	ACC
Utah Test & Training Range	Utah	ACC	AFMC
1. As part of the 2012 AFMC reorganization, the Arnold Engineering Development Complex, the 412th Test Wing, and the 96th Test Wing were placed under the Air Force Test Center as one of the Centers under AFMC.			

Table A2.3. Test Sites.

Test Site Name	Range Location	Operator	Owner
Air Force Research Laboratory Laser –1 Kilometer and 2 Kilometer Laser Targets	Kirtland Air Force Base, NewMexico	AFGSC	AFGSC
Air Force Research LaboratoryChestnut Test Site	Kirtland Air Force Base, NewMexico	AFGSC	AFGSC
Air Force Research LaboratoryHigh Energy Research Test Facility Microwave	Kirtland Air Force Base, NewMexico	AFGSC	AFGSC
Starfire Optical Range	Kirtland Air Force Base, NewMexico	AFGSC	AFGSC
Aerospace Vehicle SurvivabilityFacility	Wright-Patterson Air Force Base, Ohio	AFMC	AFMC

Attachment 3

RANGE SUPPORTED MISSION DESIGN SERIES TRAINING EVENTS

Table A3.1. A-10 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
A-10	1	Mission Profiles (General)	RAP[1]	Range/airspace access-scheduling/operating hours
A-10	2	Airborne Laser Designation	RAP	Laser capable
A-10	3	Air Strike Control Actual Ordnance	RAP	Live or inert capable
A-10	4	CAS in Urban Terrain	RAP	Constructed urban village
A-10	5	Flare	RAP	Capable
A-10	6	Full Scale Weapons Delivery/Heavy Weight	RAP	Live or inert capable
A-10	7	Laser Spot Search/Track	RAP	Laser capable
A-10	8	Laser Guided Bomb	RAP	or inert capable,laser capable
A-10	9	Maverick Event	RAP	Live or inert capable
A-10	10	Night Strafe	RAP	Strafe pit open during nighttime
A-10	11	pression of Enemy Air Defenses-Contested	RAP	Live or inert capable, visual and/or electronic threat replication
A-10	12	Self-Mark Strafe (Night)	RAP	Strafe pit with scoring
A-10	13	Strafe	RAP	Strafe pit with scoring
A-10	14	Target Mark	RAP	Multiple targets and strafe pit, flare capable, laser capable, or live or inert capable
A-10	15	Targeting Pod Event	RAP	Multiple targets
A-10	16	Chaff	RAP	Capable
A-10	17	Electronic Warfare Event Air-to-Ground	RAP	Threat emitter
A-10	18	Moving Target Attack	RAP	Remote-controlled vehicle capable of 25 miles per hour
A-10	19	Degraded/Denied Communications	RAP	Communications jammer
A-10	20	Degraded/Denied Data Link	RAP	Data link jammer

A-10	21	Degraded/Denied GPS	RAP	GPS jammer
A-10	22	Large Force/Flag Exercise	RAP	Scenario-based threat replication
1. A-10 RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1), (Effective 01 Apr 17).				

Table A3.2. AC-130 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
AC-130	1	Live Fire	RAP	High explosive impact area
AC-130	2	on Guided Munition Employment	Theater Deployment	Capable of laser and GPS guided precision-guided munition
AC-130	3	Air-to-Ground Gunnery	Theater Deployment	Live capable
AC-130	4	Chaff	RAP	Capable
AC-130	5	Flare	RAP	Capable
AC-130	6	Threat Event	RAP	Emitters
AC-130	7	Call-For-Fire	RAP	Live capable
AC-130	8	CAS	Theater Deployment	Live capable
AC-130	9	Airborne Laser Designation	RAP	Laser capable

Table A3.3. B-1 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
B-1	1	Veapon Release(Inert)	RAP	nert capable with separate cluster-bomb targets set
B-1	2	d Corrected Munitions Dispenser Release	RAP	Live or inert capable and separate cluster-bomb targetset
B-1	3	Actual Flare Event	RAP	Capable
B-1	4	Actual Chaff Event	RAP	Capable
B-1	5	ic Attack ThreatActivity	RAP	Electronic threat emitter
B-1	6	Targeting Pod Operations	RAP	Tactical targets, laser capable
B-1	7	ested/DegradedOperations	RAP	communicationsjammer
B-1	8	Large Force/Flag Exercise	RAP	ario-based threatreplication

Table A3.4. B-2 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
B-2	1	Air-to-Ground Missile-158 Bomb Run	RAP	Emitters
B-2	2	B61/83 Actual Release	RAP	Inert capable
B-2	3	Guided Bomb Unit-31/28 Actual Release	RAP	Live or inert capable
B-2	4	Guided Bomb Unit-38 Actual Release	RAP	Live or inert capable
B-2	5	Guided Bomb Unit-57 Bomb Run	RAP	Emitters
B-2	6	Conventional Unguided	RAP	Live or inert capable
B-2	7	Electronic Threat Activity	RAP	Electronic threat emitter
B-2	8	Large Force/Flag Exercise	RAP	ario-based threatreplication

Table A3.5. B-52 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
B-52	1	Actual Joint Direct Attack Munition/Wind Corrected Munitions Dispenser Release	RAP	Live or inert capable and separate cluster-bomb target set
B-52	2	Actual Weapon Release	RAP	Live or inert capable
B-52	3	Joint Direct Attack Munition/Wind Corrected Munitions Dispenser Jettison	RAP	Live or inert capable
B-52	4	Laser Spot Search/Track	RAP	Laser capable
B-52	5	Guided Bomb ActualWeapon Release	RAP	rt capable and lasercapable
B-52	6	Chaff/Flare Exercise	RAP	Chaff/flare capable
B-52	7	Targeting Pod employment with Inertially Aided Munitions	RAP	Live or inert capable
B-52	8	ic Attack ThreatActivity	RAP	Threat emitter
	9	Large Force/Flag Exercise	RAP	ario-based threatreplication

Table A3.6. C-17 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
C-17	1	p Container DeliverySystem Actual	Volume1	Drop zone
C-17	2	Airdrop Equipment Actual	Volume	Drop zone
C-17	3	Airdrop Personnel Actual	Volume	Drop zone
C-17	4	lare/Laser Infrared Countermeasures	Volume1	Simulator capable
C-17	5	Precision Airdrop System Airdrop Event	Volume1	Capable
C-17	6	Tactical Airdrop Night	Volume	Drop zone-night hours
C-17	7	Threat Event	Volume	Emitters
C-17	8	Electronic Warfare Event	Volume1	Emitters
C-17	9	Contested/Degraded Operations	Volume1	Emitters, GPS jammer, communications jammer, data link jammer
C-17	10	Large Force/Flag Exercise	Volume1	ario-based threatreplication
C-17	11	Assault Landing (Day)	Volume	Assault strip
C-17	12	Assault Landing (Semi-Prepared Runway Operations)	Volume1	Dirt assault strip
C-17	13	Assault Landing (Night Vision Goggle Operations)	Volume1	Assault strip—night hours

Table A3.7. C-130 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
C-130	1	Airdrop Container Delivery System Actual	Volume 1	Drop zone
C-130	2	Air Drop Equip Actual	Volume	Drop zone
C-130	3	Airdrop Personnel Actual	Volume	Drop zone
C-130	4	Tactical Airdrop Night	Volume	Drop zone-night hours
C-130	5	Precision Airdrop System Airdrop Event	Volume 1	Capable
C-130	6	Flare	Volume	Capable
C-130	7	Chaff	Volume	Capable
C-130	8	Threat Event	Volume	Emitters
C-130	9	Contested/Degraded Operations	Volume1	Emitters, GPS jammer, communications jammer, data link jammer
C-130	10	Large Force/Flag Exercise	Volume1	Scenario-based threat replication
C-130	11	Assault Landing (Day)	Volume1	Capable range and capable airspace (allow tactical arrival/departure)
C-130	12	Assault Landing (Semi-Prepared Runway	Volume1	ange and capableairspace
C-130	13	Assault Landing (Night Vision Goggle Operations)	Volume1	Capable range, capable airspace, and special tactics support

Table A3.8. C-145 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
C-145	1	Low Cost Low Altitude Airdrop	RAP	Drop zone day/night
C-145	2	Night Vision Goggle Airdrop	RAP	Drop zone day/night
C-145	3	Night Vision Goggle Insertion/Extraction	RAP	Blacked out landing zone
C-145	4	Semi-Prepared Surface Landing/Takeoff	RAP	Dirt landing zone
C-145	5	Military Free Fall	Theater Deployment	Drop zone day/night
C-145	6	Live Fire	Theater Deployment	Impact area day/night
C-145	7	Airland - Improved Surface	Theater Deployment	Hard surface landing zone
C-145	8	Airland - UnimprovedSurface	Theater Deploym ent	Dirt/clay surface landingzone
C-145	9	Airdrop (Personnel/Container Delivery System/Equipment)	Theater Deployment	Day/night drop zone
C-145	10	Air-to-Ground Gunnery	Theater Deployment	Live capable

Table A3.9. C-146 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
C-146	1	Night Vision Goggle Insertion/Extraction		Blacked out landing zone
C-146	2	Semi-Prepared Surface Landing/Takeoff		Dirt landing zone

Table A3.10. CV-22 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
CV-22	1	Insertion/Extraction Training	RAP	Landing zone
CV-22	2	Alternate Insertion/Extraction	RAP	Landing zone/drop zone
CV-22	3	Chaff	RAP	Capable
CV-22	4	Flare	RAP	Capable
CV-22	5	Electronic Warfare Event/Degraded Operations	RAP	Emitters, GPS jammer, communications jammer, data link jammer
CV-22	6	Hoist Training	RAP	Landing zone, drop zone, helicopter landing zone
CV-22	7	Airland - Improved Surface	RAP	Hard surface helicopter landing zone
CV-22	8	- UnimprovedSurface	RAP	Dirt/clay surface helicopter landing zone
CV-22	9	Air-to-Ground gunnery	RAP	Live capable
CV-22	10	Military Free Fall	RAP	Drop zone

Table A3.11. E-3 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
E-3	1	Electronic Warfare Event	RAP	Emitters
E-3	2	Large Force/Flag Exercise	RAP	ario-based threatreplication

Table A3.12. E-8 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
E-8	1	Large Force/Flag Exercise	RAP	Scenario-based threat replication
E-8	2	Moving Target	RAP	Capable
E-8	3	Electronic Warfare Event	RAP	Emitters

Table A3.13. EC-130H and EC-37B Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
EC-130H	1	Smokey Surface-to-Air Missile Event	RTM	Capable
EC-130H	2	Threat Emitters		Live Jam Capable EW Feedback Medium

Table A3.14. F-15C Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
F-15C	1	Aerial Gunnery	RAP[1]/ Syllabus	able, airbornetarget
F-15C	2	Chaff	RAP/ Syllabus	Capable
F-15C	3	Flare	RAP/ Syllabus	Capable
F-15C	4	Electronic Warfare Event	RAP/ Syllabus	Emitters
F-15C	5	Large Force/Flag Exercise	RAP	Scenario-based threat replication
F-15C	6	High Angle Strafe	RAP	Controlled overlandrange

1. F-15C RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1),(Effective 01 Apr 17).

Table A3.15. F-15E Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
F-15E	1	Aerial Gunnery	RAP[1]	ble and airbornetarget
F-15E	2	Chaff	RAP	Capable
F-15E	3	Flare	RAP	Capable
F-15E	4	Electronic Warfare Event	RAP	Emitters
F-15E	5	Aided MunitionActual	RAP	Live or inert capable
F-15E	6	Target Laser Guided Bomb/Strafe	RAP	Laser capable and remote-controlled vehicle capable of25 miles per hour
F-15E	7	Strafe	RAP	Strafe pit with scoring
F-15E	8	Guided MunitionActual	RAP	Live or inert capable and laser capable
F-15E	9	Large Force/Flag Exercise	RAP	Scenario-based threat replication

1. F-15E RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1),(Effective 01 Apr 17).

Table A3.16. F-16 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
F-16	1	Aerial Gunnery	RAP[1,2]	ble and airbornetarget
F-16	2	Chaff	RAP/ Syllabus	Capable
F-16	3	CAS in Urban Terrain	RAP	Constructed urban village
F-16	4	Full-Scale Weapons Delivery/Heavy Weight	RAP/ Syllabus	Live or inert capable
F-16	5	Flare	RAP/ Syllabus	Capable
F-16	6	Aided MunitionActual	RAP/ Syllabus	Live or inert capable
F-16	7	Laser Guided Bomb Actual	RAP/ Syllabus	Live or inert capable and laser capable
F-16	8	Strafe	RAP/ Syllabus	Strafe pit with scoring
F-16	9	Night Strafe	RAP/ Syllabus	Strafe pit open during nighttime
F-16	10	Electronic Warfare Event	RAP	Emitters
F-16	11	Moving Target Laser Guided Bomb Attack	RAP	Laser capable and remote-controlled vehicle capable of 25 miles per hour
F-16	12	Moving Target Strafe	RAP/ Syllabus	Remote-controlled vehicle capable of 25 miles per hour
F-16	13	Large Force/Flag Exercise	RAP/ Syllabus	Scenario-based threat replication
1. F-16 Blk 25-42 RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1), (Effective 01 Apr 17).				
2. F-16CM Blk 50/52 RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1), (Effective 01 Apr17).				

Table A3.17. F-22 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
F-22	1	Aerial Gunnery	RAP[1]	Live capable and airborne target
F-22	2	Chaff	RAP	Capable
F-22	3	Electronic Warfare Event	RAP	Emitters
F-22	4	Flare	RAP	Capable
F-22	5	Large Force/Flag Exercise	RAP	Scenario-based threat replication

1. F-22A RAP Tasking Memorandum, Aviation Schedule 2017 Change 1, (AS-17 Ch 1),(Effective 01 Apr 17).

Table A3.18. F-35 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
F-35	1	Aerial Gunnery	RAP[1]	ble and airbornetarget
F-35	2	CAS in Urban Terrain	RAP/ Syllabus	Constructed urban village
F-35	3	Electronic Warfare Event	RAP/ Syllabus	Emitters
F-35	4	Full-Scale Weapons Delivery/Heavy Weight	RAP/ Syllabus	Live or inert capable
F-35	5	Aided MunitionActual	RAP/ Syllabus	Live or inert capable
F-35	6	Laser Guided Bomb Actual	RAP/ Syllabus	rt capable and lasercapable
F-35	7	Moving Target Laser Guided Bomb Attack	RAP/ Syllabus	Laser capable and remote-controlled vehicle capable of25 miles per hour
F-35	8	Moving Target Strafe	RAP/ Syllabus	Remote-controlled vehicle capable of 25 miles per hour
F-35	9	Night Strafe	RAP/ Syllabus	Strafe pit open duringnighttime
F-35	10	Target Mark	RAP	Multiple targets and strafe pit, flare capable, laser capable, or live or inert capable
F-35	11	Large Force/Flag Exercise	RAP/ Syllabus	Scenario-based threat replication
F-35	12	Flare	RAP	Capable

Table A3.19. Ground Combat Trainer Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
Ground Combat Trainer	1	Ground Combat Training	Theater Deployment	

Table A3.20. HC-130 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
HC-130	1	p Container DeliverySystem Actual	RAP	Drop zone
HC-130	2	Heavy Equipment Airdrop	RAP	Drop zone
HC-130	3	Airdrop Personnel Actual	RAP	Drop zone
HC-130	4	Tactical Airdrop Night	RAP	Drop zone – night hours
HC-130	5	sion Airdrop SystemAirdrop Event	RAP	Capable
HC-130	6	Flare	RAP	Capable
HC-130	7	Chaff	RAP	Capable
HC-130	8	Threat Event	RAP	Emitters
HC-130	9	Max Effort Landing	RAP	Capable range and capable airspace (allow tactical arrival/departure)
HC-130	10	Max Effort Night Landing	RAP	Capable range and capable airspace
HC-130	11	ested/DegradedOperations	RAP	Emitters, GPS jammer, communications jammer, data link jammer

Table A3.21. HH-60 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
HH-60	1	Chaff	RAP	Capable
HH-60	2	Flare	RAP	Capable
HH-60	3	Electronic Warfare Event	RAP	Emitters
HH-60	4	ested/DegradedOperations	RAP	Emitters, GPS jammer, communications jammer
HH-60	5	Helicopter Gunnery Event	RAP	Multiple targets and strafe pit

Table A3.22. JTAC Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
JTAC	1	al Attack Controllermission-Day	13-112V1	Day range access for participating aircraft
JTAC	2	nal Attack Controllermission-Night	13-112V1	Night range access for participating aircraft
JTAC	3	erminal Attack Controller Mission-Live Ordnance	13-112V1	Live or inert capable
JTAC	4	Control Type 1 Terminal Attack Controller Mission	13-112V1	Range access for participating aircraft
JTAC	5	Control Type 2 Terminal Attack Controller Mission	13-112V1	Range access for participating aircraft
JTAC	6	Laser Target Designation	13-112V1	Laser capable
JTAC	7	Control CAS Mission	13-112V1	Range access for participating aircraft
JTAC	8	Degraded Operations	13-112V1	Emitters, GPS jammer, communications jammer, data link jammer

Table A3.23. KC-10 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
KC-10	1	Contested/Degraded Operations	Volume 1	Emitters, GPS jammer, communications jammer, data link jammer
KC-10	2	Large Force/Flag Exercise	RAP	Scenario-based threat replication

Table A3.24. KC-46 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
KC-46	1	Laser Infrared Countermeasures/Laser	Volume 1	Simulator capable
KC-46	2	Threat Event	Volume 1	Emitters
KC-46	3	Electronic Warfare Event	RAP	Emitters
KC-46	4	Contested/Degraded Operations	Volume 1	Emitters, GPS jammer, communications jammer, data link jammer
KC-46	5	Large Force/Flag Exercise	RAP	Scenario-based threat replication

Table A3.25. KC-135 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
KC-135	1	ested/DegradedOperations	Vol 1	Emitters, GPS jammer, communications jammer, data link jammer
KC-135	2	Large Force/Flag Exercise	RAP	ario-based threatreplication

Table A3.26. MC-130 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
MC-130	1	Airdrop Heavy equipment	RAP	Drop zone day/night
MC-130	2	Airdrop High Speed	RAP	Drop zone day/night
MC-130	3	Airdrop High Altitude	RAP	Drop zone day/night
MC-130	4	Airdrop Visual	RAP	Drop zone day/night
MC-130	5	Airdrop Personnel	RAP	Drop zone day/night
MC-130	6	Airdrop Container Delivery System	RAP	Drop zone day/night
MC-130	7	Airdrop Low Cost Low Altitude Airdrop	RAP	Drop zone day/night
MC-130	8	Airdrop Joint Precision Airdrop System	RAP	Drop zone day/night
MC-130	9	Forward Area Refueling Point	RAP	Landing zone day/night/blacked out
MC-130	10	Insertion/Extraction	RAP	Landing zone day/night/blacked out
MC-130	11	Max Effort Takeoff	RAP	Landing zone day/night/blacked out
MC-130	12	Assault Landing	RAP	Landing zone day/night/blacked out
MC-130	13	Self-Contained Approach Training	RAP	Landing zone day/night/blacked out
MC-130	14	Receiver Air-to-Air Refueling	RAP	ıir refueling trackday/night
MC-130	15	Tanker Air-to-Air Refueling	RAP	Helicopter air-to-air refueling/tanker air-to-air refueling tracks day/night
MC-130	16	Air Intercept Training	RAP	Large block altitude day/night
MC-130	17	Surface Radar	RAP	Emitters
MC-130	18	Night Vision Goggle Landing/Takeoff	RAP	Blacked out landing zone
MC-130	19	Airland - Improved surface	RAP	Hard surface landing zone
MC-130	20	Airland - Unimproved surface	RAP	Dirt/clay surface landing zone
MC-130	21	Chaff	RAP	Capable

MC-130	22	Flare	RAP	Capable
MC-130	23	Event/DegradedOperations	RAP	Emitters, GPS jammer, communications jammer, data link jammer

Table A3.27. MQ-9 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
MQ-9	1	Buddy Lase	RAP	Laser capable
MQ-9	2	Hellfire	RAP	Live or inert capable and laser capable
MQ-9	3	Inertially Aided Munition	RAP	Live or inert capable
MQ-9	4	Laser Guided Bomb	RAP	Laser capable

Table A3.28. PC-12 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
PC-12	1	Short Field Procedures	RAP	Landing zone day/night
PC-12	2	Night Vision GoggleAir/Land	RAP	Landing zone AMP3/4
PC-12	3	Semi-Prepared Surface Landing/Takeoff	RAP	Dirt landing zone
PC-12	4	Low-Cost Low Altitude Airdrop	RAP	Drop zone day/night
PC-12	5	Night Vision Goggle Airdrop	RAP	Drop zone day/night
PC-12	6	Night Vision Goggle Insertion/Extraction	RAP	Blacked out landing zone

Table A3.29. RC-135 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
RC-135	1	Contingency		Scenario-based threat replication
RC-135	2	Sensitive Recon Operation		Scenario-based threat replication
RC-135	3	ppression of Enemy Air Defenses Integration		Scenario-based threat replication
RC-135	4	Contested/Degraded Operations		GPS and communications jammer
RC-135	5	Collection		Electronic emitter
RC-135	6	Annual Combat Search and Rescue Integration Scenario		Scenario-based threat replication
RC-135	7	Electronic Warfare Activity		Multiple emitters

Table A3.30. Special Tactics Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
Special Tactics	1	Parachute Insertion	Theater Deployment	Static line/high altitude low open personnel drop zone/day/night/water
Special Tactics	2	Parachute Insertion	Theater Deployment	Static line/ high altitude low open personnel drop zone/day/night
Special Tactics	3	Insertion/Extraction Operations	Theater Deployment	Rotary wing/tiltrotor drop/landing zone/day/night
Special Tactics	4	ain/Tactical Vehicle Operations	Theater Deployment	vehicle practicecourse
Special Tactics	5	Terminal Control Operations	Theater Deployment	Live or inert capable
Special Tactics	6	Demolition	Theater Deployment	Demolition capable
Special Tactics	7	Combat/Tactical Operations	Theater Deployment	Light/heavy weapons, military operation in urban terrain

Special Tactics	8	Small Unmanned Aerial System Operations	Theater Deployment	Small Unmanned Aerial System capable and multiple large open areas within restricted airspace with supporting frequency
Special Tactics	9	Assault Zone Operations	Theater Deployment	Capable
Special Tactics	10	Medical/Combat Search and Rescue/Personnel Recovery	Theater Deployment	Capable
Special Tactics	11	Degraded Operations	Theater Deployment	Emitters, GPS jammer, communications jammer, data link jammer

Table A3.31. Small Unmanned Aerial Systems Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
Small Unmanned Aerial System	1	Unmanned Aerial System Training	Theater Deployment	Multiple large open areas within restricted airspace and supporting frequency allocations

Table A3.32. T-38C (Introduction to Fighter Fundamentals) Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
T-38C	1	Aerial Gunnery[1]	Syllabus	Restricted area airspace
T-38C	2	10-degree Low Angle High	Syllabus	Class A service
T-38C	3	10-degree Low Angle High Drag Pop	Syllabus	Class A service
T-38C	4	Strafe (high or low angle)	Syllabus	Strafe pit
T-38C	5	Visual Level Delivery	Syllabus	Class A service

1. Aerial Gunnery is simulated but requires restricted airspace by AETC policy due to student pilots. While training events which only require airspace are generally outside the scope of this manual, an exception is made for this unique case.

Table A3.33. Tactical Air Control Party Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
Tactical Air Control Party	1	Ground Training (All)	Theater Deployment	Bivouac, land navigation, ground maneuver, vehicle convoy, survival, evasion, resistance and escape, and close quarters battle capable

Table A3.34. U-28 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
U-28	1	Short Field Procedures		Landing zone day/night
U-28	2	Night Vision Goggle Air/Land	RAP	Landing zone blacked out
U-28	3	Semi-Prepared Surface Landing/Takeoff	RAP	Dirt landing zone
U-28	4	Low-Cost Low Altitude Airdrop	RAP	
U-28	5	Night Vision Goggle Airdrop	RAP	Drop zone-night hours
U-28	6	Night Vision Goggle Insertion/Extraction	RAP	Blacked-out landing zone-night hours
U-28	7	Intelligence, Surveillance, and Reconnaissance Training	RAP	Airspace
U-28	8	Airland - ImprovedSurface	Theater Deployment	Hard surface helicopter landing zone
U-28	9	Airland - Unimproved Surface	Theater Deployment	Dirt/clay surface helicopter landing zone
U-28	10	Buddy Lase	RAP	Laser capable
U-28	11	Flare	Theater Deployment	Capable
U-28	12	Threat Event/Degraded Operations	Theater Deployment	Emitters, GPS jammer, communications jammer, data link jammer

U-28	13	Intelligence, Surveillance, and Reconnaissance Event with Ground Element	Theater Deployment	Restricted access roadwork with obscuration areas

Table A3.35. UH-1N and MH-139 Events.

Mission Design Series	Event Number	Event	Event Source	Range Capability
UH-1N	1	Helicopter Gunnery	RAP	Day/Night live capable
UH-1N	2	Aerial Interdiction	RAP	Day/Night, multiple targets
UH-1N	3	CAS	RAP	Day/Night, multiple targets, live, integrated air and ground fires, constructed urban village

Attachment 4

ASSIGNED RANGE USERS AND TRAINING EVENTS

A4.1. The range assignments in this attachment are intended to capture only the primary range used by the unit for each training event. The range assignments are not intended to limit units from using other, non-assigned ranges, as required or necessary to meet their training requirements.

Figure A4.1. A-10 Range Assignments.

WG	Unit	Location	1 Mission Profiles	2 Air-borne Laser Designation	3 Airstrike Control Actual Ordnance	4 Close Air Support in Urban Terrain	5 Flare	6 Full Scale Weapons Delivery/Heavy Weight	7 Laser Spot Search/Track	8 Laser Guided Bomb	9 Maverick Event	10 Night Strafe	11 Suppression of Enemy Air Defenses-Contested	12 Self-Mark Strafe (night)	13 Strafe	14 Target Mark	15 Targeting Pod Event	16 Chaff	17 EW Event Air-to-Ground	18 Moving Target Attack	19 Degraded/Denied Communications	20 Degraded/Denied Data Link	21 Degraded/Denied Global Positioning System	22 Large Force/Flag Exercise
23 WG	75 FS	Georgia	Grand Bay								Avon Park		Grand Bay						Grand Bay	Nevada Test and Training Range				
23 WG	74 FS	Georgia	Grand Bay								Avon Park		Grand Bay						Grand Bay	Nevada Test and Training Range				
51 FW	25 FS	Korea	Joint Pacific Air Range Complex																				Joint Pacific Air Range Complex	
53 WG	422 TES	Nevada	Nevada Test and Training Range																					
57 WG	548 CTS, Det 1	Louisiana	Claiborne			Claiborne					Claiborne			Claiborne										Claiborne
96 TW	40 TS	Florida	96 TW (Eglin)																					
122 FW	163 FS	Indiana	Indiana Air Range Complex										Indiana Air Range Complex											Indiana Air Range Complex
124 FW	190 FS	Idaho	Mountain Home - events listed as TBD																					
127 FW	107 FS	Michigan	Indiana Air Range Complex																					
175 WG	104 FS	Maryland	Bollen																Bollen					
355 FW	354 FS	Arizona	Barry M. Goldwater Range																				Nevada Test and Training Range	
355 FW	357 FS	Arizona	Barry M. Goldwater Range																				Nevada Test and Training Range	
442 FW	47 FS	Arizona	Barry M. Goldwater Range																					
442 FW	76 FS	Georgia	Grand Bay																					
442 FW	303 FS	Missouri	Cannon																					
USAFWS (57 WG)	66 WPS	Nevada	Nevada Test and Training Range																					

Figure A4.2. AC-130 Range Assignments.

WG	Unit	Location	1 Live Fire	2 Precision Guided Munition Employment	3 Live Fire Radar Profile (AC-130U only)	4 Air-to-Ground Gunnery	5 Chaff	6 Flare	7 Threat Event	8 Call-For-Fire	9 Close Air Support	10 Airborne Laser Designation
1 SOW	4 SOS	Florida	96 TW (Eglin)		96 TW (Eglin)							
27 SOW	16 SOS	New Mexico	Melrose		Melrose							
AFSOAWC	19 SOS	Florida	96 TW (Eglin)		96 TW (Eglin)							
AFSOAWC	551 SOS	New Mexico	Melrose		Melrose							
USAFWS	14 WPS	Florida	96 TW (Eglin)		96 TW (Eglin)							

Figure A4.3. B-1 Range Assignments.

Unit	Location	Actual Weapon Release (Inert)	Wind Corrected Munitions Dispenser Release	Actual Flare Event	Actual Chaff Event	Electronic Attack Threat Activity	Targeting Pod Operations	Contested/Degraded Operations	Large Force/Flag Exercise
		1	2	3	4	5	6	7	8
9 BS	Texas	Melrose	Utah Test and Training Range	Melrose		Snyder		Utah Test and Training Range	Nevada Test and Training Range
28 BS	Texas	Smoky Hill	Utah Test and Training Range			Snyder		Utah Test and Training Range	Nevada Test and Training Range
34 BS	South Dakota	Utah Test and Training Range				Belle Fourche		Utah Test and Training Range	Nevada Test and Training Range
37 BS	South Dakota	Nevada Test and Training Range				Smoky Hill		Utah Test and Training Range	Nevada Test and Training Range
337 TES	Texas	Melrose	Utah Test and Training Range			Snyder		Nevada Test and Training Range	
419 FLTS	California	412 TW (Edwards)							
77 WPS	Texas	Utah Test and Training Range		Nevada Test and Training Range		Snyder		Nevada Test and Training Range	

Figure A4.4. B-2 Range Assignments.

WG	Unit	Location	Air-to-Ground Missile 158 Bomb Run	B61/83 Actual Release	Guided Bomb Unit-31/28 Actual Release	Guided Bomb Unit-31/28 Actual Release	Guided Bomb Unit-52 Bomb Run	Conventional Unguided	Electronic Threat Activity	Large Force/Flag Exercise
			1	2	3	4	5	6	7	8
53 WG	72 TES	Missouri				Smoky Hill			Nevada Test and Training Range	
131 BW	110 BS	Missouri	Utah Test and Training Range		Smoky Hill		Utah Test and Training Range	Smoky Hill	Utah Test and Training Range	
509 BW	13 BS	Missouri	Utah Test and Training Range		Utah Test and Training Range				Smoky Hill	
509 BW	393 BS	Missouri	Utah Test and Training Range		Utah Test and Training Range				Utah Test and Training Range	
509 BW	394 CTS	Missouri		Smoky Hill						
USAFWS	325 WPS	Missouri	Nevada Test and Training Range		Nevada Test and Training Range				Nevada Test and Training Range	

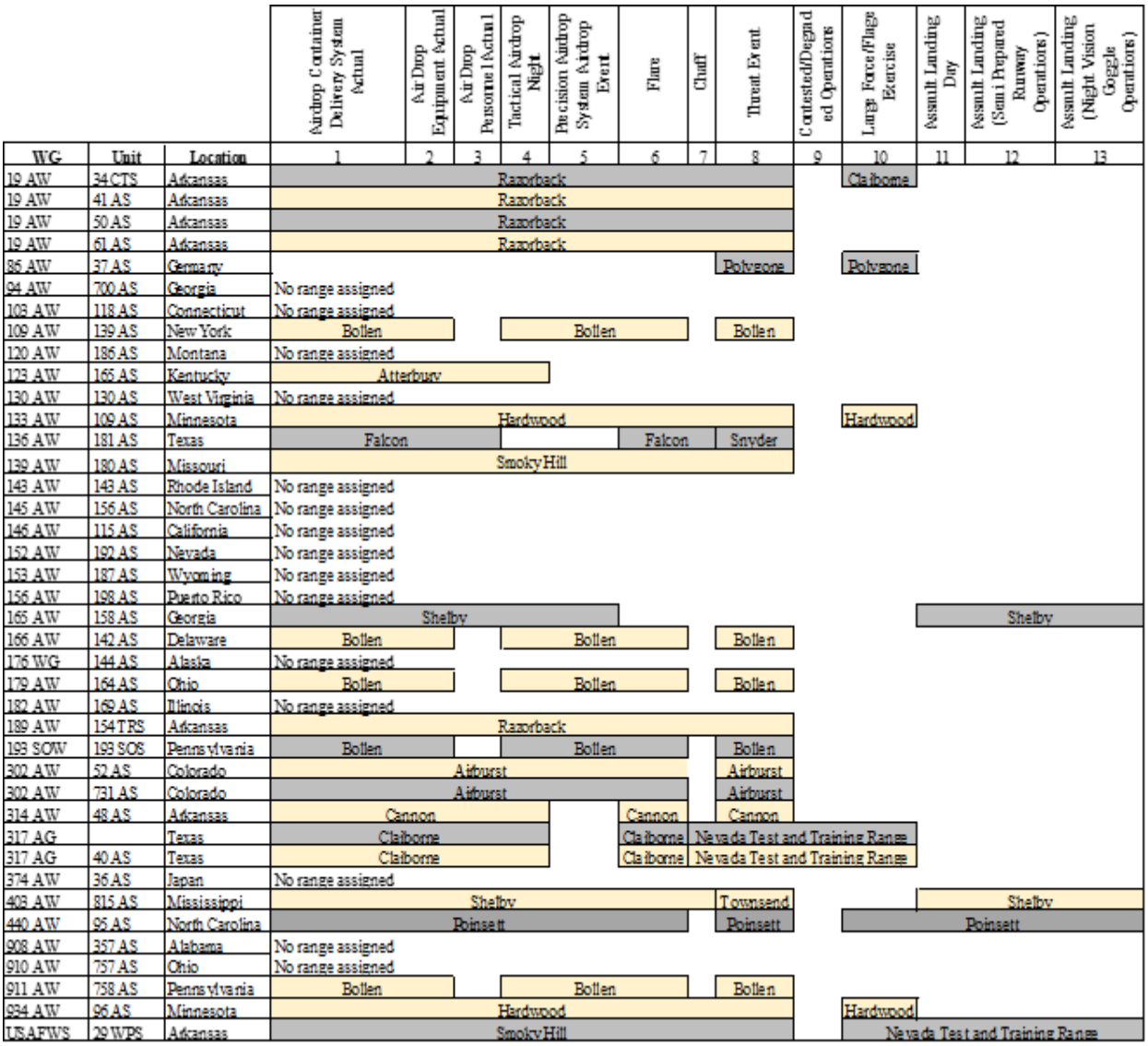
Figure A4.5. B-52 Range Assignments.

Unit	Location	Actual Joint Direct Attack Munition/Wind Corrected Munitions Dispenser Release	Actual Weapon Release	Joint Direct Attack Munition/Wind Corrected Munitions Dispenser Jettison	Laser Spot Search/Track	Laser Guided Bomb Actual Weapon Release	Chaff/Flare Exercise	Targeting Pod Employment with Internally Aided Munitions	Electronic Attack Threat Activity	Large Force/Flag Exercise
		1	2	3	4	5	6	7	8	9
11 BS	Louisiana	Utah Test and Training Range	Melrose	Utah Test and Training Range	Razorback		Utah Test and Training Range	Smoky Hill	Melrose	Nevada Test and Training Range
20 BS	Louisiana	Smoky Hill					Nevada Test and Training Range	Smoky Hill		Nevada Test and Training Range
96 BS	Louisiana	Nevada Test and Training Range			Utah Test and Training Range				Snyder	Nevada Test and Training Range
23 BS	North Dakota				Mountain Home	Smoky Hill	Mountain Home	Smoky Hill	Belle Fourche	Mountain Home
69 BS	North Dakota	Nevada Test and Training Range	Utah Test and Training Range		Belle Fourche	Grayling	Nevada Test and Training Range	Grayling	Belle Fourche	Nevada Test and Training Range
49 TES	Louisiana		Smoky Hill		Smoky Hill				Claiborne	
93 BS	Louisiana	Smoky Hill	Utah Test and Training Range	Smoky Hill	Claiborne	Smoky Hill	Nevada Test and Training Range	Smoky Hill	Snyder	Nevada Test and Training Range
343 BS	Louisiana	Utah Test and Training Range	Utah Test and Training Range	Utah Test and Training Range	Claiborne	Smoky Hill	Utah Test and Training Range	Smoky Hill		Nevada Test and Training Range
419 FLTS	California	412 TW (Edwards)								
340 WPS	Louisiana		Utah Test and Training Range		Utah Test and Training Range				Snyder	

Figure A4.6. C-17 Range Assignments.

WG	Unit	Location	1 Airdrop Container Delivery System Actual	2 Airdrop Equipment Actual	3 Airdrop Personnel Actual	4 Flare/Laser Infrared Countermeasures	5 Precision Airdrop System Airdrop Event	6 Tactical Airdrop Night	7 Threat Event	8 Electronic Warfare Event	9 Contested/Degraded Operations	10 Large Force/Flag Exercise	11 Assault Landing (Day)	12 Assault Landing (Semi Prepared Runway Operations)	13 Assault Landing (Night Vision Goggles Operations)
3 WG	517 AS	Alaska	Joint Pacific Air Range Complex												
15 WG	535 AS	Hawaii	No range assigned												
60 AW	21 ALS	California	No range assigned												
62 AW	4 AS	Washington	Joint Pacific Air Range Complex			Nevada Test and Training Range	Pacific Air Range Com		Nevada Test and Training Range						
62 AW	7 AS	Washington	Joint Pacific Air Range Complex			Nevada Test and Training Range	Pacific Air Range Com		Nevada Test and Training Range						
62 AW	8 AS	Washington	Joint Pacific Air Range Complex			Nevada Test and Training Range	Pacific Air Range Com		Nevada Test and Training Range						
97 AMW	58 AS	Oklahoma	Falcon												
105 AW	137 AS	New York	Adirondack					Adirondack					Adirondack		
145 AW	156 AS	North Carolina	No range assigned												
164 AW	155 AS	Tennessee	No range assigned												
167 AW	167 AS	West Virginia	Bollen			Bollen									
172 AW	183 AS	Mississippi	Shelby												
176 AW	249 AS	Alaska	Joint Pacific Air Range Complex									Joint Pacific Air Range Complex			
305 AMW	6 AS	New Jersey	Bollen			Bollen				Nevada Test and Training Range					
436 AW	3 AS	Delaware	Bollen			Bollen				Nevada Test and Training Range					
437 AW	14 AS	South Carolina	Avon Park					Avon Park	Nevada Test and Training Range				Avon Park	Nevada Test and Training Range	Avon Park
437 AW	15 AS	South Carolina	Avon Park					Avon Park	Nevada Test and Training Range				Avon Park	Nevada Test and Training Range	Avon Park
437 AW	16 AS	South Carolina	Avon Park					Avon Park	Nevada Test and Training Range				Avon Park	Nevada Test and Training Range	Avon Park
445 AW	89 AS	Ohio	No range assigned												
452 AMW	729 AS	California	No range assigned												
512 AW	326 AS	Delaware	Bollen			Bollen									
USAFWS	57 WPS	New Jersey	Nevada Test and Training Range			Bollen	da Test and Training R		Bollen	Nevada Test and Training Range					

Figure A4.7. C-130 Range Assignments.



WG	Unit	Location	1 Airdrop Container Delivery System Actual	2 Air Drop Equipment Actual	3 Air Drop Personnel Actual	4 Tactical Airdrop Night	5 Precision Airdrop System Airdrop Event	6 Flare	7 Chaff	8 Threat Event	9 Contested/Degraded Operations	10 Large Force/Flags Exercise	11 Assault Landing Day	12 Assault Landing (Semi Prepared Runway Operations)	13 Assault Landing (Night Vision Goggle Operations)
19 AW	34 CTS	Arkansas	Razorback									Claiborne			
19 AW	41 AS	Arkansas	Razorback												
19 AW	50 AS	Arkansas	Razorback												
19 AW	61 AS	Arkansas	Razorback												
86 AW	37 AS	Germany								Polvogne		Polvogne			
94 AW	700 AS	Georgia	No range assigned												
103 AW	118 AS	Connecticut	No range assigned												
109 AW	139 AS	New York	Bollen			Bollen				Bollen					
120 AW	186 AS	Montana	No range assigned												
123 AW	165 AS	Kentucky	Atterbury												
130 AW	130 AS	West Virginia	No range assigned												
133 AW	109 AS	Minnesota	Hardwood									Hardwood			
136 AW	181 AS	Texas	Falcon					Falcon		Snyder					
139 AW	180 AS	Missouri	Smoky Hill												
143 AW	143 AS	Rhode Island	No range assigned												
145 AW	156 AS	North Carolina	No range assigned												
146 AW	115 AS	California	No range assigned												
152 AW	192 AS	Nevada	No range assigned												
153 AW	187 AS	Wyoming	No range assigned												
156 AW	198 AS	Puerto Rico	No range assigned												
165 AW	158 AS	Georgia	Shelby										Shelby		
166 AW	142 AS	Delaware	Bollen			Bollen				Bollen					
176 WG	144 AS	Alaska	No range assigned												
179 AW	164 AS	Ohio	Bollen			Bollen				Bollen					
182 AW	169 AS	Illinois	No range assigned												
189 AW	154 TRS	Arkansas	Razorback												
193 SOW	193 SOS	Pennsylvania	Bollen			Bollen				Bollen					
302 AW	52 AS	Colorado	Airburst							Airburst					
302 AW	731 AS	Colorado	Airburst							Airburst					
314 AW	48 AS	Arkansas	Cannon					Cannon		Cannon					
317 AG		Texas	Claiborne					Claiborne		Nevada Test and Training Range					
317 AG	40 AS	Texas	Claiborne					Claiborne		Nevada Test and Training Range					
374 AW	36 AS	Japan	No range assigned												
403 AW	815 AS	Mississippi	Shelby							Townsend			Shelby		
440 AW	95 AS	North Carolina	Poinsett							Poinsett		Poinsett			
908 AW	357 AS	Alabama	No range assigned												
910 AW	757 AS	Ohio	No range assigned												
911 AW	758 AS	Pennsylvania	Bollen			Bollen				Bollen					
934 AW	96 AS	Minnesota	Hardwood									Hardwood			
USAFWS	29 WPS	Arkansas	Smoky Hill									Nevada Test and Training Range			

Figure A4.8. C-145 Range Assignments.

WG	Unit	Location	1 Low Cost Low Altitude Airdrop	2 Night Vision Goggle Airdrop	3 Night Vision Goggle Insertion/Extraction	4 Semi-Prepared Surface Landing/Takeoff	5 Military Free Fall	6 Live Fire	7 Airland - Improved Surface	8 Airland - Unimproved Surface	9 Airdrop (Personnel/Container Delivery System/Equipment)	10 Air-to-Ground Gunnery
AFSOAWC	5 SOS	Florida	96 TW (Eglin)						96 TW (Eglin)			
AFSOAWC	6 SOS	Florida	96 TW (Eglin)						96 TW (Eglin)			
919 SOW	711 SOS	Florida	96 TW (Eglin)						96 TW (Eglin)			

Figure A4.9. C-146 Range Assignments.

WG	Unit	Location	Night Vision Goggle Insertion/Extraction	Semi-Prepared Surface Landing/Takeoff
			1	2
27 SOW	524 SOS	New Mexico	Melrose	
919 SOW	859 SOS	Florida	96 TW (Eglin)	

Figure A4.10. CV-22 Range Assignments.

Unit	Location	Insertion/Extraction Training	Alternate Insertion/Extraction	Chaff	Flare	Electronic Warfare Event/Degraded Operations	Hoist Training	Airland - Improved Surface	Airland - Unimproved Surface	Air-to-Ground gunnery	Military Free Fall
		1	2	3	4	5	6	7	8	9	10
8 SOS	Florida	96 TW (Eglin)									
20 SOS	New Mexico	Melrose									
71 SOS	New Mexico	Holloman									
67 SOS	England	No range assigned									
7 SOS	England	No range assigned									
1 SOS	Japan	No range assigned									

Figure A4.11. E-3 Range Assignments.

WG	Unit	Location	Electronic Warfare Event	Large Force/Flag Exercise
			1	2
3 WG	962 AACS	Alaska	Joint Pacific Air Range Complex	
18 WG	961 AACS	Japan	Joint Pacific Air Range Complex	
552 ACW	960 AACS	Oklahoma	Poinsett	Nevada Test and Training Range
552 ACW	963 AACS	Oklahoma	Poinsett	Nevada Test and Training Range
552 ACW	964 AACS	Oklahoma	Poinsett	Nevada Test and Training Range
552 ACW	965 AACS	Oklahoma	Poinsett	Nevada Test and Training Range
552 ACW	966 AACS	Oklahoma	Poinsett	

Figure A4.12. E-8 Range Assignments.

WG	Unit	Location	Large Force/Flag Exercise	Moving Target	Electronic Warfare Event
			1	2	3
116 ACW	128 AC2S	Georgia	Nevada Test and Training Range		
461 ACW	12 AC2S	Georgia	Nevada Test and Training Range		
461 ACW	16 AC2S	Georgia	Nevada Test and Training Range		

Figure A4.13. EC-130 / EC-37B Range Assignments.

WG	Unit	Location	Smokey Surface-to-Air Missile Event	Large Force/Flag Exercise	Threat Emitters
			1	2	3
55 WG	41 ECS	Arizona	Barry M. Goldwater Range	Nevada Test and Training Range	Playas
55 WG	43 ECS	Arizona	Barry M. Goldwater Range	Nevada Test and Training Range	Playas
193 SOW	193 SOS	Pennsylvania	Bollen	Nevada Test and Training Range	

Figure A4.14. F-15C/D Range Assignments.

WG	Unit	Location	1 Aerial Gunnery	2 Chaff	3 Flare	4 Electronic Warfare Event	5 Large Force/Flag Exercise	6 High Angle Strafe
18 WG	44 FS	Japan	Joint Pacific Air Range Complex					
18 WG	67 FS	Japan	Joint Pacific Air Range Complex					
48 FW	493 FS	England	No range assigned					
53 WG	85 TES	Florida	No range assigned					
53 WG	422 TES	Nevada	No range assigned					
96 TW	40 TS	Florida	No range assigned					
104 FW	131 FS	Massachusetts		Adirondack				Adirondack
125 FW	159 FS	Florida	No range assigned					
142 FW	123 FS	Oregon	No range assigned					
144 FW	194 FS	California	No range assigned					
159 FW	122 FS	Lousiana		Claiborne				
173 FW	114 FS	Oregon	No range assigned					
USAFWS	433 WPS	Nevada	No range assigned					

Figure A4.15. F-15E Range Assignments.

Unit	Location	1 Aerial Gunnery	2 Chaff	3 Flare	4 Electronic Warfare Event	5 Inertially Aided Munition Actual	6 Moving Target Laser Guided Bomb/Strafe	7 Strafe	8 Precision Guided Munition Actual	9 Large Force/Flag Exercise
333 FS	North Carolina			Dare County						
334 FS	North Carolina			Dare County						Nevada Test and Training Range
335 FS	North Carolina			Dare County						Nevada Test and Training Range
336 FS	North Carolina			Dare County						Nevada Test and Training Range
492 FS	England	No range assigned								
494 FS	England	No range assigned								
85 TES	Florida		96 TW (Eglin)					6 TW (Eglin		Nevada Test and Training Range
422 TES	Nevada		Nevada Test and Training Range							
40 TS	Florida		96 TW (Eglin)							
428 FS*	Idaho			Mountain Home						
389 FS	Idaho			Mountain Home						Nevada Test and Training Range
391 FS	Idaho			Mountain Home						Nevada Test and Training Range
307 FS	North Carolina			Dare County						
17 WPS	Nevada		Nevada Test and Training Range							

*F-15SG – Singapore

Figure A4.16. F-16 Range Assignments.

WG	Unit	Location	1 Aerial Gunnery	2 Chaff	3 Close Air Support in Urban Terrain	4 Full-Scale Weapons Delivery/ Heavy Weight	5 Flare	6 Inertially Aided Munition Actual	7 Laser Guided Bomb Actual	8 Strafe	9 Night Strafe	10 Electronic Warfare Event	11 Moving Target Laser Guided Bomb Attack	12 Moving Target Strafe	13 Large Force/Flag Exercise
8 FW	35 FS	Korea		Joint Pacific Air Range Complex										Joint Pacific Air Range Complex	
8 FW	80 FS	Korea		Joint Pacific Air Range Complex										Joint Pacific Air Range Complex	
20 FW	55 FS	South Carolina			Poinsett										Nevada Test and Training Range
20 FW	77 FS	South Carolina			Poinsett										Nevada Test and Training Range
20 FW	79 FS	South Carolina			Poinsett										
31 FW	510 FS	Italy	No range assigned												
31 FW	555 FS	Italy	No range assigned												
35 FW	13 FS	Japan		Draughon								Draughon		Draughon	
35 FW	14 FS	Japan		Draughon								Draughon		Draughon	
49 WG	8 FS	New Mexico		Holloman											Holloman
49 WG	311 FS	New Mexico		Holloman											Holloman
49 WG	314 FS	New Mexico		Holloman											Holloman
51 FW	36 FS	Korea		Joint Pacific Air Range Complex										Joint Pacific Air Range Complex	
52 FW	480 FS	Germany										Polygone			Polygone
53 FW	85 TES	Florida		96 TW (Eglin)			96 TW (Eglin)								
53 FW	422 TES	Nevada		Nevada Test and Training Range											
56 FW	21 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
56 FW	309 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
56 FW	310 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
56 FW	425 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
57 WG	64 AGS	Nevada		Nevada Test and Training Range											
57 WG	548 CTS, Det 1	Louisiana								Claiborne		Claiborne			Claiborne
96 TW	40 TS	Florida		96 TW (Eglin)											
113 FW	121 FS	Maryland			Warren Grove									Warren Grove	
114 FW	175 FS	South Dakota		Smokey Hill											
115 FW	176 FS	Wisconsin		Hardwood											
122 FW	163 FS	Indiana	No range assigned												
138 FW	125 FS	Oklahoma		Smokey Hill											
140 FW	120 FS	Colorado	Airburst		Airburst										
148 FW	179 FS	Minnesota		Hardwood											Hardwood
149 FW	182 FS	Texas		McMullen								McMullen			
158 FW	134 FS	Vermont		Adirondack											
162 WG	148 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
162 WG	152 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
162 WG	195 FS	Arizona		Barry M. Goldwater Range										Barry M. Goldwater Range	
162 WG	AATC	Arizona		Barry M. Goldwater Range											
169 FW	157 FS	South Carolina		Townsend										Townsend	
177 FW	119 FS	New Jersey		Adirondack		Warren Grove								Warren Grove	
180 FW	112 FS	Ohio		Indiana Air Range Complex											
187 FW	100 FS	Alabama			Shelby										
301 FW	457 FS	Texas		Falcon											
354 FW	18 AGS	Alaska		Joint Pacific Air Range Complex			Joint Pacific Air Range Complex					Joint Pacific Air Range Complex			
412 FW	416 FLTS	California		412 TW (Edwards)											
442 FW	303 FS	Missouri	No range assigned												
482 FW	95 FS	Florida		Avon Park											
USAFWS	16 WPS	Nevada		Nevada Test and Training Range											

Figure A4.17. F-22 Range Assignments.

WG	Unit	Location	1 Aerial Gunnery	2 Chaff	3 Electronic Warfare Event	4 Flare	5 Large Force/Flag Exercise
1 FW	27 FS	Virgina					Nevada Test and Training Range
1FW	94 FS	Virgina					Nevada Test and Training Range
3 WG	90 FS	Alaska	Joint Pacific Air Range Complex				
3 WG	525 FS	Alaska	Joint Pacific Air Range Complex				
53 WG	422 TES	Nevada			Nevada Test and Training Range		
154 WG	199 FS	Hawaii	No range assigned				
325 FW	43 FS	Florida		96 TW (Eglin)			Nevada Test and Training Range
325 FW	95 FS	Florida		96 TW (Eglin)			Nevada Test and Training Range
412 TW	411 FTS	California	No range assigned				

Figure A4.18. F-35 Range Assignments.

WG	Unit	Base	Location	Aerial Gunnery 1	Close Air Support 2	Electronic Warfare Event 3	Full Scale Weapons Event 4	Inertially Aided Munition 5	Laser Guided Bomb (LGB) 6	Moving Target LGB Attack 7	Moving Target Strafe 8	Night Strafe 9	Large Force/FLAG Exercise 10	Flare 11
33 FW	58 FS	Eglin AFB	FL		96 TW (Eglin)									
	60 FS		FL		96 TW (Eglin)									
56 FW	61 FS	Luke AFB	AZ		Barry M. Goldwater Range (BMGR)									
	62 FS		AZ		BMGR									
	63 FS		AZ		BMGR									
	308 FS		AZ		BMGR									
53 WG	59 TES	Nellis AFB	NV		Nevada Test and Training Range (NTTR)									
57 WG	USAFWS		NV		NTTR									
	422 TES		NV		NTTR									
	65 AGRS		NV		NTTR									
412 WG	416 FLTS	Edwards AFB	CA		412 TW (Edwards)									
158 FW	134 FS	Burlington ANGB	VT		Adirondack									
388 FW	4 FS	Hill AFB	UT		Utah Test and Training Range (UTTR)								NTTR	
	34 FS		UT		UTTR								NTTR	
	421 FS		UT		UTTR								NTTR	
354 FW	355 FS	Eielson AFB	AK		JPARC									
	356 FS		AK		JPARC									
48 FW	495 FS	RAF Lakenheath	UK		No Range Assigned									
	493 FS		UK (FY23)		No Range Assigned									
115 FW	176 FS	Truax Field ANGB	WI (FY23)		Hardwood									
187 FW	100 FS	Dannelly Field	AL (FY23)		Shelby	Eglin	Shelby						NTTR	
301 FW	457 FS	NAS JRB Ft Worth	TX (FY24)		Falcon	Snyder	Falcon						NTTR	
325 FW	95 FS	Tyndall AFB	FL (FY23)		Eglin								NTTR	
125 FW	159 FS	Jacksonville ANGB	FL (FY24)		Avon Park	Eglin	Avon Park						NTTR	

Figure A4.19. Ground Combat Trainer Range Assignments.

WG	Unit	Location	Ground Combat Training 1
AFSOAWC	371 SOCTS	Florida	96 TW (Eglin)

Figure A4.20. HC-130 Range Assignments.

WG	Unit	Location	Airdrop Container Delivery System Actual 1	Heavy Equipment Airdrop 2	Airdrop Personnel Actual 3	Tactical Airdrop Night 4	Precision Airdrop System Airdrop Event 5	Flare 6	Chaff 7	Threat Event 8	Max Effort Landing 9	Max Effort Night Landing 10	Contested/Degraded Operations 11
23 WG	71 RQS	Georgia	Avon Park					Grand Bay	Avon Park	Grand Bay			Nevada Test and Training Range
23 WG	79 RQS	Arizona	Barry M. Goldwater Range										Nevada Test and Training Range
53 WG	53 TES	Arizona	Barry M. Goldwater Range										
58 SOW	550 SOS	New Mexico	Holloman										Melrose
106 RQW	102 RQS	New York	Bollen			Bollen				Bollen			
176 WG	210 RQS	Alaska	Joint Pacific Air Range Complex										
920 RW	39 RQS	Florida	No range assigned										

Figure A4.21. HH-60 Range Assignments.

WG	Unit	Location	Chaff	Flare	Electronic Warfare Event	Contested/Degraded Operations	Helicopter Gunnery Event
			1	2	3	4	5
18 WG	33 RQS	Japan	No range assigned				
23 WG	41 RQS	Georgia	Avon Park	Grand Bay		Nevada Test and Training Range	
23 WG	55 RQS	Arizona	Barry M. Goldwater Range			Nevada Test and Training Range	
23 WG	66 RQS	Nevada	Nevada Test and Training Range				
48 FW	56 RQS	England	No range assigned				
53 WG	88 TES	Nevada	Nevada Test and Training Range				
57 Wg	548 CTS, Det 1	Lousiana					
58 SOW	512 RQS	New Mexic	Holloman				
106 RW	101 RQS	New York	Adirondack				
129 RW	129 RQS	California	No range assigned				
176 WG	210 RQS	Alaska	No range assigned				
920 RW	301 RQS	Florida	No range assigned				
920 RW	305 RQS	Arizona	Barry M. Goldwater Range				
USAFWS	34 WPS	Nevada	No range assigned				

Figure A4.22. JTAC Range Assignments.

WG	Unit	Location	1 Terminal Attack Controller mission-Day	2 Terminal Attack Controller mission-Night	3 Terminal Attack Controller Mission Live Ordnance	4 Control Type 1 Terminal Attack Controller Mission	5 Control Type 2 Terminal Attack Controller Mission	6 Laser Target Designation	7 Control Close Air Support Mission	8 Degraded Operations
1 ASOG	5 ASOS	Washington	Shelby							
24 SOW	21 STS	North Carolina	Dare County							
24 SOW	22 STS	Washington	Mountain Home							
24 SOW	24 STS	North Carolina	Avon Park							
24 SOW	23 STS	Florida	Avon Park							
93 AGOW	7 ASOS	Texas	Holloman							
93 AGOW	9 ASOS	Texas	Claiborne				Claiborne			
93 AGOW	10 ASOS	Kansas	Smoky Hill							
93 AGOW	11 ASOS	Texas	Falcon							
93 AGOW	13 ASOS	Colorado	Airburst							
93 AGOW	14 ASOS	North Carolina	Dare County							
93 AGOW	15 ASOS	Georgia	Dare County							
93 AGOW	19 ASOS	Kentucky	Smoky Hill							
93 AGOW	20 ASOS	New York	Adirondack							
93 AGOW	21 ASOS	Texas	Claiborne				Claiborne			
93 AGOW	682 ASOS	South Carolina	Poinsett							
93 AGOW	712 ASOS	Texas	Claiborne				Claiborne			
124 FW	124 ASOS	Idaho	Mountain Home							
137 SOW	146 ASOS	Oklahoma	Falcon							
142 FW	125 STS	Oregon	Mountain Home							
145 AW	118 ASOS	North Carolina	Bollen							
159 FW	122 ASOS	Lousiana	Claiborne				Claiborne			
165 AW	165 ASOS	Georgia	Bollen							
174 ATKW	274 ASOS	New York	Adirondack							
177 FW	227 ASOS	New Jersey	Warren Grove							
184 IW	284 ASOS	Kansas	Smoky Hill							
186 ARW	238 ASOS	Mississippi	Shelby							
193 SOW	148 ASOS	Pennsylvania	Bollen							
354 FW	3 ASOS	Alaska	Joint Pacific Air Range Complex							
354 FW	25 ASOS	Hawaii	Bollen							
USAFWS	66 WPS	Nevada	Nevada Test and Training Range							

Figure A4.23. KC-10 Range Assignments.

WG	Unit	Location	1 Contested/Degraded Operations	2 Large Force/Flag Exercise
60 AMW	6 ARS	California	No range assigned	
60 AMW	9 ARS	California	No range assigned	
305 AMW	2 ARS	New Jersey	No range assigned	
305 AMW	32 ARS	New Jersey	No range assigned	

Figure A4.24. KC-46 Range Assignments.

WG	Unit	Location	Laser Infrared Countermeasures/ Laser 1	Threat Event 2	Electronic Warfare Event 3	Contested/Degraded Operations 4	Large Force/Flag Exercise 5
22 ARW	350 ARS	Kansas	No range assigned				
22 ARW	384 ARS	Kansas	No range assigned				
97 AMW	54 ARS	Oklahoma	No range assigned				
157 ARW	133 ARS	New Hampshire	No range assigned				

Figure A4.25. KC-135 Range Assignments.

WG	Unit	Location	Contested/Degraded Operations 1	Large Force/Flag Exercise 2
6 AMW	91 ARS	Florida	No range assigned	
18 WG	909 ARS	Japan	No range assigned	
22 ARW	344 ARS	Kansas	No range assigned	
22 ARW	349 ARS	Kansas	No range assigned	
22 ARW	350 ARS	Kansas	No range assigned	
22 ARW	384 ARS	Kansas	No range assigned	
43 ARW	72 ARS	Indiana	No range assigned	
43 ARW	74 ARS	Indiana	No range assigned	
92 ARW	92 ARS	Washington	No range assigned	
92 ARW	93 ARS	Washington	No range assigned	
97 AMW	54 ARS	Oklahoma	No range assigned	
100 ARW	351 ARS	England	No range assigned	
101 ARW	132 ARS	Maine	No range assigned	
108 ARW	141 ARS	New Jersey	No range assigned	
117 ARW	106 ARS	Alabama	No range assigned	
121 ARW	145 ARS	Ohio	No range assigned	
121 ARW	166 ARS	Ohio	No range assigned	
126 ARW	108 ARS	Illinois	No range assigned	
127 ARW	171 ARS	Michigan	No range assigned	
128 ARW	126 ARS	Wisconsin	No range assigned	
134 ARW	151 ARS	Tennessee	No range assigned	
151 ARW	191 ARS	Utah	No range assigned	
154 ARW	203 ARS	Hawaii	No range assigned	
155 ARW	173 ARS	Nebraska	No range assigned	
157 ARW	133 ARS	New Hampshire	No range assigned	
161 ARW	197 ARS	Arizona	Barry M. Goldwater Range	
168 ARW	168 ARS	Alaska	Joint Pacific Air Range Complex	
171 ARW	146 ARS	Pennsylvania	No range assigned	
171 ARW	147 ARS	Pennsylvania	No range assigned	
185 ARW	174 ARS	Iowa	No range assigned	
186 ARW	153 ARS	Mississippi	No range assigned	
190 ARW	117 ARS	Kansas	No range assigned	
412 TW	418 FLTS	California	No range assigned	
452 AMW	336 ARS	California	No range assigned	
459 ARW	756 ARS	Maryland	No range assigned	
507 ARW	465 ARS	Oklahoma	No range assigned	
914 AW	328 AS	New York	No range assigned	
916 ARW	77 ARS	North Carolina	No range assigned	
940 ARW	314 ARS	California	No range assigned	
USAFWS	509 WPS	Washington	Nevada Test and Training Range	

Figure A4.26. MC-130 Range Assignments.

Location	1 Airdrop Heavy equipment	2 Airdrop High Speed	3 Airdrop High Altitude	4 Airdrop Visual	5 Airdrop Personnel	6 Airdrop Container Delivery System	7 Airdrop Low Cost Low Altitude Airdrop	8 Airdrop Joint Precision Airdrop System	9 Forward Area Refueling Point	10 Insertion/Extraction	11 Max Effort Takeoff	12 Assault Landing	13 Self-Contained Approach Training	14 Receiver Air-to-Air Refueling	15 Tanker Air-to-Air Refueling	16 Air Intercept Training	17 Surface Radar	18 Night Vision Goggle Landing/Takeoff	19 Airland - Improved surface	20 Airland - Unimproved surface	21 Chaff	22 Flare	23 Threat Event/Degraded Operations
Florida	96 TW (Eglin)																						
New Mexico	Melrose																						
New Mexico	Holloman								Melrose														
California	No range assigned																						
England	No range assigned																						
Japan	No range assigned																						
Japan	No range assigned																						
Florida	96 TW (Eglin)																						
Florida	96 TW (Eglin)																						

Figure A4.27. MQ-9 Range Assignments.

			Buddy Lase	Hellfire	Inertially Aided Munition	Laser Guided Bomb
WG	Unit	Location	1	2	3	4
27 SOW	3 SOS	NM	Melrose	Holloman/WSMR		
27 SOW	12 SOS	NM	Melrose	Holloman/WSMR		
27 SOW	33 SOS	NM	Melrose	Holloman/WSMR		
27 SOW	65 ATKS	FL	Melrose	Holloman/WSMR		
49 WG	6 ATKS	NM	Holloman/WSMR			
49 WG	9 ATKS	NM	Holloman/WSMR			
49 WG	29 ATKS	NM	Holloman/WSMR			
49 WG	491 ATKS	NY	Misty, Adirondack			
49 WG	492 ATKS	CA	China Lake, Point Magu, San Clemente			
53 WG	556 TES	NV	NTTR and UTTR			
57 WG	26 WPS	NV	NTTR and UTTR			
107 ATKW	136 ATKS	NY	No Range Assigned			
110 ATKW	110 ATKS	MI	No Range Assigned			
111 ATKW	103 ATKS	PA	No Range Assigned			
118 WG	105 ATKS	TN	No Range Assigned			
119 WG	178 ATKS	ND	Hardwood, Smoky Hill, Grayling			
132 WG	124 ATKS	IA	No Range Assigned			
147 WG	111 ATKS	TX	Gulf of Mexico Range Complex			
162 WG	214 ATKS	AZ	Goldwater, Playas, Melrose			
163 ATKW	184 ATKS	CA	China Lake, Point Magu, San Clemente			
163 ATKW	196 ATKS	CA	China Lake, Point Magu, San Clemente			
174 ATKW	108 ATKS	NY	Misty, Adirondack			
174 ATKW	138 ATKS	NY	Misty, Adirondack			
188 WG	184 ATKS	AR	No Range Assigned			
432 WG	11 ATKS	NV	Nevada Test and Training Range			
432 WG	15 ATKS	NV	Nevada Test and Training Range			
432 WG	17 ATKS	NV	Nevada Test and Training Range			
432 WG	18 ATKS	NV	Nevada Test and Training Range			
432 WG	20 ATKS	MO	Nevada Test and Training Range			
432 WG	22 ATKS	NV	Nevada Test and Training Range			
432 WG	42 ATKS	NV	Nevada Test and Training Range			
432 WG	50 ATKS	SC	Nevada Test and Training Range			
432 WG	89 ATKS	SD	Nevada Test and Training Range			
432 WG	482 ATKS	SC	Nevada Test and Training Range			
432 WG	489 ATKS	NV	Nevada Test and Training Range			
432 WG	867 ATKS	NV	Nevada Test and Training Range			

Figure A4.28. PC-12 Range Assignments.

WG	Unit	Location	Shortfield Procedures	Night Vision Goggle Air/Land	Semi-Prepared Surface Landing/Takeoff	Low Cost Low Altitude Airdrop	Night Vision Goggle Airdrop	Night Vision Goggle Insertion/Extraction
			1	2	3	4	5	6
1 SOW	34 SOS	Florida	96 TW (Eglin)					
1 SOW	319 SOS	Florida	96 TW (Eglin)					
AFSOAWC	19 SOS	Florida	96 TW (Eglin)					

Figure A4.29. RC-135 Range Assignments.

WG	Unit	Location	Contingency	Sensitive Recon Operation	Suppression of Enemy Air Defenses Integration	Contested/Degraded Operations	Collection	Annual Combat Search and Rescue Integration Scenario	Electronic Warfare Activity
			1	2	3	4	5	6	7
55 WG	38 RS	Nebraska	Nevada Test and Training Range				Belle Fourche	Nevada Test and Training Range	Belle Fourche
55 WG	45 RS	Nebraska	Nevada Test and Training Range				Belle Fourche	Nevada Test and Training Range	Belle Fourche
55 WG	338 CTS	Nebraska	Nevada Test and Training Range				Belle Fourche	Nevada Test and Training Range	Belle Fourche
55 WG	343 RS	Nebraska	Nevada Test and Training Range				Belle Fourche	Nevada Test and Training Range	Belle Fourche

Figure A4.30. Small Unmanned Aerial System Range Assignments.

WG	Unit	Location	Unmanned Aerial System Training
			1
AFSOAWC	Det 1, 371 SOCTS	Florida	96 TW (Eglin)

Figure A4.31. Special Tactics Range Assignments.

WG	Unit	Location	1 Parachute Insertion	2 Parachute Insertion	3 Alternate Insertion/Extraction Operations	4 All Terrain/Tactical Vehicle Operations	5 Terminal Control Operations	6 Demolition	7 Combat/Tactical Operations	8 Small Unmanned Aerial System Operations	9 Assault Zone Operations	10 Medical/Combat Search and Rescue/Personnel Recovery	11 Degraded Operations
23 WG	58 RQS	Nevada	Nevada Test and Training Range										
24 SOW	23 STS	Florida	96 TW (Eglin)										
24 SOW	26 STS	New Mexico	Melrose										
24 SOW	720 STG	Florida	96 TW (Eglin)										
24 SOW	STTS	Florida	96 TW (Eglin)										
24 SOW	21 STS	North Carolina	Airburst										
123 AW	123 STS	Kentucky	No range assigned										

Figure A4.32. T-38C Range Assignments.

WG	Unit	Location	1 Aerial Gunnery	2 10-degree Low Angle High Drag	3 10-degree Low Angle High Drag Pop	4 Strafe (high or low angle)	5 Visual Level Delivery
12 FTW	435 FTS	Texas	McMullen				
12 FTW	560 FTS	Texas	No range assigned				
14 FTW	49 FTS	Mississippi	No range assigned				
14 FTW	50 FTS	Mississippi	No range assigned				
47 FTW	87 FTS	Texas	No range assigned				
71 FTW	25 FTS	Oklahoma	No range assigned				
80 FTW	88 FTS	Texas	Falcon				
80 FTW	90 FTS	Texas	No range assigned				
96 TW	586 FLTS	New Mexico	No range assigned				
412 TW	418 FLTS	California	No range assigned				

Figure A4.33. Tactical Air Control Party Range Assignments.

WG	Unit	Location	1 Ground Training (All)
AETC	Det 3, 342 TRS	Florida	96 TW (Eglin)

Figure A4.34. U-28 Range Assignments.

WG	Unit	Location	1 Shortfield procedures	2 Night Vision Goggle Air Land	3 Semi-Prepared Surface Landing/Takeoff	4 Low Cost Low Altitude Airdrop	5 Night Vision Goggle Airdrop	6 Night Vision Goggle Insertion/Extraction	7 Intelligence, Surveillance, and Reconnaissance Training	8 Airland - Improved Surface	9 Airland - Unimproved Surface	10 Buddy Lase	11 Flare	12 Threat Event/Degraded Operations	13 Intelligence, Surveillance, and Reconnaissance Event with Ground Element
1SOW	34 SOS	Florida	96 TW (Eglin)												
1SOW	319 SOS	Florida	96 TW (Eglin)												
27SOW	318 SOS	New Mexico	Melrose												
AFSOAWC	5 SOS	Florida	96 TW (Eglin)												
AFSOAWC	19 SOS	Florida	96 TW (Eglin)												
USAFWS	14 WPS	Florida	96 TW (Eglin)												

Figure A4.35. UH-1N and MH-139 Range Assignments.

WG	Unit	Location	1 Helicopter Gunnery	2 Aerial Interdiction	3 Close Air Support
11 WG	1 HS	Maryland	No range assigned		
58 SOW	512 RQS	New Mexico	Holloman		
582 HG	37 HS	Wyoming	No range assigned		
582 HG	54 HS	North Dakota	No range assigned		
96 TW	413 FLTS	Florida	No range assigned		
582 HG	40 HS	Montana	No range assigned		
374 AW	459 AS	Japan	No range assigned		
SURVIVAL SCHOOL	36 RS	Washington	No range assigned		

Attachment 5

RANGE PERSONNEL TRAINING

A5.1. General Training. All personnel assigned to the range who perform some, or all, of their duties on the range will be trained in or demonstrate adequate knowledge of the following subjects:

A5.1.1. Local range operating procedures. **(T-1)**

A5.1.2. Maintenance procedures and issues. **(T-1)**

A5.1.3. Range stewardship (environmental awareness and protection). **(T-1)**

A5.1.4. Hazardous materials/hazardous waste and local environmental procedures. **(T-1)**

A5.1.5. Poisonous or dangerous fauna and flora. **(T-1)**

A5.1.6. Local weather hazards. **(T-1)**

A5.1.7. Range fire protection/fire suppression procedures. **(T-1)**

A5.1.8. Aircraft crash procedures. **(T-1)**

A5.1.9. Local safety, emergency, and contingency procedures. **(T-1)**

A5.1.10. Basic first aid procedures. **(T-1)**

A5.1.11. Explosive and other ordnance hazards (EOD safety briefing). **(T-1)**

A5.1.12. Range access control and security procedures. **(T-1)**

A5.1.13. Laser safety (only for ranges that have been certified for laser operations). **(T-1)**

A5.2. Range Operating Authority Training. As a minimum, the range operating authority will be trained in or demonstrate adequate knowledge of the following publications or subjects:

A5.2.1. National Environmental Policy Act. **(T-1)**

A5.2.2. Risk communication. **(T-1)**

A5.2.3. Public affairs. **(T-1)**

A5.2.4. WDZ and SDZ Program. **(T-1)**

A5.2.5. EOD briefing on the proper handling of training munitions. **(T-1)**

A5.2.6. Overall range safety. **(T-1)**

A5.2.7. AFMAN 13-212V1, as supplemented.

A5.2.8. Intermediate command instructions and manuals applicable to range "operations." **(T-1)**

A5.2.9. AFMAN 11-202V3, *Flight Operations*. **(T-1)**

A5.2.10. AFI 11-214. **(T-1)**

A5.3. Range Operations Officer Training. As a minimum, the range operations officer will be trained in or demonstrate adequate knowledge of the following publications or subjects (test sites not conducting air operations are exempt for the items indicated by *):

A5.3.1. Wing scheduling. **(T-1)**

A5.3.2. Defense Acquisition University's Contracting Officer's Representative Course 222 (not applicable for the ANG). **(T-1)**

A5.3.3. National Environmental Policy Act. **(T-1)**

A5.3.4. Risk communication. **(T-1)**

A5.3.5. AFMAN 11-202V3, *Flight Operations*. * **(T-1)**

A5.3.6. AFI 11-214. * **(T-1)**

A5.3.7. Aircraft specific Air Force 11-2MDS operating procedures. * **(T-1)**

A5.3.8. Unimproved landing zone and drop zone operations, as applicable. * **(T-1)**

A5.3.9. Local airspace configuration. **(T-1)**

A5.3.10. AFMAN 13-212V1, as supplemented. **(T-1)**

A5.3.11. Approved range construction and maintenance methods. **(T-1)**

A5.3.12. Intermediate command instructions and manuals applicable to range "operations". **(T-1)**

A5.3.13. Range control officer authority and responsibilities. **(T-1)**

A5.3.14. Range utilization report/range record keeping. **(T-1)**

A5.3.15. EOD briefing on the proper handling of training munitions. **(T-1)**

A5.3.16. Unit Safety Officer or designated Laser Safety Officer training. **(T-1)**

A5.3.17. WDZ and SDZ Program. **(T-1)**

A5.3.18. Proper use and handling of applicable ground launched visual threats. **(T-1)**

A5.4. Range Control Officer / Test Area Control Officer Training. As a minimum, the range control officer/test area control officer will be trained in or demonstrate adequate knowledge of the following publications or subjects (test sites not conducting air operations are exempt for the items indicated by *):

A5.4.1. Wing scheduling. **(T-1)**

A5.4.2. Public affairs. **(T-1)**

A5.4.3. AFMAN 11-202V3, *Flight Operations*. *

A5.4.4. AFI 11-214. * **(T-1)**

A5.4.5. Aircraft specific Air Force 11-2MDS operating procedures. * **(T-1)**

A5.4.6. Unimproved landing zone and drop zone operations, as applicable. * **(T-1)**

A5.4.7. Local airspace configuration. **(T-1)**

A5.4.8. AFMAN 13-212V1, as supplemented. **(T-1)**

A5.4.9. Intermediate command instructions and manuals applicable to range "operations." **(T-1)**

A5.4.10. Range Control Officer/Test Area Control Officer authority and responsibilities. **(T-1)**

A5.4.11. Day/night aircraft ordnance delivery patterns for all aircraft using the range. * **(T-1)**

A5.4.12. Obtain and interpret weather observations. **(T-1)**

A5.4.13. Required weather minimums for each event. **(T-1)**

A5.4.14. Use of the Avian Hazard Advisory System (Avian Hazard Advisory System, **http://www.usahas.com**).* **(T-1)**

A5.4.15. Foul criteria. **(T-1)**

A5.4.16. Communications procedures. **(T-1)**

A5.4.17. Capabilities and limitations of range facilities. **(T-1)**

A5.4.18. Hazard areas, pattern safety, WDZ, SDZ, and overall range safety. **(T-1)**

A5.4.19. Range utilization report/range record keeping. **(T-1)**

A5.4.20. EOD briefing on the proper handling of training munitions. **(T-1)**

A5.4.21. Night operations. **(T-1)**

A5.4.22. NVD training (if applicable) to include human visual system, physiological issues, NVD adjustment, care, use and limitations, disorientation, and aircrew limitations as a minimum. **(T-1)**

A5.4.23. Proper use and handling of applicable ground launched visual threats. **(T-1)**

A5.4.24. On-Range Training. A qualified range control officer/range operations officer will supervise on-range range control officer training. **(T-1)** A qualified Test Area Control Officer/range control officer/range operations officer will supervise test site Test Area Control Officer training. **(T-1)** The range operating authority will develop a checklist to ensure complete and professional training. **(T-1)** Emphasize the use of sound judgment and common sense while controlling both aircraft and personnel during range operations. The on- range training should include the following items as a minimum:

A5.4.24.1. Range hazard areas. **(T-1)**

A5.4.24.2. Inspection of strafe impact area impact areas. * **(T-1)**

A5.4.24.3. Range fire protection/fire suppression procedures. **(T-1)**

A5.4.24.4. Aircraft crash procedures. * **(T-1)**

A5.4.24.5. First aid and evacuation of injured personnel. **(T-1)**

A5.4.24.6. Traffic conflict with other ranges in the area. **(T-1)**

A5.4.24.7. Range pattern spacing. * **(T-1)**

A5.4.24.8. Minimum altitude measuring devices. * **(T-1)**

A5.4.24.9. Cease fire distance estimation for low angle strafe. * **(T-1)**

A5.4.24.10. Foul criteria and procedures. **(T-1)**

A5.4.24.11. Bomb plotting and electronic strafe scoring equipment. * **(T-1)**

A5.4.24.12. Radio, other communications, and tape recorder operation. **(T-1)**

A5.4.24.13. Lost communications procedures. **(T-1)**

A5.4.24.14. Overall range safety. **(T-1)**

A5.4.24.15. Night, laser, and tactical range operations (if applicable). **(T-1)**

A5.4.24.16. Training on WDZ and SDZ tool applications. **(T-1)**

A5.4.24.17. Proper use and handling of applicable ground launched visual threats. **(T-1)**

A5.5. Range Training Officer Training. As a minimum, the Range Training Officer will be trained in or demonstrate adequate knowledge of the following publications or subjects:

A5.5.1. AFMAN 11-202V3, *Flight Operations*. **(T-1)**

A5.5.2. AFI 11-214. **(T-1)**

A5.5.3. Aircraft specific Air Force 11-2MDS operating procedures. **(T-1)**

A5.5.4. AFMAN 13-212V1, as supplemented. **(T-1)**

A5.5.5. Intermediate command instructions and manuals applicable to range "operations." **(T-1)**

A5.5.6. Range Training Officer authority and responsibilities. **(T-1)**

A5.5.7. Capabilities and limitations of Air CTS facilities and systems. **(T-1)**

A5.6. Electronic Warfare Personnel Training. As a minimum, the electronic warfare personnel (government or contracted support) will be trained in or demonstrate adequate knowledge of the following publications or subjects:

A5.6.1. AFMAN 11-202V3, *Flight Operations*. **(T-1)**

A5.6.2. AFI 11-214. **(T-1)**

A5.6.3. Aircraft specific Air Force 11-2MDS operating procedures. **(T-1)**

A5.6.4. Aircraft specific RAP tasking messages or Volume 1 training requirements. **(T-1)**

A5.6.5. AFMAN 13-212V1, as supplemented (applicable sections). **(T-1)**

A5.6.6. Intermediate command instructions and manuals applicable to range "operations." **(T-1)**

A5.6.7. Local electronic warfare equipment operating restrictions. **(T-1)**

A5.6.8. Capabilities and limitations of Air CTS facilities. **(T-1)**

A5.6.9. Range safety. **(T-1)**

A5.6.10. Electronic warfare equipment operating procedures for the applicable range equipment. **(T-1)**

A5.6.11. Current scenarios/equipment uses and aircrew-developed timelines as directed by the range operating authority. **(T-1)**